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<!DOCTYPE html>
<html lang="id">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>didiQCsyst v9.12</title>
<link rel="preconnect" href="https://fonts.googleapis.com">
<link
href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700;800;900&display=swap" rel="stylesheet">
<link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/chart.js@4.4.0/dist/chart.umd.min.js"></script>
<script src="https://cdn.jsdelivr.net/npm/chartjs-plugin-annotation@3.0.1/dist/chartjs-plugin-annotation.min.js"></script>
<script
src="https://cdn.jsdelivr.net/npm/html2canvas@1.4.1/html2canvas.min.js"></script>
<script src="https://cdn.jsdelivr.net/npm/jspdf@2.5.1/jspdf.umd.min.js"></script>
<script src="https://cdn.jsdelivr.net/npm/qrcodejs@1.0.0/qrcode.min.js"></script>
<style>
:root {
--primary: #2563eb; --primary-dark: #1d4ed8; --primary-light: #dbeafe; --primary-50: #eff6ff;
--success: #10b981; --success-light: #d1fae5;
--warning: #f59e0b; --warning-light: #fff9c4;
--danger: #ef4444; --danger-light: #ffe2e2;
--info: #06b6d4; --info-light: #cfe2f3;
--bg: #f8f9fa; --bg-soft: #f1f3f4;
--card-bg: #ffffff;
--text: #000000; --text-secondary: #6c757d; --text-muted: #6c757d;
--border: #dee2e6; --border-soft: #f1f3f4;
--radius: 14px; --radius-sm: 10px; --radius-xs: 6px;
--shadow-sm: 0 1px 2px 0 rgba(0,0,0,0.05);
--shadow: 0 4px 6px -1px rgba(0,0,0,0.07), 0 2px 4px -2px rgba(0,0,0,0.05);
--shadow-lg: 0 10px 25px -3px rgba(0,0,0,0.1), 0 4px 6px -4px rgba(0,0,0,0.05);
--shadow-xl: 0 20px 40px -10px rgba(0,0,0,0.15);
--sidebar-width: 270px; --sidebar-width-collapsed: 76px;
--sidebar-bg: #000000; --sidebar-bg-2: #1e293b;
--sidebar-bg-rgb: 15, 23, 42;
--sidebar-bg-2-rgb: 30, 41, 59;
--sb-font-size: 14px; --sb-font-weight: 500; --sb-font-color: #f1f3f4;
--topbar-h: 60px;
--transition: .25s cubic-bezier(0.4, 0, 0.2, 1);
--login-brightness: 0.8;
--login-bg-color: rgba(0,0,0,0.45);
--login-font-size: 16px;
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--login-font-color: #ffffff;
--login-font-weight: 400;
}
*, ::before, ::after { box-sizing: border-box; margin: 0; padding: 0; }
html { font-size: 14px; scroll-behavior: smooth; }
body { font-family: 'Inter', sans-serif; background: var(--bg); color: var(--text); line-height: 1.6;
overflow-x: hidden; -webkit-font-smoothing: antialiased; }
a { color: var(--primary); text-decoration: none; }
img { max-width: 100%; }
input, select, textarea, button { font-family: inherit; font-size: .95rem; }
input, select, textarea { width: 100%; padding: 10px 14px; border: 1.5px solid var(--border); border-
radius: var(--radius-sm); background: #fff; transition: all var(--transition); color: var(--text); }
input:focus, select:focus, textarea:focus { outline: none; border-color: var(--primary); box-shadow: 0
0 3px rgba(37,99,235,.12); }
textarea { resize: vertical; min-height: 90px; }
input:read-only, input:disabled, select:disabled { background: var(--bg-soft); cursor: not-allowed;
opacity: 0.85; }
::-webkit-scrollbar { width: 8px; height: 8px; }
::-webkit-scrollbar-track { background: transparent; }
::-webkit-scrollbar-thumb { background: rgba(148,163,184,.4); border-radius: 4px; }
::-webkit-scrollbar-thumb:hover { background: rgba(148,163,184,.6); }
.btn { display: inline-flex; align-items: center; gap: 7px; padding: 10px 20px; border: none; border-
radius: var(--radius-sm); font-weight: 600; cursor: pointer; transition: all var(--transition); font-size:
.9rem; text-align: center; justify-content: center; white-space: nowrap; letter-spacing: 0.1px; }
.btn i { font-size: .9rem; }
.btn-primary { background: var(--primary); color: #fff; box-shadow: 0 1px 3px rgba(37,99,235,.3); }
.btn-primary:hover { background: var(--primary-dark); transform: translateY(-1px); box-shadow: 0
4px 12px rgba(37,99,235,.4); }
.btn-success { background: var(--success); color: #fff; }
.btn-success:hover { background: #059669; transform: translateY(-1px); }
.btn-warning { background: var(--warning); color: #fff; }
.btn-warning:hover { background: #d97706; }
.btn-danger { background: var(--danger); color: #fff; }
.btn-danger:hover { background: #dc2626; }
.btn-info { background: var(--info); color: #fff; }
.btn-secondary { background: #64748b; color: #fff; }
.btn-secondary:hover { background: #475569; }
.btn-outline { background: transparent; color: var(--primary); border: 1.5px solid var(--primary); }
.btn-outline:hover { background: var(--primary-light); }
.btn-sm { padding: 6px 14px; font-size: .82rem; }
.btn-xs { padding: 4px 10px; font-size: .75rem; }
.btn-block { width: 100%; }
.btn:disabled { opacity: .5; pointer-events: none; }
.btn-group { display: flex; gap: 8px; flex-wrap: wrap; }
#loginPage { min-height: 100vh; display: flex; align-items: stretch; background: #ffffff; }

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.login-left {
flex: 1; display: flex; flex-direction: column; align-items: center; justify-content: center;
padding: 50px; position: relative; overflow: hidden;
background:
linear-gradient(135deg, var(--login-bg-color) 0%, rgba(15, 23, 42, 0.55) 45%, rgba(8, 47, 73, 0.60)
100%),
url('https://lh3.googleusercontent.com/d/1aBOPf2jTytvzcrv8hIEQcmT7GRHNOA2Y');
background-size: cover, 100%;
background-position: center, center;
background-repeat: no-repeat, no-repeat;
background-blend-mode: overlay;
color: var(--login-font-color, #fff);
font-size: var(--login-font-size, 16px);
font-weight: var(--login-font-weight, 400);
filter: brightness(var(--login-brightness));
}
.login-left::before {
content: ''; position: absolute; inset: 0;
background:
radial-gradient(ellipse at center, rgba(2, 6, 23, 0.55) 0%, rgba(2, 6, 23, 0.85) 100%),
linear-gradient(180deg, rgba(15, 23, 42, 0.4) 0%, rgba(2, 6, 23, 0.6) 100%);
pointer-events: none; z-index: 0;
}
.login-left > * { position: relative; z-index: 1; }
.login-logo-big {
width: 80px; height: 80px; background: rgba(255,255,255,0.15); backdrop-filter: blur(10px);
display: flex; align-items: center; justify-content: center;
box-shadow: 0 4px 16px rgba(0,0,0,.2), 0 0 0 4px rgba(255,255,255,.2);
padding: 8px; margin-top: 18px; margin-bottom: 10px; border: 2px solid rgba(255,255,255,.7);
position: relative; z-index: 2; border-radius: 12px;
}
.login-logo-big img { max-width: 100%; max-height: 100%; object-fit: contain; }
.login-left h1 { font-size: 3.4rem; font-weight: 900; margin-bottom: 16px; text-shadow: 0 2px 12px
rgba(0,0,0,.6), 0 0 2px rgba(255,255,255,.3); letter-spacing: -1px; position: relative; z-index: 2;
animation: fadeInDown 1s ease; color: #ffffff; }
.login-left p { font-size: 1.12rem; max-width: 480px; text-align: center; margin-bottom: 28px; line-
height: 1.7; opacity: 1; position: relative; z-index: 2; animation: fadeInUp 1s ease 0.2s both; color:
#ffffff; text-shadow: 0 1px 6px rgba(0,0,0,.5); }
.login-features-grid { display: grid; grid-template-columns: 1fr 1fr; gap: 14px; max-width: 540px;
width: 100%; position: relative; z-index: 2; }
.login-feature {
background: rgba(255,255,255,.13); backdrop-filter: blur(12px);
border-radius: 14px; padding: 16px 20px;
display: flex; align-items: center; gap: 14px;
border: 1px solid rgba(255,255,255,.18);

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transition: all .25s; animation: fadeInUp 1s ease both;
}
.login-feature:nth-child(1) { animation-delay: 0.3s; }
.login-feature:nth-child(2) { animation-delay: 0.4s; }
.login-feature:nth-child(3) { animation-delay: 0.5s; }
.login-feature:nth-child(4) { animation-delay: 0.6s; }
.login-feature:hover { background: rgba(255,255,255,.22); transform: translateY(-3px); }
.login-feature i { font-size: 1.5rem; color: #fff; flex-shrink: 0; animation: pulse 2.5s infinite; }
.login-feature .ft-title { font-weight: 700; font-size: 1.02rem; }
.login-feature .ft-desc { font-size: .88rem; opacity: .88; }
.login-right { flex: 0 0 480px; display: flex; flex-direction: column; align-items: center; justify-content: center; padding: 40px 50px; background: #fff; }
.login-right-logo {
width: 110px; height: 110px; background: #fff; border-radius: 14px;
display: flex; align-items: center; justify-content: center;
padding: 14px; margin-bottom: 18px;
box-shadow: 0 0 4px var(--primary), 0 8px 24px rgba(37,99,235,.25);
}
.login-right-logo img { max-width: 100%; max-height: 100%; object-fit: contain; }
.login-right h2 { font-size: 2.3rem; font-weight: 900; color: var(--primary); margin-bottom: 8px;
letter-spacing: -.5px; animation: fadeInDown 1s ease 0.3s both; }
.login-right .subtitle { font-size: 1.05rem; color: var(--text-secondary); margin-bottom: 26px; text-align: center; animation: fadeInUp 1s ease 0.4s both; }
.login-tabs { display: flex; width: 100%; max-width: 400px; background: var(--bg-soft); border-radius: 12px; padding: 5px; margin-bottom: 24px; }
.login-tab { flex: 1; padding: 13px; text-align: center; font-weight: 600; cursor: pointer; transition: all var(--transition); border: none; color: var(--text-secondary); background: transparent; font-size: 1rem; border-radius: 8px; }
.login-tab.active { background: #fff; color: var(--primary); box-shadow: var(--shadow-sm); }
.login-form-wrap { width: 100%; max-width: 380px; }
.login-form { display: none; }
.login-form.active { display: block; animation: fadeIn .4s; }
@keyframes fadeIn { from{opacity:0;transform:translateY(8px)}
to{opacity:1;transform:translateY(0)} }
.form-group { margin-bottom: 16px; }
.form-group label { display: block; margin-bottom: 8px; font-weight: 600; font-size: 1rem; color: var(--text); }
.form-group .input-icon { position: relative; }
.form-group .input-icon i { position: absolute; left: 14px; top: 50%; transform: translateY(-50%); color: var(--text-muted); }
.form-group .input-icon input { padding-left: 44px; font-size: 1.08rem; height: 52px; }
.form-group .input-icon .toggle-pwd { position: absolute; right: 14px; top: 50%; transform: translateY(-50%); color: var(--text-muted); cursor: pointer; font-size: 1rem; }
.login-btn-big {
width: 100%; padding: 16px; font-size: 1.2rem; font-weight: 800;

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background: linear-gradient(135deg, var(--primary), #6366f1);
color: #fff; border: none; border-radius: 14px; cursor: pointer;
box-shadow: 0 6px 20px rgba(37,99,235,.4);
transition: all .3s; margin-top: 12px;
animation: fadeInUp 1s ease 0.6s both;
}
.login-btn-big:hover { transform: translateY(-3px); box-shadow: 0 10px 30px rgba(37,99,235,.5); }
.login-footer-text { margin-top: 20px; text-align: center; font-size: 1rem; color: var(--text-secondary); }
}
.login-footer-text a { color: var(--primary); font-weight: 700; }
.login-copyright { margin-top: 30px; font-size: .82rem; color: var(--text-muted); text-align: center; }
@keyframes fadeInDown { from { opacity: 0; transform: translateY(-30px); } to { opacity: 1; transform: translateY(0); } }
@keyframes fadeInUp { from { opacity: 0; transform: translateY(30px); } to { opacity: 1; transform: translateY(0); } }

@media (max-width: 968px) {
#loginPage { flex-direction: column; }
.login-left { padding: 35px 24px; min-height: 340px; }
.login-logo-big { width: 70px; height: 70px; }
.login-left h1 { font-size: 2.3rem; }
.login-left p { font-size: 1rem; }
.login-features-grid { grid-template-columns: 1fr; }
.login-right { flex: 1; padding: 35px 26px; }
.login-right-logo { width: 100px; height: 100px; }
.login-right h2 { font-size: 1.8rem; }
.login-right .subtitle { font-size: .95rem; }
}
#appPage { display: none; }
#appPage.show { display: flex; min-height: 100vh; }
.sidebar { width: var(--sidebar-width); background: linear-gradient(180deg, var(--sidebar-bg) 0%, var(--sidebar-bg-2) 100%); position: fixed; top: 0; left: 0; height: 100vh; display: flex; flex-direction: column; z-index: 1000; transition: width var(--transition), transform var(--transition); overflow-y: auto; overflow-x: hidden; box-shadow: 4px 0 24px rgba(0,0,0,.08); }
.sidebar.collapsed { width: var(--sidebar-width-collapsed); }
.sidebar.collapsed .brand-text, .sidebar.collapsed .user-info, .sidebar.collapsed .nav-section, .sidebar.collapsed .nav-item span:not(.nav-badge), .sidebar.collapsed .nav-group-toggle span:not(.nav-badge), .sidebar.collapsed .nav-group-toggle i.fa-chevron-down { display: none; }
.sidebar.collapsed .sidebar-brand { justify-content: center; padding: 18px 8px; }
.sidebar.collapsed .sidebar-user { justify-content: center; padding: 12px 8px; }
.sidebar.collapsed .nav-item, .sidebar.collapsed .nav-group-toggle { justify-content: center; padding: 12px 8px; }
.sidebar.collapsed .nav-item i, .sidebar.collapsed .nav-group-toggle i { margin: 0; font-size: 1.15rem; }
.sidebar.collapsed .nav-submenu { display: none; }

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.sidebar-brand { padding: 18px; display: flex; align-items: center; gap: 12px; border-bottom: 1px solid
rgba(255,255,255,.08); }
.brand-icon-img { width: 44px; height: 44px; border-radius: 12px; background: #fff; padding: 5px;
display: flex; align-items: center; justify-content: center; flex-shrink: 0; box-shadow: 0 2px 8px
rgba(0,0,0,.2); overflow: hidden; }
.brand-icon-img img { width: 100%; height: 100%; object-fit: contain; }
.brand-text { color: var(--sb-font-color); font-weight: 800; font-size: 1.1rem; letter-spacing: -.3px; }
.brand-text small { display: block; font-weight: 500; font-size: .68rem; opacity: .65; margin-top: 1px;
letter-spacing: 0; }
.sidebar-user { padding: 14px 18px; border-bottom: 1px solid rgba(255,255,255,.08); display: flex;
align-items: center; gap: 12px; }
.user-avatar { width: 38px; height: 38px; background: linear-gradient(135deg, var(--primary),
#6366f1); border-radius: 50%; display: flex; align-items: center; justify-content: center; color: #fff;
font-weight: 700; font-size: 1rem; flex-shrink: 0; box-shadow: 0 2px 8px rgba(37,99,235,.3); }
.user-info { color: var(--sb-font-color); min-width: 0; }
.user-name { font-weight: 600; font-size: var(--sb-font-size); white-space: nowrap; overflow: hidden;
text-overflow: ellipsis; }
.user-role { font-size: .68rem; opacity: .65; text-transform: uppercase; letter-spacing: .8px; font-
weight: 600; }
.sidebar-nav { flex: 1; padding: 10px 0; }
.nav-section { padding: 14px 18px 6px; font-size: .68rem; text-transform: uppercase; letter-spacing:
1.4px; color: rgba(255,255,255,.35); font-weight: 700; }
.nav-item { display: flex; align-items: center; gap: 12px; padding: 11px 18px; color: var(--sb-font-
color); cursor: pointer; transition: all var(--transition); font-size: var(--sb-font-size); font-weight: var(-
sb-font-weight); border: none; background: none; width: 100%; text-align: left; position: relative;
border-left: 3px solid transparent; }
.nav-item:hover { background: rgba(255,255,255,.06); }
.nav-item.active { background: rgba(37,99,235,.15); border-left-color: var(--primary); color: #fff; }
.nav-item i { width: 20px; text-align: center; font-size: 1rem; opacity: .85; flex-shrink: 0; }
.nav-item .nav-badge { margin-left: auto; background: var(--danger); color: #fff; font-size: .65rem;
padding: 2px 8px; border-radius: 10px; font-weight: 700; }
.nav-divider { height: 1px; background: rgba(255,255,255,.06); margin: 8px 18px; }
.nav-group { position: relative; }
.nav-group-toggle { display: flex; align-items: center; gap: 12px; padding: 11px 18px; color: var(--sb-
font-color); cursor: pointer; transition: all var(--transition); font-size: var(--sb-font-size); font-weight:
var(--sb-font-weight); border: none; background: none; width: 100%; text-align: left; border-left: 3px
solid transparent; }
.nav-group-toggle:hover { background: rgba(255,255,255,.06); }
.nav-group-toggle i.fa-chevron-down { margin-left: auto; transition: transform .25s ease; font-size:
.72rem; opacity: .55; }
.nav-group.open .nav-group-toggle i.fa-chevron-down { transform: rotate(180deg); }
.nav-group.open .nav-group-toggle { background: rgba(255,255,255,.04); }
.nav-submenu { max-height: 0; overflow: hidden; transition: max-height .35s ease; background:
rgba(0,0,0,.2); }
.nav-group.open .nav-submenu { max-height: 600px; }

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.nav-submenu .nav-item { padding-left: 50px; font-size: calc(var(--sb-font-size) - 1px); opacity: .85; }
.nav-submenu .nav-item.active { border-left-color: var(--warning); opacity: 1; }
/* === FUTURISTIC SIDEBAR v9.7 === */
.sidebar{border-right:1px solid rgba(255,255,255,0.08);box-shadow:4px 0 30px rgba(0,0,0,0.15);}
.sidebar::after{content:"";position:absolute;top:0;right:0;width:2px;height:100%;background:linear-
gradient(180deg,var(--primary),transparent,var(--primary));opacity:0.5;pointer-events:none;}
.nav-item{position:relative;transition:all .2s ease;}
.nav-item::before{content:"";position:absolute;left:0;top:0;width:3px;height:100%;background:var(--
primary);border-radius:0 4px 4px 0;transform:scaleY(0);transition:transform .25s ease;}
.nav-item.active::before,.nav-item:hover::before{transform:scaleY(1);}
.nav-item.active{background:rgba(255,255,255,0.08);box-shadow:inset 3px 0 0 var(--primary);}
.nav-item:hover{background:rgba(255,255,255,0.05);transform:translateX(2px);}
.nav-item i{transition:text-shadow .2s ease;}
.nav-item:hover i{text-shadow:0 0 12px rgba(99,179,237,0.5);}
.sidebar-brand{position:relative;overflow:hidden;}
.sidebar-brand::after{content:"";position:absolute;bottom:-50%;left:-
50%;width:200%;height:200%;background:radial-gradient(circle,rgba(99,179,237,0.05)
0%,transparent 70%);animation:sidebarGlow 8s ease-in-out infinite alternate;}
@keyframes sidebarGlow{0%{transform:translate(0,0)}100%{transform:translate(10%,10%)}}
.sidebar.collapsed .nav-item span{opacity:0;transform:translateX(-10px);}
.sidebar.collapsed .nav-item:hover span{opacity:1;transform:translateX(0);}
/* Mobile slide-in */
@media(max-width:768px){.sidebar{transform:translateX(-100%);transition:transform .3s cubic-
bezier(.4,0,.2,1);}.sidebar.show-mobile{transform:translateX(0);}}
.main-content { flex: 1; margin-left: var(--sidebar-width); min-height: 100vh; transition: margin-left
var(--transition); }
.main-content.expanded { margin-left: var(--sidebar-width-collapsed); }
.topbar { height: var(--topbar-h); background: var(--card-bg); border-bottom: 1px solid var(--border);
display: flex; align-items: center; padding: 0 24px; position: sticky; top: 0; z-index: 99; box-shadow: 0
1px 3px rgba(0,0,0,.04); }
.topbar-left { display: flex; align-items: center; gap: 14px; }
.topbar-toggle { background: none; border: none; font-size: 1.25rem; color: var(--text-secondary);
cursor: pointer; padding: 8px; border-radius: 8px; transition: all var(--transition); }
.topbar-toggle:hover { background: var(--bg-soft); color: var(--primary); }
.topbar-title { font-weight: 700; font-size: 1.15rem; color: var(--text); }
.topbar-right { margin-left: auto; display: flex; align-items: center; gap: 12px; }
.view-as-wrap { display: flex; align-items: center; gap: 8px; font-size: .85rem; padding: 4px 12px;
background: var(--bg-soft); border-radius: 20px; }
.view-as-wrap select { width: auto; padding: 4px 8px; font-size: .8rem; border: none; background:
transparent; cursor: pointer; }
.topbar-btn { background: none; border: none; cursor: pointer; padding: 8px; color: var(--text-
secondary); font-size: 1.1rem; border-radius: 8px; transition: all var(--transition); }
.topbar-btn:hover { background: var(--bg-soft); color: var(--primary); }
.content { padding: 24px; }
.page { display: none; }

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.page.active { display: block; animation: pageIn .35s cubic-bezier(0.4,0,0.2,1); }
@keyframes pageIn { from{opacity:0;transform:translateY(10px)}
to{opacity:1;transform:translateY(0)} }
.card { background: var(--card-bg); border-radius: var(--radius); box-shadow: var(--shadow-sm);
margin-bottom: 20px; overflow: hidden; border: 1px solid var(--border-soft); }
.card-header { padding: 18px 22px; border-bottom: 1px solid var(--border-soft); display: flex; align-
items: center; justify-content: space-between; flex-wrap: wrap; gap: 10px; background: linear-
gradient(180deg, #fff 0%, #fafbfc 100%); }
.card-header h3 { font-size: 1.02rem; font-weight: 700; color: var(--text); letter-spacing: -.2px; }
.card-header h3 i { color: var(--primary); margin-right: 6px; }
.card-body { padding: 22px; }
.stat-grid { display: grid; grid-template-columns: repeat(auto-fill, minmax(210px, 1fr)); gap: 18px;
margin-bottom: 26px; }
.stat-card { background: var(--card-bg); border-radius: var(--radius); padding: 20px; box-shadow:
var(--shadow-sm); display: flex; align-items: center; gap: 16px; transition: all var(--transition); border:
1px solid var(--border-soft); position: relative; overflow: hidden; }
.stat-card::before { content: ""; position: absolute; top: 0; left: 0; width: 4px; height: 100%;
background: var(--primary); }
.stat-card:hover { transform: translateY(-4px); box-shadow: var(--shadow-lg); }
.stat-card .stat-icon { width: 52px; height: 52px; border-radius: 14px; display: flex; align-items:
center; justify-content: center; font-size: 1.4rem; flex-shrink: 0; background: var(--primary-light);
color: var(--primary); }
.stat-card .stat-info .stat-value { font-size: 1.75rem; font-weight: 800; line-height: 1.1; letter-spacing:
-.5px; }
.stat-card .stat-info .stat-label { font-size: .78rem; color: var(--text-secondary); font-weight: 600;
margin-top: 2px; text-transform: uppercase; letter-spacing: .4px; }
.stat-card.success::before { background: var(--success); } .stat-card.success .stat-icon { background:
var(--success-light); color: var(--success); }
.stat-card.warning::before { background: var(--warning); } .stat-card.warning .stat-icon { background:
var(--warning-light); color: var(--warning); }
.stat-card.danger::before { background: var(--danger); } .stat-card.danger .stat-icon { background:
var(--danger-light); color: var(--danger); }
.stat-card.info::before { background: var(--info); } .stat-card.info .stat-icon { background: var(--info-
light); color: var(--info); }
.table-wrap { overflow-x: auto; border-radius: var(--radius-sm); }
table { width: 100%; border-collapse: collapse; font-size: .87rem; }
thead th { background: linear-gradient(180deg, var(--primary) 0%, var(--primary-dark) 100%); color:
#fff; padding: 12px 14px; text-align: left; font-weight: 600; white-space: nowrap; position: sticky;
top: 0; z-index: 2; font-size: .82rem; }
tbody td { padding: 11px 14px; border-bottom: 1px solid var(--border-soft); vertical-align: middle; }
tbody tr:nth-child(even) { background: #fafbfc; }
tbody tr:hover { background: var(--primary-50); }
.table-actions { display: flex; gap: 5px; }
.badge { display: inline-block; padding: 4px 11px; border-radius: 20px; font-size: .72rem; font-weight:
700; white-space: nowrap; }

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.badge-success { background: var(--success-light); color: #047857; }
.badge-danger { background: var(--danger-light); color: #b91c1c; }
.badge-warning { background: var(--warning-light); color: #b45309; }
.badge-info { background: var(--info-light); color: #0e7490; }
.badge-secondary { background: var(--bg-soft); color: var(--text-secondary); }
.badge-primary { background: var(--primary); color: #fff; }
.badge-random { background: #fce7f3; color: #be185d; }
.badge-systematic { background: #ddd6fe; color: #5b21b6; }
.modal-overlay { position: fixed; inset: 0; background: rgba(15,23,42,.6); z-index: 2000; display: none;
align-items: center; justify-content: center; padding: 16px; backdrop-filter: blur(4px); }
.modal-overlay.show { display: flex; }
.modal { position: fixed; top: 50%; left: 50%; transform: translate(-50%, -50%); background: var(--card-bg); border-radius: var(--radius); box-shadow: var(--shadow-xl); z-index: 1100; width: 520px;
max-width: 94vw; max-height: 90vh; display: flex; flex-direction: column; animation: modalIn .3s ease; }
.modal.modal-lg { width: 720px; max-width: 96vw; max-height: 90vh; display: flex; flex-direction: column; }
.modal.modal-xl { width: 900px; max-width: 96vw; max-height: 90vh; display: flex; flex-direction: column; }
.modal-header { padding: 18px 22px; border-bottom: 1px solid var(--border-soft); display: flex; align-items: center; justify-content: space-between; background: linear-gradient(135deg, var(--primary) 0%, #6366f1 100%); color: #fff; border-radius: var(--radius) var(--radius) 0 0; position: sticky; top: 0; z-index: 1; }
.modal-header h3 { font-size: 1.05rem; font-weight: 700; }
.modal-close { background: rgba(255,255,255,.18); border: none; color: #fff; width: 32px; height: 32px; border-radius: 50%; cursor: pointer; display: flex; align-items: center; justify-content: center; font-size: 1.1rem; }
.modal-close:hover { background: rgba(255,255,255,.3); }
.modal-body { padding: 22px; overflow-y: auto; flex: 1; -webkit-overflow-scrolling: touch; }
.modal-footer { padding: 16px 22px; border-top: 1px solid var(--border-soft); display: flex; justify-content: flex-end; gap: 10px; position: sticky; bottom: 0; background: var(--card-bg); border-radius: 0 0 var(--radius) var(--radius); }
.form-row { display: grid; grid-template-columns: 1fr 1fr; gap: 14px; margin-bottom: 14px; }
.form-row-3 { grid-template-columns: repeat(3, 1fr); }
.form-row-4 { grid-template-columns: repeat(4, 1fr); }
#toastContainer { position: fixed; top: 24px; right: 24px; z-index: 9999; display: flex; flex-direction: column; gap: 10px; }
.toast { padding: 14px 22px; border-radius: var(--radius-sm); color: #fff; font-size: .88rem; font-weight: 600; min-width: 300px; max-width: 420px; box-shadow: var(--shadow-xl); display: flex; align-items: center; gap: 12px; animation: toastIn .35s cubic-bezier(0.4,0,0.2,1); cursor: pointer; }
.toast.out { animation: toastOut .3s ease forwards; }
@keyframes toastIn { from{transform:translateX(120%);opacity:0} to{transform:translateX(0);opacity:1} }
@keyframes toastOut { from{opacity:1} to{opacity:0;transform:translateX(120%)} }
.toast-success { background: linear-gradient(135deg, var(--success), #059669); }

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.toast-error { background: linear-gradient(135deg, var(--danger), #dc2626); }
.toast-warning { background: linear-gradient(135deg, var(--warning), #d97706); }
.toast-info { background: linear-gradient(135deg, var(--info), #0891b2); }
#loadingOverlay { position: fixed; inset: 0; background: rgba(255,255,255,.92); z-index: 9998; display: none; align-items: center; justify-content: center; flex-direction: column; gap: 16px; backdrop-filter: blur(4px); }
#loadingOverlay.show { display: flex; }
.spinner { width: 50px; height: 50px; border: 4px solid var(--border); border-top-color: var(--primary); border-radius: 50%; animation: spin .8s linear infinite; }
@keyframes spin { to{transform:rotate(360deg)} }
#loadingText { font-weight: 600; color: var(--text-secondary); font-size: .95rem; }
#confirmOverlay { position: fixed; inset: 0; background: rgba(15,23,42,.6); z-index: 9997; display: none; align-items: center; justify-content: center; padding: 16px; }
#confirmOverlay.show { display: flex; }
.confirm-box { background: var(--card-bg); border-radius: var(--radius); padding: 32px; max-width: 440px; width: 95%; text-align: center; box-shadow: var(--shadow-xl); }
.confirm-box .confirm-icon { font-size: 3.5rem; margin-bottom: 16px; }
.confirm-box .confirm-title { font-size: 1.15rem; font-weight: 700; margin-bottom: 8px; }
.confirm-box .confirm-msg { font-size: .9rem; color: var(--text-secondary); margin-bottom: 22px; line-height: 1.6; white-space: pre-wrap; }
.confirm-box .confirm-btns { display: flex; gap: 10px; justify-content: center; }
#expiredModal { position: fixed; inset: 0; background: rgba(15,23,42,.8); z-index: 10000; display: none; align-items: center; justify-content: center; padding: 16px; }
#expiredModal.show { display: flex; }
.expired-box { background: #fff; border-radius: 18px; padding: 44px; text-align: center; max-width: 420px; width: 95%; box-shadow: var(--shadow-xl); }
.expired-box .exp-icon { font-size: 4.5rem; color: var(--danger); margin-bottom: 18px; }
.expired-box h2 { font-size: 1.35rem; color: var(--danger); margin-bottom: 12px; font-weight: 800; }
.expired-box p { font-size: .9rem; color: var(--text-secondary); margin-bottom: 22px; }
.filter-bar { display: flex; flex-wrap: wrap; gap: 12px; align-items: flex-end; margin-bottom: 18px; padding: 16px 18px; background: var(--card-bg); border-radius: var(--radius-sm); box-shadow: var(--shadow-sm); border: 1px solid var(--border-soft); }
.filter-bar .filter-group { display: flex; flex-direction: column; gap: 5px; }
.filter-bar .filter-group label { font-size: .76rem; font-weight: 700; color: var(--text-secondary); text-transform: uppercase; letter-spacing: .4px; }
.filter-bar .filter-group input, .filter-bar .filter-group select { min-width: 150px; padding: 8px 11px; font-size: .85rem; }
.filter-bar .filter-actions { display: flex; gap: 7px; align-items: flex-end; margin-top: auto; }
.collapse-toggle { display: flex; align-items: center; justify-content: space-between; padding: 13px 18px; cursor: pointer; background: var(--bg-soft); border: 1px solid var(--border); border-radius: var(--radius-sm); margin-bottom: 8px; font-weight: 600; }
.collapse-toggle:hover { background: var(--primary-light); }
.collapse-toggle .arrow { transition: transform .2s; font-size: .85rem; }
.collapse-toggle.open .arrow { transform: rotate(180deg); }
.collapse-content { display: none; padding: 14px 0; }
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.collapse-content.show { display: block; }
.tab-bar { display: flex; border-bottom: 2px solid var(--border-soft); margin-bottom: 18px; overflow-x: auto; }
.tab-btn { padding: 11px 20px; border: none; background: none; cursor: pointer; font-weight: 600; color: var(--text-secondary); border-bottom: 3px solid transparent; white-space: nowrap; font-size: .9rem; }
.tab-btn.active { color: var(--primary); border-bottom-color: var(--primary); }
.tab-btn:hover { color: var(--primary); background: var(--primary-light); }
.tab-pane { display: none; }
.tab-pane.active { display: block; }
.chart-box { background: var(--card-bg); border-radius: var(--radius); padding: 18px; box-shadow: var(--shadow-sm); margin-bottom: 20px; border: 1px solid var(--border-soft); }
.chart-box canvas { max-height: 500px; }
.chart-grid { display: grid; grid-template-columns: 1fr 1fr; gap: 20px; }
.chart-grid-4 { display: grid; grid-template-columns: repeat(2, 1fr); gap: 16px; }
.qgi-mini-grid { display: grid; grid-template-columns: repeat(auto-fill, minmax(220px, 1fr)); gap: 12px; margin: 12px 0; }
.qgi-mini-box { background: linear-gradient(135deg, #fff7ed 0%, #fef3c7 100%); border: 1.5px solid var(--warning); border-radius: 10px; padding: 12px 14px; }
.qgi-mini-box h5 { color: var(--warning); margin-bottom: 4px; font-size: .85rem; font-weight: 700; }
.qgi-mini-box .qgi-mini-value { font-size: 1.3rem; font-weight: 800; line-height: 1.1; }
.qgi-mini-box .qgi-mini-interp { font-size: .72rem; margin-top: 4px; color: var(--text-secondary); line-height: 1.4; }
.qgi-mini-box .qgi-mini-bar { height: 6px; background: #fde68a; border-radius: 4px; margin: 6px 0; overflow: hidden; }
.qgi-mini-box .qgi-mini-bar-fill { height: 100%; border-radius: 4px; transition: width .5s ease; }
.qgi-conclusion { background: var(--primary-50); border-left: 4px solid var(--primary); padding: 12px 14px; border-radius: 8px; margin-top: 10px; font-size: .85rem; }
.qgi-conclusion strong { color: var(--primary); }
.sigma-info-box { background: linear-gradient(135deg, #eff6ff 0%, #ede9fe 100%); border-left: 4px solid var(--primary); padding: 18px; border-radius: var(--radius); margin-bottom: 16px; }
.sigma-info-box h4 { color: var(--primary); margin-bottom: 10px; font-size: 1.02rem; }
.sigma-info-box .sigma-mode { display: inline-block; background: var(--primary); color: #fff; padding: 5px 14px; border-radius: 20px; font-size: .8rem; font-weight: 700; margin-right: 8px; }
.sigma-info-box .sigma-warning { color: var(--danger); font-weight: 700; margin-top: 10px; padding: 10px 14px; background: var(--danger-light); border-radius: 8px; }
.sigma-info-box .rules-active { margin-top: 8px; font-size: .85rem; }
.sigma-info-box .rules-active code { background: rgba(255,255,255,.7); padding: 3px 9px; border-radius: 5px; margin-right: 4px; font-size: .8rem; color: var(--primary-dark); font-weight: 600; }
.opspecs-chart-wrap canvas { max-height: 320px; height: 100%; max-width: 100%; }
.zone-legend { display: flex; gap: 16px; margin-top: 10px; flex-wrap: wrap; font-size: .8rem; }
.zone-legend .zone-item { display: flex; align-items: center; gap: 6px; }
.zone-legend .zone-dot { width: 14px; height: 14px; border-radius: 4px; }
.zone-aman { background: rgba(16,185,129,.3); border: 1px solid var(--success); }
.zone-kritis { background: rgba(245,158,11,.3); border: 1px solid var(--warning); }

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.zone-bahaya { background: rgba(239,68,68,.3); border: 1px solid var(--danger); }
.ops-explain-box { background: linear-gradient(135deg, #eff6ff 0%, #f0fdfa 100%); border-left: 4px solid var(--info); padding: 16px 18px; border-radius: 10px; margin-bottom: 16px; font-size: .88rem; }
.ops-explain-box h4 { color: var(--info); margin-bottom: 10px; font-size: 1rem; }
.ops-explain-box dl { display: grid; grid-template-columns: auto 1fr; gap: 8px 14px; }
.ops-explain-box dt { font-weight: 700; color: var(--primary); }
.ops-explain-box dd { color: var(--text); }
.ops-explain-box .interp-zone { margin-top: 10px; padding-top: 10px; border-top: 1px dashed var(--border); }
.checklist-grid { display: grid; grid-template-columns: repeat(auto-fill, minmax(180px, 1fr)); gap: 8px; max-height: 260px; overflow-y: auto; padding: 12px; border: 1px solid var(--border); border-radius: var(--radius-sm); margin-bottom: 12px; background: var(--bg-soft); }
.checklist-item { display: flex; align-items: center; gap: 8px; padding: 6px 12px; border-radius: var(--radius-xs); cursor: pointer; font-size: .85rem; background: #fff; }
.checklist-item:hover { background: var(--primary-light); }
.checklist-item input[type="checkbox"] { width: auto; margin: 0; accent-color: var(--primary); }
.checklist-actions { display: flex; gap: 8px; margin-bottom: 8px; }
.rank-card { display: flex; align-items: center; gap: 16px; padding: 18px; background: var(--card-bg); border-radius: var(--radius); box-shadow: var(--shadow-sm); margin-bottom: 12px; border: 1px solid var(--border-soft); }
.rank-medal { font-size: 2.4rem; }
.rank-bar-wrap { flex: 1; }
.rank-bar { height: 12px; background: var(--bg-soft); border-radius: 6px; overflow: hidden; }
.rank-bar-fill { height: 100%; border-radius: 6px; transition: width .8s ease; }
.rank-sigma { font-weight: 800; font-size: 1.3rem; margin-left: 12px; }
.gate-overlay { display: flex; flex-direction: column; align-items: center; justify-content: center; padding: 50px 20px; min-height: 340px; }
.gate-overlay .gate-icon { font-size: 4.5rem; color: var(--danger); margin-bottom: 18px; }
.gate-overlay .gate-form { max-width: 380px; width: 100%; }
.img-result-header { text-align: center; border-bottom: 3px double #333; padding-bottom: 16px; margin-bottom: 22px; }
.img-result-header img { max-height: 80px; margin-bottom: 10px; }
.img-result-header h2 { font-size: 1.3rem; text-transform: uppercase; font-weight: 800; }
.img-result-header .header-line { font-size: .85rem; color: var(--text-secondary); margin-top: 3px; }
.img-result-body table { font-size: .88rem; }
.img-result-body table td { padding: 6px 10px; }
.img-result-body table td:first-child { font-weight: 600; width: 170px; white-space: nowrap; }
.img-result-footer { margin-top: 32px; display: flex; justify-content: space-between; }
.img-result-footer .ttd-col { text-align: center; width: 45%; }
.img-result-footer .ttd-col .ttd-name { font-weight: 700; margin-top: 4px; }
.img-electronic { text-align: center; font-size: .72rem; color: var(--text-secondary); margin-top: 20px; font-style: italic; border-top: 1px solid var(--border); padding-top: 8px; }
.img-mikroskopis { max-width: 100%; max-height: 280px; border-radius: var(--radius-sm); border: 1px solid var(--border); margin: 8px 4px; }
.sigma-world { color: #047857; font-weight: 800; }

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.sigma-excellent { color: var(--primary); font-weight: 800; }
.sigma-good { color: #7c3aed; font-weight: 800; }
.sigma-marginal { color: var(--warning); font-weight: 800; }
.sigma-poor { color: var(--danger); font-weight: 800; }
.sigma-unacceptable { color: #991b1b; font-weight: 800; }
.z-ok { color: var(--success); font-weight: 700; }
.z-warn { color: var(--warning); font-weight: 700; }
.z-reject { color: var(--danger); font-weight: 700; }
.wg-panel { max-height: 360px; overflow-y: auto; }
.wg-item { display: flex; align-items: center; gap: 12px; padding: 12px 16px; border-bottom: 1px solid var(--border-soft); font-size: .87rem; }
.wg-rule-badge { padding: 4px 11px; border-radius: 5px; font-weight: 700; font-size: .75rem; background: var(--danger-light); color: #b91c1c; flex-shrink: 0; }
.wg-rule-badge.warning { background: var(--warning-light); color: #b45309; }
.wg-rule-badge.ignored { background: var(--bg-soft); color: var(--text-muted); text-decoration: line-through; }
.smart-table { border: 1px solid var(--border); border-radius: var(--radius-sm); overflow: auto; max-height: 520px; }
.smart-table table { margin: 0; }
.smart-table input { border: none; border-bottom: 1px solid var(--border); border-radius: 0; padding: 7px 9px; font-size: .85rem; width: 100%; min-width: 80px; }
.smart-table input:focus { border-bottom-color: var(--primary); box-shadow: none; }
.smart-table .col-existing { background: #ecfdf5; }
.smart-table .col-existing input { background: transparent; }
.upload-zone { border: 2px dashed var(--border); border-radius: var(--radius); padding: 44px; text-align: center; cursor: pointer; }
.upload-zone:hover { border-color: var(--primary); background: var(--primary-light); }
.upload-zone i { font-size: 3rem; color: var(--primary); margin-bottom: 10px; }
.multi-img-grid { display: grid; grid-template-columns: repeat(3, 1fr); gap: 12px; margin-top: 10px; }
.img-slot { border: 2px dashed var(--border); border-radius: 10px; padding: 12px; text-align: center; cursor: pointer; min-height: 140px; display: flex; flex-direction: column; align-items: center; justify-content: center; transition: all .25s; position: relative; }
.img-slot:hover { border-color: var(--primary); background: var(--primary-light); }
.img-slot.has-image { padding: 4px; border-style: solid; border-color: var(--success); }
.img-slot i { font-size: 1.8rem; color: var(--text-muted); margin-bottom: 6px; }
.img-slot p { font-size: .8rem; color: var(--text-secondary); margin-top: 4px; }
.img-slot .slot-label { font-size: .72rem; font-weight: 700; color: var(--primary); margin-bottom: 4px; }
.img-slot img { max-width: 100%; max-height: 120px; border-radius: 6px; object-fit: contain; }
.img-slot .remove-img { position: absolute; top: 4px; right: 4px; background: var(--danger); color: #fff; border: none; width: 24px; height: 24px; border-radius: 50%; cursor: pointer; font-size: .75rem; display: none; }
.img-slot.has-image .remove-img { display: block; }
.backup-info { display: grid; grid-template-columns: 1fr 1fr; gap: 18px; }
.backup-card { padding: 22px; border-radius: var(--radius); border: 1px solid var(--border); text-align: center; }
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.backup-card i { font-size: 2.6rem; margin-bottom: 10px; }
.color-presets { display: flex; flex-wrap: wrap; gap: 8px; margin-top: 8px; }
.color-preset { width: 38px; height: 38px; border-radius: 50%; border: 3px solid transparent; cursor: pointer; box-shadow: 0 2px 4px rgba(0,0,0,.1); }
.color-preset:hover { transform: scale(1.15); }
.restore-folder { border: 1px solid var(--border); border-radius: var(--radius-sm); margin-bottom: 8px; overflow: hidden; }
.restore-folder-header { padding: 12px 16px; background: var(--bg-soft); cursor: pointer; font-weight: 600; display: flex; justify-content: space-between; }
.restore-folder-files { display: none; padding: 10px; background: #fff; }
.restore-folder-files.show { display: block; }
.restore-file-item { display: flex; justify-content: space-between; padding: 9px 12px; border-bottom: 1px solid #eee; font-size: .85rem; }
.histori-edit { background: var(--warning-light); color: #b45309; }
.histori-delete { background: var(--danger-light); color: #b91c1c; }
.histori-restored { background: var(--success-light); color: #047857; }
.laporan-interp { padding: 12px 16px; border-radius: var(--radius-sm); margin-top: 12px; border-left: 4px solid var(--primary); background: var(--primary-50); }
.laporan-interp.success { background: var(--success-light); border-left-color: var(--success); }
.laporan-interp.warning { background: var(--warning-light); border-left-color: var(--warning); }
.laporan-interp.danger { background: var(--danger-light); border-left-color: var(--danger); }
.laporan-interp h5 { font-size: .92rem; font-weight: 700; margin-bottom: 6px; }
.laporan-interp ul { margin-left: 18px; font-size: .82rem; }
.wg-cell-flag { display: inline-block; padding: 1px 6px; border-radius: 4px; font-size: .65rem; font-weight: 700; margin-left: 4px; }
.wg-cell-flag.random { background: #fce7f3; color: #be185d; }
.wg-cell-flag.systematic { background: #ddd6fe; color: #5b21b6; }
.wg-cell-flag.warning-flag { background: #fef3c7; color: #b45309; }
/* print-options consolidated below */
.trend-interp-box { background: var(--primary-50); border-left: 4px solid var(--primary); padding: 14px 18px; border-radius: 10px; margin-top: 12px; }
.trend-interp-box h5 { color: var(--primary); margin-bottom: 8px; font-size: .95rem; font-weight: 700; }
.trend-interp-box ul { margin-left: 20px; font-size: .85rem; line-height: 1.6; }
.trend-interp-box .error-estimate { display: inline-block; background: linear-gradient(135deg, var(--warning), var(--danger)); color: #fff; padding: 6px 14px; border-radius: 20px; font-weight: 700; font-size: .85rem; margin-top: 6px; }
.lap-kop-block { page-break-after: avoid !important; break-after: avoid !important; }
.lap-first-section { page-break-before: avoid !important; break-before: avoid !important; }
.catatan-dokter-box { background: linear-gradient(135deg, #fef3c7 0%, #fff 100%); border-left: 4px solid var(--warning); padding: 14px 18px; border-radius: 10px; margin-top: 12px; }
.catatan-dokter-box h5 { color: var(--warning); margin-bottom: 8px; font-size: .95rem; font-weight: 700; }
.catatan-dokter-box textarea { min-height: 80px; margin-top: 8px; }
@media print {

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body { background: #fff !important; font-size: 11pt; }
.sidebar, .topbar, .topbar-toggle, .no-print, .btn, .filter-bar, .tab-bar, #toastContainer,
#loadingOverlay, #confirmOverlay, .modal-overlay, .nav-item, .sidebar-brand, .sidebar-user, .nav-
group, .print-options-wrap, .export-panel-overlay, #sidebarOverlay { display: none !important; }
.main-content { margin-left: 0 !important; }
.content { padding: 0 !important; }
.page { display: none !important; }
.page.print-active { display: block !important; }
.card, .chart-box, .qgi-mini-box, .sigma-info-box, .laporan-interp, .trend-interp-box, .ops-explain-box,
.stat-card, .rank-card, .restore-folder, .backup-card, .filter-bar, .modal { box-shadow: none
!important; border: none !important; background: #fff !important; }
.card-header { background: #fff !important; border-bottom: 1px solid #ddd !important; padding: 8px
0 !important; }
.card-body { padding: 8px 0 !important; }
table { font-size: 9pt; }
thead th { background: #1e3a8a !important; color: #fff !important; -webkit-print-color-adjust: exact;
print-color-adjust: exact; border: none !important; }
td { border-bottom: 1px solid #eee !important; }
.img-result-header { border-bottom: 3px double #000 !important; }
.hide-on-print-markers .wg-cell-flag { display: none !important; }
@page { margin: 12mm; size: A4; }
}
/* — v9.8 ENHANCEMENTS — */
.page{animation:fadeInPage .3s ease}
@keyframes
fadeInPage{from{opacity:0;transform:translateY(8px)}to{opacity:1;transform:translateY(0)}}
.btn{transition:all .2s ease}
.btn:hover{transform:translateY(-1px);box-shadow:var(--shadow)}
.btn:active{transform:translateY(0)}
.card{transition:box-shadow .25s ease}
.card:hover{box-shadow:var(--shadow-lg)}
input,select,textarea{transition:border-color .2s ease,box-shadow .2s ease}
input:focus,select:focus,textarea:focus{border-color:var(--primary)!important;box-shadow:0 0 0 3px
var(--primary-light)!important}
.table-wrap{overflow-x:auto;-webkit-overflow-scrolling:touch}
.table-wrap::-webkit-scrollbar{height:6px}
.table-wrap::-webkit-scrollbar-track{background:var(--bg-soft);border-radius:3px}
.table-wrap::-webkit-scrollbar-thumb{background:var(--text-muted);border-radius:3px}
.sidebar{scrollbar-width:thin;scrollbar-color:rgba(255,255,255,.15) transparent}
.sidebar::-webkit-scrollbar{width:4px}
.sidebar::-webkit-scrollbar-thumb{background:rgba(255,255,255,.15);border-radius:2px}
@media(max-width:768px){.filter-bar{flex-direction:column;gap:8px!important}.filter-bar
.fg{width:100%!important}.filter-bar select,.filter-bar input{width:100%!important}}
.badge{font-size:.72rem;padding:2px 8px;border-radius:20px;font-weight:600;letter-spacing:.3px}
.modal-overlay .modal{animation:modalIn .25s ease}

```

```

@keyframes modalIn{from{opacity:0;transform:scale(.95)
translateY(10px)}to{opacity:1;transform:scale(1) translateY(0)}}
.nav-item{position:relative;transition:all .2s ease}
.nav-item::after{content:"";position:absolute;right:0;top:50%;transform:translateY(-50%)
scaleY(0);width:3px;height:60%;background:var(--primary);border-radius:3px 0 0
3px;transition:transform .25s ease}
.nav-item.active::after,.nav-item.hover::after{transform:translateY(-50%) scaleY(1)}
.view-as-wrap{animation:fadeInPage .3s ease}
.view-as-wrap select{border-radius:20px;padding:4px 12px;font-size:.8rem;border:1px solid var(--
border);background:var(--card-bg);color:var(--text);min-width:140px}
.view-as-wrap select:focus{outline:none;border-color:var(--primary)}
@media (max-width: 768px) {
.sidebar { transform: translateX(-100%); width: 280px; }
.sidebar.show-mobile { transform: translateX(0); }
.sidebar.collapsed { width: 280px; transform: translateX(-100%); }
.sidebar.collapsed.show-mobile { transform: translateX(0); }
.sidebar.collapsed .brand-text, .sidebar.collapsed .user-info, .sidebar.collapsed .nav-section,
.sidebar.collapsed .nav-item span:not(.nav-badge), .sidebar.collapsed .nav-group-toggle
span:not(.nav-badge), .sidebar.collapsed .nav-group-toggle i.fa-chevron-down { display: block; }
.sidebar.collapsed .nav-item, .sidebar.collapsed .nav-group-toggle { justify-content: flex-start;
padding: 11px 18px; }
.sidebar.collapsed .nav-submenu { display: block; }
.sidebar.collapsed.open .nav-submenu { max-height: 600px; }
.sidebar-overlay { position: fixed; inset: 0; background: rgba(0,0,0,.5); z-index: 999; display: none; }
.sidebar-overlay.show { display: block; }
.main-content, .main-content.expanded { margin-left: 0 !important; }
.stat-grid { grid-template-columns: repeat(auto-fill, minmax(150px, 1fr)); }
.form-row, .form-row-3, .form-row-4 { grid-template-columns: 1fr; }
.chart-grid, .chart-grid-4 { grid-template-columns: 1fr; }
.filter-bar { flex-direction: column; }
.filter-bar .filter-group { width: 100%; }
.filter-bar .filter-group input, .filter-bar .filter-group select { min-width: unset; width: 100%; }
.backup-info { grid-template-columns: 1fr; }
.multi-img-grid { grid-template-columns: 1fr; }
.qgi-mini-grid { grid-template-columns: 1fr; }
}
body.dark-mode { --bg: #0f172a; --bg-soft: #1e293b; --card-bg: #1e293b; --text: #f1f5f9; --text-
secondary: #94a3b8; --text-muted: #64748b; --border: #334155; --border-soft: #1e293b; }
body.dark-mode .card { background: var(--card-bg); border: 1px solid var(--border); }
body.dark-mode .card-header { background: linear-gradient(180deg, var(--card-bg) 0%, var(--bg-soft)
100%); border-bottom-color: var(--border); }
body.dark-mode tbody tr:nth-child(even) { background: rgba(255,255,255,0.02); }
body.dark-mode tbody tr:hover { background: rgba(37,99,235,0.1); }
body.dark-mode input, body.dark-mode select, body.dark-mode textarea { background: var(--bg-
soft); border-color: var(--border); color: var(--text); }

```



```

body.dark-mode .filter-bar { background: var(--card-bg); border-color: var(--border); }
body.dark-mode .modal { background: var(--card-bg); }
body.dark-mode .modal-header { background: linear-gradient(135deg, var(--primary), #6366f1); }
body.dark-mode .login-right { background: var(--bg); }
body.dark-mode .topbar { background: var(--card-bg); border-bottom-color: var(--border); }
body.dark-mode .sidebar { background: linear-gradient(180deg, var(--sidebar-bg) 0%, var(--sidebar-
bg-2) 100%); }
body.dark-mode .nav-submenu { background: rgba(0,0,0,0.3); }
body.light-mode { --bg: #f8fafc; --bg-soft: #f1f5f9; --card-bg: #ffffff; --text: #0f172a; --text-secondary:
#64748b; --text-muted: #94a3b8; --border: #e2e8f0; --border-soft: #f1f5f9; }
/* Pastikan container grafik terlihat */
#ljChartWrap{
  position: relative;
  min-height: 520px;
  background: #fff;
  border-radius: 8px;
  padding: 10px;
}

#ljChartUnified{
  width: 100% !important;
  height: 500px !important;
  max-height: 500px !important;
}

.chart-error{
  background: #fee2e2;
  border: 1px solid #ef4444;
  color: #b91c1c;
  padding: 20px;
  border-radius: 8px;
  text-align: center;
  margin: 20px 0;
}

.print-options-wrap{position:relative;z-index:1000;display:inline-block}
.print-options-wrap .print-options-
menu{display:none;position:absolute;top:110%;right:0;background:var(--card-bg);border:2px solid
var(--primary);border-radius:var(--radius-sm);box-shadow:var(--shadow-xl);z-index:9999;min-
width:280px;overflow:visible!important;padding:4px 0}
.print-options-wrap .print-options-menu.show{display:block}
.print-options-wrap .print-options-menu button{display:flex;align-
items:center;gap:8px;width:100%;padding:10px
16px;border:none;background:none;cursor:pointer;font-size:.88rem;color:var(--text);text-
align:left;transition:background .15s;font-weight:500}

```

```

.print-options-wrap .print-options-menu button:hover{background:var(--primary-light);color:var(--primary)}
.filter-actions{flex-wrap:wrap;gap:6px!important}
.filter-actions .print-options-wrap .btn{white-space:nowrap;font-weight:700;box-shadow:0 2px 8px rgba(0,0,0,.12);min-height:36px;border-radius:var(--radius-xs)}
/* ===== EXPORT PANEL (Form Pilihan Cetak) v10.0 ===== */
.export-panel-overlay{position:fixed;inset:0;background:rgba(0,0,0,.5);z-index:10000;display:flex;align-items:center;justify-content:center;padding:16px;opacity:0;pointer-events:none;transition:opacity .2s}
.export-panel-overlay.show{opacity:1;pointer-events:auto}
.export-panel{background:var(--card-bg);border-radius:var(--radius);box-shadow:var(--shadow-xl);width:100%;max-width:540px;max-height:85vh;overflow-y:auto;transform:scale(.9);transition:transform .2s}
.export-panel-overlay.show .export-panel{transform:scale(1)}
.export-panel-header{display:flex;align-items:center;justify-content:space-between;padding:18px 22px 14px 22px;border-bottom:2px solid var(--primary)}
.export-panel-header h3{font-size:1.1rem;font-weight:800;color:var(--primary);display:flex;align-items:center;gap:8px}
.export-panel-close{width:36px;height:36px;border:none;background:var(--bg-soft);border-radius:50%;cursor:pointer;font-size:1.2rem;display:flex;align-items:center;justify-content:center;color:var(--text-secondary);transition:all .15s}
.export-panel-close:hover{background:var(--danger-light);color:var(--danger)}
.export-panel-body{padding:16px 22px 22px 22px}
.export-option-card{display:flex;align-items:flex-start;gap:14px;padding:14px 16px;border:2px solid var(--border);border-radius:var(--radius-sm);margin-bottom:10px;cursor:pointer;transition:all .15s;background:var(--card-bg)}
.export-option-card:hover{border-color:var(--primary);background:var(--primary-50)}
.export-option-card.selected{border-color:var(--primary);background:var(--primary-light)}
.export-option-radio{width:22px;height:22px;border-radius:50%;border:2px solid var(--border);flex-shrink:0;margin-top:2px;position:relative;transition:all .15s}
.export-option-card.selected .export-option-radio{border-color:var(--primary);background:var(--primary)}
.export-option-card.selected .export-option-radio::after{content:"";position:absolute;inset:4px;background:#fff;border-radius:50%}
.export-option-label{font-weight:700;font-size:.92rem;color:var(--text);line-height:1.3}
.export-option-desc{font-size:.8rem;color:var(--text-secondary);margin-top:3px;line-height:1.3}
/* ===== RESPONSIVE MOBILE v9.9 ===== */
@media(max-width:768px){
.main-area{margin-left:0!important;padding:12px!important;padding-bottom:80px!important}
.sidebar{transform:translateX(-100%);width:var(--sidebar-width)!important}
.sidebar.open{transform:translateX(0)}
.sidebar-overlay.show{display:block!important}
.topbar{left:0!important;padding:0 12px!important}
.topbar .topbar-right{gap:8px!important}
.filter-bar{flex-direction:column!important;gap:8px!important;padding:12px!important}

```

```

.filter-group{width:100%!important;min-width:0!important}
.filter-actions{width:100%;justify-content:flex-start!important}
.form-row{flex-direction:column!important;gap:8px!important}
.form-row-3,.form-row-4{flex-direction:column!important}
.card{margin-bottom:12px!important;border-radius:var(--radius-sm)!important}
.card-header{padding:12px 14px!important;flex-
direction:column!important;gap:8px!important;align-items:flex-start!important}
.card-header .btn-group{flex-wrap:wrap;width:100%}
.modal{width:96vw!important;max-width:96vw!important;max-height:92vh!important;margin:4vh
auto!important}
.modal-lg,.modal-xl{width:96vw!important;max-width:96vw!important;max-height:92vh!important}
.modal-body{max-height:65vh!important;overflow-y:auto!important;-webkit-overflow-
scrolling:touch!important;padding:16px!important}
.modal-footer{flex-wrap:wrap;gap:6px!important;padding:12px!important}
.login-left{display:none!important}
.login-right{width:100%!important;min-height:auto!important}
#loginPage{min-height:auto!important}
.table-wrap{overflow-x:auto!important;-webkit-overflow-scrolling:touch!important}
.smart-table{max-height:400px!important}
.stat-card{flex-direction:column!important;text-align:center!important;gap:8px!important}
.stat-card .stat-icon{width:40px!important;height:40px!important}
.checklist-grid{grid-template-columns:1fr!important;max-height:200px!important}
.tab-bar{overflow-x:auto!important;-webkit-overflow-scrolling:touch!important}
.tab-btn{white-space:nowrap!important;padding:8px 12px!important;font-size:.8rem!important}
.print-options-wrap .print-options-menu{min-width:220px!important;right:-8px!important}
.print-options-wrap .print-options-menu button{padding:10px 14px!important;font-
size:.85rem!important}
.img-result-footer{flex-direction:column!important;gap:30px!important}
.img-result-footer .ttd-col{width:100%!important}
.wg-panel{max-height:280px!important}
.chart-grid{grid-template-columns:1fr!important}
.multi-img-grid{grid-template-columns:1fr!important}
.login-features-grid{display:none!important}
.opspecs-chart-wrap{height:250px!important}
#ljChartWrap{min-height:350px!important}
#ljChartUnified{height:350px!important;max-height:350px!important}
}
@media(max-width:480px){
.filter-actions .btn{font-size:.75rem!important;padding:6px 10px!important}
.filter-actions .btn i{font-size:.7rem!important}
.card-header h3{font-size:.95rem!important}
.page{padding:0!important}
}
</style>
</head>

```

```

<body>
<div id="toastContainer"></div>
<div id="loadingOverlay"><div class="spinner"></div><div
id="loadingText">Memproses...</div></div>
<div id="confirmOverlay">
<div class="confirm-box">
<div class="confirm-icon" id="confirmIcon">❌</div>
<div class="confirm-title" id="confirmTitle">Konfirmasi</div>
<div class="confirm-msg" id="confirmMsg">Apakah Anda yakin?</div>
<div class="confirm-btns">
<button class="btn btn-secondary" onclick="clConfirm(false)">Batal</button>
<button class="btn btn-danger" id="confirmYesBtn" onclick="clConfirm(true)">Ya</button>
</div>
</div>
</div>
</div>
<div id="expiredModal">
<div class="expired-box">
<div class="exp-icon"><i class="fas fa-exclamation-triangle"></i></div>
<h2>Akun Kedaluwarsa</h2>
<p id="expiredMsg">Masa aktif akun Anda telah berakhir.</p>
<button class="btn btn-primary" onclick="doLogout()">Kembali ke Login</button>
</div>
</div>
<div class="sidebar-overlay" id="sidebarOverlay" onclick="closeMobileSidebar()"></div>
<div id="loginPage">
<div class="login-left">
<h1>didiQCsys</h1>
<p>Sistem Manajemen Quality Control Laboratorium Modern dengan analisis Sigma Metric,
Westgard Rules & Multi-User.</p>
<div class="login-logo-big">
<rect width=%22100%22 height=%22100%22 fill=%22%232563eb%22
rx=%2220%22/><text x=%2250%22 y=%2265%22 font-size=%2245%22 fill=%22white%22 text-
anchor=%22middle%22 font-family=%22Arial%22 font-weight=%22bold%22>QC</text></svg>'"/>
</div>
<div class="login-features-grid">
<div class="login-feature"><i class="fas fa-chart-line"></i><div><div class="ft-title">Grafik
LJ</div><div class="ft-desc">Levey-Jennings real-time</div></div></div>
<div class="login-feature"><i class="fas fa-shield-alt"></i><div><div class="ft-title">Westgard
Rules</div><div class="ft-desc">Auto detection</div></div></div>
<div class="login-feature"><i class="fas fa-bullseye"></i><div><div class="ft-title">QGI
Analysis</div><div class="ft-desc">Quality Goal Index</div></div></div>

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<div class="login-feature"><i class="fas fa-users"></i></div><div class="ft-title">Compare-
Instrument</div><div class="ft-desc">Performa Alat berdasarkan Sigma</div></div></div>
</div>
<div class="login-right">
<div class="login-right-logo">
<rect width=%22100%22 height=%22100%22 fill=%22%232563eb%22
rx=%2218%22/><text x=%2250%22 y=%2265%22 font-size=%2240%22 fill=%22white%22 text-
anchor=%22middle%22 font-family=%22Arial%22 font-weight=%22bold%22>QC</text></svg>'"/>
</div>
<h2>Selamat Datang</h2>
<p class="subtitle">didik-digital Quality Control System</p>
<div class="login-tabs">
<button class="login-tab active" onclick="switchLoginTab('login')"><i class="fas fa-sign-in-alt"></i>
Login</button>
<button class="login-tab" onclick="switchLoginTab('register')"><i class="fas fa-user-plus"></i>
Daftar</button>
</div>
<div class="login-form-wrap">
<form class="login-form active" id="loginForm" onsubmit="doLogin(event)">
<div class="form-group"><label>Username</label><div class="input-icon"><i class="fas fa-
user"></i></div><input type="text" id="loginUsername" required autocomplete="username"></div></div>
<div class="form-group"><label>Password</label><div class="input-icon"><i class="fas fa-
lock"></i></div><input type="password" id="loginPassword" required autocomplete="current-
password"><i class="fas fa-eye toggle-pwd"
onclick="togglePwdVisibility('loginPassword',this)"></i></div></div>
<button class="login-btn-big" type="submit" id="loginBtn"><i class="fas fa-sign-in-alt"></i>
Masuk</button>
<div class="login-footer-text">Belum punya akun? <a href="javascript:void(0)"
onclick="switchLoginTab('register')">Daftar di sini</a></div>
</form>
<form class="login-form" id="registerForm" onsubmit="doRegister(event)">
<div class="form-group"><label>Username</label><div class="input-icon"><i class="fas fa-
user"></i></div><input type="text" id="regUsername" required></div></div>
<div class="form-group"><label>Nama Lengkap</label><div class="input-icon"><i class="fas fa-id-
card"></i></div><input type="text" id="regFullName" required></div></div>
<div class="form-group"><label>Email</label><div class="input-icon"><i class="fas fa-
envelope"></i></div><input type="email" id="regEmail" required></div></div>
<div class="form-group"><label>Password</label><div class="input-icon"><i class="fas fa-
lock"></i></div><input type="password" id="regPassword" required minlength="6"><i class="fas fa-eye
toggle-pwd" onclick="togglePwdVisibility('regPassword',this)"></i></div></div>

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```

<button class="login-btn-big" type="submit" id="regBtn"><i class="fas fa-user-plus"></i>
Daftar</button>
</form>
</div>
<div class="login-copyright">&copy; 2026 didiQCsys v9.12</div>
</div>
</div>
<div id="appPage">
<nav class="sidebar" id="sidebar">
<div class="sidebar-brand">
<div class="brand-icon-img">
<rect width=%22100%22 height=%22100%22 fill=%22%232563eb%22
rx=%2218%22/><text x=%2250%22 y=%2265%22 font-size=%2240%22 fill=%22white%22 text-
anchor=%22middle%22 font-family=%22Arial%22 font-weight=%22bold%22>QC</text></svg>'"/>
</div>
<div class="brand-text">didiQCsys<small>v9.12 Lab QC System</small></div>
</div>
<div class="sidebar-user">
<div class="user-avatar" id="sbUserAvatar">U</div>
<div class="user-info"><div class="user-name" id="sbUserName">User</div><div class="user-role"
id="sbUserRole">user</div></div>
</div>
<div class="sidebar-nav" id="sidebarNav">
<div class="nav-section">MENU UTAMA</div>
<button class="nav-item active" data-page="dashboard" onclick="goPage('dashboard')"><i
class="fas fa-tachometer-alt"></i> <span>Dashboard</span></button>
<div class="nav-group" id="grpDaftarParam">
<button class="nav-group-toggle" onclick="toggleNavGroup('grpDaftarParam')"><i class="fas fa-
folder-open"></i> <span>Daftar Parameter</span> <i class="fas fa-chevron-down"></i></button>
<div class="nav-submenu">
<button class="nav-item" data-page="parameters" onclick="goPage('parameters')"><i class="fas fa-
list-ul"></i> <span>Parameter</span></button>
<button class="nav-item" data-page="lotqc" onclick="goPage('lotqc')"><i class="fas fa-box"></i>
<span>Lot QC</span></button>
<button class="nav-item" data-page="biaspme" onclick="goPage('biaspme')"><i class="fas fa-
exchange-alt"></i> <span>Bias PME</span></button>
<button class="nav-item" data-page="calcstats" onclick="goPage('calcstats')"><i class="fas fa-
calculator"></i> <span>Calc Stats</span></button>
<button class="nav-item" data-page="sigmacvopt" onclick="goPage('sigmacvopt')"><i class="fas fa-
percentage"></i> <span>% Sigma CV</span></button>
<button class="nav-item" data-page="daftartea" onclick="goPage('daftartea')"><i class="fas fa-ruler-
combined"></i> <span>Daftar TEa</span></button>

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</div>
</div>
<button class="nav-item" data-page="inputqc" onclick="goPage('inputqc')"><i class="fas fa-keyboard"></i> <span>Input QC</span></button>
<button class="nav-item" data-page="grafik" onclick="goPage('grafik')"><i class="fas fa-chart-line"></i> <span>Grafik & Analisis</span></button>
<button class="nav-item" data-page="laporan" onclick="goPage('laporan')"><i class="fas fa-file-alt"></i> <span>Laporan</span></button>
<div class="nav-divider"></div>
<div class="nav-group" id="grpDashAnalysis">
<button class="nav-group-toggle" onclick="toggleNavGroup('grpDashAnalysis')"><i class="fas fa-chart-pie"></i> <span>Dashboard Analisis</span> <i class="fas fa-chevron-down"></i></button>
<div class="nav-submenu">
<button class="nav-item" data-page="trendanalysis" onclick="goPage('trendanalysis')"><i class="fas fa-chart-area"></i> <span>Trend Analisis</span></button>
<button class="nav-item" data-page="instrcompare" onclick="goPage('instrcompare')"><i class="fas fa-balance-scale"></i> <span>Instrument Compare</span></button>
<button class="nav-item" data-page="tabulasi" onclick="goPage('tabulasi')"><i class="fas fa-table"></i> <span>Tabulasi Rekap</span></button>
<button class="nav-item" data-page="opspecs" onclick="goPage('opspecs')"><i class="fas fa-bullseye"></i> <span>Analisis OPSpecs</span></button>
</div>
</div>
<div class="nav-divider"></div>
<div class="nav-section">DATA</div>
<button class="nav-item" data-page="validasi" onclick="goPage('validasi')"><i class="fas fa-check-double"></i> <span>Validasi QC</span><span class="nav-badge" id="badgePending" style="display:none">0</span></button>
<button class="nav-item" data-page="histori" onclick="goPage('histori')"><i class="fas fa-history"></i> <span>Histori QC</span></button>
<button class="nav-item" data-page="smartimport" onclick="goPage('smartimport')"><i class="fas fa-file-import"></i> <span>Smart Import</span></button>
<div class="nav-divider"></div>
<div class="nav-group" id="grpImageAnalysis" style="display:none">
<button class="nav-group-toggle" onclick="toggleNavGroup('grpImageAnalysis')"><i class="fas fa-microscope"></i> <span>Image Analysis</span> <i class="fas fa-chevron-down"></i></button>
<div class="nav-submenu">
<button class="nav-item" data-page="imghemato" onclick="goPage('imghemato')"><i class="fas fa-tint"></i> <span>Hematologi Sel</span></button>
<button class="nav-item" data-page="imgurin" onclick="goPage('imgurin')"><i class="fas fa-flask"></i> <span>Urin Sedimen</span></button>
<button class="nav-item" data-page="imgmalaria" onclick="goPage('imgmalaria')"><i class="fas fa-bug"></i> <span>Malaria</span></button>
<button class="nav-item" data-page="imgbta" onclick="goPage('imgbta')"><i class="fas fa-bacteria"></i> <span>BTA</span></button>

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<button class="nav-item" data-page="imgpatologi" onclick="goPage('imgpatologi')"><i class="fas fa-microscope"></i> <span>Patologi Anatomi</span></button>
<button class="nav-item" data-page="imglain" onclick="goPage('imglain')"><i class="fas fa-image"></i> <span>Analisis Lain</span></button>
</div>
</div>
<div class="nav-divider"></div>
<div class="nav-section">PENGATURAN</div>
<button class="nav-item" data-page="hapusdata" onclick="goPage('hapusdata')"><i class="fas fa-trash-alt"></i> <span>Hapus Data</span></button>
<button class="nav-item" data-page="users" id="navUsers" style="display:none"
onclick="goPage('users')"><i class="fas fa-users-cog"></i> <span>Users</span><span class="nav-badge" id="badgePendingUsers" style="display:none">0</span></button>
<button class="nav-item" data-page="kopsurat" onclick="goPage('kopsurat')"><i class="fas fa-stamp"></i> <span>Kop Surat</span></button>
<button class="nav-item" data-page="settings" id="navSettings" style="display:none"
onclick="goPage('settings')"><i class="fas fa-cog"></i> <span>Pengaturan</span></button>
<button class="nav-item" data-page="logactivity" onclick="goPage('logactivity')"><i class="fas fa-clipboard-list"></i> <span>Log Aktivitas</span></button>
<div class="nav-divider"></div>
<button class="nav-item" onclick="doLogout()" style="color:#fca5a5"><i class="fas fa-sign-out-alt"></i> <span>Logout</span></button>
</div>
</nav>
<div class="main-content" id="mainContent">
<header class="topbar">
<div class="topbar-left">
<button class="topbar-toggle" onclick="toggleSidebarMode()"><i class="fas fa-bars"></i></button>
<span class="topbar-title" id="topbarTitle">Dashboard</span>
</div>
<div class="topbar-right">
<div id="loginAsWrap" style="display:none;margin-right:8px"><span style="font-size:.78rem;color:var(--success);font-weight:600"><i class="fas fa-user-check"></i> <span id="loginAsName"></span></span></div>
<div class="view-as-wrap" id="viewAsWrap" style="display:none">
<label style="font-size:.78rem;color:var(--text-secondary);font-weight:600">View As:</label>
<select id="viewAsSelect" onchange="onViewAsChange()"><option value="">— Diri Sendiri —</option></select>
</div>
<button class="topbar-btn" onclick="loadCurrentPage()" title="Refresh"><i class="fas fa-sync-alt"></i></button>
<button class="topbar-btn" onclick="doLogout()" title="Logout"><i class="fas fa-sign-out-alt"></i></button>
</div>
</header>

```



```

<div class="content" id="contentArea">
<div class="page active" id="pageDashboard">
<div class="stat-grid" id="dashStatGrid"></div>
<div class="chart-grid">
<div class="chart-box"><h4 style="margin-bottom:12px"><i class="fas fa-chart-bar"></i> Sigma per
Level per Bidang</h4><canvas id="chartSigmaBidang"></canvas></div>
<div class="chart-box"><h4 style="margin-bottom:12px"><i class="fas fa-chart-bar"></i> CV & Bias
per Bidang</h4><canvas id="chartCVBiasBidang"></canvas></div>
</div>
<div class="card"><div class="card-header"><h3><i class="fas fa-chart-line"></i> Trend Detail Bulan
Berjalan</h3></div><div class="card-body" id="dashTrendDetail"></div></div>
<div class="card"><div class="card-header"><h3><i class="fas fa-exclamation-triangle"></i>
Pelanggaran Westgard (30 Hari)</h3></div><div class="card-body wg-panel" id="dashWGPanel"><p
style="color:var(--text-secondary)">Memuat...</p></div></div>
<div style="display:grid;grid-template-columns:1fr 1fr;gap:16px;margin-top:16px">
<div class="card"><div class="card-body" style="text-align:center"><div style="font-
size:.85rem;color:var(--text-secondary)">QC Minggu Ini</div><div id="dashWeekly" style="font-
size:2.2rem;font-weight:800;color:var(--primary)">0</div></div></div>
<div class="card"><div class="card-body" style="text-align:center"><div style="font-
size:.85rem;color:var(--text-secondary)">QC Bulan Ini</div><div id="dashMonthly" style="font-
size:2.2rem;font-weight:800;color:var(--success)">0</div></div></div>
</div>
</div>
<div class="page" id="pageParameters">
<div class="card">
<div class="card-header"><h3><i class="fas fa-list-ul"></i> Daftar Parameter</h3>
<div class="btn-group"><select id="filterBidangParam" onchange="renderParamTable()"
style="width:auto;padding:6px 10px;font-size:.85rem"><option value="">Semua
Bidang</option></select><button class="btn btn-primary btn-sm" onclick="openParamModal()"><i
class="fas fa-plus"></i> Tambah</button></div>
</div>
<div class="card-body"><div class="table-
wrap"><table><thead><tr><th>No</th><th>Parameter</th><th>Bidang</th><th>Dibuat</th><th>A
ksi</th></tr></thead><tbody id="paramTableBody"></tbody></table></div></div>
</div>
</div>
<div class="page" id="pageLotqc">
<div class="card">
<div class="card-header"><h3><i class="fas fa-box"></i> Lot QC</h3>
<div class="btn-group"><select id="filterParamLot" onchange="renderLotTable()"
style="width:auto;padding:6px 10px;font-size:.85rem"><option value="">Semua
Parameter</option></select><button class="btn btn-primary btn-sm" onclick="openLotModal()"><i
class="fas fa-plus"></i> Tambah</button></div>
</div>

```

```

<div class="card-body"><div class="table-
wrap"><table><thead><tr><th>No</th><th>Parameter</th><th>No.
Lot</th><th>Alat</th><th>Metode</th><th>Satuan</th><th>Sumber</th><th>Expired</th><th>TE
a</th><th>Status</th><th>Aksi</th></tr></thead><tbody
id="lotTableBody"></tbody></table></div></div>
</div>
</div>
<div class="page" id="pageInputqc">
<div class="card">
<div class="card-header"><h3><i class="fas fa-keyboard"></i> Input QC Harian</h3></div>
<div class="card-body">
<div class="tab-bar">
<button class="tab-btn active" onclick="switchInputTab('single')">Single Input</button>
<button class="tab-btn" onclick="switchInputTab('bulk')">Bulk Input</button>
<button class="tab-btn" onclick="switchInputTab('smart')">Smart Input</button>
<div style="margin-left:auto;display:flex;gap:6px">
<button class="btn btn-danger btn-sm" onclick="resetInputQCTabs()"><i class="fas fa-eye-
slash"></i> Reset</button>
<button class="btn btn-success btn-sm" onclick="tampilkanInputQCTabs()"><i class="fas fa-
eye"></i> Tampilkan</button>
</div>
</div>
</div>
<div class="tab-pane active" id="tabSingle">
<div class="filter-bar" style="box-shadow:none;padding:0;margin-bottom:16px;border:none">
<div class="filter-group"><label>Bidang</label><select id="inputBidang"
onchange="filterParamByBidang('inputBidang','inputParam')"><option
value="">Semua</option></select></div>
<div class="filter-group"><label>Parameter</label><select id="inputParam"
onchange="onInputParamChange()"><option value="">Pilih</option></select></div>
<div class="filter-group"><label>Lot QC</label><select id="inputLot"><option
value="">Pilih</option></select></div>
</div>
<input type="hidden" id="inputEditQcID">
<div class="form-row form-row-4">
<div class="form-group"><label>Tanggal</label><input type="date" id="inputTanggal"></div>
<div class="form-group"><label>Level 1</label><input type="number" step="any"
id="inputL1"></div>
<div class="form-group"><label>Level 2</label><input type="number" step="any"
id="inputL2"></div>
<div class="form-group"><label>Level 3</label><input type="number" step="any"
id="inputL3"></div>
</div>
<div class="btn-group">
<button class="btn btn-primary" onclick="saveInputQC()" id="btnSaveInput"><i class="fas fa-
save"></i> Simpan</button>

```

```

<button class="btn btn-secondary" onclick="cancelEditInput()" id="btnCancelEdit"
style="display:none"><i class="fas fa-times"></i> Batal Edit</button>
</div>
</div>
<div class="tab-pane" id="tabBulk">
<div class="filter-bar" style="box-shadow:none;padding:0;margin-bottom:16px;border:none">
<div class="filter-group"><label>Bidang</label><select id="bulkBidang"
onchange="filterParamByBidang('bulkBidang','bulkParam')"><option
value="">Semua</option></select></div>
<div class="filter-group"><label>Parameter</label><select id="bulkParam"
onchange="onBulkParamChange()"><option value="">Pilih</option></select></div>
<div class="filter-group"><label>Lot QC</label><select id="bulkLot"><option
value="">Pilih</option></select></div>
<div class="filter-group"><label>Tgl Mulai</label><input type="date" id="bulkStart"></div>
<div class="filter-group"><label>Tgl Akhir</label><input type="date" id="bulkEnd"></div>
<div class="filter-group"><label>Password</label><input type="password" id="bulkPwd"></div>
<div class="filter-actions"><button class="btn btn-primary btn-sm" onclick="generateBulkRows()"><i
class="fas fa-table"></i> Generate</button></div>
</div>
<div id="bulkTableWrap"></div>
<button class="btn btn-success" onclick="submitBulk()" id="btnSubmitBulk"
style="display:none;margin-top:12px"><i class="fas fa-save"></i> Simpan</button>
</div>
<div class="tab-pane" id="tabSmart">
<div class="filter-bar" style="box-shadow:none;padding:0;margin-bottom:16px;border:none">
<div class="filter-group"><label>Bidang</label><select id="smartBidang"
onchange="filterParamByBidang('smartBidang','smartParam')"><option
value="">Semua</option></select></div>
<div class="filter-group"><label>Parameter</label><select id="smartParam"
onchange="onSmartParamChange()"><option value="">Pilih</option></select></div>
<div class="filter-group"><label>Lot QC</label><select id="smartLot"><option
value="">Pilih</option></select></div>
<div class="filter-group"><label>Tgl Mulai</label><input type="date" id="smartStart"></div>
<div class="filter-group"><label>Tgl Akhir</label><input type="date" id="smartEnd"></div>
<div class="filter-group"><label>Password</label><input type="password" id="smartPwd"></div>
<div class="filter-actions"><button class="btn btn-primary btn-sm"
onclick="generateSmartTable()"><i class="fas fa-table"></i> Tampilkan</button></div>
</div>
<div class="smart-table" id="smartTableWrap"><p style="padding:24px;text-align:center;color:var(--
text-secondary)">Pilih dulu lalu klik Tampilkan</p></div>
<div class="btn-group" style="margin-top:12px">
<button class="btn btn-success btn-sm" onclick="submitSmart()" id="btnSubmitSmart"
style="display:none"><i class="fas fa-save"></i> Simpan</button>
<button class="btn btn-outline btn-sm" onclick="clearSmartTable()" id="btnClearSmart"
style="display:none"><i class="fas fa-eraser"></i> Bersihkan</button>

```

```

</div>
</div>
</div>
</div>
<div class="card">
<div class="card-header"><h3>Data Input QC</h3></div>
<div class="card-body">
<div class="filter-bar" style="box-shadow:none;padding:0;margin-bottom:14px;border:none">
<div class="filter-group"><label>Dari Tanggal</label><input type="date" id="qcFilterStart"></div>
<div class="filter-group"><label>Sampai Tanggal</label><input type="date" id="qcFilterEnd"></div>
<div class="filter-group"><label>Bidang</label><select id="qcFilterBidang"
onchange="filterParamByBidang('qcFilterBidang','qcFilterParam')"><option
value="">Semua</option></select></div>
<div class="filter-group"><label>Parameter</label><select id="qcFilterParam"
onchange="onQCFilterParamChange()"><option value="">Semua</option></select></div>
<div class="filter-group"><label>Lot QC</label><select id="qcFilterLot"><option
value="">Semua</option></select></div>
<div class="filter-actions">
<button class="btn btn-primary btn-sm" onclick="loadInputQCTable()"><i class="fas fa-search"></i>
Cari</button>
<button class="btn btn-secondary btn-sm" onclick="resetQCFilter()"><i class="fas fa-undo"></i>
Reset</button>
</div>
</div>
<div class="table-
wrap"><table><thead><tr><th>No</th><th>Tanggal</th><th>Parameter</th><th>Alat</th><th>No.
Lot</th><th>L1</th><th>L2</th><th>L3</th><th>Status</th><th>Aksi</th></tr></thead><tbody
id="inputQCTableBody"></tbody></table></div>
</div>
</div>
</div>
<div class="page" id="pageGrafik">
<div class="card">
<div class="card-header"><h3><i class="fas fa-chart-line"></i> Levey-Jennings & Analisis</h3></div>
<div class="card-body">
<div class="filter-bar" style="box-shadow:none;padding:0;margin-bottom:16px;border:none">
<div class="filter-group"><label>Bidang</label><select id="grafBidang"
onchange="filterParamByBidang('grafBidang','grafParam')"><option
value="">Semua</option></select></div>
<div class="filter-group"><label>Parameter</label><select id="grafParam"
onchange="onGrafParamChange()"><option value="">Pilih</option></select></div>
<div class="filter-group"><label>Lot QC</label><select id="grafLot"><option
value="">Pilih</option></select></div>
<div class="filter-group"><label>Dari Tanggal</label><input type="date" id="grafStart"></div>
<div class="filter-group"><label>Sampai Tanggal</label><input type="date" id="grafEnd"></div>

```

```

<div class="filter-group"><label>Sumber Mean/SD</label><select id="grafSumber"><option
value="Manufaktur">Manufaktur</option><option value="Terhitung berjalan">Terhitung
Berjalan</option><option value="Terhitung Fix">Terhitung Fix</option></select></div>
<div class="filter-group"><label>Sumber Sigma</label><select id="grafSumberSigma"><option
value="Sigma Terkecil Terhitung">Sigma Terkecil Terhitung</option><option value="Sigma Terkecil
PME">Sigma Terkecil PME</option><option value="Sigma Terkecil PME CV">Sigma Terkecil PME
CV</option><option value="Sigma Terkecil CV Optional">Sigma Terkecil CV
Optional</option></select></div>
<!-- FILTER TAMBAHAN v9.6 (Permintaan #2,3,4,5) -->
<div class="filter-group"><label>Siklus PME</label><select id="grafSiklusPME"><option
value="">Semua</option></select></div>
<div class="filter-group"><label>Tahun Siklus</label><select id="grafTahunSiklus"><option
value="">Semua</option></select></div>
<div class="filter-group"><label>Periode Calc Stats</label><select id="grafPeriodeCS"><option
value="">Semua</option></select></div>
<div class="filter-group"><label>Periode % CV Optional</label><select
id="grafPeriodeCVOpt"><option value="">Semua</option></select></div>
<div class="filter-actions">
<button class="btn btn-primary btn-sm" onclick="loadGrafik()"><i class="fas fa-chart-line"></i>
Tampilkan</button>
<button class="btn btn-secondary btn-sm" onclick="resetGrafikPage()" title="Bersihkan semua
tampilan grafik"><i class="fas fa-redo"></i> Refresh</button>
<button class="btn btn-warning btn-sm" onclick="cekSigmaWestgard()" id="btnCekSigma"
style="display:none"><i class="fas fa-check-circle"></i> Cek Sigma</button>
<button class="btn btn-primary btn-sm" onclick="openExportPanel('grafik','print')" style="white-
space:nowrap;font-weight:700;box-shadow:0 2px 8px rgba(0,0,0,.12);min-height:36px;border-
radius:var(--radius-xs)"><i class="fas fa-print"></i> Print</button>
<button class="btn btn-danger btn-sm" onclick="openExportPanel('grafik','pdf')" style="white-
space:nowrap;font-weight:700;box-shadow:0 2px 8px rgba(0,0,0,.12);min-height:36px;border-
radius:var(--radius-xs)"><i class="fas fa-file-pdf"></i> PDF</button>
<button class="btn btn-success btn-sm" onclick="openExportPanel('grafik','ss')" style="white-
space:nowrap;font-weight:700;box-shadow:0 2px 8px rgba(0,0,0,.12);min-height:36px;border-
radius:var(--radius-xs)"><i class="fas fa-camera"></i> SS</button>
</div>
</div>
</div>
</div>
</div>
<div id="grafikResult" style="display:none">
<div id="grafKopContainer" style="display:none"></div>
<div class="card" id="grafLJCard"><div class="card-header"><h3><i class="fas fa-chart-line"></i>
Grafik Levey-Jennings (3 Level)</h3><div id="grafLJDetailInfo" style="font-size:.82rem;color:var(--
text-secondary);font-weight:500;margin-top:4px"></div></div><div class="card-body"><div
id="ljChartWrap" style="height:520px"><canvas id="ljChartUnified"></canvas></div><p
style="margin-top:2px;font-size:.8rem;color:var(--text-secondary)"
id="grafSumberInfo"></p></div></div>

```

```

<div class="card" id="ljSigmaCard" style="display:none"><div class="card-header"
style="background:linear-gradient(180deg,#fff7ed 0%,#ffffff 100%)><h3><i class="fas fa-
bullseye"></i> Grafik LJ Sigma-Based QC Selection</h3></div><div class="card-body"><div
id="sigmaInfoBox" class="sigma-info-box"></div><div id="ljSigmaChartWrap"
style="height:520px"><canvas id="ljChartSigma"></canvas></div><p style="margin-top:2px;font-
size:.8rem;color:var(--text-secondary)" id="grafSigmaSourceInfo"></p></div></div>
<div class="card" id="grafStatsCard"><div class="card-header"><h3>Statistik QC</h3></div><div
class="card-body" id="grafStatsBody"></div></div>
<div class="card" id="grafSigmaMetricCard"><div class="card-header"><h3>Sigma Metric per
Level</h3></div><div class="card-body"><canvas id="chartSigmaMetric" style="max-
height:280px"></canvas></div></div>
<div class="card" id="grafQGICard"><div class="card-header"><h3><i class="fas fa-crosshairs"></i>
Quality Goal Index (QGI)</h3></div><div class="card-body" id="grafQGIbody"></div></div>
<div class="card" id="cardInterpSigmaBased" style="display:none"><div class="card-header"><h3><i
class="fas fa-info-circle"></i> Interpretasi Sigma-Based QC Selection</h3></div><div class="card-
body" id="interpSigmaBasedBody"></div></div>
<div class="card" id="grafWGCard"><div class="card-header"><h3><i class="fas fa-shield-alt"></i>
Analisis Westgard Rules</h3></div><div class="card-body" id="grafWGBody"></div></div>
<div class="card" id="grafCatatanCard"><div class="card-header"><h3><i class="fas fa-sticky-
note"></i> Catatan</h3></div><div class="card-body"><textarea id="grafCatatan"
placeholder="Catatan grafik..."></textarea><button class="btn btn-primary btn-sm"
onclick="saveGrafCatatan()" style="margin-top:8px"><i class="fas fa-save"></i>
Simpan</button></div></div>
<div class="card" id="grafTTDCard"><div class="card-header"><h3><i class="fas fa-signature"></i>
Tanda Tangan</h3></div><div class="card-body" id="grafTTDBody"></div></div>

</div>
</div>

```

```

<div class="page" id="pageLaporan">
<div class="card"><div class="card-header"><h3><i class="fas fa-file-alt"></i> Laporan
QC</h3></div>
<div class="card-body">
<div class="filter-bar" style="box-shadow:none;padding:0;margin-bottom:16px;border:none">
<div class="filter-group"><label>Dari</label><input type="date" id="lapStart"></div>
<div class="filter-group"><label>Sampai</label><input type="date" id="lapEnd"></div>
<div class="filter-group"><label>Bidang</label><select id="lapBidang"
onchange="filterParamByBidang('lapBidang','lapParam')"><option
value="">Semua</option></select></div>
<div class="filter-group"><label>Parameter</label><select id="lapParam"
onchange="onLapParamChange()"><option value="">Semua</option></select></div>
<div class="filter-group"><label>Lot</label><select id="lapLot"><option
value="">Semua</option></select></div>
<div class="filter-group"><label>Alat</label><input type="text" id="lapAlat"></div>

```

```
<div class="filter-actions"><button class="btn btn-primary btn-sm" onclick="loadLaporan()"><i
class="fas fa-search"></i> Tampilkan</button>
<button class="btn btn-primary btn-sm" onclick="openExportPanel('laporan','print')"
style="display:none;white-space:nowrap;font-weight:700;box-shadow:0 2px 8px
rgba(0,0,0,.12);min-height:36px;border-radius:var(--radius-xs)" id="lapPrintBtn2"><i class="fas fa-
print"></i> Print</button>
<button class="btn btn-danger btn-sm" onclick="openExportPanel('laporan','pdf')"
style="display:none;white-space:nowrap;font-weight:700;box-shadow:0 2px 8px
rgba(0,0,0,.12);min-height:36px;border-radius:var(--radius-xs)" id="lapPdfBtn2"><i class="fas fa-file-
pdf"></i> PDF</button>
<button class="btn btn-success btn-sm" onclick="openExportPanel('laporan','ss')"
style="display:none;white-space:nowrap;font-weight:700;box-shadow:0 2px 8px
rgba(0,0,0,.12);min-height:36px;border-radius:var(--radius-xs)" id="lapSSBtn2"><i class="fas fa-
camera"></i> SS</button></div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
<div id="laporanResult" style="display:none">
```

```
<div id="laporanContent"></div>
```

```
<!--  CATATAN DOKTER v9.6 (Permintaan #9) -->
```

```
<div class="card" id="laporanCatatanDokterCard" style="display:none">
```

```
<div class="card-header"><h3><i class="fas fa-user-md"></i> Catatan Dokter</h3></div>
```

```
<div class="card-body">
```

```
<div class="catatan-dokter-box">
```

```
<h5><i class="fas fa-sticky-note"></i> Catatan Tambahan untuk Parameter & Lot Ini</h5>
```

```
<textarea id="laporanCatatanDokter" placeholder="Tulis catatan dokter di sini..."
```

```
rows="4"></textarea>
```

```
<button class="btn btn-primary btn-sm" onclick="saveLaporanCatatanDokter()" style="margin-
top:10px"><i class="fas fa-save"></i> Simpan Catatan</button>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
<div class="page" id="pageTrendanalysis">
```

```
<div class="card"><div class="card-header"><h3><i class="fas fa-chart-area"></i> Trend
Analisis</h3></div>
```

```
<div class="card-body">
```

```
<div class="filter-bar" style="box-shadow:none;padding:0;margin-bottom:16px;border:none">
```

```
<div class="filter-group"><label>Tahun</label><select id="trendTahun"></select></div>
```

```
<div class="filter-group"><label>Bulan Awal</label><select id="trendBulanAwal"></select></div>
```

```
<div class="filter-group"><label>Bulan Akhir</label><select id="trendBulanAkhir"></select></div>
```

```

<div class="filter-group"><label>Bidang</label><select id="trendBidang"
onchange="filterParamByBidang('trendBidang','trendParam')"><option
value="">Semua</option></select></div>
<div class="filter-group"><label>Parameter</label><select id="trendParam"
onchange="onTrendParamChange()"><option value="">Semua</option></select></div>
<div class="filter-group"><label>Lot</label><select id="trendLot"><option
value="">Semua</option></select></div>
<div class="filter-group"><label>Siklus PME</label><select id="trendSiklusPME"><option
value="">Semua</option></select></div>
<div class="filter-group"><label>Tahun Siklus</label><select id="trendTahunSiklus"><option
value="">Semua</option></select></div>
<div class="filter-actions">
<button class="btn btn-primary btn-sm" onclick="loadTrendAnalisis()"><i class="fas fa-chart-
area"></i> Tampilkan</button>
<button class="btn btn-secondary btn-sm" onclick="resetTrendFilter()"><i class="fas fa-undo"></i>
Reset</button>
<div class="print-options-wrap" style="display:none" id="trendExportBtns">
<button class="btn btn-primary btn-sm" onclick="toggleTrendExportMenu('print')"><i class="fas fa-
print"></i> Print ▼</button><div class="print-options-menu" id="trendPrintMenu"></div>
</div>
<div class="print-options-wrap" style="display:none" id="trendExportBtns2">
<button class="btn btn-danger btn-sm" onclick="toggleTrendExportMenu('pdf')"><i class="fas fa-file-
pdf"></i> PDF ▼</button><div class="print-options-menu" id="trendPdfMenu"></div>
</div>
<div class="print-options-wrap" style="display:none" id="trendExportBtns3">
<button class="btn btn-success btn-sm" onclick="toggleTrendExportMenu('ss')"><i class="fas fa-
camera"></i> SS ▼</button><div class="print-options-menu" id="trendSSMenu"></div>
</div>
</div>
</div>
</div>
</div>
</div>
<div id="trendResult" style="display:none">
<div id="trendContent"></div>

</div>
</div>

```

```

<div class="page" id="pageInstrcompare">
<div class="card"><div class="card-header"><h3><i class="fas fa-balance-scale"></i> Instrument
Compare</h3>
<div class="btn-group"><select id="icParamFilter" style="width:auto;padding:6px 10px;font-
size:.85rem"><option value="">Semua</option></select><button class="btn btn-primary btn-sm"
onclick="loadInstrCompare()"><i class="fas fa-balance-scale"></i> Compare</button></div>

```



```
</div><div class="card-body" id="instrCompareBody"><p style="color:var(--text-secondary)">Klik
Compare</p></div>
</div>
</div>
```

```
<div class="page" id="pageTabulasi">
<div class="card"><div class="card-header"><h3><i class="fas fa-table"></i> Tabulasi
Rekap</h3></div>
<div class="card-body">
<div class="filter-bar" style="box-shadow:none;padding:0;margin-bottom:16px;border:none">
<div class="filter-group"><label>Tahun</label><select id="tabTahun"></select></div>
<div class="filter-group"><label>Bulan Awal</label><select id="tabBulanAwal"><option
value="01">Jan</option><option value="02">Feb</option><option
value="03">Mar</option><option value="04">Apr</option><option
value="05">Mei</option><option value="06">Jun</option><option
value="07">Jul</option><option value="08">Agu</option><option
value="09">Sep</option><option value="10">Okt</option><option
value="11">Nov</option><option value="12">Des</option></select></div>
<div class="filter-group"><label>Bulan Akhir</label><select id="tabBulanAkhir"><option
value="01">Jan</option><option value="02">Feb</option><option
value="03">Mar</option><option value="04">Apr</option><option
value="05">Mei</option><option value="06">Jun</option><option
value="07">Jul</option><option value="08">Agu</option><option
value="09">Sep</option><option value="10">Okt</option><option
value="11">Nov</option><option value="12">Des</option></select></div>
<div class="filter-group"><label>Bidang</label><select id="tabBidangFilter"
onchange="filterTabParams()"><option value="">Semua</option></select></div>
</div>
<div style="margin-bottom:12px"><label style="font-weight:600;font-size:.9rem">Pilih
Parameter:</label>
<div class="checkboxlist-actions"><button class="btn btn-xs btn-outline"
onclick="tabCheckAll(true)">Pilih Semua</button><button class="btn btn-xs btn-outline"
onclick="tabCheckAll(false)">Hapus Semua</button></div>
<div class="checkboxlist-grid" id="tabParamChecklist"></div>
</div>
<button class="btn btn-primary" onclick="loadTabulasi()"><i class="fas fa-table"></i>
Tampilkan</button>
<button class="btn btn-primary" style="display:none" id="tabPrintBtn" onclick="printTabulasi()"><i
class="fas fa-print"></i> Print</button>
<button class="btn btn-danger" style="display:none" id="tabPdfBtn" onclick="pdfTabulasi()"><i
class="fas fa-file-pdf"></i> PDF</button>
<button class="btn btn-success" style="display:none" id="tabSSBtn"
onclick="screenshotTabulasi()"><i class="fas fa-camera"></i> SS</button>
</div>
</div>
```

```

<div id="tabulasiResult" style="display:none">
<div class="card"><div class="card-body" id="tabulasiKopContainer"></div></div>
<div class="card"><div class="card-header"><h3>Rekap Statistik</h3></div><div class="card-
body"><div class="table-wrap" id="tabRekapTable"></div></div></div>
<div class="card" id="tabPMECard" style="display:none"><div class="card-header"><h3>Rekap
PME</h3></div><div class="card-body"><div class="table-wrap"
id="tabPMETable"></div></div></div>
<div class="card"><div class="card-header"><h3><i class="fas fa-lightbulb"></i>
Interpretasi</h3></div><div class="card-body" id="tabInterpretasi"></div></div>
<div class="card"><div class="card-header"><h3><i class="fas fa-sticky-note"></i>
Catatan</h3></div><div class="card-body"><textarea id="tabCatatan"
rows="4"></textarea><button class="btn btn-primary btn-sm" onclick="saveTabCatatan()"
style="margin-top:8px"><i class="fas fa-save"></i> Simpan</button></div></div>

</div>
</div>

```

```

<div class="page" id="pageOpspecs">
<div class="card"><div class="card-header"><h3><i class="fas fa-bullseye"></i> Analisis
OPSpecs</h3></div>
<div class="card-body">
<div class="filter-bar" style="box-shadow:none;padding:0;margin-bottom:16px;border:none">
<div class="filter-group"><label>Tahun</label><select id="opsTahun"><option
value="">Semua</option></select></div>
<div class="filter-group"><label>Bidang</label><select id="opsBidangFilter"
onchange="filterParamByBidang('opsBidangFilter','opsParamFilter')"><option
value="">Semua</option></select></div>
<div class="filter-group"><label>Parameter</label><select id="opsParamFilter"><option
value="">Semua</option></select></div>
<div class="filter-group"><label>Tgl Mulai</label><input type="date" id="opsStartDate"></div>
<div class="filter-group"><label>Tgl Akhir</label><input type="date" id="opsEndDate"></div>
<div class="filter-actions">
<button class="btn btn-primary btn-sm" onclick="loadOPSpecs()"><i class="fas fa-bullseye"></i>
Analisis</button>
<button class="btn btn-primary btn-sm" style="display:none" id="opsPrintBtn"
onclick="printOPSpecs()"><i class="fas fa-print"></i> Print</button>
<button class="btn btn-danger btn-sm" style="display:none" id="opsPdfBtn"
onclick="pdfOPSpecs()"><i class="fas fa-file-pdf"></i> PDF</button>
<button class="btn btn-success btn-sm" style="display:none" id="opsSSBtn"
onclick="screenshotOPSpecs()"><i class="fas fa-camera"></i> SS</button>
</div>
</div>
</div>
</div>
</div>
<div id="opspecsResult" style="display:none">

```

```

<div class="card"><div class="card-body" id="opspecsKopContainer"></div></div>
<div class="ops-explain-box">
<h4><i class="fas fa-info-circle"></i> Apa itu Ped dan Pfr?</h4>
<dl>
<dt>Ped (Probability of Error Detection)</dt>
<dd>Probabilitas QC mendeteksi error analitik saat benar-benar terjadi. Semakin tinggi Ped, semakin efektif. Target: <strong>≥ 90%</strong>.</dd>
<dt>Pfr (Probability of False Rejection)</dt>
<dd>Probabilitas QC menolak run yang sebenarnya valid (false alarm). Target: <strong>≤ 5%</strong>.</dd>
</dl>
<div class="interp-zone">
<strong><i class="fas fa-chart-pie"></i> Interpretasi:</strong><br>
• <strong>Ped tinggi + Pfr rendah</strong> ( $\sigma \geq 4$ ): QC optimal<br>
• <strong>Ped sedang + Pfr sedang</strong> ( $3 \leq \sigma < 4$ ): Perlu multi-rule<br>
• <strong>Ped rendah + Pfr tinggi</strong> ( $\sigma < 3$ ): QC tidak reliable
</div>
</div>
<div class="chart-grid">
<div class="chart-box"><h4 style="margin-bottom:12px">OPSspecs Chart</h4><div class="opspecs-chart-wrap"><canvas id="chartOPSspecs"></canvas></div>
<div class="zone-legend"><div class="zone-item"><div class="zone-dot zone-aman"></div> AMAN</div><div class="zone-item"><div class="zone-dot zone-kritis"></div> KRITIS</div><div class="zone-item"><div class="zone-dot zone-bahaya"></div> BAHAYA</div></div>
<div class="chart-box"><h4 style="margin-bottom:12px">Critical-Error</h4><div class="opspecs-chart-wrap"><canvas id="chartCritError"></canvas></div>
<div class="zone-legend"><div class="zone-item"><div class="zone-dot zone-aman"></div> AMAN</div><div class="zone-item"><div class="zone-dot zone-kritis"></div> KRITIS</div><div class="zone-item"><div class="zone-dot zone-bahaya"></div> BAHAYA</div></div>
</div>
</div>
<div class="card"><div class="card-header"><h3>Tabel Specs</h3></div><div class="card-body"><div class="table-wrap" id="opsTableWrap"></div></div></div>
<div class="card"><div class="card-header"><h3>Analisis Detail</h3></div><div class="card-body" id="opsAnalisisDetail"></div></div>
<div class="card"><div class="card-header"><h3><i class="fas fa-lightbulb"></i> Kesimpulan</h3></div><div class="card-body" id="opsKesimpulan"></div></div>

</div>
</div>

<div class="page" id="pageValidasi">
<div class="card"><div class="card-header"><h3><i class="fas fa-check-double"></i> Validasi QC</h3>

```

```

<div class="btn-group"><select id="valParamFilter" onchange="loadValidasi()"
style="width:auto;padding:6px 10px;font-size:.85rem"><option
value="">Semua</option></select><button class="btn btn-outline btn-sm"
onclick="loadValidasi()"><i class="fas fa-sync"></i></button></div>
</div>
<div class="card-body">
<div class="val-bulk-bar" id="valBulkBar" style="display:none;margin-
bottom:12px;padding:12px;background:var(--primary-50,#eff6ff);border:1px solid var(--primary-
light,#dbeafe);border-radius:var(--radius-sm,10px);flex-wrap:wrap;gap:10px;align-items:center">
<div style="display:flex;align-items:center;gap:8px;flex:1;min-width:200px">
<label style="font-weight:600;font-size:.85rem;white-space:nowrap"><i class="fas fa-clipboard-
check"></i> Catatan Validasi Masal:</label>
<textarea id="valBulkCatatan" rows="1" placeholder="Catatan opsional untuk semua data terpilih..."
style="flex:1;min-width:150px;padding:6px 10px;border:1px solid var(--border);border-radius:var(--
radius-xs,6px);font-size:.85rem;resize:vertical;min-height:36px;max-height:80px"></textarea>
</div>
<div style="display:flex;gap:8px;align-items:center">
<span id="valBulkCount" style="font-size:.85rem;font-weight:600;color:var(--primary)">0
terpilih</span>
<button class="btn btn-success btn-sm" onclick="submitBulkValidasi()"><i class="fas fa-check-
double"></i> Validasi Masal</button>
<button class="btn btn-secondary btn-sm" onclick="toggleAllValChecks(true)"><i class="fas fa-check-
square"></i> Pilih Semua</button>
<button class="btn btn-secondary btn-sm" onclick="toggleAllValChecks(false)"><i class="fas fa-
square"></i> Batal Pilih</button>
</div>
</div>
<div class="table-wrap"><table><thead><tr><th style="width:36px"><input type="checkbox"
id="valCheckAll" onchange="toggleAllValChecks(this.checked)"
title="Pilih/Semua"></th><th>Tgl</th><th>Parameter</th><th>Alat</th><th>L1</th><th>Z1</th><th>
L2</th><th>Z2</th><th>L3</th><th>Z3</th><th>Status</th><th>Aksi</th></tr></thead><tbody
id="valTableBody"></tbody></table></div></div>
</div>
</div>
<div class="page" id="pageHistori">
<div class="card"><div class="card-header"><h3><i class="fas fa-history"></i> Histori QC</h3></div>
<div class="card-body">
<div class="filter-bar" style="box-shadow:none;padding:0;margin-bottom:16px;border:none">
<div class="filter-group"><label>Dari</label><input type="date" id="histStart"></div>
<div class="filter-group"><label>Sampai</label><input type="date" id="histEnd"></div>
<div class="filter-group"><label>Bidang</label><select id="histBidang"
onchange="filterParamByBidang('histBidang','histParam')"><option
value="">Semua</option></select></div>

```

```

<div class="filter-group"><label>Parameter</label><select id="histParam"
onchange="onHistParamChange()"><option value="">Semua</option></select></div>
<div class="filter-group"><label>Lot</label><select id="histLot"><option
value="">Semua</option></select></div>
<div class="filter-group"><label>Tipe</label><select id="histFilter"><option
value="">Semua</option><option value="EDIT_QC">Edit</option><option value="DATA
DIHAPUS">Dihapus</option><option value="RESTORED">Restored</option></select></div>
<div class="filter-actions"><button class="btn btn-primary btn-sm" onclick="loadHistori()"><i
class="fas fa-search"></i> Cari</button><button class="btn btn-secondary btn-sm"
onclick="resetHistFilter()"><i class="fas fa-undo"></i></button></div>
</div>
<div class="table-wrap"><table><thead><tr><th>Tgl
Aksi</th><th>Tipe</th><th>Bidang</th><th>Parameter</th><th>No.Lot</th><th>Tgl
QC</th><th>L1</th><th>L2</th><th>L3</th><th>Oleh</th><th>Detail</th><th>Aksi</th></tr></th
ead><tbody id="histTableBody"></tbody></table></div>
</div>
</div>
</div>

```

```

<div class="page" id="pageDaftartea"><div class="card"><div class="card-header"><h3><i class="fas
fa-percentage"></i> Daftar TEa</h3><button class="btn btn-primary btn-sm"
onclick="openTeaModal()"><i class="fas fa-plus"></i> Tambah</button></div><div class="card-
body"><div class="table-
wrap"><table><thead><tr><th>No</th><th>Parameter</th><th>TEa(%)</th><th>Referensi</th><th
>Aksi</th></tr></thead><tbody id="teaTableBody"></tbody></table></div></div></div><div
class="card" style="margin-top:16px">
<div class="card-header"><h3><i class="fas fa-external-link-alt"></i> Referensi Nilai TEa</h3></div>
<div class="card-body">
<p style="color:var(--text-secondary);margin-bottom:12px;font-size:.9rem">Klik link di bawah untuk
browsing atau download nilai TEa dari sumber terpercaya:</p>
<div style="display:grid;grid-template-columns:repeat(auto-fill,minmax(280px,1fr));gap:10px">
<a href="https://datainnovations.com/allowable-total-error-table/" target="_blank" rel="noopener
norereferrer" style="display:flex;align-items:center;gap:10px;padding:12px 16px;background:var(--bg-
soft);border:1px solid var(--border);border-radius:10px;text-decoration:none;color:var(--
text);transition:all .2s" onmouseover="this.style.borderColor='var(--
primary)';this.style.background='var(--primary-50,#eff6ff)'"
onmouseout="this.style.borderColor='var(--border)';this.style.background='var(--bg-soft)'"><i
class="fas fa-globe" style="color:var(--primary);font-size:1.2rem"></i><div><div style="font-
weight:600;font-size:.9rem">Data Innovations - CLIA TEa Table</div><div style="font-
size:.78rem;color:var(--text-secondary)">Allowable Total Error berdasarkan CLIA</div></div></a>
<a href="https://www.westgard.com/clia.htm" target="_blank" rel="noopener norereferrer"
style="display:flex;align-items:center;gap:10px;padding:12px 16px;background:var(--bg-
soft);border:1px solid var(--border);border-radius:10px;text-decoration:none;color:var(--
text);transition:all .2s" onmouseover="this.style.borderColor='var(--
primary)';this.style.background='var(--primary-50,#eff6ff)'"

```

onmouseout="this.style.borderColor='var(--border)';this.style.background='var(--bg-soft)'"><i class="fas fa-globe" style="color:var(--primary);font-size:1.2rem"></i><div><div style="font-weight:600;font-size:.9rem">Westgard QC - CLIA Requirements</div><div style="font-size:.78rem;color:var(--text-secondary)">Daftar CLIA proficiency testing criteria</div></div>
<i class="fas fa-globe" style="color:var(--primary);font-size:1.2rem"></i><div><div style="font-weight:600;font-size:.9rem">Westgard - Biological Variation Database</div><div style="font-size:.78rem;color:var(--text-secondary)">Ricos Goals / Biological Variation (Desirable Specs)</div></div>
<i class="fas fa-globe" style="color:var(--primary);font-size:1.2rem"></i><div><div style="font-weight:600;font-size:.9rem">Ricos / EFLM Biological Variation</div><div style="font-size:.78rem;color:var(--text-secondary)">European Federation of Clinical Chemistry specifications</div></div>
<i class="fas fa-file-pdf" style="color:var(--danger);font-size:1.2rem"></i><div><div style="font-weight:600;font-size:.9rem">NPAAC - QA Programmes (PDF)</div><div style="font-size:.78rem;color:var(--text-secondary)">Allowable Total Error & Performance Goals</div></div>
</div>
</div></div></div>

<div class="page" id="pageBiaspme"><div class="card"><div class="card-header"><h3><i class="fas fa-exchange-alt"></i> Bias PME</h3><div class="btn-group"><select id="pmeParamFilter" onchange="loadBiasPME()" style="width:auto;padding:6px 10px;font-size:.85rem"><option value="">Semua</option></select><button class="btn btn-primary btn-sm" onclick="openPMEModal()"><i class="fas fa-plus"></i> Tambah</button></div></div><div class="card-body"><div class="table-wrap"><table><thead><tr><th>Siklus</th><th>Tahun</th><th>Parameter</th><th>Alat</th><th>Lv

</th><th>Mean</th><th>SD</th><th>CV%</th><th>Bias%</th><th>TE%</th><th>TEa%</th><th> σ
 </th><th>Aksi</th></tr></thead><tbody
 id="pmeTableBody"></tbody></table></div></div></div></div>

<div class="page" id="pageCalcstats"><div class="card"><div class="card-header"><h3><i class="fas
 fa-calculator"></i> Calc Stats</h3><div class="btn-group"><select id="csParamFilter"
 onchange="loadCalcStats()" style="width:auto;padding:6px 10px;font-size:.85rem"><option
 value="">Semua</option></select><button class="btn btn-primary btn-sm"
 onclick="openCalcStatsModal()"><i class="fas fa-plus"></i> Tambah</button></div></div><div
 class="card-body"><div class="table-
 wrap"><table><thead><tr><th>Parameter</th><th>No.Lot</th><th>Lvl</th><th>Mean</th><th>SD
 </th><th>CV%</th><th>Bias%</th><th>TE%</th><th>TEa%</th><th> σ </th><th>N</th><th>Period
 e</th><th>Aksi</th></tr></thead><tbody
 id="csTableBody"></tbody></table></div></div></div></div>

<div class="page" id="pageSigmacvopt"><div class="card"><div class="card-header"><h3><i
 class="fas fa-percentage"></i> % Sigma CV</h3><div class="btn-group"><select id="scvParamFilter"
 onchange="loadSigmaCVOpt()" style="width:auto;padding:6px 10px;font-size:.85rem"><option
 value="">Semua</option></select><button class="btn btn-primary btn-sm"
 onclick="openSigmaCVOptModal()"><i class="fas fa-plus"></i> Tambah</button></div></div><div
 class="card-body"><div class="table-
 wrap"><table><thead><tr><th>Parameter</th><th>No.Lot</th><th>Alat</th><th>Periode</th><th>
 >Siklus
 PME</th><th>Tahun</th><th>Lvl</th><th>Mean</th><th>SD</th><th>CV%</th><th>Bias%</th><t
 h>TE%</th><th>TEa%</th><th> σ </th><th>Avg σ </th><th>Aksi</th></tr></thead><tbody
 id="scvTableBody"></tbody></table></div></div></div></div>

<div class="page" id="pageSmartimport"><div class="card"><div class="card-header"><h3><i
 class="fas fa-file-import"></i> Smart Import</h3></div><div class="card-body"><div class="form-
 group"><label>Password</label><input type="password" id="importPwd"></div><div
 class="upload-zone" onclick="G('importFileInput').click()"><i class="fas fa-cloud-upload-
 alt"></i><p>Klik atau seret file</p><input type="file" id="importFileInput" accept=".csv,.txt"
 style="display:none" onchange="handleImportFile(event)"></div><div id="importPreview"
 style="display:none;margin-top:16px"></div><button class="btn btn-success" id="btnImportSubmit"
 onclick="submitImport()" style="display:none;margin-top:12px"><i class="fas fa-upload"></i>
 Import</button></div></div></div>

<div class="page" id="pageHapusdata">
 <div id="hapusGate"><div class="gate-overlay"><div class="gate-icon"><i class="fas fa-shield-
 alt"></i></div><h3>Akses Terbatas</h3><div class="gate-form"><div class="form-group"><input
 type="password" id="hapusGatePwd"></div><button class="btn btn-danger btn-block"
 onclick="unlockHapusData()"><i class="fas fa-lock-open"></i> Buka</button></div></div></div>
 <div id="hapusContent" style="display:none">
 <div class="card"><div class="card-header"><h3><i class="fas fa-trash-alt"></i> Hapus Data
 Privat</h3></div>

```

<div class="card-body">
<div class="form-group"><label>Alasan (min 5 karakter)</label><textarea id="hapusAlasan"
rows="3"></textarea></div>
<div class="tab-bar"><button class="tab-btn active" onclick="switchHapusTab('inputqc')">Input
QC</button><button class="tab-btn" onclick="switchHapusTab('lotqc')">Lot QC</button><button
class="tab-btn" onclick="switchHapusTab('parameter')">Parameter</button></div>
<div class="tab-pane active" id="hapusTabInputqc">
<div class="filter-bar" style="box-shadow:none;padding:0;margin-bottom:12px;border:none">
<div class="filter-group"><label>Dari</label><input type="date" id="hapusQCStart"></div>
<div class="filter-group"><label>Sampai</label><input type="date" id="hapusQCEnd"></div>
<div class="filter-actions"><button class="btn btn-primary btn-sm" onclick="loadHapusInputQC()"><i
class="fas fa-search"></i> Cari</button></div>
</div>
<div style="margin-bottom:10px"><label style="font-weight:600;font-size:.82rem">Pilih
Parameter:</label><div class="checklist-grid" id="hapusParamChecklist" style="max-
height:140px"></div></div>
<div class="table-wrap" id="hapusInputQCTable"></div>
</div>
<div class="tab-pane" id="hapusTabLotqc"><div class="table-wrap"
id="hapusLotQCTable"></div></div>
<div class="tab-pane" id="hapusTabParameter"><div class="table-wrap"
id="hapusParamTable"></div></div>
<button class="btn btn-danger" onclick="submitHapusData()" style="margin-top:12px"><i class="fas
fa-trash"></i> Hapus Terpilih</button>
</div>
</div>
</div>
</div>

```

```

<div class="page" id="pageUsers"><div class="card"><div class="card-header"><h3><i class="fas fa-
users-cog"></i> Users</h3><button class="btn btn-primary btn-sm" onclick="openUserModal()"><i
class="fas fa-plus"></i> Tambah</button></div><div class="card-body"><div
id="pendingUsersWrap" style="display:none;margin-bottom:20px"><h4><i class="fas fa-clock"></i>
Menunggu Persetujuan</h4><div id="pendingUsersList"></div></div><div class="table-
wrap"><table><thead><tr><th>Username</th><th>Nama</th><th>Email</th><th>Role</th><th>St
atus</th><th>Masa
Aktif</th><th>ImgAccess</th><th>Password</th><th>Aksi</th></tr></thead><tbody
id="usersTableBody"></tbody></table></div></div></div></div>

```

```

<div class="page" id="pageKopsurat"><div class="card"><div class="card-header"><h3><i class="fas
fa-stamp"></i> Kop Surat</h3></div><div class="card-body"><div class="form-row"><div
class="form-group"><label>Nama RS/Lab</label><input type="text" id="kopNamaRS"></div><div
class="form-group"><label>Alamat</label><input type="text" id="kopAlamat"></div></div><div
class="form-row"><div class="form-group"><label>Telepon</label><input type="text"
id="kopTelepon"></div><div class="form-group"><label>Email Lab</label><input type="email"

```



```

id="kopEmail"></div></div><div class="form-row"><div class="form-group"><label>Dokter
PJ</label><input type="text" id="kopDokterPJ"></div><div class="form-group"><label>Email Dokter
PJ</label><input type="email" id="kopEmailDokterPJ"></div></div><div class="form-row"><div
class="form-group"><label>Kepala Ruangan</label><input type="text"
id="kopKepalaRuangan"></div><div class="form-group"><label>Email Kep.Ruangan</label><input
type="email" id="kopEmailKepalaRuangan"></div></div><div class="form-group"><label>PEMDA
(Pemerintah Daerah)</label><input type="text" id="kopPEMDA" placeholder="Nama PEMDA /
Pemerintah Daerah"></div>
<div class="form-row"><div class="form-group"><label>Logo Lab (Kiri)</label><input type="file"
id="kopLogoFile" accept="image/*" onchange="handleLogoUpload(event)"><div
id="kopLogoPreview" style="margin-top:10px"></div><input type="hidden"
id="kopLogo"></div><div class="form-group"><label>Logo Lab Tambahan (Kanan)</label><input
type="file" id="kopLogo2File" accept="image/*" onchange="handleLogo2Upload(event)"><div
id="kopLogo2Preview" style="margin-top:10px"></div><input type="hidden"
id="kopLogo2"></div></div><button class="btn btn-primary" onclick="saveKopSuratForm()"><i
class="fas fa-save"></i> Simpan</button></div></div></div>

```

```

<div class="page" id="pageSettings">
<div class="card"><div class="card-header"><h3><i class="fas fa-cog"></i> Pengaturan</h3></div>
<div class="card-body">
<h4>Warna Tema</h4>
<div class="form-row">
<div class="form-group"><label>Primary</label><input type="color" id="setThemeColor"
value="#2563eb" style="height:40px"><div class="color-presets" id="themePresets"></div></div>
<div class="form-group"><label>Sidebar</label><input type="color" id="setSidebarColor"
value="#0f172a" style="height:40px"><div class="color-presets" id="sidebarPresets"></div></div>
</div>
<h4 style="margin:16px 0 8px">Font Sidebar</h4>
<div class="form-row form-row-3">
<div class="form-group"><label>Warna</label><input type="color" id="setSbFontColor"
value="#f1f5f9" style="height:40px"></div>
<div class="form-group"><label>Ukuran</label><input type="number" id="setSbFontSize"
value="14" min="10" max="20"></div>
<div class="form-group"><label>Ketebalan</label><select id="setSbFontWeight"><option
value="400">400</option><option value="500" selected>500</option><option
value="600">600</option><option value="700">700</option></select></div>
</div>
<h4 style="margin:16px 0 8px">Auto Logout</h4>
<div class="form-group" style="max-width:300px"><label>Timer (detik)</label><input
type="number" id="setAutoLogout" value="180" min="30" max="3600"></div>
<h4 style="margin:16px 0 8px">Tema Aplikasi</h4>
<div class="form-group" style="max-width:300px">
<label>Mode Tampilan</label>
<select id="setAppTheme" style="width:100%">
<option value="light">Light Mode (Terang)</option>

```

```

<option value="dark">Dark Mode (Gelap)</option>
<option value="system">Ikuti Sistem</option>
</select>
<small style="color:var(--text-secondary);display:block;margin-top:4px">Pengaturan ini bersifat
privat untuk akun Anda</small>
</div>
<button class="btn btn-primary" onclick="saveAppSettings()"><i class="fas fa-save"></i>
Simpan</button>
<h4 style="margin:16px 0 8px"><i class="fas fa-palette"></i> Pengaturan Form Login (Kiri)</h4>
<div class="form-row form-row-3">
<div class="form-group"><label>Ukuran Font</label><input type="number" id="setLoginFontSize"
value="16" min="10" max="28" style="width:100%"><small>px</small></div>
<div class="form-group"><label>Warna Font</label><input type="color" id="setLoginFontColor"
value="#ffffff" style="height:40px;width:100%"></div>
<div class="form-group"><label>Ketebalan Font</label><select id="setLoginFontWeight"
style="width:100%"><option value="300">300 - Light</option><option value="400" selected>400 -
Normal</option><option value="500">500 - Medium</option><option value="600">600 - Semi
Bold</option><option value="700">700 - Bold</option><option value="800">800 - Extra
Bold</option></select></div>
</div>
<h4 style="margin:16px 0 8px"><i class="fas fa-sun"></i> Tingkat Kecerahan Background Login</h4>
<div class="brightness-levels" id="brightnessLevels" style="display:flex;gap:8px;flex-
wrap:wrap;margin-bottom:8px">
<button class="btn btn-outline btn-sm brightness-btn" data-val="0.4" data-label="Sangat Gelap"
onclick="setLoginBrightness(0.4,this)"><i class="fas fa-moon"></i> 1. Sangat Gelap</button>
<button class="btn btn-outline btn-sm brightness-btn" data-val="0.6" data-label="Gelap"
onclick="setLoginBrightness(0.6,this)"><i class="fas fa-cloud-moon"></i> 2. Gelap</button>
<button class="btn btn-outline btn-sm brightness-btn active" data-val="0.8" data-label="Sedang"
onclick="setLoginBrightness(0.8,this)"><i class="fas fa-cloud-sun"></i> 3. Sedang</button>
<button class="btn btn-outline btn-sm brightness-btn" data-val="1" data-label="Cerah"
onclick="setLoginBrightness(1,this)"><i class="fas fa-sun"></i> 4. Cerah</button>
<button class="btn btn-outline btn-sm brightness-btn" data-val="1.3" data-label="Cerah Sekali"
onclick="setLoginBrightness(1.3,this)"><i class="fas fa-sun" style="color:#f59e0b"></i> 5. Cerah
Sekali</button>
</div>
<small style="color:var(--text-secondary);display:block;margin-bottom:8px">Kecerahan saat ini:
<strong id="loginBrightVal">Sedang</strong></small>
<div class="form-group" style="max-width:300px"><label>Warna Layer Background</label><input
type="color" id="setLoginBgColor" value="#1e293b" style="height:40px"><div class="color-presets"
id="loginBgPresets"></div></div>
<hr style="margin:24px 0">
<h4>Advanced</h4>
<div class="btn-group">
<button class="btn btn-warning btn-sm" onclick="triggerAutoArchive()"><i class="fas fa-
archive"></i> Archive</button>

```

```

<button class="btn btn-info btn-sm" onclick="initSheetsManual()"><i class="fas fa-database"></i>
Init</button>
<button class="btn btn-danger btn-sm" onclick="confirmResetDB()"><i class="fas fa-bomb"></i>
Reset DB</button>
</div>
<hr style="margin:24px 0">
<h4><i class="fas fa-cloud"></i> Backup & Restore</h4>
<div class="backup-info">
<div class="backup-card"><i class="fas fa-download" style="color:var(--
primary)"></i><h5>Backup</h5><div class="btn-group" style="justify-content:center;margin-
top:8px"><button class="btn btn-primary btn-sm" onclick="doBackupSheets()"><i class="fas fa-
table"></i> Sheets</button><button class="btn btn-success btn-sm"
onclick="doBackupAppProject()"><i class="fas fa-code"></i> App</button></div></div>
<div class="backup-card"><i class="fas fa-upload" style="color:var(--
warning)"></i><h5>Restore</h5><button class="btn btn-warning btn-sm"
onclick="openRestoreModal()" style="margin-top:8px"><i class="fas fa-history"></i>
Buka</button></div>
</div>
<div class="form-group" style="margin-top:12px;max-width:250px"><label><input type="checkbox"
id="setAutoBackup" style="width:auto;margin-right:6px"> Auto Backup (02:00)</label></div>
<button class="btn btn-outline btn-sm" onclick="setupBackupTrigger()"><i class="fas fa-clock"></i>
Setup Trigger</button>
<hr style="margin:24px 0">
<h4><i class="fas fa-file-export"></i> Export Sheet</h4>
<div class="form-row">
<div class="form-group"><label>Sheet</label><select id="exportSheetName"><option
value="Users">Users</option><option value="Parameters">Parameters</option><option
value="LotQC">LotQC</option><option value="InputQC">InputQC</option><option
value="HistoriQC">HistoriQC</option><option value="DaftarTEa">DaftarTEa</option><option
value="BiasPME">BiasPME</option><option
value="CalculatedStats">CalculatedStats</option><option
value="SigmaCVOpt">SigmaCVOpt</option><option value="Settings">Settings</option><option
value="LogActivity">LogActivity</option></select></div>
<div class="form-group" style="display:flex;align-items:flex-end;gap:8px"><button class="btn btn-
info btn-sm" onclick="doExportSheet('json')"><i class="fas fa-download"></i>
JSON</button><button class="btn btn-info btn-sm" onclick="doExportSheet('csv')"><i class="fas fa-
download"></i> CSV</button></div>
</div>
<hr style="margin:24px 0">
<h4><i class="fas fa-envelope"></i> Email</h4>
<div class="form-row">
<div class="form-group"><label>Kuota</label><div id="emailQuotaDisplay" style="font-
size:1.3rem;font-weight:700;color:var(--primary)">—</div></div></div>

```

```

<div class="form-group"><label>Test</label><div style="display:flex;gap:8px"><input type="email"
id="testEmailTo"><button class="btn btn-primary btn-sm" onclick="doTestEmail()"><i class="fas fa-
paper-plane"></i></button></div></div>
</div>
</div>
</div>
</div>

```

```

<div class="page" id="pageLogactivity"><div class="card"><div class="card-header"><h3><i
class="fas fa-clipboard-list"></i> Log Aktivitas <small style="color:var(--text-secondary)">(Hari
Ini)</small></h3><div class="btn-group"><button class="btn btn-outline btn-sm"
onclick="loadLogActivity()"><i class="fas fa-sync"></i></button><button class="btn btn-danger btn-
sm" onclick="clearLog()"><i class="fas fa-trash"></i> Bersihkan</button></div></div><div
class="card-body"><div class="table-wrap" style="max-height:500px;overflow-
y:auto"><table><thead><tr><th>Waktu</th><th>User</th><th>Aksi</th><th>Detail</th></tr></thead><tbody id="logTableBody"></tbody></table></div></div></div></div>

```

```

<div class="page" id="pageImghemato"><div class="card"><div class="card-header"><h3><i
class="fas fa-tint"></i> Hematologi Sel</h3><button class="btn btn-primary btn-sm"
onclick="openImgModal('hemato')"><i class="fas fa-plus"></i></button><div class="print-options-
wrap" style="position:relative;display:inline-block">
<button class="btn btn-info btn-sm" onclick="toggleImgPrintMenu(this)" title="Print"><i class="fas
fa-print"></i></button>
<div class="print-options-menu" id="printMenu_hemato">
<button onclick="printImgFromList('hemato','withImg')"><i class="fas fa-image"></i> Hasil + Gambar
+ TTD</button>
<button onclick="printImgFromList('hemato','noImg')"><i class="fas fa-file-alt"></i> Hasil +
TTD</button>
</div>
</div>
<div class="print-options-wrap" style="position:relative;display:inline-block">
<button class="btn btn-danger btn-sm" onclick="toggleImgPrintMenu(this)" title="PDF"><i class="fas
fa-file-pdf"></i></button>
<div class="print-options-menu" id="pdfMenu_hemato">
<button onclick="pdfImgFromList('hemato','withImg')"><i class="fas fa-image"></i> Hasil + Gambar
+ TTD</button>
<button onclick="pdfImgFromList('hemato','noImg')"><i class="fas fa-file-alt"></i> Hasil +
TTD</button>
</div>
</div></div><div class="card-body"><div class="filter-bar" style="box-
shadow:none;padding:0;margin-bottom:12px;border:none"><div class="filter-
group"><label>Dari</label><input type="date" id="imgHematoStart"></div><div class="filter-
group"><label>Sampai</label><input type="date" id="imgHematoEnd"></div><div class="filter-
group"><label>No.RM</label><input type="text" id="imgHematoRM"></div><div class="filter-
group"><label>Nama</label><input type="text" id="imgHematoNama"></div><div class="filter-

```

```

actions"><button class="btn btn-primary btn-sm" onclick="IdImgData('hemato')"><i class="fas fa-
search"></i></button><button class="btn btn-secondary btn-sm"
onclick="resetImgFilter('hemato')"><i class="fas fa-undo"></i></button></div></div><div
class="table-
wrap"><table><thead><tr><th>No</th><th>Tgl</th><th>No.RM</th><th>Nama</th><th>JK</th><t
h>Dokter</th><th>Kesan</th><th>Aksi</th></tr></thead><tbody
id="imgHematoBody"></tbody></table></div></div></div></div>

```

```

<div class="page" id="pageImgurin"><div class="card"><div class="card-header"><h3><i class="fas
fa-flask"></i> Urin Sedimen</h3><button class="btn btn-primary btn-sm"
onclick="openImgModal('urin')"><i class="fas fa-plus"></i></button><div class="print-options-wrap"
style="position:relative;display:inline-block">
<button class="btn btn-info btn-sm" onclick="toggleImgPrintMenu(this)" title="Print"><i class="fas
fa-print"></i></button>
<div class="print-options-menu" id="printMenu_urin">
<button onclick="printImgFromList('urin','withImg')"><i class="fas fa-image"></i> Hasil + Gambar +
TTD</button>
<button onclick="printImgFromList('urin','noImg')"><i class="fas fa-file-alt"></i> Hasil +
TTD</button>
</div>
</div>
<div class="print-options-wrap" style="position:relative;display:inline-block">
<button class="btn btn-danger btn-sm" onclick="toggleImgPrintMenu(this)" title="PDF"><i class="fas
fa-file-pdf"></i></button>
<div class="print-options-menu" id="pdfMenu_urin">
<button onclick="pdfImgFromList('urin','withImg')"><i class="fas fa-image"></i> Hasil + Gambar +
TTD</button>
<button onclick="pdfImgFromList('urin','noImg')"><i class="fas fa-file-alt"></i> Hasil + TTD</button>
</div>
</div></div><div class="card-body"><div class="filter-bar" style="box-
shadow:none;padding:0;margin-bottom:12px;border:none"><div class="filter-
group"><label>Dari</label><input type="date" id="imgUrinStart"></div><div class="filter-
group"><label>Sampai</label><input type="date" id="imgUrinEnd"></div><div class="filter-
group"><label>No.RM</label><input type="text" id="imgUrinRM"></div><div class="filter-
group"><label>Nama</label><input type="text" id="imgUrinNama"></div><div class="filter-
actions"><button class="btn btn-primary btn-sm" onclick="IdImgData('urin')"><i class="fas fa-
search"></i></button><button class="btn btn-secondary btn-sm" onclick="resetImgFilter('urin')"><i
class="fas fa-undo"></i></button></div></div><div class="table-
wrap"><table><thead><tr><th>No</th><th>Tgl</th><th>No.RM</th><th>Nama</th><th>JK</th><t
h>Dokter</th><th>Interpretasi</th><th>Aksi</th></tr></thead><tbody
id="imgUrinBody"></tbody></table></div></div></div></div>

```

```

<div class="page" id="pageImgmalaria"><div class="card"><div class="card-header"><h3><i
class="fas fa-bug"></i> Malaria</h3><button class="btn btn-primary btn-sm"

```

```

onclick="openImgModal('malaria')"><i class="fas fa-plus"></i></button><div class="print-options-wrap" style="position:relative;display:inline-block">
<button class="btn btn-info btn-sm" onclick="toggleImgPrintMenu(this)" title="Print"><i class="fas fa-print"></i></button>
<div class="print-options-menu" id="printMenu_malaria">
<button onclick="printImgFromList('malaria','withImg')"><i class="fas fa-image"></i> Hasil + Gambar + TTD</button>
<button onclick="printImgFromList('malaria','noImg')"><i class="fas fa-file-alt"></i> Hasil + TTD</button>
</div>
</div>
<div class="print-options-wrap" style="position:relative;display:inline-block">
<button class="btn btn-danger btn-sm" onclick="toggleImgPrintMenu(this)" title="PDF"><i class="fas fa-file-pdf"></i></button>
<div class="print-options-menu" id="pdfMenu_malaria">
<button onclick="pdfImgFromList('malaria','withImg')"><i class="fas fa-image"></i> Hasil + Gambar + TTD</button>
<button onclick="pdfImgFromList('malaria','noImg')"><i class="fas fa-file-alt"></i> Hasil + TTD</button>
</div>
</div></div><div class="card-body"><div class="filter-bar" style="box-shadow:none;padding:0;margin-bottom:12px;border:none"><div class="filter-group"><label>Dari</label><input type="date" id="imgMalariaStart"></div><div class="filter-group"><label>Sampai</label><input type="date" id="imgMalariaEnd"></div><div class="filter-group"><label>No.RM</label><input type="text" id="imgMalariaRM"></div><div class="filter-group"><label>Nama</label><input type="text" id="imgMalariaNama"></div><div class="filter-actions"><button class="btn btn-primary btn-sm" onclick="IdImgData('malaria')"><i class="fas fa-search"></i></button><button class="btn btn-secondary btn-sm" onclick="resetImgFilter('malaria')"><i class="fas fa-undo"></i></button></div></div><div class="table-wrap"><table><thead><tr><th>No</th><th>Tgl</th><th>No.RM</th><th>Nama</th><th>JK</th><th>Dokter</th><th>Hasil</th><th>Aksi</th></tr></thead><tbody id="imgMalariaBody"></tbody></table></div></div></div></div>

<div class="page" id="pageImgBta"><div class="card"><div class="card-header"><h3><i class="fas fa-bacteria"></i> BTA</h3><button class="btn btn-primary btn-sm" onclick="openImgModal('bta')"><i class="fas fa-plus"></i></button><div class="print-options-wrap" style="position:relative;display:inline-block">
<button class="btn btn-info btn-sm" onclick="toggleImgPrintMenu(this)" title="Print"><i class="fas fa-print"></i></button>
<div class="print-options-menu" id="printMenu_bta">
<button onclick="printImgFromList('bta','withImg')"><i class="fas fa-image"></i> Hasil + Gambar + TTD</button>
<button onclick="printImgFromList('bta','noImg')"><i class="fas fa-file-alt"></i> Hasil + TTD</button>
</div>
</div>

```

```

</div>
<div class="print-options-wrap" style="position:relative;display:inline-block">
<button class="btn btn-danger btn-sm" onclick="toggleImgPrintMenu(this)" title="PDF"><i class="fas
fa-file-pdf"></i></button>
<div class="print-options-menu" id="pdfMenu_bta">
<button onclick="pdfImgFromList('bta','withImg')"><i class="fas fa-image"></i> Hasil + Gambar +
TTD</button>
<button onclick="pdfImgFromList('bta','noImg')"><i class="fas fa-file-alt"></i> Hasil + TTD</button>
</div>
</div></div><div class="card-body"><div class="filter-bar" style="box-
shadow:none;padding:0;margin-bottom:12px;border:none"><div class="filter-
group"><label>Dari</label><input type="date" id="imgBTASStart"></div><div class="filter-
group"><label>Sampai</label><input type="date" id="imgBTAEnd"></div><div class="filter-
group"><label>No.RM</label><input type="text" id="imgBTARM"></div><div class="filter-
group"><label>Nama</label><input type="text" id="imgBTANama"></div><div class="filter-
actions"><button class="btn btn-primary btn-sm" onclick="IdImgData('bta')"><i class="fas fa-
search"></i></button><button class="btn btn-secondary btn-sm" onclick="resetImgFilter('bta')"><i
class="fas fa-undo"></i></button></div></div><div class="table-
wrap"><table><thead><tr><th>No</th><th>Tgl</th><th>No.RM</th><th>Nama</th><th>JK</th><th>
Dokter</th><th>Hasil</th><th>Aksi</th></tr></thead><tbody
id="imgBTABody"></tbody></table></div></div></div></div>

```

```

<div class="page" id="pageImgpatologi"><div class="card"><div class="card-header"><h3><i
class="fas fa-microscope"></i> Patologi Anatomi</h3><button class="btn btn-primary btn-sm"
onclick="openImgPatologiModal()"><i class="fas fa-plus"></i> Tambah</button><div class="print-
options-wrap" style="position:relative;display:inline-block">
<button class="btn btn-info btn-sm" onclick="toggleImgPrintMenu(this)" title="Print"><i class="fas
fa-print"></i></button>
<div class="print-options-menu" id="printMenu_patologi">
<button onclick="printImgFromList('patologi','withImg')"><i class="fas fa-image"></i> Hasil +
Gambar + TTD</button>
<button onclick="printImgFromList('patologi','noImg')"><i class="fas fa-file-alt"></i> Hasil +
TTD</button>
</div>
</div>
<div class="print-options-wrap" style="position:relative;display:inline-block">
<button class="btn btn-danger btn-sm" onclick="toggleImgPrintMenu(this)" title="PDF"><i class="fas
fa-file-pdf"></i></button>
<div class="print-options-menu" id="pdfMenu_patologi">
<button onclick="pdfImgFromList('patologi','withImg')"><i class="fas fa-image"></i> Hasil + Gambar
+ TTD</button>
<button onclick="pdfImgFromList('patologi','noImg')"><i class="fas fa-file-alt"></i> Hasil +
TTD</button>
</div>

```

```

</div></div><div class="card-body"><div class="filter-bar" style="box-
shadow:none;padding:0;margin-bottom:12px;border:none"><div class="filter-
group"><label>Dari</label><input type="date" id="imgPatologiStart"></div><div class="filter-
group"><label>Sampai</label><input type="date" id="imgPatologiEnd"></div><div class="filter-
group"><label>No.RM</label><input type="text" id="imgPatologiRM"></div><div class="filter-
group"><label>Nama</label><input type="text" id="imgPatologiNama"></div><div class="filter-
group"><label>No.PA</label><input type="text" id="imgPatologiPA"></div><div class="filter-
actions"><button class="btn btn-primary btn-sm" onclick="IdImgPatologiData()"><i class="fas fa-
search"></i></button><button class="btn btn-secondary btn-sm"
onclick="resetImgPatologiFilter()"><i class="fas fa-undo"></i></button></div></div><div
class="table-
wrap"><table><thead><tr><th>No</th><th>TglTerima</th><th>No.RM</th><th>No.PA</th><th>Na
ma</th><th>JK</th><th>JenisPemeriksaan</th><th>Diagnosis</th><th>Aksi</th></tr></thead><tb
ody id="imgPatologiBody"></tbody></table></div></div></div></div>
<div class="page" id="pageImglain"><div class="card"><div class="card-header"><h3><i class="fas
fa-image"></i> Analisis Lain</h3><button class="btn btn-primary btn-sm"
onclick="openImgModal('lain')"><i class="fas fa-plus"></i></button><div class="print-options-wrap"
style="position:relative;display:inline-block">
<button class="btn btn-info btn-sm" onclick="toggleImgPrintMenu(this)" title="Print"><i class="fas
fa-print"></i></button>
<div class="print-options-menu" id="printMenu_lain">
<button onclick="printImgFromList('lain','withImg')"><i class="fas fa-image"></i> Hasil + Gambar +
TTD</button>
<button onclick="printImgFromList('lain','noImg')"><i class="fas fa-file-alt"></i> Hasil +
TTD</button>
</div>
</div>
<div class="print-options-wrap" style="position:relative;display:inline-block">
<button class="btn btn-danger btn-sm" onclick="toggleImgPrintMenu(this)" title="PDF"><i class="fas
fa-file-pdf"></i></button>
<div class="print-options-menu" id="pdfMenu_lain">
<button onclick="pdfImgFromList('lain','withImg')"><i class="fas fa-image"></i> Hasil + Gambar +
TTD</button>
<button onclick="pdfImgFromList('lain','noImg')"><i class="fas fa-file-alt"></i> Hasil + TTD</button>
</div>
</div></div><div class="card-body"><div class="filter-bar" style="box-
shadow:none;padding:0;margin-bottom:12px;border:none"><div class="filter-
group"><label>Dari</label><input type="date" id="imgLainStart"></div><div class="filter-
group"><label>Sampai</label><input type="date" id="imgLainEnd"></div><div class="filter-
group"><label>No.RM</label><input type="text" id="imgLainRM"></div><div class="filter-
group"><label>Nama</label><input type="text" id="imgLainNama"></div><div class="filter-
actions"><button class="btn btn-primary btn-sm" onclick="IdImgData('lain')"><i class="fas fa-
search"></i></button><button class="btn btn-secondary btn-sm" onclick="resetImgFilter('lain')"><i
class="fas fa-undo"></i></button></div></div><div class="table-
wrap"><table><thead><tr><th>No</th><th>Tgl</th><th>No.RM</th><th>Nama</th><th>JK</th><t

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h>Dokter</th><th>Interpretasi</th><th>Aksi</th></tr></thead><tbody
id="imgLainBody"></tbody></table></div></div></div></div>
</div>
</div>
</div>

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<!-- MODALS -->

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<div class="modal-overlay" id="modalParam"><div class="modal"><div class="modal-header"><h3
id="modalParamTitle">Parameter</h3><button class="modal-close"
onclick="clM('modalParam')">&times;</button></div><div class="modal-body"><input
type="hidden" id="mParamID"><div class="form-group"><label>Nama Parameter</label><input
type="text" id="mParamName"></div><div class="form-group"><label>Bidang</label><input
type="text" id="mParamBidang" list="dlBidang"><datalist id="dlBidang"><option value="Kimia
Klinik"><option value="Hematologi"><option value="Imunologi"><option
value="Koagulasi"><option value="Urinalisis"><option value="Serologi"><option
value="Elektrolit"><option value="Blood Gas"><option value="Lainnya"></datalist></div></div><div
class="modal-footer"><button class="btn btn-secondary"
onclick="clM('modalParam')">Batal</button><button class="btn btn-primary"
onclick="saveParam()"><i class="fas fa-save"></i> Simpan</button></div></div></div>

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<div class="modal-overlay" id="modalLot"><div class="modal modal-lg"><div class="modal-
header"><h3 id="modalLotTitle">Lot QC</h3><button class="modal-close"
onclick="clM('modalLot')">&times;</button></div><div class="modal-body"><input type="hidden"
id="mLotID"><div class="form-row"><div class="form-group"><label>Parameter</label><select
id="mLotParam" onchange="autoFillTEa()"><option value="">Pilih</option></select></div><div
class="form-group"><label>No. Lot</label><input type="text" id="mLotNoLot"></div></div><div
class="form-row form-row-3"><div class="form-group"><label>Alat</label><input type="text"
id="mLotAlat"></div><div class="form-group"><label>Metode</label><input type="text"
id="mLotMethode"></div><div class="form-group"><label>Satuan</label><input type="text"
id="mLotSatuan"></div></div><div class="form-row form-row-3"><div class="form-
group"><label>Expired</label><input type="date" id="mLotExpired"></div><div class="form-
group"><label>TEa(%)</label><input type="number" step="any" id="mLotTEa"></div><div
class="form-group"><label>Bias(%)</label><input type="number" step="any"
id="mLotBias"></div></div><div class="form-group"><label>Sumber</label><select
id="mLotSumber"><option value="Manufaktur">Manufaktur</option><option value="Terhitung
berjalan">Terhitung berjalan</option><option value="Terhitung Fix">Terhitung
Fix</option></select></div><h4 style="margin:16px 0 8px;color:var(--primary)">Level 1</h4><div
class="form-row form-row-3"><div class="form-group"><label>Mean L1</label><input
type="number" step="any" id="mLotMeanL1"></div><div class="form-group"><label>SD
L1</label><input type="number" step="any" id="mLotSDL1"></div><div class="form-
group"><label>Target L1</label><input type="number" step="any"
id="mLotTargetL1"></div></div><h4 style="margin:16px 0 8px;color:var(--warning)">Level
2</h4><div class="form-row form-row-3"><div class="form-group"><label>Mean L2</label><input
type="number" step="any" id="mLotMeanL2"></div><div class="form-group"><label>SD
L2</label><input type="number" step="any" id="mLotSDL2"></div><div class="form-

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group"><label>Target L2</label><input type="number" step="any"
id="mLotTargetL2"></div></div><h4 style="margin:16px 0 8px;color:var(--danger)">Level
3</h4><div class="form-row form-row-3"><div class="form-group"><label>Mean L3</label><input
type="number" step="any" id="mLotMeanL3"></div><div class="form-group"><label>SD
L3</label><input type="number" step="any" id="mLotSDL3"></div><div class="form-
group"><label>Target L3</label><input type="number" step="any"
id="mLotTargetL3"></div></div></div><div class="modal-footer"><button class="btn btn-
secondary" onclick="cIM('modalLot')">Batal</button><button class="btn btn-primary"
onclick="saveLot()"><i class="fas fa-save"></i> Simpan</button></div></div></div>

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<div class="modal-overlay" id="modalTEa"><div class="modal"><div class="modal-header"><h3
id="modalTeaTitle">TEa</h3><button class="modal-close"
onclick="cIM('modalTEa')">&times;</button></div><div class="modal-body"><input type="hidden"
id="mTeaID"><div class="form-group"><label>Parameter</label><select id="mTeaParam"><option
value="">Pilih</option></select></div><div class="form-group"><label>TEa(%)</label><input
type="number" step="any" id="mTeaValue"></div><div class="form-
group"><label>Referensi</label><input type="text" id="mTeaRef"></div></div><div class="modal-
footer"><button class="btn btn-secondary" onclick="cIM('modalTEa')">Batal</button><button
class="btn btn-primary" onclick="saveTea()"><i class="fas fa-save"></i>
Simpan</button></div></div></div>

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<div class="modal-overlay" id="modalPME"><div class="modal modal-lg"><div class="modal-
header"><h3 id="modalPMETitle">Bias PME</h3><button class="modal-close"
onclick="cIM('modalPME')">&times;</button></div><div class="modal-body"><input type="hidden"
id="mPMEID"><div class="form-row"><div class="form-group"><label>Parameter*</label><select
id="mPMEParam" onchange="onPMEParamChange()"><option
value="">Pilih</option></select></div><div class="form-group"><label>Lot QC*</label><select
id="mPMELot" onchange="onPMELotChange()"><option
value="">Pilih</option></select></div></div><div class="form-row form-row-4"><div class="form-
group"><label>Siklus*</label><select id="mPMESiklus"><option value="">Pilih</option><option
value="Siklus 1">Siklus 1</option><option value="Siklus 2">Siklus 2</option><option value="Siklus
3">Siklus 3</option><option value="Siklus 4">Siklus 4</option><option value="Siklus 5">Siklus
5</option><option value="Siklus 6">Siklus 6</option><option value="Siklus 7">Siklus
7</option><option value="Siklus 8">Siklus 8</option><option value="Siklus 9">Siklus
9</option><option value="Siklus 10">Siklus 10</option><option value="Siklus 11">Siklus
11</option><option value="Siklus 12">Siklus 12</option></select></div><div class="form-
group"><label>Tahun</label><input type="number" id="mPMETahun" value="2025"></div><div
class="form-group"><label>Alat(auto)</label><input type="text" id="mPMEAlat"
readonly></div><div class="form-group"><label>Metode(auto)</label><input type="text"
id="mPMEMetode" readonly></div></div><div class="form-row form-row-3"><div class="form-
group"><label>Satuan(auto)</label><input type="text" id="mPMESatuan" readonly></div><div
class="form-group"><label>TEa(auto)</label><input type="number" step="any" id="mPMETea"
readonly></div></div></div><h4 style="margin:12px 0 6px">Hasil & Mean Peserta</h4><div
class="form-row form-row-3"><div class="form-group"><label>Hasil L1</label><input
type="number" step="any" id="mPMEHasilL1"></div><div class="form-group"><label>Hasil

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L2</label><input type="number" step="any" id="mPMEHasilL2"></div><div class="form-group"><label>Hasil L3</label><input type="number" step="any" id="mPMEHasilL3"></div></div><div class="form-row form-row-3"><div class="form-group"><label>Mean P L1</label><input type="number" step="any" id="mPMEMPL1"></div><div class="form-group"><label>Mean P L2</label><input type="number" step="any" id="mPMEMPL2"></div><div class="form-group"><label>Mean P L3</label><input type="number" step="any" id="mPMEMPL3"></div></div><h4 style="margin:12px 0 6px">CV</h4><div class="form-row form-row-4"><div class="form-group"><label>CV Start</label><input type="date" id="mPMECVStart"></div><div class="form-group"><label>CV End</label><input type="date" id="mPMECVEnd"></div><div class="form-group"><label>CV L1</label><input type="number" step="any" id="mPMECVL1"></div><div class="form-group"><label>CV L2</label><input type="number" step="any" id="mPMECVL2"></div></div><div class="form-row form-row-3"><div class="form-group"><label>CV L3</label><input type="number" step="any" id="mPMECVL3"></div></div></div></div></div><div class="modal-footer"><button class="btn btn-secondary" onclick="cIM('modalPME')">Batal</button><button class="btn btn-primary" onclick="savePME()"><i class="fas fa-save"></i> Simpan</button></div></div></div>

<div class="modal-overlay" id="modalCalcStats"><div class="modal"><div class="modal-header"><h3 id="modalCalcStatsTitle">Calc Stats</h3><button class="modal-close" onclick="cIM('modalCalcStats')">×</button></div><div class="modal-body"><input type="hidden" id="mCSID"><div class="form-row"><div class="form-group"><label>Parameter*</label><select id="mCSParam" onchange="onCSParamChange()"><option value="">Pilih</option></select></div><div class="form-group"><label>Lot QC*</label><select id="mCSLot"><option value="">Pilih</option></select></div></div><div class="form-row"><div class="form-group"><label>Dari*</label><input type="date" id="mCSStart"></div><div class="form-group"><label>Sampai*</label><input type="date" id="mCSEnd"></div></div><div style="padding:12px;background:var(--primary-50);border-radius:8px;border-left:4px solid var(--primary);font-size:.85rem"><i class="fas fa-info-circle" style="color:var(--primary)"></i> Menghitung semua level (L1,L2,L3) sekaligus.</div></div><div class="modal-footer"><button class="btn btn-secondary" onclick="cIM('modalCalcStats')">Batal</button><button class="btn btn-primary" onclick="saveCalcStatsForm()"><i class="fas fa-calculator"></i> Simpan & Hitung</button></div></div></div>

<!--  MODAL SIGMA CV OPT v9.6 (Permintaan #6,7) -->

<div class="modal-overlay" id="modalSigmaCVOpt"><div class="modal modal-lg"><div class="modal-header"><h3 id="modalSCVTitle">% Sigma CV</h3><button class="modal-close" onclick="cIM('modalSigmaCVOpt')">×</button></div><div class="modal-body"><input type="hidden" id="mSCVID"><div class="form-row"><div class="form-group"><label>Parameter*</label><select id="mSCVParam" onchange="onSCVParamChange()"><option value="">Pilih</option></select></div><div class="form-group"><label>Lot QC*</label><select id="mSCVLot" onchange="onSCVLotChange()"><option value="">Pilih</option></select></div></div><div class="form-row form-row-3"><div class="form-group"><label>Alat(auto)</label><input type="text" id="mSCVAlat" readonly></div><div class="form-group"><label>TEa(auto)</label><input

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type="number" step="any" id="mSCVTEa" readonly></div><div class="form-
group"><label>NQC(auto)</label><input type="number" id="mSCVNQC" readonly></div></div>
<!--  SIKLUS & TAHUN PME -->
<div class="form-row form-row-3">
<div class="form-group"><label>Siklus Bias PME</label><select id="mSCVSiklusPME"
onchange="autoFillBiasPME()"><option value="">Pilih</option></select></div>
<div class="form-group"><label>Tahun Siklus</label><select id="mSCVTahunSiklus"
onchange="autoFillBiasPME()"><option value="">Pilih</option></select></div>
<div class="form-group"><label>&nbsp;</label><button class="btn btn-info btn-sm btn-block"
onclick="autoFillBiasPME()"><i class="fas fa-magic"></i> Auto Fill Bias</button></div>
</div>
<!--  BIAS PME L1/L2/L3 -->
<div class="form-row form-row-3">
<div class="form-group"><label>Bias PME L1 (%)</label><input type="number" step="any"
id="mSCVBiasPME1" readonly style="background:var(--bg-soft)"></div>
<div class="form-group"><label>Bias PME L2 (%)</label><input type="number" step="any"
id="mSCVBiasPME2" readonly style="background:var(--bg-soft)"></div>
<div class="form-group"><label>Bias PME L3 (%)</label><input type="number" step="any"
id="mSCVBiasPME3" readonly style="background:var(--bg-soft)"></div>
</div>
<div class="form-row"><div class="form-group"><label>Dari*</label><input type="date"
id="mSCVStart"></div><div class="form-group"><label>Sampai*</label><input type="date"
id="mSCVEnd"></div></div><div class="form-group"><label>Catatan</label><textarea
id="mSCVCatatan" rows="3"></textarea></div></div><div class="modal-footer"><button class="btn
btn-secondary" onclick="cIM('modalSigmaCVOpt')>Batal</button><button class="btn btn-primary"
onclick="saveSigmaCVOptForm()"><i class="fas fa-calculator"></i> Simpan &
Hitung</button></div></div></div>

<div class="modal-overlay" id="modalValidasi"><div class="modal"><div class="modal-
header"><h3>Validasi QC</h3><button class="modal-close"
onclick="cIM('modalValidasi')>&times;</button></div><div class="modal-body"><input
type="hidden" id="mValQCID"><div id="mValDetail" style="margin-bottom:14px"></div><div
class="form-group"><label>Catatan Validasi</label><textarea
id="mValCatatan"></textarea></div></div><div class="modal-footer"><button class="btn btn-
secondary" onclick="cIM('modalValidasi')>Batal</button><button class="btn btn-success"
onclick="submitValidasi()"><i class="fas fa-check"></i> Validasi</button></div></div></div>

<div class="modal-overlay" id="modalUser"><div class="modal" style="max-height:92vh"><div
class="modal-header"><h3 id="modalUserTitle">User</h3><button class="modal-close"
onclick="cIM('modalUser')>&times;</button></div><div class="modal-body"
style="display:flex;flex-direction:column"><input type="hidden" id="mUserIsNew" value="1"><div
style="flex:1;overflow-y:auto"><div class="form-row"><div class="form-
group"><label>Username</label><input type="text" id="mUserUsername"></div><div class="form-
group"><label>Nama</label><input type="text" id="mUserFullName"></div></div><div
class="form-row"><div class="form-group"><label>Email</label><input type="email"

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id="mUserEmail"></div><div class="form-group"><label>Password</label><input type="text"
id="mUserPassword" placeholder="didikqc123"></div></div><div class="form-row form-row-
3"><div class="form-group"><label>Role</label><select id="mUserRole"><option
value="user">User</option><option value="admin">Admin</option><option
value="superadmin">Superadmin</option></select></div><div class="form-
group"><label>Status</label><select id="mUserStatus"><option
value="active">Active</option><option value="pending">Pending</option><option
value="rejected">Rejected</option></select></div><div class="form-group"><label>Masa
Aktif</label><input type="date" id="mUserExpiry"></div></div><div class="form-
group"><label><input type="checkbox" id="mUserImgAccess" style="width:auto;margin-right:6px">
Akses Image Analysis</label></div></div><div style="padding-top:16px;border-top:1px solid var(--
border-soft);display:flex;justify-content:flex-end;gap:10px;flex-shrink:0;margin-top:auto"><button
class="btn btn-secondary" onclick="clM('modalUser')">Batal</button><button class="btn btn-
primary" onclick="saveUserForm()"><i class="fas fa-save"></i>
Simpan</button></div></div></div></div>
<div class="modal-overlay" id="modalUserPwd"><div class="modal"><div class="modal-
header"><h3>Kelola Password Tambahan</h3><button class="modal-close"
onclick="clM('modalUserPwd')">&times;</button></div><div class="modal-body"
style="display:flex;flex-direction:column"><input type="hidden" id="mPwdTargetUser"><input
type="hidden" id="mPwdEditOld" value=""><div style="flex:1;overflow-y:auto"><div
style="background:var(--info-light);padding:10px;border-radius:8px;margin-bottom:14px;font-
size:.85rem"><i class="fas fa-info-circle" style="color:var(--info)"></i> User: <strong
id="mPwdTargetLabel"></strong> — Semua password di bawah sah untuk login.</div><div
class="form-row form-row-3"><div class="form-group"><label>Username*</label><input
type="text" id="mPwdUsername" placeholder="Username login"></div><div class="form-
group"><label>Password Baru*</label><input type="text" id="mPwdNew" placeholder="Min 6
karakter"></div><div class="form-group"><label>Nama*</label><input type="text"
id="mPwdNama" placeholder="Nama pemegang password"></div></div>
<div class="form-row"><div class="form-group"><label>Catatan</label><input type="text"
id="mPwdNote" placeholder="Opsional"></div><div></div></div><button class="btn btn-primary
btn-sm" id="mPwdAddBtn" onclick="addUserPwdAction()" style="margin-bottom:16px"><i
class="fas fa-plus"></i> Tambah Password</button><button class="btn btn-warning btn-sm"
id="mPwdUpdateBtn" onclick="updateUserPwdAction()" style="margin-
bottom:16px;display:none"><i class="fas fa-save"></i> Update Password</button><button
class="btn btn-secondary btn-sm" id="mPwdCancelEditBtn" onclick="cancelEditPwd()"
style="margin-bottom:16px;display:none"><i class="fas fa-times"></i> Batal</button><div
class="table-wrap" style="max-height:250px;overflow-
y:auto"><table><thead><tr><th>No</th><th>Username</th><th>Password</th><th>Nama</th><th>
Catatan</th><th>Status</th><th>Dibuat
Oleh</th><th>Tanggal</th><th>Aksi</th></tr></thead><tbody
id="userPwdTableBody"></tbody></table></div></div><div style="padding-top:16px;border-
top:1px solid var(--border-soft);display:flex;justify-content:flex-end;gap:10px;flex-shrink:0;margin-
top:auto"><button class="btn btn-secondary"
onclick="clM('modalUserPwd')">Tutup</button></div></div></div></div>

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<div class="modal-overlay" id="modallmg"><div class="modal modal-lg"><div class="modal-
header"><h3 id="modallmgTitle">Data Pasien</h3><button class="modal-close"
onclick="cIM('modallmg')">&times;</button></div><div class="modal-body"><input type="hidden"
id="mImgID"><input type="hidden" id="mImgType"><div class="form-row form-row-3"><div
class="form-group"><label>No. RM</label><input type="text" id="mImgNoRM"></div><div
class="form-group"><label>Nama</label><input type="text" id="mImgNama"></div><div
class="form-group"><label>Tgl Lahir</label><input type="date" id="mImgTglLahir"
onchange="calcImgUmur()"></div></div><div class="form-row form-row-4"><div class="form-
group"><label>Umur</label><input type="text" id="mImgUmur" placeholder="Auto"></div><div
class="form-group"><label>JK</label><select id="mImgJK"><option value=""></option><option
value="L">L</option><option value="P">P</option></select></div><div class="form-
group"><label>Tgl Periksa</label><input type="date" id="mImgTglPeriksa"></div><div class="form-
group"><label>Tgl Hasil</label><input type="date" id="mImgTglHasil"></div></div><div
class="form-row"><div class="form-group"><label>Alamat</label><input type="text"
id="mImgAlamat"></div><div class="form-group"><label>Dokter</label><input type="text"
id="mImgDokter"></div></div><div class="form-row"><div class="form-
group"><label>Diagnosis</label><input type="text" id="mImgDiagnosis"></div><div class="form-
group"><label>Asal</label><input type="text" id="mImgAsal"></div></div><div
id="imgFieldsHemato" style="display:none"><h4 style="margin:12px 0 8px;color:var(--
primary)">Hasil Hematologi</h4><div class="form-group"><label>Eritrosit</label><textarea
id="mImgEritrosit" rows="2"></textarea></div><div class="form-
group"><label>Leukosit</label><textarea id="mImgLeukosit" rows="2"></textarea></div><div
class="form-group"><label>Trombosit</label><textarea id="mImgTrombosit"
rows="2"></textarea></div><div class="form-group"><label>Kesan</label><textarea
id="mImgKesan" rows="2"></textarea></div><div class="form-
group"><label>Saran</label><textarea id="mImgSaran" rows="2"></textarea></div></div><div
id="imgFieldsUrin" style="display:none"><h4 style="margin:12px 0 8px;color:var(--primary)">Hasil
Urin</h4><div class="form-group"><label>Interpretasi</label><textarea id="mImgInterpretasiUrin"
rows="4"></textarea></div></div><div id="imgFieldsMalaria" style="display:none"><h4
style="margin:12px 0 8px;color:var(--primary)">Malaria</h4><div class="form-row form-row-
3"><div class="form-group"><label>Hasil Mikro</label><input type="text"
id="mImgHasilMikroMalaria"></div><div class="form-group"><label>Spesies</label><input
type="text" id="mImgSpesies"></div><div class="form-group"><label>Kepadatan</label><input
type="text" id="mImgKepadatanMalaria"></div></div></div><div id="imgFieldsBTA"
style="display:none"><h4 style="margin:12px 0 8px;color:var(--primary)">BTA</h4><div
class="form-row"><div class="form-group"><label>Hasil Mikro</label><input type="text"
id="mImgHasilMikroBTA"></div><div class="form-group"><label>Kepadatan</label><input
type="text" id="mImgKepadatanBTA"></div></div></div><div id="imgFieldsLain"
style="display:none"><h4 style="margin:12px 0 8px;color:var(--primary)">Analisis</h4><div
class="form-group"><label>Interpretasi</label><textarea id="mImgInterpretasiLain"
rows="4"></textarea></div></div><h4 style="margin:16px 0 8px;color:var(--primary)"><i class="fas
fa-camera"></i> Gambar (maks 3)</h4><div class="multi-img-grid"><div class="img-slot"
id="imgSlot1" onclick="G('imgFileInput1').click()"><div class="slot-label">Gambar 1</div><i
class="fas fa-image"></i><p>Klik upload</p><input type="file" id="imgFileInput1" accept="image/*"
style="display:none" onchange="handleMultiImgUpload(event,1)"><button class="remove-img"

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onclick="event.stopPropagation();removeImgSlot(1)">&times;</button></div><div class="img-slot"
id="imgSlot2" onclick="G('imgFileInput2').click()"><div class="slot-label">Gambar 2</div><i
class="fas fa-image"></i><p>Klik upload</p><input type="file" id="imgFileInput2" accept="image/*"
style="display:none" onchange="handleMultiImgUpload(event,2)"><button class="remove-img"
onclick="event.stopPropagation();removeImgSlot(2)">&times;</button></div><div class="img-slot"
id="imgSlot3" onclick="G('imgFileInput3').click()"><div class="slot-label">Gambar 3</div><i
class="fas fa-image"></i><p>Klik upload</p><input type="file" id="imgFileInput3" accept="image/*"
style="display:none" onchange="handleMultiImgUpload(event,3)"><button class="remove-img"
onclick="event.stopPropagation();removeImgSlot(3)">&times;</button></div></div><input
type="hidden" id="mImgURL"><input type="hidden" id="mImgURL2"><input type="hidden"
id="mImgURL3"></div><div class="modal-footer"><div class="print-options-wrap"
style="position:relative;display:inline-block">
<button class="btn btn-info btn-sm" onclick="toggleImgPrintMenu(this)"><i class="fas fa-
print"></i></button>
<div class="print-options-menu">
<button onclick="printImgForm('withImg')"><i class="fas fa-image"></i> Hasil + Gambar +
TTD</button>
<button onclick="printImgForm('noImg')"><i class="fas fa-file-alt"></i> Hasil + TTD</button>
</div>
</div>
<div class="print-options-wrap" style="position:relative;display:inline-block">
<button class="btn btn-danger btn-sm" onclick="toggleImgPrintMenu(this)"><i class="fas fa-file-
pdf"></i></button>
<div class="print-options-menu">
<button onclick="pdfImgForm('withImg')"><i class="fas fa-image"></i> Hasil + Gambar +
TTD</button>
<button onclick="pdfImgForm('noImg')"><i class="fas fa-file-alt"></i> Hasil + TTD</button>
</div>
</div>
<button class="btn btn-secondary" onclick="clM('modallmg')">Batal</button><button class="btn
btn-primary" onclick="saveImgForm()"><i class="fas fa-save"></i>
Simpan</button></div></div></div>

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<div class="modal-overlay" id="modallmgView"><div class="modal modal-xl"><div class="modal-
header" style="background:linear-gradient(135deg,var(--success),#059669)"><h3>Hasil
Pemeriksaan</h3><button class="modal-close"
onclick="clM('modallmgView')">&times;</button></div><div class="modal-body"
style="padding:0"><div id="imgViewContent" style="padding:24px"></div></div><div class="modal-
footer"><div class="print-options-wrap" style="position:relative;display:inline-block">
<button class="btn btn-primary" onclick="toggleImgPrintMenu(this)"><i class="fas fa-print"></i>
Print ▼</button>
<div class="print-options-menu">
<button onclick="printImgResultWithImg()"><i class="fas fa-image"></i> Hasil + Gambar +
TTD</button>
<button onclick="printImgResultNoImg()"><i class="fas fa-file-alt"></i> Hasil + TTD</button>

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</div>
</div>
<div class="print-options-wrap" style="position:relative;display:inline-block">
<button class="btn btn-danger" onclick="toggleImgPrintMenu(this)"><i class="fas fa-file-pdf"></i>
PDF ▼</button>
<div class="print-options-menu">
<button onclick="pdfImgResultWithImg()"><i class="fas fa-image"></i> Hasil + Gambar +
TTD</button>
<button onclick="pdfImgResultNoImg()"><i class="fas fa-file-alt"></i> Hasil + TTD</button>
</div>
</div>
<button class="btn btn-secondary"
onclick="clM('modallmgView')">Tutup</button></div></div></div>

```

```

<div class="modal-overlay" id="modallmgPatologi"><div class="modal modal-xl"><div class="modal-
header"><h3 id="modalPatologiTitle">Patologi Anatomi</h3><button class="modal-close"
onclick="clM('modallmgPatologi')">&times;</button></div><div class="modal-body"><input
type="hidden" id="mPatologiID"><div class="form-row form-row-3"><div class="form-
group"><label>No. RM</label><input type="text" id="mPatologiNoRM"></div><div class="form-
group"><label>No. PA</label><input type="text" id="mPatologiNoPA"></div><div class="form-
group"><label>Nama Pasien</label><input type="text" id="mPatologiNama"></div></div><div
class="form-row form-row-3"><div class="form-group"><label>Tgl Lahir</label><input type="date"
id="mPatologiTglLahir" onchange="calcPatologiUmur()"></div><div class="form-
group"><label>Umur</label><input type="text" id="mPatologiUmur"
placeholder="Auto"></div><div class="form-group"><label>JK</label><select
id="mPatologiJK"><option value="">-</option><option value="L">Laki-laki</option><option
value="P">Perempuan</option></select></div></div><div class="form-row"><div class="form-
group"><label>Alamat</label><input type="text" id="mPatologiAlamat"></div><div class="form-
group"><label>Asal Ruangan</label><input type="text"
id="mPatologiAsalRuangan"></div></div><div class="form-row form-row-3"><div class="form-
group"><label>Tgl Terima</label><input type="date" id="mPatologiTglTerima"></div><div
class="form-group"><label>Tgl Jawab</label><input type="date"
id="mPatologiTglJawab"></div><div class="form-group"><label>Asal Rujukan</label><input
type="text" id="mPatologiAsalRujukan"></div></div><div class="form-row form-row-3"><div
class="form-group"><label>Status Biaya</label><select id="mPatologiStatusBiaya"><option
value="">Pilih</option><option value="Umum">Umum</option><option
value="BPJS">BPJS</option><option value="Asuransi">Asuransi</option><option
value="Pribadi">Pribadi</option></select></div><div class="form-group"><label>Dokter
Pengirim</label><input type="text" id="mPatologiDokterPengirim"></div><div class="form-
group"><label>Jenis Pemeriksaan</label><select id="mPatologiJenisPemeriksaan"><option
value="">Pilih</option><option value="Histologi">Histologi</option><option
value="Papsmear">Papsmear</option><option value="FNAB/FNAC">FNAB/FNAC</option><option
value="Biopsi">Biopsi</option><option value="Sitologi">Sitologi</option><option value="Frozen
Section">Frozen Section</option><option value="Autopsi">Autopsi</option><option>

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value="Lainnya">Lainnya</option></select></div></div><div class="form-
group"><label>Diagnosis</label><textarea id="mPatologiDiagnosis"
rows="2"></textarea></div><h4 style="margin:16px 0 8px;color:var(--primary)"><i class="fas fa-file-
medical-alt"></i> Template Hasil</h4><div class="form-group"><label>Pilih Template</label><div
style="display:flex;flex-wrap:wrap;gap:8px;margin-bottom:12px"><label style="font-
size:.85rem;display:flex;align-items:center;gap:4px"><input type="checkbox"
id="tplPatologiMakroskopis" style="width:auto"> Makroskopis</label><label style="font-
size:.85rem;display:flex;align-items:center;gap:4px"><input type="checkbox"
id="tplPatologiMikroskopis" style="width:auto"> Mikroskopis</label><label style="font-
size:.85rem;display:flex;align-items:center;gap:4px"><input type="checkbox" id="tplPatologiKesan"
style="width:auto"> Kesan</label><label style="font-size:.85rem;display:flex;align-
items:center;gap:4px"><input type="checkbox" id="tplPatologiSaran" style="width:auto">
Saran</label><label style="font-size:.85rem;display:flex;align-items:center;gap:4px"><input
type="checkbox" id="tplPatologiCatatan" style="width:auto"> Catatan</label><label style="font-
size:.85rem;display:flex;align-items:center;gap:4px"><input type="checkbox"
id="tplPatologiTopografi" style="width:auto"> Topografi</label><label style="font-
size:.85rem;display:flex;align-items:center;gap:4px"><input type="checkbox"
id="tplPatologiMorfologi" style="width:auto"> Morfologi</label></div></div><div class="form-
group"><label>Makroskopis</label><textarea id="mPatologiMakroskopis" rows="3"
placeholder="Deskripsi makroskopis..."></textarea></div><div class="form-
group"><label>Mikroskopis</label><textarea id="mPatologiMikroskopis" rows="4"
placeholder="Deskripsi mikroskopis..."></textarea></div><div class="form-row"><div class="form-
group"><label>Topografi (ICD-O)</label><input type="text" id="mPatologiTopografi"
placeholder="C00-C99"></div><div class="form-group"><label>Morfologi (ICD-O)</label><input
type="text" id="mPatologiMorfologi" placeholder="M800-M999"></div></div><div class="form-
group"><label>Kesan</label><textarea id="mPatologiKesan" rows="3"
placeholder="Kesan/kesimpulan..."></textarea></div><div class="form-
group"><label>Saran</label><textarea id="mPatologiSaran" rows="2" placeholder="Saran tindak
lanjut..."></textarea></div><div class="form-group"><label>Catatan</label><textarea
id="mPatologiCatatan" rows="2" placeholder="Catatan tambahan..."></textarea></div><h4
style="margin:16px 0 8px;color:var(--primary)"><i class="fas fa-camera"></i> Gambar (maks
5)</h4><div class="multi-img-grid" style="grid-template-columns:repeat(3,1fr)"><div class="img-
slot" id="patologiSlot1" onclick="G('patologiFileInput1').click()"><div class="slot-label">Gambar
1</div><i class="fas fa-image"></i><p>Klik upload</p><input type="file" id="patologiFileInput1"
accept="image/*" style="display:none" onchange="handlePatologiImgUpload(event,1)"><button
class="remove-img"
onclick="event.stopPropagation();removePatologiImgSlot(1)">&times;</button></div><div
class="img-slot" id="patologiSlot2" onclick="G('patologiFileInput2').click()"><div class="slot-
label">Gambar 2</div><i class="fas fa-image"></i><p>Klik upload</p><input type="file"
id="patologiFileInput2" accept="image/*" style="display:none"
onchange="handlePatologiImgUpload(event,2)"><button class="remove-img"
onclick="event.stopPropagation();removePatologiImgSlot(2)">&times;</button></div><div
class="img-slot" id="patologiSlot3" onclick="G('patologiFileInput3').click()"><div class="slot-
label">Gambar 3</div><i class="fas fa-image"></i><p>Klik upload</p><input type="file"
id="patologiFileInput3" accept="image/*" style="display:none"

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onchange="handlePatologiImgUpload(event,3)"><button class="remove-img"
onclick="event.stopPropagation();removePatologiImgSlot(3)">&times;</button></div><div
class="img-slot" id="patologiSlot4" onclick="G('patologiFileInput4').click()"><div class="slot-
label">Gambar 4</div><i class="fas fa-image"></i><p>Klik upload</p><input type="file"
id="patologiFileInput4" accept="image/*" style="display:none"
onchange="handlePatologiImgUpload(event,4)"><button class="remove-img"
onclick="event.stopPropagation();removePatologiImgSlot(4)">&times;</button></div><div
class="img-slot" id="patologiSlot5" onclick="G('patologiFileInput5').click()"><div class="slot-
label">Gambar 5</div><i class="fas fa-image"></i><p>Klik upload</p><input type="file"
id="patologiFileInput5" accept="image/*" style="display:none"
onchange="handlePatologiImgUpload(event,5)"><button class="remove-img"
onclick="event.stopPropagation();removePatologiImgSlot(5)">&times;</button></div></div><input
type="hidden" id="mPatologiURL1"><input type="hidden" id="mPatologiURL2"><input
type="hidden" id="mPatologiURL3"><input type="hidden" id="mPatologiURL4"><input
type="hidden" id="mPatologiURL5"></div><div class="modal-footer"><div class="print-options-
wrap" style="position:relative;display:inline-block">
<button class="btn btn-info btn-sm" onclick="toggleImgPrintMenu(this)"><i class="fas fa-
print"></i></button>
<div class="print-options-menu">
<button onclick="printPatologiForm('withImg')"><i class="fas fa-image"></i> Hasil + Gambar +
TTD</button>
<button onclick="printPatologiForm('noImg')"><i class="fas fa-file-alt"></i> Hasil + TTD</button>
</div>
</div>
<div class="print-options-wrap" style="position:relative;display:inline-block">
<button class="btn btn-danger btn-sm" onclick="toggleImgPrintMenu(this)"><i class="fas fa-file-
pdf"></i></button>
<div class="print-options-menu">
<button onclick="pdfPatologiForm('withImg')"><i class="fas fa-image"></i> Hasil + Gambar +
TTD</button>
<button onclick="pdfPatologiForm('noImg')"><i class="fas fa-file-alt"></i> Hasil + TTD</button>
</div>
</div>
<button class="btn btn-secondary" onclick="clM('modalImgPatologi')">Batal</button><button
class="btn btn-primary" onclick="saveImgPatologiForm()"><i class="fas fa-save"></i>
Simpan</button></div></div></div>
<div class="modal-overlay" id="modalRestore"><div class="modal modal-lg"><div class="modal-
header" style="background:linear-gradient(135deg,var(--warning),#d97706)"><h3><i class="fas fa-
history"></i> Restore</h3><button class="modal-close"
onclick="clM('modalRestore')">&times;</button></div><div class="modal-body"><div
style="background:var(--warning-light);padding:12px;border-radius:8px;margin-
bottom:14px;border-left:4px solid var(--warning)"><strong>☐</strong> Restore menimpa
data.</div><button class="btn btn-primary btn-sm" onclick="loadBackupList()" style="margin-
bottom:12px"><i class="fas fa-sync"></i> Refresh</button><div id="restoreList" class="restore-

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section"><p>Memuat...</p></div></div><div class="modal-footer"><button class="btn btn-
secondary" onclick="clM('modalRestore')">Tutup</button></div></div></div>
<script>
var CU=null,CD={},CP='dashboard';
window.addEventListener('DOMContentLoaded',function(){loadLoginSettings();});
var chartInstances={};
var autoLogoutTimer=null,autoLogoutSec=180;
var hapusUnlocked=false,confirmCallback=null;
var importData=[];
var smartTableState={paramID:null,lotID:null,dates:[],existingMap:{}};
var lastGraphData=null,lastSigmaBasedData=null,lastLaporanData=null;
var sidebarCollapsedState=false;
var inputQCCache=[],pmeCache=[],csCache=[],scvCache=[];
var imgDataCache={};
function G(id){return document.getElementById(id);}
function Q(s){return document.querySelector(s);}
function QA(s){return document.querySelectorAll(s);}
function escH(s){if(!s&&s!=="")return "<";var
d=document.createElement('div');d.textContent=String(s);return d.innerHTML;}
function round(v,d){if(v===null || v===undefined || v==="")return "<";d=d || 2;return
parseFloat(Number(v).toFixed(d));}
function toast(msg,type,dur){type=type || 'info';dur=dur || 3500;var
c=G('toastContainer'),t=document.createElement('div');var icons={success:'fa-check-circle',error:'fa-
times-circle',warning:'fa-exclamation-triangle',info:'fa-info-circle'};t.className='toast toast-
'+type;t.innerHTML='<i class="fas '+icons[type]+' '+fa-info-
circle'+""></i><span>'+msg+'</span>';t.onclick=function(){t.classList.add('out');setTimeout(function()
){try{c.removeChild(t);}catch(e){}},300);c.appendChild(t);setTimeout(function(){t.classList.add('out')
};setTimeout(function(){try{c.removeChild(t);}catch(e){}},300);},dur);}
function
sl(m){G('loadingText').textContent=m || 'Memproses...';G('loadingOverlay').classList.add('show');}
function hl(){G('loadingOverlay').classList.remove('show');}
function opM(id){var m=G(id);if(m)m.classList.add('show');}
function clM(id){var m=G(id);if(m)m.classList.remove('show');}
function
cfm(t,m,i,y,c){G('confirmTitle').textContent=t || 'Konfirmasi';G('confirmMsg').textContent=m || 'Yaki
n?';G('confirmIcon').textContent=i || '❏';G('confirmYesBtn').textContent=y || 'Ya';confirmCallback=cb;
G('confirmOverlay').classList.add('show');}
function clConfirm(r){G('confirmOverlay').classList.remove('show');if(confirmCallback){var
c=confirmCallback;confirmCallback=null;c(r);}}
function fmtDate(d){if(!d)return '-';var dt=new Date(d);if(isNaN(dt.getTime()))return d;return
String(dt.getDate()).padStart(2,'0')+'/' +String(dt.getMonth()+1).padStart(2,'0')+'/' +dt.getFullYear();}
function fmtDateIndo(d){if(!d)return '-';var dt=new Date(d);if(isNaN(dt.getTime()))return d;var
b=['Januari','Februari','Maret','April','Mei','Juni','Juli','Agustus','September','Oktober','November','De
seMBER'];return dt.getDate()+ ' '+b[dt.getMonth()+1]+' '+dt.getFullYear();}

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function todayISO(){var d=new Date();return d.getFullYear()+'-
'+String(d.getMonth()+1).padStart(2,'0')+'-'+String(d.getDate()).padStart(2,'0');}
function sigmaClass(s){if(s===null || s===undefined || s==="")return "";s=parseFloat(s);if(s>=6)return
'sigma-world';if(s>=5)return 'sigma-excellent';if(s>=4)return 'sigma-good';if(s>=3)return 'sigma-
marginal';if(s>=2)return 'sigma-poor';return 'sigma-unacceptable';}
function sigmaLabel(s){if(s===null || s===undefined || s==="")return '-';s=parseFloat(s);if(s>=6)return
'World Class';if(s>=5)return 'Excellent';if(s>=4)return 'Good';if(s>=3)return 'Marginal';if(s>=2)return
'Poor';return 'Unacceptable';}
function sigmaColor(s){if(s===null || s===undefined)return '#94a3b8';s=parseFloat(s);if(s>=6)return
'#047857';if(s>=5)return '#2563eb';if(s>=4)return '#7c3aed';if(s>=3)return '#f59e0b';if(s>=2)return
'#ef4444';return '#991b1b';}
function zClass(z){if(z===null || z===undefined)return '';z=Math.abs(parseFloat(z));if(z>3)return 'z-
reject';if(z>2)return 'z-warn';return 'z-ok';}
function getParamName(id){if(!CD.params)return id;var p=CD.params.find(function(x){return
x.paramID===id;});return p?p.parameter:id;}
function getLotByIDLocal(id){if(!CD.lots)return null;return CD.lots.find(function(x){return
x.lotID===id;});}
function destroyChart(k){if(chartInstances[k]){chartInstances[k].destroy();delete chartInstances[k];}}
function togglePwdVisibility(inputId,iconEl){var
inp=G(inputId);if(inp.type==='password'){inp.type='text';iconEl.classList.remove('fa-
eye');iconEl.classList.add('fa-eye-slash');}else{inp.type='password';iconEl.classList.remove('fa-eye-
slash');iconEl.classList.add('fa-eye');}}
function switchLoginTab(tab){QA('.login-
tab').forEach(function(t){t.classList.remove('active');});QA('.login-
form').forEach(function(f){f.classList.remove('active');});if(tab==='login'){QA('.login-
tab')[0].classList.add('active');G('loginForm').classList.add('active');}else{QA('.login-
tab')[1].classList.add('active');G('registerForm').classList.add('active');}}
function doLogin(e){e.preventDefault();var
u=G('loginUsername').value.trim(),p=G('loginPassword').value;if(!u || !p){toast('Username dan
password
wajib','warning');return;}sl('Memverifikasi...');G('loginBtn').disabled=true;google.script.run.withSucc
essHandler(function(r){hl();G('loginBtn').disabled=false;if(!r.ok){if(r.msg==='EXPIRED'){G('expiredMsg
').textContent='Akun berakhir pada
'+fmtDate(r.expiryDate);G('expiredModal').classList.add('show');}else
toast(r.msg,'error');return;}CU={username:r.username,fullName:r.fullName,role:r.role,email:r.email,
expiryDate:r.expiryDate,imgAnalAccess:r.imgAnalAccess,loginAsName:r.loginAsName || "",loginUsern
ame:r.loginUsername || ""};if(r.loginAsName || r.loginUsername){G('loginAsWrap').style.display="";G('l
oginAsName').textContent=(r.loginUsername?'@'+r.loginUsername:'')+r.loginAsName?'
'+r.loginAsName+'':""};toast('Selamat datang,
'+(r.loginAsName || r.fullName)+'!', 'success');}else{G('loginAsWrap').style.display='none';toast('Selam
at datang,
'+r.fullName+'!', 'success');}enterApp();}).withFailureHandler(function(e){hl();G('loginBtn').disabled=f
alse;toast('Error: '+e.message || e || 'Terjadi kesalahan'),'error');}).loginUser(u,p);}
function doRegister(e){e.preventDefault();var
data={username:G('regUsername').value.trim(),password:G('regPassword').value,fullName:G('regFull

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IName').value.trim(),email:G('regEmail').value.trim());if(!data.username||!data.password||!data.fullName||!data.email){toast('Semua
wajib','warning');return;}if(data.password.length<6){toast('Password min
6','warning');return;}sl('Mendaftar...');G('regBtn').disabled=true;google.script.run.withSuccessHandle
r(function(r){hl();G('regBtn').disabled=false;if(r.ok){toast(r.msg,'success');switchLoginTab('login');G('r
egisterForm').reset();}else
toast(r.msg,'error');}).withFailureHandler(function(e){hl();G('regBtn').disabled=false;toast('Error:
'+(e.message||e)||'Terjadi kesalahan'),'error');}).registerUser(data);}

```

//  FUNGSI BARU v9.6: Reset semua UI ke state awal saat logout

```

function resetAllUI(){
try{
var dsg=G('dashStatGrid'); if(dsg) dsg.innerHTML="";
var dtd=G('dashTrendDetail'); if(dtd) dtd.innerHTML='<p style="color:var(--text-
secondary)">Memuat...</p>';
var dwp=G('dashWGPanel'); if(dwp) dwp.innerHTML='<p style="color:var(--text-
secondary)">Memuat...</p>';
var dw=G('dashWeekly'); if(dw) dw.textContent='0';
var dm=G('dashMonthly'); if(dm) dm.textContent='0';
var bp=G('badgePending'); if(bp){bp.textContent='0';bp.style.display='none';}
var bpu=G('badgePendingUsers'); if(bpu){bpu.textContent='0';bpu.style.display='none';}
['paramTableBody','lotTableBody','inputQCTableBody','valTableBody','histTableBody',
'teaTableBody','pmeTableBody','csTableBody','scvTableBody','usersTableBody','logTableBody',
'imgHematoBody','imgUrinBody','imgMalariaBody','imgBTABody','imgLainBody','imgPatologiBody',
'hapusInputQCTable','hapusLotQCTable','hapusParamTable'].forEach(function(id){
var el=G(id); if(el) el.innerHTML="";
});
['grafikResult','laporanResult','trendResult','tabulasiResult','opspecsResult'].forEach(function(id){
var el=G(id); if(el) el.style.display='none';
});
['laporan','trend','tabulasi','opspecs'].forEach(function(p){hideExportBtns(p)});
['qcFilterStart','qcFilterEnd','qcFilterBidang','qcFilterParam','qcFilterLot',
'histStart','histEnd','histBidang','histParam','histLot','histFilter',
'grafStart','grafEnd','grafBidang','grafParam','grafLot','grafSumber','grafSumberSigma',
'grafSiklusPME','grafTahunSiklus','grafPeriodeCS','grafPeriodeCVOpt',
'lapStart','lapEnd','lapBidang','lapParam','lapLot','lapAlat',
'trendStart','trendEnd','trendBidang','trendParam','trendLot','trendYear',
'hapusQCStart','hapusQCEnd','importPwd','bulkPwd','smartPwd',
'testEmailTo','hapusGatePwd','hapusAlasan','icParamFilter',
'valParamFilter','pmeParamFilter','csParamFilter','scvParamFilter',
'opsBidangFilter','filterBidangParam','filterParamLot',
'imgPatologiStart','imgPatologiEnd','imgPatologiRM','imgPatologiNama','imgPatologiPA'].forEach(fun
ction(id){
var el=G(id); if(el) el.value="";
});

```

```

['modalParam','modalLot','modalTEa','modalPME','modalCalcStats','modalSigmaCVOpt',
'modalValidasi','modalUser','modalImg','modalImgView','modalImgPatologi','modalRestore'].forEach
(function(id){
  clM(id);
});
var tabChk=G('tabParamChecklist'); if(tabChk) tabChk.innerHTML="";
var hapusChk=G('hapusParamChecklist'); if(hapusChk) hapusChk.innerHTML="";
var stw=G('smartTableWrap'); if(stw) stw.innerHTML=<p style="padding:24px;text-align:center;color:var(--text-secondary)">Pilih dulu lalu klik Tampilkan</p>;
var btw=G('bulkTableWrap'); if(btw) btw.innerHTML="";
var ip=G('importPreview'); if(ip){ip.innerHTML="";ip.style.display='none';}
var bsb=G('btnSubmitBulk'); if(bsb) bsb.style.display='none';
var bss=G('btnSubmitSmart'); if(bss) bss.style.display='none';
var bcs=G('btnClearSmart'); if(bcs) bcs.style.display='none';
var bis=G('btnImportSubmit'); if(bis) bis.style.display='none';
var ieq=G('inputEditQcID'); if(ieq) ieq.value="";
['inputL1','inputL2','inputL3'].forEach(function(id){var el=G(id);if(el)el.value="});
var bsi=G('btnSaveInput'); if(bsi) bsi.innerHTML=<i class="fas fa-save"></i> Simpan';
var bce=G('btnCancelEdit'); if(bce) bce.style.display='none';
var gc=G('grafCatatan'); if(gc) gc.value="";
var lsc=G('ljSigmaCard'); if(lsc) lsc.style.display='none';
var cisb=G('cardInterpSigmaBased'); if(cisb) cisb.style.display='none';
var bcks=G('btnCekSigma'); if(bcks) bcks.style.display='none';
var tc=G('tabCatatan'); if(tc){tc.value="";tc.dataset.periodKey="};
var puw=G('pendingUsersWrap'); if(puw) puw.style.display='none';
var pul=G('pendingUsersList'); if(pul) pul.innerHTML="";
var klp=G('kopLogoPreview'); if(klp) klp.innerHTML="";
var sat=G('setAppTheme'); if(sat) sat.value='light';
QA('.page').forEach(function(p){ p.classList.remove('active'); p.classList.remove('print-active'); });
var pd=G('pageDashboard'); if(pd) pd.classList.add('active');
QA('.nav-item').forEach(function(n){ n.classList.remove('active'); });
var dn=Q('.nav-item[data-page="dashboard"]'); if(dn) dn.classList.add('active');
var tt=G('topbarTitle'); if(tt) tt.textContent='Dashboard';
QA('.nav-group').forEach(function(g){ g.classList.remove('open'); });
var tabs=QA('#pageInputqc .tab-btn');
if(tabs.length){
  tabs.forEach(function(b,i){b.classList.toggle('active',i===0)});
  var ts=G('tabSingle'); if(ts) ts.classList.add('active');
  var tb=G('tabBulk'); if(tb) tb.classList.remove('active');
  var tsm=G('tabSmart'); if(tsm) tsm.classList.remove('active');
}
var htabs=QA('#pageHapusdata .tab-btn');
if(htabs.length){
  htabs.forEach(function(b,i){b.classList.toggle('active',i===0)});
}

```

```

var hti=G('hapusTabInputqc'); if(hti) hti.classList.add('active');
var htl=G('hapusTabLotqc'); if(htl) htl.classList.remove('active');
var http=G('hapusTabParameter'); if(http) http.classList.remove('active');
var vaw=G('viewAsWrap'); if(vaw) vaw.style.display='none';
var law=G('loginAsWrap'); if(law) law.style.display='none';
var gia=G('grpImageAnalysis'); if(gia) gia.style.display='none';
var nu=G('navUsers'); if(nu) nu.style.display='none';
var ns=G('navSettings'); if(ns) ns.style.display='none';
}catch(e){ console.warn('resetAllUI error:', e); }
}

```

function

```

loadLoginSettings(){google.script.run.withSuccessHandler(function(s){if(!s)return;try{if(s.login_font_size)document.documentElement.style.setProperty('--login-font-size',s.login_font_size+'px');if(s.login_font_color)document.documentElement.style.setProperty('--login-font-color',s.login_font_color);if(s.login_font_weight)document.documentElement.style.setProperty('--login-font-weight',s.login_font_weight);if(s.login_brightness)document.documentElement.style.setProperty('--login-brightness',s.login_brightness);if(s.login_bg_color)document.documentElement.style.setProperty('--login-bg-color',s.login_bg_color);}catch(e){}}).withFailureHandler(function({}).getLoginSettings();}
function doLogout(){
resetAllUI();
CU=null; CD={}; CP='dashboard';
inputQCCache=[]; pmeCache=[]; csCache=[]; scvCache=[];
imgDataCache={}; importData=[]; smartTableState={paramID:null,lotID:null,dates:[],existingMap:{}};
lastGraphData=null; lastSigmaBasedData=null; lastLaporanData=null;
Object.keys(chartInstances).forEach(function(k){ destroyChart(k); });
chartInstances={};
G('appPage').classList.remove('show');
G('appPage').style.display='none';
G('loginPage').style.display='flex';
G('expiredModal').classList.remove('show');
clearAutoLogout();
G('loginForm').reset();
G('registerForm').reset();
sidebarCollapsedState=false;
try{ localStorage.removeItem('didiqc_sidebar_collapsed'); }catch(e){}
hapusUnlocked=false;
if(G('hapusGate')) G('hapusGate').style.display='';
if(G('hapusContent')) G('hapusContent').style.display='none';
var vas=G('viewAsSelect'); if(vas) vas.value='';
try{
document.documentElement.style.setProperty('--primary','#2563eb');

```

```

document.documentElement.style.setProperty('--sidebar-bg', '#0f172a');
document.documentElement.style.setProperty('--sidebar-bg-2', '#1e293b');
document.documentElement.style.setProperty('--sidebar-bg-rgb', '15,23,42');
document.documentElement.style.setProperty('--sidebar-bg-2-rgb', '30,41,59');
document.documentElement.style.setProperty('--sb-font-size', '14px');
document.documentElement.style.setProperty('--sb-font-weight', '500');
document.documentElement.style.setProperty('--sb-font-color', '#f1f5f9');
document.body.classList.remove('dark-mode');
document.body.classList.add('light-mode');
} catch (e) {}
}

```

```

function enterApp(){
G('loginPage').style.display='none';
G('appPage').style.display='flex';
G('appPage').classList.add('show');
applyRole();
sl('Memuat data '+CU.fullName+'...');
loadInitData();
startAutoLogout();
try{var
saved=localStorage.getItem('didiqc_sidebar_collapsed');if(saved==='1')applySidebarCollapsed(true);}
catch(e){}
}

```

//  PERBAIKAN v9.6: applyRole dengan hide navSettings (Permintaan #12)

```

function applyRole(){
if(!CU)return;
G('sbUserName').textContent=CU.fullName;
G('sbUserRole').textContent=CU.role;
G('sbUserAvatar').textContent=CU.fullName.charAt(0).toUpperCase();
var isSA=CU.role==='superadmin';
G('navUsers').style.display=isSA?":none";
G('navSettings').style.display=isSA?":none";
G('viewAsWrap').style.display=isSA?":none";
var
imgAccess=CU.imgAnalAccess===true || CU.imgAnalAccess==='TRUE' || CU.imgAnalAccess==='true';
G('grpImageAnalysis').style.display=imgAccess?":none";
}

```

```

function
startAutoLogout(){clearAutoLogout();autoLogoutSec=CD.settings?parseInt(CD.settings.auto_logout)|
180:180;resetAutoLogout();['click','keydown','mousemove','touchstart'].forEach(function(ev){docu
ment.addEventListener(ev,resetAutoLogout,{passive:true});});}

```



```

function
resetAutoLogout(){if(autoLogoutTimer)clearTimeout(autoLogoutTimer);autoLogoutTimer=setTimeo
ut(function(){toast('Sesi berakhir','warning');doLogout();},autoLogoutSec*1000);}
function
clearAutoLogout(){if(autoLogoutTimer){clearTimeout(autoLogoutTimer);autoLogoutTimer=null;}}

```

```

function loadInitData(){
sl('Memuat data...');
Object.keys(chartInstances).forEach(function(k){ destroyChart(k); });
google.script.run.withSuccessHandler(function(r){
hl();
if(!r || !r.ok){ toast('Gagal memuat data: '+{r.msg || 'Unknown error'},'error'); return; }
CD.params=r.params || [];
CD.lots=r.lots || [];
CD.teaList=r.teaList || [];
CD.settings=r.settings || {};
CD._backupTriggerActive=r.backupTriggerActive || false;
if(CD.settings.backup_auto==='true'&&r.backupTriggerActive===false){google.script.run.withSuccess
Handler(function({}).withFailureHandler(function({}).setupAutoBackupTrigger();}
CD.kop=r.kop || {};
CD.archiveYears=r.archiveYears || [];
CD.allUsers=r.allUsers || [];
CD.appLogo=r.appLogo || "";
if(r.userInfo){
CU.imgAnalAccess=r.userInfo.imgAnalAccess;
applyRole();
}
applyTheme();
loadUserTheme();
populateFilters();
populateViewAs();
populateTrendYear();
goPage('dashboard');
}).withFailureHandler(function(e){
hl();
toast('Error: '+{e.message || e || 'Terjadi kesalahan'}+'. Mencoba ulang...','error');
setTimeout(function(){ loadInitData(); }, 2000);
}).getInitData(CU.username, CU.role, CU.role);
}

```

```

function applyUserTheme(){
var theme = CD.settings && CD.settings.app_theme ? CD.settings.app_theme : 'light';
var body = document.body;
body.classList.remove('dark-mode', 'light-mode');
if(theme === 'dark') { body.classList.add('dark-mode'); }

```

```

else if(theme === 'light') { body.classList.add('light-mode'); }
else if(theme === 'system') {
if(window.matchMedia '&& window.matchMedia('(prefers-color-scheme: dark)').matches) {
body.classList.add('dark-mode'); }
else { body.classList.add('light-mode'); }
}
try { localStorage.setItem('didiqc_theme', theme); } catch(e) {}
}

```

```

function loadUserTheme(){
try {
var savedTheme = localStorage.getItem('didiqc_theme');
if(savedTheme && CD.settings) {
CD.settings.app_theme = savedTheme;
if(G('setAppTheme')) G('setAppTheme').value = savedTheme;
}
} catch(e) {}
}

```

```

function applyTheme(){
var s=CD.settings|{};
if(s.theme_primary){
document.documentElement.style.setProperty('--primary',s.theme_primary);
var rgb = hexToRgb(s.theme_primary);
if(rgb) document.documentElement.style.setProperty('--primary-rgb', rgb.r+', '+rgb.g+', '+rgb.b);
}
if(s.sidebar_bg){
document.documentElement.style.setProperty('--sidebar-bg',s.sidebar_bg);
var sbRgb = hexToRgb(s.sidebar_bg);
if(sbRgb){
document.documentElement.style.setProperty('--sidebar-bg-rgb', sbRgb.r+', '+sbRgb.g+', '+sbRgb.b);
var sb2Rgb = {
r: Math.min(255, sbRgb.r + 20),
g: Math.min(255, sbRgb.g + 20),
b: Math.min(255, sbRgb.b + 20)
};
document.documentElement.style.setProperty('--sidebar-bg-2-rgb',
sb2Rgb.r+', '+sb2Rgb.g+', '+sb2Rgb.b);
document.documentElement.style.setProperty('--sidebar-bg-2', rgbToHex(sb2Rgb.r, sb2Rgb.g,
sb2Rgb.b));
}
}
if(s.sb_font_size)document.documentElement.style.setProperty('--sb-font-size',s.sb_font_size+'px');
if(s.sb_font_weight)document.documentElement.style.setProperty('--sb-font-
weight',s.sb_font_weight);

```

```

if(s.sb_font_color)document.documentElement.style.setProperty('--sb-font-color',s.sb_font_color);
if(s.auto_logout)autoLogoutSec=parseInt(s.auto_logout)||180;
applyUserTheme();
if(s.login_brightness)document.documentElement.style.setProperty('--login-
brightness',s.login_brightness);
if(s.login_bg_color)document.documentElement.style.setProperty('--login-bg-
color',s.login_bg_color);
if(s.login_font_size)document.documentElement.style.setProperty('--login-font-
size',s.login_font_size+'px');
if(s.login_font_color)document.documentElement.style.setProperty('--login-font-
color',s.login_font_color);
if(s.login_font_weight)document.documentElement.style.setProperty('--login-font-
weight',s.login_font_weight);
}
function setLoginBrightness(val,btn){document.querySelectorAll('.brightness-
btn').forEach(function(b){b.classList.remove('active');});if(btn)btn.classList.add('active');G('loginBright
Val').textContent=btn?btn.getAttribute('data-
label')||val:val;document.documentElement.style.setProperty('--login-brightness',val);}

```

```

function hexToRgb(hex) {
var result = /^#?([a-f\d]{2})([a-f\d]{2})([a-f\d]{2})$/i.exec(hex);
return result ? { r: parseInt(result[1], 16), g: parseInt(result[2], 16), b: parseInt(result[3], 16) } : null;
}

```

```

function rgbToHex(r, g, b) {
return '#' + [r, g, b].map(function(x) {
var hex = x.toString(16);
return hex.length === 1 ? '0' + hex : hex;
}).join("");
}

```

```

function showExportBtns(prefix){var ids;if(prefix==='grafik')return;else
if(prefix==='laporan')ids=['lapPrintBtn2','lapPdfBtn2','lapSSBtn2'];else
if(prefix==='trend')ids=['trendExportBtns','trendExportBtns2','trendExportBtns3'];else
if(prefix==='tabulasi')ids=['tabPrintBtn','tabPdfBtn','tabSSBtn'];else
if(prefix==='opspecs')ids=['opsPrintBtn','opsPdfBtn','opsSSBtn'];else
return;ids.forEach(function(id){var el=G(id);if(el)el.style.display="";});
function hideExportBtns(prefix){var ids;if(prefix==='grafik')return;else
if(prefix==='laporan')ids=['lapPrintBtn2','lapPdfBtn2','lapSSBtn2'];else
if(prefix==='trend')ids=['trendExportBtns','trendExportBtns2','trendExportBtns3'];else
if(prefix==='tabulasi')ids=['tabPrintBtn','tabPdfBtn','tabSSBtn'];else
if(prefix==='opspecs')ids=['opsPrintBtn','opsPdfBtn','opsSSBtn'];else
return;ids.forEach(function(id){var el=G(id);if(el)el.style.display='none';});
function populateViewAs(){if(CU.role!=='superadmin')return;var
sel=G('viewAsSelect');sel.innerHTML='<option value="">— Kembali ke Akun Saya —

```

```

</option>';(CD.allUsers || []).forEach(function(u){if(u.username!==CU.username)sel.innerHTML+ '<option value="'+escH(u.username)+'">'+escH(u.fullName)+'</option>';});}
function getActiveUsername(){if(CU.role==='superadmin'){var v=G('viewAsSelect').value;if(v)return v;}return CU.username;}
function getLogUser(){return CU.loginUsername || '';}
function getActiveRole(){if(CU.role==='superadmin'&&G('viewAsSelect').value)return 'user';return CU.role;}

```

```

//  PERBAIKAN v9.5: onViewAsChange - Reset total & muat data akun baru
function onViewAsChange(){
  // 1. Reset semua UI & DOM ke state awal (bersihkan sisa akun lama)
  resetAllUI();
  // 2. Reset semua cache data
  inputQCCache=[];pmeCache=[];csCache=[];scvCache=[];
  imgDataCache={};
  lastGraphData=null;lastSigmaBasedData=null;lastLaporanData=null;
  // 3. Destroy semua chart yang aktif
  Object.keys(chartInstances).forEach(function(k){destroyChart(k)});
  chartInstances={};
  // 4. Muat data dari akun yang dipilih (getActiveUsername() sudah return akun target)
  sl('Memuat data akun...');
  google.script.run.withSuccessHandler(function(r){
    hl();
    if(!r || !r.ok){toast('Gagal memuat data','error');return;}
    // Update data global dengan data akun baru
    CD.params=r.params || [];CD.lots=r.lots || [];CD.teaList=r.teaList || [];
    CD.settings=r.settings || {};CD.kop=r.kop || {};CD.archiveYears=r.archiveYears || [];
    CD.allUsers=r.allUsers || [];CD.appLogo=r.appLogo || '';
    if(r.userInfo){CU.imgAnalAccess=r.userInfo.imgAnalAccess;applyRole();}
    // Apply tema & populate filter dengan data baru
    applyTheme();loadUserTheme();
    populateFilters();populateViewAs();populateTrendYear();
    // Muat halaman yang sedang aktif dengan data akun baru
    loadCurrentPage();
    toast('Data akun dimuat','success');
  }).withFailureHandler(function(e){
    hl();toast('Error: '+e.message || e || 'Terjadi kesalahan','error');
  }).getInitData(getActiveUsername(),getActiveRole(), CU.role);
}

```

```

function toggleSidebarMode(){var
isMobile=window.innerWidth<=768;if(isMobile){G('sidebar').classList.toggle('show-
mobile');G('sidebarOverlay').classList.toggle('show');}else{sidebarCollapsedState=!sidebarCollapsedS
tate;applySidebarCollapsed(sidebarCollapsedState);try{localStorage.setItem('didiqc_sidebar_collaps
ed',sidebarCollapsedState?'1':'0');}catch(e){}}}

```

```

function
applySidebarCollapsed(state){sidebarCollapsedState=state;if(state){G('sidebar').classList.add('collapsed');G('mainContent').classList.add('expanded');}else{G('sidebar').classList.remove('collapsed');G('mainContent').classList.remove('expanded');}}
function closeMobileSidebar(){G('sidebar').classList.remove('show-mobile');G('sidebarOverlay').classList.remove('show');}
function toggleNavGroup(id){var el=G(id);if(el)el.classList.toggle('open');}
function populateTrendYear() {
var sel = G('trendTahun');
if (!sel) return;
sel.innerHTML = "";
var now = new Date().getFullYear();
for (var y = now; y >= now - 5; y--) sel.innerHTML += '<option value="' + y + ">' + y + '</option>';
sel.value = now;

// Populate Bulan Awal & Bulan Akhir
var bulanNames = ['Januari', 'Februari', 'Maret', 'April', 'Mei', 'Juni', 'Juli', 'Agustus', 'September', 'Oktober', 'November', 'Desember'];
var selBA = G('trendBulanAwal');
var selBK = G('trendBulanAkhir');
if (selBA) {
selBA.innerHTML = "";
bulanNames.forEach(function(n, i) { selBA.innerHTML += '<option value="' + (i + 1) + ">' + n + '</option>'; });
selBA.value = '1';
}
if (selBK) {
selBK.innerHTML = "";
bulanNames.forEach(function(n, i) { selBK.innerHTML += '<option value="' + (i + 1) + ">' + n + '</option>'; });
selBK.value = '12';
}

// Populate Siklus PME & Tahun Siklus
populateTrendSiklusFilters();
}
function populateTrendSiklusFilters() {
google.script.run.withSuccessHandler(function(data) {
var sel = G('trendSiklusPME'); if (!sel) return;
sel.innerHTML = '<option value="">Semua</option>';
(data || []).forEach(function(s) { sel.innerHTML += '<option value="' + escH(s) + ">' + escH(s) + '</option>'; });
}).withFailureHandler(function(e){hl();toast('Error: '+(e.message||e)||'Terjadi kesalahan'),'error'});}).getSiklusPMEList(getActiveUsername(), getActiveRole());
google.script.run.withSuccessHandler(function(data) {

```

```

var sel = G('trendTahunSiklus'); if (!sel) return;
sel.innerHTML = '<option value="">Semua</option>';
(data || []).forEach(function(t) { sel.innerHTML += '<option value="' + escH(t) + '">' + escH(t) +
'</option>'; });
}).withFailureHandler(function(e){hl();toast('Error: '+(e.message || e) || 'Terjadi
kesalahan'),'error');}).getTahunSiklusList(getActiveUsername(), getActiveRole());
}
var
SUBMENU_MAP={parameters:'grpDaftarParam',lotqc:'grpDaftarParam',biaspme:'grpDaftarParam',c
alcstats:'grpDaftarParam',sigmacvopt:'grpDaftarParam',daftartea:'grpDaftarParam',trendanalisis:'grp
DashAnalisis',instrcompare:'grpDashAnalisis',tabulasi:'grpDashAnalisis',opspecs:'grpDashAnalisis',im
ghemato:'grplImageAnalysis',imgurin:'grplImageAnalysis',imgmalaria:'grplImageAnalysis',imgbta:'grpl
ImageAnalysis',imgpatologi:'grplImageAnalysis',imglain:'grplImageAnalysis'};

function goPage(page){
    //  PERBAIKAN: Bersihkan konten halaman sebelumnya
    clearPageContent(CP);

    if(CP==='hapusdata'&&page!=='hapusdata'){hapusUnlocked=false;G('hapusGate').style.display="";G('
hapusContent').style.display='none';}
    CP=page;QA('.page').forEach(function(p){p.classList.remove('active');p.classList.remove('print-
active');});
    var
    pageEl=G('page'+page.charAt(0).toUpperCase()+page.slice(1));if(pageEl)pageEl.classList.add('active');
    QA('.nav-item').forEach(function(n){n.classList.remove('active');});QA('.nav-item[data-
page="'+page+'"]').forEach(function(n){n.classList.add('active');});
    var parentGroup=SUBMENU_MAP[page];if(parentGroup){var
    grpEl=G(parentGroup);if(grpEl&&!grpEl.classList.contains('open'))grpEl.classList.add('open');}
    var titles={dashboard:'Dashboard',parameters:'Parameter',lotqc:'Lot QC',inputqc:'Input
QC',grafik:'Grafik & Analisis',laporan:'Laporan',trendanalisis:'Trend
Analisis',instrcompare:'Instrument Compare',tabulasi:'Tabulasi Rekap',opspecs:'Analisis
OPSpecs',validasi:'Validasi QC',histori:'Histori QC',daftartea:'Daftar TEa',biaspme:'Bias
PME',calcstats:'Calc Stats',sigmacvopt:'% Sigma CV',smartimport:'Smart Import',hapusdata:'Hapus
Data',users:'Users',kopsurat:'Kop Surat',settings:'Pengaturan',logactivity:'Log
Aktivitas',imgghemato:'Hematologi Sel',imgurin:'Urin
Sedimen',imgmalaria:'Malaria',imgbta:'BTA',imgpatologi:'Patologi Anatomi',imglain:'Analisis Lain'};

    G('topbarTitle').textContent=titles[page] || page;if(window.innerWidth<=768)closeMobileSidebar();lo
adCurrentPage();
}

//  FUNGSI BARU: Bersihkan konten halaman saat berpindah menu
function clearPageContent(page){
    if(!page) return;

```

```

try{
  switch(page){
    case 'dashboard':
      var dsg=G('dashStatGrid'); if(dsg) dsg.innerHTML="";
      var dtd=G('dashTrendDetail'); if(dtd) dtd.innerHTML='<p style="color:var(--text-secondary)">Memuat...</p>';
      var dwp=G('dashWGPanel'); if(dwp) dwp.innerHTML='<p style="color:var(--text-secondary)">Memuat...</p>';
      var dw=G('dashWeekly'); if(dw) dw.textContent='0';
      var dm=G('dashMonthly'); if(dm) dm.textContent='0';
      ['chartSigmaBidang','chartCVBiasBidang'].forEach(function(id){destroyChart(id)});
      break;
    case 'parameters':
      var ptb=G('paramTableBody'); if(ptb) ptb.innerHTML="";
      break;
    case 'lotqc':
      var ltb=G('lotTableBody'); if(ltb) ltb.innerHTML="";
      break;
    case 'inputqc':
      var iqt=G('inputQCTableBody'); if(iqt) iqt.innerHTML="";
      var btw=G('bulkTableWrap'); if(btw) btw.innerHTML="";
      var stw=G('smartTableWrap'); if(stw) stw.innerHTML='<p style="padding:24px;text-align:center;color:var(--text-secondary)">Pilih dulu lalu klik Tampilkan</p>';
      var bsb=G('btnSubmitBulk'); if(bsb) bsb.style.display='none';
      var bss=G('btnSubmitSmart'); if(bss) bss.style.display='none';
      var bcs=G('btnClearSmart'); if(bcs) bcs.style.display='none';
      break;
    case 'grafik':
      G('grafikResult').style.display='none';
      G('btnCekSigma').style.display='none';
      G('ljSigmaCard').style.display='none';
      G('cardInterpSigmaBased').style.display='none';

    if(G('grafKopContainer')){G('grafKopContainer').style.display='none';G('grafKopContainer').innerHTML="";}

    lastGraphData=null;lastSigmaBasedData=null;
    ['ljChartUnified','ljChartSigma','chartSigmaMetric'].forEach(function(id){destroyChart(id)});
    if(G('grafStatsBody'))G('grafStatsBody').innerHTML="";
    if(G('grafQGIBody'))G('grafQGIBody').innerHTML="";
    if(G('grafWGBody'))G('grafWGBody').innerHTML="";
    if(G('sigmaInfoBox'))G('sigmaInfoBox').innerHTML="";
    if(G('interpSigmaBasedBody'))G('interpSigmaBasedBody').innerHTML="";
    if(G('grafSumberInfo'))G('grafSumberInfo').innerHTML="";
    if(G('grafSigmaSourceInfo'))G('grafSigmaSourceInfo').innerHTML="";
    if(G('grafCatatan'))G('grafCatatan').value="";
  }
}

```

```

        break;
    case 'laporan':
        G('laporanResult').style.display='none';
        if(G('laporanContent'))G('laporanContent').innerHTML="";
        G('laporanCatatanDokterCard').style.display='none';
        if(G('laporanCatatanDokter'))G('laporanCatatanDokter').value="";
        lastLaporanData=null;
        break;
    case 'trendanalysis':
        G('trendResult').style.display='none';
        if(G('trendContent'))G('trendContent').innerHTML="";
        break;
    case 'instrcompare':
        if(G('instrCompareBody'))G('instrCompareBody').innerHTML='<p style="color:var(--text-secondary)">Klik Compare</p>';
        break;
    case 'tabulasi':
        G('tabulasiResult').style.display='none';
        if(G('tabulasiKopContainer'))G('tabulasiKopContainer').innerHTML="";
        if(G('tabRekapTable'))G('tabRekapTable').innerHTML="";
        if(G('tabPMETable'))G('tabPMETable').innerHTML="";
        if(G('tabInterpretasi'))G('tabInterpretasi').innerHTML="";
        if(G('tabCatatan')){G('tabCatatan').value="";G('tabCatatan').dataset.periodKey="";}
        break;
    case 'opspecs':
        G('opspecsResult').style.display='none';
        if(G('opspecsKopContainer'))G('opspecsKopContainer').innerHTML="";
        if(G('opsTableWrap'))G('opsTableWrap').innerHTML="";
        if(G('opsAnalisisDetail'))G('opsAnalisisDetail').innerHTML="";
        if(G('opsKesimpulan'))G('opsKesimpulan').innerHTML="";
        ['chartOPSspecs','chartCritError'].forEach(function(id){destroyChart(id)});
        break;
    case 'validasi':
        var vtb=G('valTableBody'); if(vtb) vtb.innerHTML="";var vbb=G('valBulkBar'); if(vbb)
vbb.style.display='none';
        break;
    case 'histori':
        var htb=G('histTableBody'); if(htb) htb.innerHTML="";
        break;
    case 'daftartea':
        var ttb=G('teaTableBody'); if(ttb) ttb.innerHTML="";
        break;
    case 'biaspme':
        var ptb=G('pmeTableBody'); if(ptb) ptb.innerHTML="";
        break;

```



```

case 'calcstats':
    var ctb=G('csTableBody'); if(ctb) ctb.innerHTML="";
    break;
case 'sigmacvopt':
    var stb=G('scvTableBody'); if(stb) stb.innerHTML="";
    break;
case 'users':
    var utb=G('usersTableBody'); if(utb) utb.innerHTML="";
    var puw=G('pendingUsersWrap'); if(puw) puw.style.display='none';
    var pul=G('pendingUsersList'); if(pul) pul.innerHTML="";
    var bpu=G('badgePendingUsers'); if(bpu){bpu.textContent='0';bpu.style.display='none';}
    break;
case 'logactivity':
    var ltb=G('logTableBody'); if(ltb) ltb.innerHTML="";
    break;
case 'hapusdata':
    hapusUnlocked=false;
    if(G('hapusGate')) G('hapusGate').style.display="";
    if(G('hapusContent')) G('hapusContent').style.display='none';
    if(G('hapusInputQCTable')) G('hapusInputQCTable').innerHTML="";
    if(G('hapusLotQCTable')) G('hapusLotQCTable').innerHTML="";
    if(G('hapusParamTable')) G('hapusParamTable').innerHTML="";
    if(G('hapusAlasan')) G('hapusAlasan').value="";
    if(G('hapusGatePwd')) G('hapusGatePwd').value="";
    break;
case 'imghemato':
    var ihb=G('imgHematoBody'); if(ihb) ihb.innerHTML="";
    break;
case 'imgurin':
    var iub=G('imgUrinBody'); if(iub) iub.innerHTML="";
    break;
case 'imgmalaria':
    var imb=G('imgMalariaBody'); if(imb) imb.innerHTML="";
    break;
case 'imgbta':
    var ibb=G('imgBTABody'); if(ibt) ibb.innerHTML="";
    break;
case 'imglain':
    var ilb=G('imgLainBody'); if(ilb) ilb.innerHTML="";
    break;
// FIX v9.8.1: Tambah clear untuk patologi
case 'imgpatologi':
    var ipb=G('imgPatologiBody'); if(ipb) ipb.innerHTML="";
    break;
case 'smartimport':

```



```

tion value="" + escH(p.paramID) + ">" + escH(p.parameter) + (p.bidang ?
['+p.bidang+'].') + "</option>";});s.value=v;
});
var hpc=G('hapusParamChecklist');
if(hpc){var hhtml="";(CD.params || []).forEach(function(p){hhtml+=" Panggil populateGrafikFilters
populateGrafikFilters();
}

//  FUNGSI BARU v9.6: Populate filter grafik (Permintaan #2,3,4,5)
function populateGrafikFilters(){
google.script.run.withSuccessHandler(function(data){
var sel=G('grafSiklusPME');if(!sel)return;
sel.innerHTML=<option value="">Semua</option>;
(data || []).forEach(function(s){sel.innerHTML+="

```

```
//  FUNGSI BARU v9.6: Populate Siklus & Tahun di modal Sigma CV Opt (Permintaan #6)
function populateSCVSiklusTahun(){
google.script.run.withSuccessHandler(function(data){
var sel=G('mSCVSiklusPME');if(!sel)return;
sel.innerHTML=<option value="">Pilih</option>;
(data || []).forEach(function(s){sel.innerHTML+=<option value="" +esch(s)+" ">+esch(s)+</option>;});
}).withFailureHandler(function(e){hl();toast('Error: '+e.message | |e | 'Terjadi
kesalahan'),'error');}).getSiklusPMEList(getActiveUsername(),getActiveRole());
google.script.run.withSuccessHandler(function(data){
var sel=G('mSCVTahunSiklus');if(!sel)return;
sel.innerHTML=<option value="">Pilih</option>;
(data || []).forEach(function(t){sel.innerHTML+=<option value="" +esch(t)+" ">+esch(t)+</option>;});
}).withFailureHandler(function(e){hl();toast('Error: '+e.message | |e | 'Terjadi
kesalahan'),'error');}).getTahunSiklusList(getActiveUsername(),getActiveRole());
}
}
```

```
// FUNGSI BARU v9.6: Auto-fill Bias PME (Permintaan #7)
function autoFillBiasPME(){
var paramID=G('mSCVParam').value;
var lotID=G('mSCVLot').value;
var siklus=G('mSCVSiklusPME').value;
var tahun=G('mSCVTahunSiklus').value;
if(!paramID || !lotID || !siklus || !tahun){
G('mSCVBiasPME1').value="";G('mSCVBiasPME2').value="";G('mSCVBiasPME3').value="";return;
}
sl('Mengambil data Bias PME...');
google.script.run.withSuccessHandler(function(r){
hl();
if(r&&r.ok){
G('mSCVBiasPME1').value=r.biasL1!==null?r.biasL1:"";
G('mSCVBiasPME2').value=r.biasL2!==null?r.biasL2:"";
G('mSCVBiasPME3').value=r.biasL3!==null?r.biasL3:"";
toast('Bias PME berhasil diisi','success',2000);
}else{
G('mSCVBiasPME1').value="";G('mSCVBiasPME2').value="";G('mSCVBiasPME3').value="";
toast(r?r.msg:'Data PME tidak ditemukan','warning');
}
}).withFailureHandler(function(e){
hl();G('mSCVBiasPME1').value="";G('mSCVBiasPME2').value="";G('mSCVBiasPME3').value="";
toast('Error: '+e.message | |e | 'Terjadi kesalahan'),'error');
}).getBiasPMEByFilter(paramID,lotID,siklus,tahun,getActiveUsername());
}
}
```

```

function filterParamByBidang(bidangSelId,paramSelId){var bidang=G(bidangSelId).value;var
s=G(paramSelId);s.innerHTML='<option
value="">Pilih</option>';(CD.params | []).forEach(function(p){if(bidang&&(p.bidang | 'Lainnya')!==bi
dang)return;s.innerHTML+='<option
value="'+esch(p.paramID)+'">'+esch(p.parameter)+'</option>'}});
function populateLotSelect(selId,paramID){var s=G(selId);s.innerHTML='<option value="">Pilih
Lot</option>';(CD.lots | []).filter(function(l){return
l.paramID===paramID;}).forEach(function(l){s.innerHTML+='<option
value="'+esch(l.lotID)+'">'+esch(l.noLot)+' — '+esch(l.namaAlat)+'</option>'}});
function onInputParamChange(){populateLotSelect('inputLot',G('inputParam').value);}
function onBulkParamChange(){populateLotSelect('bulkLot',G('bulkParam').value);}
function onSmartParamChange(){populateLotSelect('smartLot',G('smartParam').value);}
function onGrafParamChange(){populateLotSelect('grafLot',G('grafParam').value);}
function onLapParamChange(){populateLotSelect('lapLot',G('lapParam').value);}
function onCSPParamChange(){populateLotSelect('mCSLot',G('mCSPParam').value);}
function onHistParamChange(){populateLotSelect('histLot',G('histParam').value);}
function onQCFilterParamChange(){populateLotSelect('qcFilterLot',G('qcFilterParam').value);}
function onTrendParamChange(){populateLotSelect('trendLot',G('trendParam').value);}
function
onPMEParamChange(){populateLotSelect('mPMELot',G('mPMEParam').value);G('mPMEAlat').value='
';G('mPMEMethod').value="";G('mPMESatuan').value="";G('mPMETea').value="";}
function onPMELotChange(){var
lot=getLotByIDLocal(G('mPMELot').value);if(lot){G('mPMEAlat').value=lot.namaAlat | "";G('mPMEMet
hode').value=lot.methode | "";G('mPMESatuan').value=lot.satuan | "";G('mPMETea').value=lot.tea | ";
}}
function
onSCVParamChange(){populateLotSelect('mSCVLot',G('mSCVParam').value);G('mSCVAlat').value="";G
('mSCVTEa').value="";G('mSCVNQC').value="";}
function onSCVLotChange(){var
lot=getLotByIDLocal(G('mSCVLot').value);if(lot){G('mSCVAlat').value=lot.namaAlat | "";G('mSCVTEa').v
alue=lot.tea | "";G('mSCVNQC').value=3;}}
function autoFillTEa(){var p=G('mLotParam').value;if(!p)return;var
t=(CD.teaList | []).find(function(x){return x.paramID===p;});if(t)G('mLotTEa').value=t.nilaiTEa;}

function loadDashboard(){if(!CU | !CU.username){toast('Sesi belum
aktif','warning');return;}sl('Memuat
dashboard...');google.script.run.withSuccessHandler(function(r){hl();if(!r | !r.ok){toast('Gagal','error')
;return;}renderDashStats(r.stats);renderSigmaBidangChart(r.sigmaByBidang);renderCVBiasBidangCh
art(r.cvBiasByBidang);renderWGPanel(r.wgViolations);renderDashTrendDetail(r.trendDetail);G('dash
Weekly').textContent=r.weeklyQC | 0;G('dashMonthly').textContent=r.monthlyQC | 0;var
pending=r.stats.pending | 0;var
bp=G('badgePending');if(pending>0){bp.textContent=pending;bp.style.display="";}else
bp.style.display='none';}).withFailureHandler(function(e){hl();toast('Error: '+e.message | e | 'Terjadi
kesalahan'),'error');}).getDashboardData(getActiveUsername(),getActiveRole());}

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function renderDashStats(s){var cards=[{label:'Parameter',value:s.totalParam,icon:'fa-list-ul',cls:'info'},{label:'Lot QC',value:s.totalLot,icon:'fa-box',cls:'info'},{label:'Total QC',value:s.totalQC,icon:'fa-database',cls:''},{label:'Hari Ini',value:s.todayQC,icon:'fa-calendar-day',cls:'success'},{label:'Tervalidasi',value:s.validated,icon:'fa-check-circle',cls:'success'},{label:'Pending',value:s.pending,icon:'fa-clock',cls:'warning'},{label:'Expired',value:s.expired,icon:'fa-calendar-times',cls:'danger'},{label:'Near Expiry',value:s.nearExpiry,icon:'fa-exclamation-circle',cls:'warning'}];G('dashStatGrid').innerHTML=cards.map(function(c){return '<div class="stat-card '+c.cls+'"><div class="stat-icon"><i class="fas '+c.icon+'"></i></div><div class="stat-info"><div class="stat-value">'+c.value+'</div><div class="stat-label">'+c.label+'</div></div></div>'}).join("");}
function
renderSigmaBidangChart(data){destroyChart('sigmaBidang');if(!data || !Object.keys(data).length)return;var
labels=Object.keys(data),ds1=[],ds2=[],ds3=[];labels.forEach(function(b){ds1.push(data[b].sigL1);ds2.push(data[b].sigL2);ds3.push(data[b].sigL3)});chartInstances.sigmaBidang=new
Chart(G('chartSigmaBidang').getContext('2d'),{type:'bar',data:{labels:labels,datasets:[{label:'L1',data:ds1,backgroundColor:'rgba(37,99,235,.75)',borderRadius:6},{label:'L2',data:ds2,backgroundColor:'rgba(245,158,11,.75)',borderRadius:6},{label:'L3',data:ds3,backgroundColor:'rgba(239,68,68,.75)',borderRadius:6}]}},options:{responsive:true,maintainAspectRatio:false,plugins:{annotation:{annotations:[{line6:{type:'line',yMin:6,yMax:6,borderColor:'#047857',borderDash:[5,5],label:{display:true,content:'6σ',position:'end'}},line3:{type:'line',yMin:3,yMax:3,borderColor:'#f59e0b',borderDash:[5,5],label:{display:true,content:'3σ',position:'end'}}]}]}},scales:{y:{beginAtZero:true,title:{display:true,text:'Sigma'}}}}});}
function
renderCVBiasBidangChart(data){destroyChart('cvBiasBidang');if(!data || !Object.keys(data).length)return;var
labels=Object.keys(data),cvs=[],biases=[];labels.forEach(function(b){cvs.push(data[b].avgCV);biases.push(data[b].avgBias)});chartInstances.cvBiasBidang=new
Chart(G('chartCVBiasBidang').getContext('2d'),{type:'bar',data:{labels:labels,datasets:[{label:'Avg CV(%)',data:cvs,backgroundColor:'rgba(37,99,235,.7)',borderRadius:6},{label:'Avg Bias(%)',data:biases,backgroundColor:'rgba(139,92,246,.7)',borderRadius:6}]}},options:{responsive:true,maintainAspectRatio:false,scales:{y:{beginAtZero:true}}}});}
function renderWGPanel(violations){var
p=G('dashWGPanel');if(!violations || !violations.length){p.innerHTML='<p style="color:var(--success);font-weight:600"><i class="fas fa-check-circle"></i> Tidak ada pelanggaran 30 hari terakhir</p>';return;}var html="";violations.forEach(function(v){var isWarn=v.rule==='1-2s';var catBadge="";if(v.category==='Random Error')catBadge='<span class="badge badge-random">RE</span>';else if(v.category==='Systematic Error')catBadge='<span class="badge badge-systematic">SE</span>';html+='<div class="wg-item"><span class="wg-rule-badge '+isWarn?'warning':'">'+<escH(v.rule)>+</span>'+catBadge+'<div><strong>'+<escH(v.parameter)>+</strong> — '+<escH(v.namaAlat)>+ — '+<escH(v.level)>+(v.sigma!==null?'(σ='+v.sigma+'):'')>+<br><small>'+<escH(v.desc)>+</small></div><small style="margin-left:auto;color:var(--text-secondary)">'+<fmtDate(v.tanggal)>+</small></div>');p.innerHTML=html;}
function renderDashTrendDetail(data){var
w=G('dashTrendDetail');if(!data || !Object.keys(data).length){w.innerHTML='<p style="color:var(--

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text-secondary)">Belum ada data bulan ini</p>;return;}var
html=";Object.keys(data).sort().forEach(function(bidang){var
items=data[bidang];if(!items || !items.length)return;var uid='dt_'+bidang.replace(/[^a-zA-Z0-9]/g, '_');html+=<div class="collapse-toggle" onclick="toggleCollapse(\'"+uid+"\',this)"><span><i
class="fas fa-folder"></i> '+escH(bidang)+' ('+items.length+')</span><span class="arrow"><i
class="fas fa-chevron-down"></i></span></div>;html+=<div class="collapse-content"
id="'+uid+'"><div class="table-
wrap"><table><thead><tr><th>Parameter</th><th>Alat</th><th>Level</th><th>N</th><th>CV%</t
h><th>Bias%</th><th>TE%</th><th> $\sigma$ </th><th>TEa</th><th>WG</th></tr></thead><tbody>;items
.forEach(function(item){[1,2,3].forEach(function(lv){var
d=item.levels['L'+lv];if(!d)return;html+=<tr><td>'+escH(item.parameter)+'</td><td>'+escH(item.na
maAlat)+'</td><td>'+lv+'</td><td>'+d.n+'</td><td>'+round(d.cv)+'</td><td>'+round(d.bias)+'</td>
<td>'+round(d.te)+'</td><td class="'+sigmaClass(d.sigma)+'">'+(d.sigma!==null?round(d.sigma):-
')+</td><td>'+(d.tea || '-')+'</td><td>'+(d.violations>0?'<span class="badge badge-
danger">'+d.violations+'</span>':'<span class="badge badge-
success">OK</span>')+'</td></tr>;});html+=</tbody></table></div></div>;});w.innerHTML=html
;}}
function toggleCollapse(id,btn){var
el=G(id);if(!el)return;el.classList.toggle('show');if(btn)btn.classList.toggle('open');}
function renderParamTable(){var bidang=G('filterBidangParam')?G('filterBidangParam').value:";var
params=(CD.params || []).filter(function(p){if(bidang&&(p.bidang || 'Lainnya')!==bidang)return
false;return true;});var
html=";params.forEach(function(p,i){html+=<tr><td>'+(i+1)+'</td><td>'+escH(p.parameter)+'</td><
td><span class="badge badge-
info">'+escH(p.bidang || 'Lainnya')+'</span></td><td>'+escH(p.createdDate)+'</td><td class="table-
actions"><button class="btn btn-warning btn-xs" onclick="editParam(\'"+p.paramID+"\')"><i
class="fas fa-edit"></i></button> <button class="btn btn-danger btn-xs"
onclick="delParam(\'"+p.paramID+"\')"><i class="fas fa-
trash"></i></button></td></tr>;});if(!params.length)html=<tr><td colspan="5" style="text-
align:center;color:var(--text-secondary)">Belum ada
parameter</td></tr>;G('paramTableBody').innerHTML=html;}
function
openParamModal(data){G('mParamID').value=data?data.paramID:";G('mParamName').value=data?
data.parameter:";G('mParamBidang').value=data?(data.bidang || ""):";G('modalParamTitle').textCont
ent=data?'Edit Parameter':'Tambah Parameter';opM('modalParam');}
function editParam(id){var p=(CD.params || []).find(function(x){return
x.paramID===id;});if(p)openParamModal(p);}
function saveParam(){var
payload={paramID:G('mParamID').value || null,parameter:G('mParamName').value.trim(),bidang:G('
mParamBidang').value.trim();if(!payload.parameter){toast('Nama
wajib','warning');return;}sl('Menyimpan...');google.script.run.withSuccessHandler(function(r){hl();if(r.
ok){toast('Tersimpan','success');clM('modalParam');refreshParams();}else
toast(r.msg,'error');}).withFailureHandler(function(e){hl();toast('Error: '+e.message || e || 'Terjadi
kesalahan'),'error');}).saveParameter(payload,getActiveUsername(),getLogUser());}

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function delParam(id){cfm('Hapus
Parameter','Lanjutkan?', '❑', 'Hapus',function(yes){if(!yes)return;sl('Menghapus...');google.script.run.
withSuccessHandler(function(r){hl();if(r.ok){toast('Dihapus', 'success');refreshParams();}else
toast(r.msg,'error');}).withFailureHandler(function(e){hl();toast('Error: '+e.message| |e| | 'Terjadi
kesalahan'),'error');}).deleteParameter(id,getActiveUsername(),getLogUser());});}
function
refreshParams(){google.script.run.withSuccessHandler(function(r){CD.params=r| |[];populateFilters()
;renderParamTable();}).withFailureHandler(function(e){toast('Error: '+e.message| |e| | 'Terjadi
kesalahan'),'error');}).getParameters(getActiveUsername(),getActiveRole());}
function renderLotTable(){var pf=G('filterParamLot')?G('filterParamLot').value:'';var
lots=(CD.lots| |[]).filter(function(l){if(pf&&l.paramID!=pf)return false;return true;});var now=new
Date(),d30=new Date(now.getTime()+30*86400000);var html="";lots.forEach(function(l,i){var
exp=l.expiredDate?new Date(l.expiredDate):null;var sb=exp&&exp<now?'<span class="badge badge-
danger">Expired</span>':exp&&exp<=d30?'<span class="badge badge-warning">Near
Exp</span>':<span class="badge badge-success">Aktif</span>;var
sumberBadge="";if(l.sumber==='Manufaktur')sumberBadge='<span class="badge badge-
info">Manufaktur</span>';else if(l.sumber==='Terhitung berjalan')sumberBadge='<span
class="badge badge-warning">Berjalan</span>';else if(l.sumber==='Terhitung
Fix')sumberBadge='<span class="badge badge-success">Fix</span>';else sumberBadge='<span
class="badge badge-secondary">'+escH(l.sumber| |'-
')+</span>';html+=<tr><td>'+(i+1)+'</td><td>'+escH(getParamName(l.paramID))+</td><td>'+escH(
l.noLot)+</td><td>'+escH(l.namaAlat)+</td><td>'+escH(l.methode)+</td><td>'+escH(l.satuan)+</
td><td>'+sumberBadge+'</td><td>'+fmtDate(l.expiredDate)+'</td><td>'+(l.tea| |'-
')+</td><td>'+sb+'</td><td class="table-actions"><button class="btn btn-warning btn-xs"
onclick="editLot(\''+l.lotID+'\'")><i class="fas fa-edit"></i></button> <button class="btn btn-danger
btn-xs" onclick="delLot(\''+l.lotID+'\'")><i class="fas fa-
trash"></i></button></td></tr>');if(!lots.length)html=<tr><td colspan="11" style="text-
align:center;color:var(--text-secondary)">Belum ada
lot</td></tr>;G('lotTableBody').innerHTML=html;}
function
openLotModal(data){['LotID','NoLot','Alat','Methode','Satuan','Expired','TEa','Bias','MeanL1','SDL1','T
argetL1','MeanL2','SDL2','TargetL2','MeanL3','SDL3','TargetL3'].forEach(function(f){var
el=G('mLot'+f);if(el)el.value="";});G('mLotParam').value="";G('mLotSumber').value='Manufaktur';if(dat
a){G('mLotID').value=data.lotID;G('mLotParam').value=data.paramID;G('mLotNoLot').value=data.noL
ot;G('mLotAlat').value=data.namaAlat;G('mLotMethode').value=data.methode;G('mLotSatuan').valu
e=data.satuan;G('mLotExpired').value=data.expiredDate;G('mLotTEa').value=data.tea;G('mLotBias').
value=data.biasPct;G('mLotSumber').value=data.sumber| |'Manufaktur';G('mLotMeanL1').value=dat
a.meanL1;G('mLotSDL1').value=data.sdL1;G('mLotTargetL1').value=data.targetL1;G('mLotMeanL2').v
alue=data.meanL2;G('mLotSDL2').value=data.sdL2;G('mLotTargetL2').value=data.targetL2;G('mLotM
eanL3').value=data.meanL3;G('mLotSDL3').value=data.sdL3;G('mLotTargetL3').value=data.targetL3;G
('modalLotTitle').textContent='Edit Lot';}else G('modalLotTitle').textContent='Tambah
Lot';opM('modalLot');}
function editLot(id){var l=(CD.lots| |[]).find(function(x){return x.lotID===id;});if(l)openLotModal(l);}
function saveLot(){var
p={lotID:G('mLotID').value| |null,paramID:G('mLotParam').value,noLot:G('mLotNoLot').value.trim(),n

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amaAlat:G('mLotAlat').value.trim(),methode:G('mLotMethode').value.trim(),satuan:G('mLotSatuan').
value.trim(),expiredDate:G('mLotExpired').value,tea:G('mLotTEa').value,biasPct:G('mLotBias').value,s
umber:G('mLotSumber').value | 'Manufaktur',meanL1:G('mLotMeanL1').value,sdL1:G('mLotSDL1').v
alue,targetL1:G('mLotTargetL1').value,meanL2:G('mLotMeanL2').value,sdL2:G('mLotSDL2').value,tar
getL2:G('mLotTargetL2').value,meanL3:G('mLotMeanL3').value,sdL3:G('mLotSDL3').value,targetL3:G(
'mLotTargetL3').value;if(!p.paramID || !p.noLot){toast('Parameter dan No.Lot
wajib','warning');return;}sl('Menyimpan...');google.script.run.withSuccessHandler(function(r){hl();if(r.
ok){toast('Lot disimpan','success');clM('modalLot');refreshLots();}else
toast(r.msg,'error');}).withFailureHandler(function(e){hl();toast('Error: '+e.message | | e | 'Terjadi
kesalahan'),'error');}).saveLotQC(p,getActiveUsername(),getLogUser());}
function delLot(id){cfm('Hapus
Lot','Lanjutkan?','☐','Hapus',function(yes){if(!yes)return;sl('Menghapus...');google.script.run.withSu
ccessHandler(function(r){hl();if(r.ok){toast('Dihapus','success');refreshLots();}else
toast(r.msg,'error');}).withFailureHandler(function(e){hl();toast('Error: '+e.message | | e | 'Terjadi
kesalahan'),'error');}).deleteLotQC(id,getActiveUsername(),getLogUser());});}
function
refreshLots(){google.script.run.withSuccessHandler(function(r){CD.lots=r | | [];populateFilters();rende
rLotTable();}).withFailureHandler(function(e){toast('Error: '+e.message | | e | 'Terjadi
kesalahan'),'error');}).getLotQC(getActiveUsername(),getActiveRole());}
function switchInputTab(tab){QA('#pageInputqc .tab-
btn').forEach(function(b,i){b.classList.toggle('active',(tab==='single'&&i===0) || (tab==='bulk'&&i===1
) || (tab==='smart'&&i===2));});G('tabSingle').classList.toggle('active',tab==='single');G('tabBulk').class
List.toggle('active',tab==='bulk');G('tabSmart').classList.toggle('active',tab==='smart');}
function initInputQCFilter(){var
today=todayISO();if(!G('qcFilterStart').value)G('qcFilterStart').value=today;if(!G('qcFilterEnd').value)G
('qcFilterEnd').value=today;loadInputQCTable();}
function
resetQCFilter(){G('qcFilterStart').value=todayISO();G('qcFilterEnd').value=todayISO();G('qcFilterBidan
g').value="";G('qcFilterParam').value="";G('qcFilterLot').innerHTML=<option
value="">Semua</option>;loadInputQCTable();}
function saveInputQC(){var
p={qcID:G('inputEditQcID').value | null,paramID:G('inputParam').value,lotID:G('inputLot').value,tang
gal:G('inputTanggal').value,level1:G('inputL1').value,level2:G('inputL2').value,level3:G('inputL3').valu
e};if(!p.paramID || !p.lotID){toast('Parameter dan Lot
wajib','warning');return;}if(!p.tanggal){toast('Tanggal wajib','warning');return;}var
isEdit=!p.qcID;sl(isEdit?'Memperbarui...':'Menyimpan...');google.script.run.withSuccessHandler(func
tion(r){hl();if(r.ok){toast(isEdit?'Data diperbarui':'Data
disimpan','success');cancelEditInput();G('inputL1').value="";G('inputL2').value="";G('inputL3').value="";
G('inputTanggal').value=todayISO();loadInputQCTable();}else
toast(r.msg,'error');}).withFailureHandler(function(e){hl();toast('Error: '+e.message | | e | 'Terjadi
kesalahan'),'error');}).saveInputQC(p,getActiveUsername(),getLogUser());}
function cancelEditInput(){G('inputEditQcID').value="";G('btnSaveInput').innerHTML=<i class="fas fa-
save"></i> Simpan';G('btnCancelEdit').style.display='none';}
function loadInputQCTable(){if(!CU || !CU.username){toast('Sesi belum aktif','warning');return;}var
filter={startDate:G('qcFilterStart').value | null,endDate:G('qcFilterEnd').value | null,bidang:G('qcFilter

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Bidang').value || null,paramID:G('qcFilterParam').value || null,lotID:G('qcFilterLot').value || null);sl('Me
muat QC...');google.script.run.withSuccessHandler(function(data){hl();inputQCCache=data || [];var
html='';(data || []).reverse().forEach(function(q,i){var sb=q.validated?'<span class="badge badge-
success">Valid</span>':'<span class="badge badge-
warning">Pending</span>';html+='<tr><td>'+(i+1)+'</td><td>'+fmtDate(q.tanggal)+'</td><td>'+esc
H(q.parameter)+'</td><td>'+escH(q.namaAlat)+'</td><td>'+escH(q.noLot)+'</td><td>'+(q.level1 || '-'
)+'</td><td>'+(q.level2 || '-'+'</td><td>'+(q.level3 || '-'+'</td><td>'+sb+'</td><td class="table-
actions"><button class="btn btn-warning btn-xs" onclick="editInputQC(\''+q.qcID+\'"\><i class="fas
fa-edit"></i></button> <button class="btn btn-danger btn-xs"
onclick="delInputQC(\''+q.qcID+\'"\><i class="fas fa-
trash"></i></button></td></tr>');});if(!data || !data.length)html='<tr><td colspan="10" style="text-
align:center;color:var(--text-secondary)">Tidak ada
data</td></tr>';G('inputQCTableBody').innerHTML=html;}).withFailureHandler(function(e){hl();toast
('Error: '+e.message || e || 'Terjadi
kesalahan'),'error');}).getInputQC(getActiveUsername(),getActiveRole(),filter);}
function editInputQC(id){var q=inputQCCache.find(function(x){return
x.qcID===id;});if(!q){sl('Memuat...');google.script.run.withSuccessHandler(function(qq){hl();if(qq)doE
ditInput(qq);else toast('Tidak ditemukan','error');}).withFailureHandler(function(e){hl();toast('Error:
'+e.message || e || 'Terjadi
kesalahan'),'error');}).getInputQCById(id,getActiveUsername(),getActiveRole());return;}doEditInput(q
);}
function doEditInput(q){switchInputTab('single');G('inputEditQcID').value=q.qcID;var
param=(CD.params || []).find(function(p){return
p.paramID===q.paramID;});if(param){G('inputBidang').value=param.bidang || '';filterParamByBidang('
inputBidang','inputParam');G('inputParam').value=q.paramID;onInputParamChange();setTimeout(fu
nction(){G('inputLot').value=q.lotID;G('inputTanggal').value=q.tanggal;G('inputL1').value=q.level1!
==null&&q.level1!=="?q.level1:";G('inputL2').value=q.level2!==null&&q.level2!=="?q.level2:";G('inputL
3').value=q.level3!==null&&q.level3!=="?q.level3:";G('btnSaveInput').innerHTML='<i class="fas fa-
save"></i>
Update';G('btnCancelEdit').style.display="";G('pageInputqc').scrollIntoView({behavior:'smooth',block:
'start'});toast('Mode Edit: Ubah data lalu klik Update','info');},200);}
function delInputQC(id){cfm('Hapus
QC','Lanjutkan?','🗑️','Hapus',function(yes){if(!yes)return;sl('Menghapus...');google.script.run.withSu
ccessHandler(function(r){hl();if(r.ok){toast('Dihapus','success');loadInputQCTable();}else
toast(r.msg,'error');}).withFailureHandler(function(e){hl();toast('Error: '+e.message || e || 'Terjadi
kesalahan'),'error');}).deleteInputQC(id,getActiveUsername(),'Dihapus via Input
QC',getLogUser());});}
function generateBulkRows(){var s=G('bulkStart').value,e=G('bulkEnd').value;if(!s || !e){toast('Tanggal
wajib','warning');return;}var d=new Date(s),de=new Date(e);if(d>de){toast('Tanggal mulai harus
sebelum akhir','warning');return;}var
html='<table><thead><tr><th>Tanggal</th><th>L1</th><th>L2</th><th>L3</th></tr></thead><tbo
dy>';var n=0;while(d<=de){var iso=d.getFullYear()+'-'+String(d.getMonth()+1).padStart(2,'0')+'-
'+String(d.getDate()).padStart(2,'0');html+='<tr><td><input type="date" value="'+iso+'" class="bulk-
date" readonly></td><td><input type="number" step="any" class="bulk-l1"></td><td><input
type="number" step="any" class="bulk-l2"></td><td><input type="number" step="any" class="bulk-

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l3"></td></tr>';d.setDate(d.getDate()+1);n++;}html+='</tbody></table>';G('bulkTableWrap').innerH
TML='<div class="smart-table">'+html+'</div>';G('btnSubmitBulk').style.display=n>0?'':'none';}
function submitBulk(){var
pwd=G('bulkPwd').value,paramID=G('bulkParam').value,lotID=G('bulkLot').value;if(!paramID||!lotID)
{toast('Parameter dan Lot wajib','warning');return;}if(!pwd){toast('Password
wajib','warning');return;}var rows=[];QA('.bulk-
date').forEach(function(el,i){rows.push({tanggal:el.value,level1:QA('.bulk-
l1')[i].value,level2:QA('.bulk-l2')[i].value,level3:QA('.bulk-l3')[i].value});});if(!rows.length){toast('Tidak
ada
data','warning');return;}sl('Menyimpan...');google.script.run.withSuccessHandler(function(r){hl();if(r.
ok){toast(r.count+'
disimpan','success');G('bulkTableWrap').innerHTML="";G('btnSubmitBulk').style.display='none';loadIn
putQCTable();}else toast(r.msg,'error');}).withFailureHandler(function(e){hl();toast('Error:
'+(e.message||e)||'Terjadi
kesalahan'),'error');}).bulkInputQC({paramID:paramID,lotID:lotID,rows:rows,smartPassword:pwd,get
ActiveUsername(),getLogUser()});}
function generateSmartTable(){var
paramID=G('smartParam').value,lotID=G('smartLot').value,s=G('smartStart').value,e=G('smartEnd').v
alue;if(!paramID||!lotID){toast('Pilih parameter dan lot','warning');return;}if(!s||!e){toast('Pilih
rentang tanggal','warning');return;}var sd=new Date(s),ed=new Date(e);if(sd>ed){toast('Tanggal
mulai harus sebelum
akhir','warning');return;}smartTableState.paramID=paramID;smartTableState.lotID=lotID;smartTable
State.dates=[];var cur=new Date(sd);while(cur<=ed){smartTableState.dates.push(cur.getFullYear()+'-
'+String(cur.getMonth()+1).padStart(2,'0')+'-
'+String(cur.getDate()).padStart(2,'0'));cur.setDate(cur.getDate()+1);}sl('Memuat...');google.script.ru
n.withSuccessHandler(function(r){hl();smartTableState.existingMap={};if(r.&&r.ok&&r.data){r.data.fo
rEach(function(q){smartTableState.existingMap[q.tanggal]={level1:q.level1,level2:q.level2,level3:q.le
vel3,qcID:q.qcID};});renderSmartTable();}).withFailureHandler(function(e){hl();toast('Error:
'+(e.message||e)||'Terjadi
kesalahan'),'error');renderSmartTable();}).getQCByDateRange(paramID,lotID,s,e,getActiveUsername(
));}
function renderSmartTable(){var
html='<table><thead><tr><th>#</th><th>Tanggal</th><th>L1</th><th>L2</th><th>L3</th></tr></t
head><tbody>';smartTableState.dates.forEach(function(date,idx){var
ex=smartTableState.existingMap[date];var hasExisting=!ex;var rowCls=hasExisting?' class="col-
existing":""';html+='<tr>'+rowCls+'><td>'+(idx+1)+(hasExisting?' <i class="fas fa-database"
style="color:var(--success)"></i>':"')+'</td><td><input type="date" value="'+date+'" class="smart-
date" readonly></td><td><input type="number" step="any" class="smart-l1"
value="'+(ex&&ex.level1?ex.level1:'")+'></td><td><input type="number" step="any" class="smart-
l2" value="'+(ex&&ex.level2?ex.level2:'")+'></td><td><input type="number" step="any"
class="smart-l3"
value="'+(ex&&ex.level3?ex.level3:'")+'></td></tr>');html+='</tbody></table>';G('smartTableWra
p').innerHTML=html;G('btnSubmitSmart').style.display="";G('btnClearSmart').style.display="";G('smart
TableWrap').addEventListener('paste',handleSmartPaste,true);}

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function handleSmartPaste(e){var target=e.target;if(!target || target.tagName!=='INPUT')return;var
txt=(e.clipboardData || window.clipboardData).getData('text');if(!txt)return;e.preventDefault();var
lines=txt.split(/\r?\n/).filter(function(l){return l.trim();});var tr=target.closest('tr');var
tbody=tr.parentElement;var startIdx=Array.from(tbody.children).indexOf(tr);var
td=target.closest('td');var startCol=Array.from(tr.children).indexOf(td);var
pasted=0,filled=0;lines.forEach(function(line,li){var cells=line.split(/\t|;|,/);var
rowIdx=startIdx+li;if(rowIdx>=tbody.children.length)return;var
row=tbody.children[rowIdx];cells.forEach(function(val,ci){var
colIdx=startCol+ci;if(colIdx>=row.children.length)return;var
input=row.children[colIdx].querySelector('input');if(input&&input.type!=='date'){input.value=val.tri
m();filled++;});pasted++;});toast(pasted+' baris di-paste','info');}
function
clearSmartTable(){cfm('Bersihkan','Lanjutkan?',' ','Bersihkan',function(yes){if(!yes)return;QA('#smart
TableWrap input.smart-l1,#smartTableWrap input.smart-l2,#smartTableWrap input.smart-
l3').forEach(function(i){i.value="";});})
function submitSmart(){var pwd=G('smartPwd').value;if(!pwd){toast('Password
wajib','warning');return;}if(!smartTableState.paramID || !smartTableState.lotID){toast('Generate
tabel dulu','warning');return;}var rows=[];QA('#smartTableWrap tbody tr').forEach(function(tr){var
date=tr.querySelector('.smart-date').value;var l1=tr.querySelector('.smart-l1').value;var
l2=tr.querySelector('.smart-l2').value;var l3=tr.querySelector('.smart-
l3').value;if(date&&(l1 || l2 || l3))rows.push({tanggal:date,level1:l1,level2:l2,level3:l3});});if(!rows.leng
th){toast('Tidak ada data','warning');return;}cfm('Simpan '+rows.length+'
baris','Lanjutkan?','','Simpan',function(yes){if(!yes)return;sl('Menyimpan...');google.script.run.with
SuccessHandler(function(r){hl();if(r.ok){toast(r.count+'
disimpan','success');generateSmartTable();loadInputQCTable();}else
toast(r.msg,'error');}).withFailureHandler(function(e){hl();toast('Error: '+e.message || e || 'Terjadi
kesalahan'),'error');}).bulkInputQC({paramID:smartTableState.paramID,lotID:smartTableState.lotID,r
ows:rows,smartPassword:pwd},getActiveUsername(),getLogUser());});}
function loadValidasi(){if(!CU || !CU.username){toast('Sesi belum
aktif','warning');return;}sl('Memuat...');var
pf=G('valParamFilter').value;google.script.run.withSuccessHandler(function(data){hl();var html="";var
hasPending=false;(data || []).reverse().forEach(function(q){var sb=q.validated?'<span class="badge
badge-success">Valid</span>':'<span class="badge badge-
warning">Pending</span>';if(!q.validated)hasPending=true;html+=<tr><td>'+(q.validated?'<input
type="checkbox" class="val-chk" value="" +escH(q.qcID)+"
onchange="updateValBulkBar()">'+</td><td>'+fmtDate(q.tanggal)+</td><td>'+escH(q.parameter)
+'</td><td>'+escH(q.namaAlat)+</td><td>'+(q.level1 || '-')+'</td><td>
class="" +zClass(q.zL1)+">'+(q.zL1!==null?round(q.zL1):-')+'</td><td>'+(q.level2 || '-')+'</td><td>
class="" +zClass(q.zL2)+">'+(q.zL2!==null?round(q.zL2):-')+'</td><td>'+(q.level3 || '-')+'</td><td>
class="" +zClass(q.zL3)+">'+(q.zL3!==null?round(q.zL3):-
')+'</td><td>'+sb+'</td><td>'+(q.validated?'<small>'+escH(q.validatedBy)+'</small>':'<button
class="btn btn-success btn-xs" onclick="openValidasiModal(\'"+q.qcID+\')"><i class="fas fa-
check"></i></button>'+</td></tr>');});if(!data || !data.length)html=<tr><td colspan="12"
style="text-align:center;color:var(--text-secondary)">Tidak ada
data</td></tr>';G('valTableBody').innerHTML=html;G('valBulkBar').style.display=hasPending?'flex':'n

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one';G('valCheckAll').checked=false;updateValBulkBar();}).withFailureHandler(function(e){hl();toast('
Error: '+e.message+' | e | 'Terjadi
kesalahan'),'error');}).getValidasiData(getActiveUsername(),getActiveRole(),{paramID:pf});}
function
openValidasiModal(id){G('mValQCID').value=id;G('mValCatatan').value="";G('mValDetail').innerHTML
='<p>QC ID: <strong>'+id+'</strong></p>';opM('modalValidasi');}
function submitValidasi(){var
id=G('mValQCID').value,c=G('mValCatatan').value;sl('Memvalidasi...');google.script.run.withSuccessH
andler(function(r){hl();if(r.ok){toast('Tervalidasi','success');clM('modalValidasi');loadValidasi();}else
toast(r.msg,'error');}).withFailureHandler(function(e){hl();toast('Error: '+e.message+' | e | 'Terjadi
kesalahan'),'error');}).validateQC(id,c,CU.username,getActiveUsername(),getLogUser());}
function getSelectedValIDs(){var ids=[];QA('.val-
chk:checked').forEach(function(c){ids.push(c.value);});return ids;}
function updateValBulkBar(){var chks=QA('.val-chk');var checked=QA('.val-
chk:checked');G('valBulkCount').textContent=checked.length+'
terpilih';G('valCheckAll').checked=chks.length>0&&checked.length===chks.length;}
function toggleAllValChecks(checked){QA('.val-
chk').forEach(function(c){c.checked=checked;});G('valCheckAll').checked=checked;updateValBulkBar(
);}
function submitBulkValidasi(){var ids=getSelectedValIDs();if(!ids.length){toast('Pilih data yang akan
divalidasi','warning');return;}var catatan=G('valBulkCatatan').value;cfm('Validasi Masal','Validasi
'+ids.length+' data QC?',true,'Validasi',function(yes){if(!yes)return;sl('Memvalidasi '+ids.length+'
data...');google.script.run.withSuccessHandler(function(r){hl();if(r.ok){toast(r.count+' data
tervalidasi','success');loadValidasi();}else
toast(r.msg+' | 'Gagal','error');}).withFailureHandler(function(e){hl();toast('Error:
'+e.message+' | e | 'Terjadi
kesalahan'),'error');}).validateQCBulk(ids,catatan,CU.username,getActiveUsername(),getLogUser());}
});
function loadHistori(){if(!CU || !CU.username){toast('Sesi belum
aktif','warning');return;}sl('Memuat...');var
filter={actionType:G('histFilter').value || null,startDate:G('histStart').value || null,endDate:G('histEnd').
value || null,bidang:G('histBidang').value || null,paramID:G('histParam').value || null,lotID:G('histLot').v
alue || null};google.script.run.withSuccessHandler(function(data){hl();var
html="";(data || []).reverse().forEach(function(h){var b="";if(h.actionType==='EDIT_QC')b='<span
class="badge histori-edit">EDIT</span>';else if(h.actionType==='DATA DIHAPUS')b='<span
class="badge histori-delete">HAPUS</span>';else if(h.actionType==='RESTORED')b='<span
class="badge histori-restored">RESTORED</span>';else b='<span class="badge badge-
secondary">'+escH(h.actionType)+'</span>';var aksiHtml="";if(h.actionType==='DATA
DIHAPUS')aksiHtml='<button class="btn btn-success btn-xs"
onclick="confirmRestoreHistori(\''+h.hqcID+'\')"' title="Restore data"><i class="fas fa-undo"></i>
Restore</button> <button class="btn btn-danger btn-xs" onclick="deleteHistori(\''+h.hqcID+'\')"'
title="Hapus permanen dari histori"><i class="fas fa-trash"></i> Hapus</button>;else
aksiHtml='<button class="btn btn-danger btn-xs" onclick="deleteHistori(\''+h.hqcID+'\')"'
title="Hapus permanen dari histori"><i class="fas fa-trash"></i>
Hapus</button>';html+='<tr><td>'+escH(h.deletedDate)+'</td><td>'+b+'</td><td><span

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class="badge badge-info">'+escH(h.bidang| |'-
')+</span></td><td>'+escH(h.parameter)+'</td><td>'+escH(h.noLot)+'</td><td>'+fmtDate(h.tangga
l)+'</td><td>'+(h.level1| |'-')+</td><td>'+(h.level2| |'-')+</td><td>'+(h.level3| |'-
')+</td><td>'+escH(h.deletedBy)+'</td><td><small>'+escH(h.changeDetail| |'')+</small></td><td
class="table-actions">'+aksiHtml+'</td></tr>');});if(!data| |!data.length)html=<tr><td colspan="12"
style="text-align:center;color:var(--text-secondary)">Tidak ada
histori</td></tr>;G('histTableBody').innerHTML=html;}).withFailureHandler(function(e){hl();toast('E
rror: '+e.message| |e| |'Terjadi
kesalahan'),'error');}).getHistoriQC(getActiveUsername(),getActiveRole(),filter);}
function
resetHistFilter(){G('histStart').value="";G('histEnd').value="";G('histBidang').value="";G('histParam').val
ue="";G('histLot').innerHTML=<option
value="">Semua</option>;G('histFilter').value="";loadHistori();}
function confirmRestoreHistori(hqcID){cfm('Restore Data QC','Data yang dihapus akan dikembalikan.
Lanjutkan?','<input type="checkbox">','Restore',function(yes){if(!yes)return;sl('Merestore...');google.script.run.withSucces
sHandler(function(r){hl();if(r.ok){toast(r.msg,'success',4000);loadHistori();}else
toast(r.msg,'error');}).withFailureHandler(function(e){hl();toast('Error: '+e.message| |e| |'Terjadi
kesalahan'),'error');}).restoreHistoriQC(hqcID,getActiveUsername());});}
function deleteHistori(hqcID){cfm('Hapus Histori Permanen','Data histori ini akan dihapus
PERMANEN dan tidak dapat dikembalikan. Lanjutkan?','<input type="checkbox">','Hapus
Permanen',function(yes){if(!yes)return;sl('Menghapus
histori...');google.script.run.withSuccessHandler(function(r){hl();if(r.ok){toast(r.msg,'success',4000);lo
adHistori();}else toast(r.msg,'error');}).withFailureHandler(function(e){hl();toast('Error:
'+e.message| |e| |'Terjadi
kesalahan'),'error');}).deleteHistoriQC(hqcID,getActiveUsername(),getLogUser());});}
function renderTeaTable(){var
html="";(CD.teaList| |[]).forEach(function(t,i){html+=<tr><td>'+(i+1)+'</td><td>'+escH(t.parameter)+'
</td><td>'+t.nilaiTEa+'</td><td>'+escH(t.referensi| |'-')+</td><td class="table-actions"><button
class="btn btn-warning btn-xs" onclick="editTea(\''+t.teaID+\'"\)><i class="fas fa-edit"></i></button>
<button class="btn btn-danger btn-xs" onclick="delTea(\''+t.teaID+\'"\)><i class="fas fa-
trash"></i></button></td></tr>');});if(!CD.teaList| |!CD.teaList.length)html=<tr><td colspan="5"
style="text-align:center;color:var(--text-secondary)">Belum ada
data</td></tr>;G('teaTableBody').innerHTML=html;}
function
openTeaModal(d){G('mTeaID').value=d?d.teaID:"";G('mTeaParam').value=d?d.paramID:"";G('mTeaVal
ue').value=d?d.nilaiTEa:"";G('mTeaRef').value=d?d.referensi:"";G('modalTeaTitle').textContent=d?'Edit
TEa':'Tambah TEa';opM('modalTEa');}
function editTea(id){var t=(CD.teaList| |[]).find(function(x){return
x.teaID===id;});if(t)openTeaModal(t);}
function saveTea(){var pid=G('mTeaParam').value;var
p={teaID:G('mTeaID').value| |null,paramID:pid,parameter:getParamName(pid),nilaiTEa:G('mTeaValu
e').value,referensi:G('mTeaRef').value.trim();if(!p.paramID| |!p.nilaiTEa){toast('Parameter dan nilai
wajib','warning');return;}sl('Menyimpan...');google.script.run.withSuccessHandler(function(r){hl();if(r.
ok){toast('TEa disimpan','success');clM('modalTEa');refreshTEa();}else

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toast(r.msg,'error'));}).withFailureHandler(function(e){hl();toast('Error: '+e.message | | e | | 'Terjadi
kesalahan'),'error');}).saveDaftarTEa(p,getActiveUsername(),getLogUser());}
function delTea(id){cfm('Hapus
TEa','Lanjutkan?','','Hapus',function(yes){if(!yes)return;sl('Menghapus...');google.script.run.withS
uccessHandler(function(r){hl();if(r.ok){toast('Dihapus','success');refreshTEa();}else
toast(r.msg,'error');}).withFailureHandler(function(e){hl();toast('Error: '+e.message | | e | | 'Terjadi
kesalahan'),'error');}).deleteDaftarTEa(id,getActiveUsername(),getLogUser());});}
function
refreshTEa(){google.script.run.withSuccessHandler(function(r){CD.teaList=r | | [];renderTeaTable();}).
withFailureHandler(function(e){toast('Error: '+e.message | | e | | 'Terjadi
kesalahan'),'error');}).getDaftarTEa(getActiveUsername(),getActiveRole());}
</script>
<script>
function loadBiasPME(){
sl('Memuat PME...');var pf=G('pmeParamFilter').value;
google.script.run.withSuccessHandler(function(data){
hl();pmeCache=data | | [];var html="";
(data | | []).forEach(function(p){
var first=true;
var rowCount=['L1','L2','L3'].filter(function(l){return p.details&& p.details[l];}).length | | 1;
['L1','L2','L3'].forEach(function(lv){
var d=p.details? p.details[lv]:null;
if(!d&&lv!=='L1')return;
if(!d)d={mean:'-',sd:'-',cv:'-',bias:'-',te:'-',tea:p.tea,sigma:'-'};
html+<tr>;
if(first){
html+<td rowspan="">+rowCount+</td>;
html+<td rowspan="">+rowCount+</td>;
html+<td rowspan="">+rowCount+</td>;
html+<td rowspan="">+rowCount+</td>;
first=false;
}
html+<td><strong>+lv+</strong></td>;
html+<td>+(d.mean!=='null&&d.mean!=='-'?d.mean:'-')+</td>;
html+<td>+(d.sd!=='null&&d.sd!=='-'?d.sd:'-')+</td>;
html+<td>+(d.cv!=='null&&d.cv!=='-'?d.cv:'-')+</td>;
html+<td>+(d.bias!=='null&&d.bias!=='-'?d.bias:'-')+</td>;
html+<td>+(d.te!=='null&&d.te!=='-'?d.te:'-')+</td>;
html+<td>+(d.tea!=='null&&d.tea!=='-'?d.tea:(p.tea | | '-'))+</td>;
html+<td class="">+sigmaClass(d.sigma)+</td>+(d.sigma!=='null&&d.sigma!=='-'?d.sigma:'-')+</td>;
if(lv==='L1'){
html+<td rowspan="">+rowCount+</td> class="table-actions">;
html+<button class="btn btn-warning btn-xs" onclick="editPMEFast(\"+p.pmeID+"\")><i class="fas
fa-edit"></i></button>';

```

```

html+="

```



```

G('mPMEHasilL3').value=d.hasilL3 || '';
G('mPMEMPL1').value=d.meanPesertaL1 || '';
G('mPMEMPL2').value=d.meanPesertaL2 || '';
G('mPMEMPL3').value=d.meanPesertaL3 || '';
G('mPMECVStart').value=d.cvStartDate || '';
G('mPMECVEnd').value=d.cvEndDate || '';
G('mPMECVL1').value=d.cvL1 || '';
G('mPMECVL2').value=d.cvL2 || '';
G('mPMECVL3').value=d.cvL3 || '';
},80);
G('modalPMETitle').textContent='Edit PME';
}else{
G('mPMETahun').value=new Date().getFullYear();
G('modalPMETitle').textContent='Tambah PME';
}
opM('modalPME');
}
function savePME(){
var p={
pmeID:G('mPMEID').value || null,
paramID:G('mPMEParam').value,lotID:G('mPMELot').value,
namaAlat:G('mPMEAlat').value,metode:G('mPMEMetode').value,satuan:G('mPMESatuan').value,
siklus:G('mPMESiklus').value,tahun:G('mPMETahun').value,tea:G('mPMETEa').value,
hasilL1:G('mPMEHasilL1').value,hasilL2:G('mPMEHasilL2').value,hasilL3:G('mPMEHasilL3').value,
meanPesertaL1:G('mPMEMPL1').value,meanPesertaL2:G('mPMEMPL2').value,meanPesertaL3:G('mP
MEMPL3').value,
cvStartDate:G('mPMECVStart').value,cvEndDate:G('mPMECVEnd').value,
cvL1:G('mPMECVL1').value,cvL2:G('mPMECVL2').value,cvL3:G('mPMECVL3').value
};
if(!p.paramID || !p.lotID){toast('Parameter dan Lot wajib','warning');return;}
if(!p.siklus){toast('Siklus wajib','warning');return;}
sl('Menyimpan...');
google.script.run.withSuccessHandler(function(r){hl();if(r.ok){toast('PME
disimpan','success');clM('modalPME');loadBiasPME();}else toast(r.msg,'error');})
.withFailureHandler(function(e){hl();toast('Error: '+e.message || e || 'Terjadi
kesalahan'),'error');}).saveBiasPME(p,getActiveUsername(),getLogUser());
}
function delPME(id){
cfm('Hapus PME','Lanjutkan?','🗑️','Hapus',function(yes){
if(!yes)return;sl('Menghapus...');
google.script.run.withSuccessHandler(function(r){hl();if(r.ok){toast('Dihapus','success');loadBiasPME(
)};else toast(r.msg,'error');})
.withFailureHandler(function(e){hl();toast('Error: '+e.message || e || 'Terjadi
kesalahan'),'error');}).deleteBiasPME(id,getActiveUsername(),getLogUser());
});
}

```

```

}
function loadCalcStats(){
sl('Memuat...');var pf=G('csParamFilter').value;
google.script.run.withSuccessHandler(function(data){
hl();csCache=data || [];var html="";
var groups={};
(data || []).forEach(function(s){
var key=s.paramID+'_'+s.lotID;
if(!groups[key])groups[key]=[];
groups[key].push(s);
});
Object.keys(groups).forEach(function(key){
var items=groups[key].sort(function(a,b){
var lvA=parseInt(String(a.level).replace('L',''));
var lvB=parseInt(String(b.level).replace('L',''));
return lvA-lvB;
});
var first=true;
items.forEach(function(s){
html+='\n<tr>';
if(first){
html+='\n<td rowspan="'+items.length+'">'+escH(s.parameter || '-')+'</td>';
html+='\n<td rowspan="'+items.length+'">'+escH(s.noLot || '-')+'</td>';
first=false;
}
html+='\n<td><strong>'+escH(s.level)+'</strong></td>';
html+='\n<td>'+round(s.calcMean,4)+'</td>';
html+='\n<td>'+round(s.calcSD,4)+'</td>';
html+='\n<td>'+round(s.calcCV)+'%</td>';
html+='\n<td>'+(s.bias!==undefined?round(s.bias):'-')+'</td>';
html+='\n<td>'+(s.te!==undefined?round(s.te):'-')+'</td>';
html+='\n<td>'+(s.tea || '-')+'</td>';
html+='\n<td'
class="'+sigmaClass(s.sigma)+'">'+(s.sigma!==undefined&& s.sigma!==null?round(s.sigma):'-')+'</td>';
html+='\n<td>'+s.n+'</td>';
html+='\n<td>'+fmtDate(s.startDate)+'<br>s/d<br>'+fmtDate(s.endDate)+'</td>';
html+='\n<td class="table-actions">';
html+='\n<button class="btn btn-warning btn-xs" onclick="editCalcStatsFast(\''+s.statID+'\'")><i'
class="fas fa-edit"></i></button> ';
html+='\n<button class="btn btn-danger btn-xs" onclick="delCalcStats(\''+s.statID+'\'")><i class="fas'
fa-trash"></i></button>';
html+='\n</td></tr>';
});
});
}

```

```

if(!data || !data.length)html=<tr><td colspan="13" style="text-align:center;color:var(--text-secondary)">Tidak ada data</td></tr>;
G('csTableBody').innerHTML=html;
}).withFailureHandler(function(e){hl();toast('Error: '+e.message || e || 'Terjadi kesalahan'),'error');}).getCalcStats(getActiveUsername(),getActiveRole(),{paramID:pf});
}
function openCalcStatsModal(){
G('mCSID').value="";G('mCSPParam').value="";
G('mCSLot').innerHTML=<option value="">Pilih</option>;
G('mCSStart').value="";G('mCSEnd').value="";
G('modalCalcStatsTitle').textContent='Tambah Calc Stats — Hitung Semua Level';
opM('modalCalcStats');
}
function editCalcStatsFast(id){
var s=csCache.find(function(x){return x.statID===id;});
if(s){doEditCalcStats(s);}
else{
sl('Memuat...');
google.script.run.withSuccessHandler(function(ss){hl();if(ss)doEditCalcStats(ss);else toast('Tidak ditemukan'),'error');}).withFailureHandler(function(e){hl();toast('Error: '+e.message || e || 'Terjadi kesalahan'),'error');}).getCalcStatById(id,getActiveUsername(),getActiveRole());
}
}
function doEditCalcStats(s){
G('mCSID').value=s.statID || "";
G('mCSPParam').value=s.paramID || "";
onCSPParamChange();
setTimeout(function(){
G('mCSLot').value=s.lotID || "";
G('mCSStart').value=s.startDate || "";
G('mCSEnd').value=s.endDate || "";
},80);
G('modalCalcStatsTitle').textContent='Edit Calc Stats — Hitung Ulang Semua Level';
opM('modalCalcStats');
}
function saveCalcStatsForm(){
var p={
paramID:G('mCSPParam').value,lotID:G('mCSLot').value,
startDate:G('mCSStart').value,endDate:G('mCSEnd').value
};
if(!p.paramID || !p.lotID){toast('Parameter dan Lot wajib','warning');return;}
if(!p.startDate || !p.endDate){toast('Rentang tanggal wajib','warning');return;}
sl('Menghitung semua level...');
google.script.run.withSuccessHandler(function(r){

```

```

hl();
if(r.ok){
var detail=[];
['L1','L2','L3'].forEach(function(lv){
if(r.results&&r.results[lv])detail.push(lv+' Mean='+r.results[lv].mean+', SD='+r.results[lv].sd+',
CV='+r.results[lv].cv+'%');
});
toast(r.msg+(detail.length?'\\n'+detail.join('\\n'):'),'success',6000);
cIM('modalCalcStats');loadCalcStats();
}else toast(r.msg,'error');
}).withFailureHandler(function(e){hl();toast('Error: '+e.message || e || 'Terjadi
kesalahan'),'error');}).saveCalcStatsAllLevels(p,getActiveUsername(),getLogUser());
}
function delCalcStats(id){
cfm('Hapus','Lanjutkan?','☐','Hapus',function(yes){
if(!yes)return;sl('Menghapus...');
google.script.run.withSuccessHandler(function(r){hl();if(r.ok){toast('Dihapus','success');loadCalcStats
()};else toast(r.msg,'error');}).
.withFailureHandler(function(e){hl();toast('Error: '+e.message || e || 'Terjadi
kesalahan'),'error');}).deleteCalcStats(id,getActiveUsername(),getLogUser());
});
}
function loadSigmaCVOpt(){
sl('Memuat...');var pf=G('scvParamFilter').value;
google.script.run.withSuccessHandler(function(data){
hl();scvCache=data || [];var html='';
(data || []).forEach(function(s){
var hasDetails=s.details&&(s.details.L1 || s.details.L2 || s.details.L3);
if(!hasDetails){
html+='<tr>';
html+='<td>'+escH(s.parameter || '-')+'</td>';
html+='<td>'+escH(s.noLot || '-')+'</td>';
html+='<td>'+escH(s.namaAlat || '-')+'</td>';
html+='<td>'+fmtDate(s.startDate)+'<br>s/d<br>'+fmtDate(s.endDate)+'</td>';
html+='<td>'+escH(s.siklusPME || '-')+'</td>';
html+='<td>'+escH(s.tahunSiklus || '-')+'</td>';
html+='<td colspan="7" style="text-align:center;color:var(--text-secondary)">Tidak ada data periode
ini</td>';
html+='<td class="'+sigmaClass(s.avgSigma)+'">'+round(s.avgSigma)+'</td>';
html+='<td class="table-actions">';
html+='<button class="btn btn-warning btn-xs" onclick="editSigmaCVOptFast(\'+s.cvOptID+\')"><i
class="fas fa-edit"></i></button> ';
html+='<button class="btn btn-danger btn-xs" onclick="delSigmaCVOpt(\'+s.cvOptID+\')"><i
class="fas fa-trash"></i></button>';
html+='</td></tr>';

```

```

return;
}
var levels=['L1','L2','L3'].filter(function(l){return s.details[l]});
var first=true;
levels.forEach(function(lv){
var d=s.details[lv];
html+<tr>;
if(first){
html+<td rowspan="" +levels.length+"">'+escH(s.parameter | | '-')+'</td>;
html+<td rowspan="" +levels.length+"">'+escH(s.noLot | | '-')+'</td>;
html+<td rowspan="" +levels.length+"">'+escH(s.namaAlat | | '-')+'</td>;
html+<td
rowspan="" +levels.length+"">'+fmtDate(s.startDate)+'<br>s/d<br>'+fmtDate(s.endDate)+'</td>;
html+<td rowspan="" +levels.length+"">'+escH(s.siklusPME | | '-')+'</td>;
html+<td rowspan="" +levels.length+"">'+escH(s.tahunSiklus | | '-')+'</td>;
first=false;
}
html+<td><strong>'+lv+'</strong></td>;
html+<td>'+(d.mean!==null?d.mean:'-')+'</td>;
html+<td>'+(d.sd!==null?d.sd:'-')+'</td>;
html+<td>'+(d.cv!==null?d.cv:'-')+'</td>;
html+<td>'+(d.bias!==null?d.bias:'-')+'</td>;
html+<td>'+(d.te!==null?d.te:'-')+'</td>;
html+<td>'+(d.tea | | s.tea | | '-')+'</td>;
html+<td class="" +sigmaClass(d.sigma)+"">'+(d.sigma!==null?d.sigma:'-')+'</td>;
if(lv===levels[0]){
html+<td rowspan="" +levels.length+""
class="" +sigmaClass(s.avgSigma)+""><strong>'+round(s.avgSigma)+'</strong></td>;
html+<td rowspan="" +levels.length+"" class="table-actions">;
html+<button class="btn btn-warning btn-xs" onclick="editSigmaCVOptFast(\"'+s.cvOptID+'\"")><i
class="fas fa-edit"></i></button> ';
html+<button class="btn btn-danger btn-xs" onclick="delSigmaCVOpt(\"'+s.cvOptID+'\"")><i
class="fas fa-trash"></i></button>;
html+</td>;
}
html+</tr>;
});
});
if(!data || !data.length)html+<tr><td colspan="16" style="text-align:center;color:var(--text-secondary)">Tidak ada data</td></tr>;
G('scvTableBody').innerHTML=html;
}).withFailureHandler(function(e){hl();toast('Error: '+e.message | | e | | 'Terjadi
kesalahan'),'error');}).getSigmaCVOpt(getActiveUsername(),getActiveRole(),{paramID:pf});
}

```

```
//  PERBAIKAN v9.6: openSigmaCVOptModal dengan populateSCVSiklusTahun
function openSigmaCVOptModal(){
G('mSCVID').value="";G('mSCVParam').value="";
G('mSCVLot').innerHTML='<option value="">Pilih</option>';
G('mSCVAlat').value="";G('mSCVTEa').value="";G('mSCVNQC').value="";
G('mSCVStart').value="";G('mSCVEnd').value="";G('mSCVCatatan').value="";
G('mSCVSiklusPME').innerHTML='<option value="">Pilih</option>';
G('mSCVTahunSiklus').innerHTML='<option value="">Pilih</option>';
G('mSCVBiasPME1').value="";G('mSCVBiasPME2').value="";G('mSCVBiasPME3').value="";
G('modalSCVTitle').textContent='% Sigma CV (Optional)';
populateSCVSiklusTahun();
opM('modalSigmaCVOpt');
}
```

```
function editSigmaCVOptFast(id){
var s=scvCache.find(function(x){return x.cvOptID===id;});
if(s){doEditSCV(s);}
else{
sl('Memuat...');
google.script.run.withSuccessHandler(function(ss){hl();if(ss)doEditSCV(ss);else toast('Tidak
ditemukan','error');})
.withFailureHandler(function(e){hl();toast('Error: '+(e.message|'|e|'|'Terjadi
kesalahan'),'error');}).getSigmaCVOptById(id,getActiveUsername(),getActiveRole());
}
}
```

```
// PERBAIKAN v9.6: doEditSCV dengan load data siklus, tahun, bias PME
function doEditSCV(s){
G('mSCVID').value=s.cvOptID|'|';
G('mSCVParam').value=s.paramID|'|';
onSCVParamChange();
setTimeout(function(){
G('mSCVLot').value=s.lotID|'|';
G('mSCVAlat').value=s.namaAlat|'|';
G('mSCVTEa').value=s.tea|'|';
G('mSCVNQC').value=s.nqc|'|';
G('mSCVStart').value=s.startDate|'|';
G('mSCVEnd').value=s.endDate|'|';
G('mSCVCatatan').value=s.catatan|'|';
G('mSCVSiklusPME').value=s.siklusPME|'|';
G('mSCVTahunSiklus').value=s.tahunSiklus|'|';
G('mSCVBiasPME1').value=s.biasPME1|'|';
G('mSCVBiasPME2').value=s.biasPME2|'|';
G('mSCVBiasPME3').value=s.biasPME3|'|';
},80);
```

```

G('modalSCVTitle').textContent='Edit % Sigma CV';
populateSCVSiklusTahun();
opM('modalSigmaCVOpt');
}

// PERBAIKAN v9.6: saveSigmaCVOptForm dengan field baru
function saveSigmaCVOptForm(){
var p={
cvOptID:G('mSCVID').value || null,
paramID:G('mSCVParam').value,lotID:G('mSCVLot').value,
namaAlat:G('mSCVAlat').value,tea:G('mSCVTEa').value,nqc:G('mSCVNQC').value,
startDate:G('mSCVStart').value,endDate:G('mSCVEnd').value,catatan:G('mSCVCatatan').value,
siklusPME:G('mSCVSiklusPME').value,tahunSiklus:G('mSCVTahunSiklus').value,
biasPME1:G('mSCVBiasPME1').value,biasPME2:G('mSCVBiasPME2').value,biasPME3:G('mSCVBiasPM
E3').value
};
if(!p.paramID || !p.lotID){toast('Parameter dan Lot wajib','warning');return;}
if(!p.startDate || !p.endDate){toast('Rentang tanggal wajib','warning');return;}
sl('Menghitung & menyimpan...');
google.script.run.withSuccessHandler(function(r){
hl();
if(r.ok){
toast('Disimpan! Avg  $\sigma$ : '+round(r.avgSigma)+' , Avg  $\sigma$  L1-2: '+round(r.avgSigmaL12),'success',5000);
clM('modalSigmaCVOpt');loadSigmaCVOpt();
}else toast(r.msg,'error');
}}.withFailureHandler(function(e){hl();toast('Error: '+e.message || e || 'Terjadi
kesalahan'),'error');}).saveSigmaCVOpt(p,getActiveUsername(),getLogUser());
}

function delSigmaCVOpt(id){
cfm('Hapus','Lanjutkan?','☐','Hapus',function(yes){
if(!yes)return;sl('Menghapus...');
google.script.run.withSuccessHandler(function(r){hl();if(r.ok){toast('Dihapus','success');loadSigmaCV
Opt();}else toast(r.msg,'error');})
.withFailureHandler(function(e){hl();toast('Error: '+e.message || e || 'Terjadi
kesalahan'),'error');}).deleteSigmaCVOpt(id,getActiveUsername(),getLogUser());
});
}
</script>
<script>
function buildKopHTML(kop,dualLogo){kop=kop || CD.kop || {};var html='<div class="img-result-
header" style="text-align:center;border-bottom:3px double #333;padding-bottom:10px;margin-
bottom:12px">';if(dualLogo&&kop.logo&&kop.logo2){html+=''<div
style="display:table;width:100%;margin-bottom:6px"><div style="display:table-cell;vertical-
align:middle;width:90px;text-align:center"></div><div style="display:table-cell;vertical-align:middle;text-align:center">';if(kop.pemda)html+=<p style="font-size:.82rem;font-weight:600;margin:0 0 2px">'+escH(kop.pemda).toUpperCase()+'</p>';html+=<h2 style="margin:0">'+escH(kop.namaRS|'|LABORATORIUM').toUpperCase()+'</h2>';if(kop.alamat)html+=<div style="font-size:.9rem">'+escH(kop.alamat)+'</div>';var cts=[];if(kop.telepon)cts.push('Telp: '+kop.telepon);if(kop.email)cts.push('Email: '+kop.email);if(cts.length)html+=<div style="font-size:.82rem;color:#64748b">'+cts.join(' | ')+</div>';html+=</div><div style="display:table-cell;vertical-align:middle;width:90px;text-align:center"></div></div>';}else{if(kop.logo)html+=<div style="margin-bottom:10px"></div>';if(kop.pemda)html+=<p style="font-size:.82rem;font-weight:600;margin:0 0 2px">'+escH(kop.pemda).toUpperCase()+'</p>';html+=<h2 style="margin:0">'+escH(kop.namaRS|'|LABORATORIUM').toUpperCase()+'</h2>';if(kop.alamat)html+=<div class="header-line" style="font-size:.9rem">'+escH(kop.alamat)+'</div>';var cts=[];if(kop.telepon)cts.push('Telp: '+kop.telepon);if(kop.email)cts.push('Email: '+kop.email);if(cts.length)html+=<div class="header-line" style="font-size:.82rem;color:var(--text-secondary)">'+cts.join(' | ')+</div>';}html+=</div>';return html;}

function buildTTDHTML(kop,qrIdPrefix){kop=kop|'|CD.kop|'|{};qrIdPrefix=qrIdPrefix|'|qr';var html=<div class="img-result-footer" style="display:flex;justify-content:space-between;align-items:flex-start;margin-top:40px;gap:20px"><div class="ttd-col" style="text-align:center;width:45%;flex-shrink:0"><p style="font-weight:600;margin-bottom:4px">Kepala Ruangan Lab</p><div id="'+qrIdPrefix+'1" style="width:80px;height:80px;margin:8px auto"></div><p class="ttd-name" style="margin-top:40px;border-top:1px solid #333;padding-top:4px;font-weight:700">'+escH(kop.kepalaRuangan|'|'____)'+</p></div><div class="ttd-col" style="text-align:center;width:45%;flex-shrink:0"><p style="font-weight:600;margin-bottom:4px">Dokter PJ</p><div id="'+qrIdPrefix+'2" style="width:80px;height:80px;margin:8px auto"></div><p class="ttd-name" style="margin-top:40px;border-top:1px solid #333;padding-top:4px;font-weight:700">'+escH(kop.dokterPJ|'|'____)'+</p></div></div><p class="img-electronic" style="text-align:center;font-size:.75rem;color:#666;margin-top:10px">Dokumen ini dihasilkan secara elektronik oleh didiQCsys</p>';return html;}

function renderQR(elId,text){var el=G(elId);if(!el)return;el.innerHTML="";try{new QRCode(el,{text:text,width:80,height:80,colorDark:'#0f172a',colorLight:'#fff',correctLevel:QRCode.CorrectLevel.L});setTimeout(function(){var cv=el.querySelector('canvas');if(cv)cv.remove();},100);}catch(e){}}

function loadGrafik(){
  var paramID=G('grafParam').value,lotID=G('grafLot').value;
  if(!paramID|'|lotID){toast('Pilih parameter dan lot','warning');return;}
  if(!G('grafStart').value|'|G('grafEnd').value){
    var endD=new Date();
    var startD=new Date();
    startD.setDate(startD.getDate()-29);
    if(!G('grafStart').value)G('grafStart').value=startD.getFullYear()+'-'+String(startD.getMonth()+1).padStart(2,'0')+'-'+String(startD.getDate()).padStart(2,'0');
    if(!G('grafEnd').value)G('grafEnd').value=endD.getFullYear()+'-'+String(endD.getMonth()+1).padStart(2,'0')+'-'+String(endD.getDate()).padStart(2,'0');
  }
}

```



```

}
sl('Memuat grafik...');
google.script.run.withSuccessHandler(function(r){
  hl();
  if(!r || !r.ok){
    toast(r?r.msg:'Gagal','error');
    return;
  }
  lastGraphData=r;

  // Tampilkan hasil
  G('grafikResult').style.display="";
  G('btnCekSigma').style.display="";
  // PERBAIKAN: Baris G('btnResetGrafik').style.display=""; DIHAPUS karena elemen tidak ada di
  HTML
  G('ljSigmaCard').style.display='none';
  G('cardInterpSigmaBased').style.display='none';

  // Render kop surat
  G('grafKopContainer').innerHTML=buildKopHTML(CD.kop,true);
  G('grafKopContainer').style.display="";

  // Tunggu sebentar lalu draw chart
  setTimeout(function(){
    try{
      var canvas = G('ljChartUnified');
      if(!canvas){
        toast('Canvas grafik tidak ditemukan','error');
        return;
      }
      destroyChart('ljChartUnified');
      drawLJUnifiedChart('ljChartUnified',r,false,false);

      // Render komponen lain
      drawSigmaMetricChart(r.stats);
      renderGrafStats(r.stats);
      renderQGIBoxes(r.qgiData,r.stats);
      renderGrafWG(r,false);
      renderGrafTTD();

      // Sumber info
      var sumberInfoTexts=[];
      ['L1','L2','L3'].forEach(function(lv){
        if(r.levelMeta[lv])sumberInfoTexts.push(lv+' : Mean='+r.levelMeta[lv].mean+',
SD='+r.levelMeta[lv].sd+' ('+r.levelMeta[lv].sumberInfo+');

```

```

});
G('grafSumberInfo').innerHTML=<strong>Sumber:</strong> '+escH(r.sumber)+' —
'+sumberInfoTexts.join(' | ');
var periodeStr=(r.startDate&&r.endDate)?r.startDate+' s/d '+r.endDate:'-';
G('grafLJDetailInfo').innerHTML='Periode QC: '+escH(periodeStr)+' | Parameter:
'+escH(r.parameter)+' | Alat: '+escH(r.namaAlat)+' | ED: '+escH(r.expiredDate | '-'');

// Load catatan
loadGrafCatatanByKey(paramID+'_'+lotID);
toast('Grafik berhasil ditampilkan','success');
}catch(e){
    toast('Error menggambar grafik: '+e.message,'error');
}
}, 300);
}).withFailureHandler(function(e){
    hl();
    toast('Error: '+e.message | |e | |'Terjadi kesalahan'),'error');
}).getGraphData({
    paramID:paramID,
    lotID:lotID,
    startDate:G('grafStart').value,
    endDate:G('grafEnd').value,
    sumber:G('grafSumber').value | |'Manufaktur'
},getActiveUsername(),getActiveRole());
}
function loadGrafCatatanByKey(key){google.script.run.withSuccessHandler(function(data){var
found=(data | []).find(function(c){return
c.filterKey===key;});G('grafCatatan').value=found?(found.catatan | |'');}).withFailureHandler(function(e){hl();toast('Error: '+e.message | |e | |'Terjadi
kesalahan'),'error');}).getCatatanLaporan(getActiveUsername(),key);}
function drawLJUnifiedChart(canvasId, data, isSigmaBased, hideMarkers){
destroyChart(canvasId);
var ljData = data.ljDataUnified;
var levelMeta = data.levelMeta;
var colors = {L1:'#2563eb', L2:'#f59e0b', L3:'#ef4444'};
var xDates = [];
if(data.startDate && data.endDate){
var cur = new Date(data.startDate + 'T00:00:00');
var endD = new Date(data.endDate + 'T00:00:00');
while(cur <= endD){
xDates.push(cur.getFullYear()+'-'+String(cur.getMonth()+1).padStart(2,'0')+'-'
'+String(cur.getDate()).padStart(2,'0'));
cur.setDate(cur.getDate()+1);
}
}
}

```

```

var xLabels = xDates.map(function(d){
var dt = new Date(d+'T00:00:00');
return String(dt.getDate()).padStart(2,'0')+'/'+String(dt.getMonth()+1).padStart(2,'0');
});
var datasets = [];
['L1','L2','L3'].forEach(function(lv){
var meta = levelMeta[lv];
if(!meta || !meta.mean || !meta.sd) return;
var pts = (ljData[lv] || []);
if(!pts.length) return;
var dateGroups = {};
pts.forEach(function(p){
if(!dateGroups[p.date]) dateGroups[p.date] = [];
dateGroups[p.date].push(p);
});
var sortedDates = Object.keys(dateGroups).sort();
var dataPoints = [];
var pointStyles = [];
var pointColors = [];
var pointRadius = [];
var pointBorderColors = [];
var pointLabels = [];
var pointHoverRadii = [];
sortedDates.forEach(function(dateStr){
var dayPts = dateGroups[dateStr];
var xIdx = xDates.indexOf(dateStr);
if(xIdx < 0) return;
dayPts.forEach(function(p, subIdx){
var z = (p.value - meta.mean) / meta.sd;
var yOffset = (subIdx > 0) ? (subIdx * 0.03) : 0;
dataPoints.push({
x: xIdx,
y: parseFloat((z + yOffset).toFixed(3)),
rawValue: p.value,
qcID: p.qcID,
dateLabel: p.date,
_originalZ: parseFloat(z.toFixed(3))
});
});
var wg = meta.westgard[xIdx] || [];
var hasRejection = wg.some(function(v){ return v.type==='rejection' && (!isSigmaBased ? true : lv.ignored); });
var hasWarning = wg.some(function(v){ return v.type==='warning' && (!isSigmaBased ? true : lv.ignored); });
var firstReject = wg.find(function(v){ return v.type==='rejection' && (!isSigmaBased ? true : lv.ignored); });

```

```

if(hasRejection && !hideMarkers){
  pointStyles.push('triangle');
  pointColors.push(isSigmaBased ? '#9333ea' : '#ef4444');
  pointBorderColors.push(isSigmaBased ? '#581c87' : '#991b1b');
  pointRadius.push(13);
  pointLabels.push(firstReject ? firstReject.rule + (isSigmaBased ? ' ( $\sigma$ )' : '') : 'R');
  pointHoverRadii.push(16);
} else if(hasWarning && !hideMarkers){
  // PERBAIKAN: Tampilkan warning di mode sigma-based juga (jika tidak di-ignore)
  var firstWarn = wg.find(function(v){ return v.type==='warning' && (!isSigmaBased ? true :
!v.ignored); });
  if (firstWarn) {
    pointStyles.push('rectRot');
    pointColors.push(isSigmaBased ? '#f59e0b' : '#f59e0b');
    pointBorderColors.push('#b45309');
    pointRadius.push(10);
    pointLabels.push(firstWarn.rule + (isSigmaBased ? ' ( $\sigma$ )' : ''));
    pointHoverRadii.push(13);
  } else {
    pointStyles.push('circle');
    pointColors.push(colors[lv]);
    pointBorderColors.push(colors[lv]);
    pointRadius.push(7);
    pointLabels.push('');
    pointHoverRadii.push(10);
  }
} else {
  pointStyles.push('circle');
  pointColors.push(colors[lv]);
  pointBorderColors.push(colors[lv]);
  pointRadius.push(7);
  pointLabels.push('');
  pointHoverRadii.push(10);
}
});
});
dataPoints.sort(function(a,b){ return a.x - b.x; });
datasets.push({
label: lv + ' ( $\mu$ = ' + meta.mean + ',  $\sigma$ = ' + meta.sd + ')',
data: dataPoints,
borderColor: colors[lv],
backgroundColor: colors[lv] + '33',
pointBackgroundColor: pointColors,
pointBorderColor: pointBorderColors,
pointStyle: pointStyles,

```

```

pointRadius: pointRadius,
pointHoverRadius: pointHoverRadii,
tension: 0,
fill: false,
borderWidth: 2,
showLine: true,
parsing: false,
spanGaps: false,
_labels: pointLabels,
_dateIndices: sortedDates.map(function(d){ return xDates.indexOf(d); })
});
});
var annotations = {
mean: {type:'line', yMin:0, yMax:0, borderColor:'#0f172a', borderWidth:2, label:{display:true,
content:'Mean', position:'start', font:{size:10}}},
p1: {type:'line', yMin:1, yMax:1, borderColor:'#10b981', borderDash:[4,4], label:{display:true,
content:'+1SD', position:'end', font:{size:9}}},
m1: {type:'line', yMin:-1, yMax:-1, borderColor:'#10b981', borderDash:[4,4], label:{display:true,
content:'-1SD', position:'end', font:{size:9}}},
p2: {type:'line', yMin:2, yMax:2, borderColor:'#f59e0b', borderDash:[6,4], label:{display:true,
content:'+2SD', position:'end', font:{size:9}}},
m2: {type:'line', yMin:-2, yMax:-2, borderColor:'#f59e0b', borderDash:[6,4], label:{display:true,
content:'-2SD', position:'end', font:{size:9}}},
p3: {type:'line', yMin:3, yMax:3, borderColor:'#ef4444', borderDash:[8,4], label:{display:true,
content:'+3SD', position:'end', font:{size:9}}},
m3: {type:'line', yMin:-3, yMax:-3, borderColor:'#ef4444', borderDash:[8,4], label:{display:true,
content:'-3SD', position:'end', font:{size:9}}}
};
var ruleLabelPlugin = {
id: 'ruleLabel_' + canvasId,
afterDatasetsDraw: function(chart){
if(!hideMarkers) return;
var ctx = chart.ctx;
chart.data.datasets.forEach(function(ds, dsi){
if(!ds._labels) return;
var meta = chart.getDatasetMeta(dsi);
if(!meta || !meta.data) return;
meta.data.forEach(function(pt, idx){
var label = ds._labels[idx];
if(label && pt){
ctx.save();
var isWarn = label === '1-2s';
ctx.font = 'bold 10px Inter,sans-serif';
ctx.textAlign = 'center';
var txtW = ctx.measureText(label).width + 8;

```

```

ctx.fillStyle = isWarn ? 'rgba(254,243,199,0.98)' : 'rgba(254,226,226,0.98)';
ctx.fillRect(pt.x - txtW/2, pt.y - 28, txtW, 16);
ctx.strokeStyle = isWarn ? '#b45309' : '#991b1b';
ctx.lineWidth = 1;
ctx.strokeRect(pt.x - txtW/2, pt.y - 28, txtW, 16);
ctx.fillStyle = isWarn ? '#b45309' : '#991b1b';
ctx.fillText(label, pt.x, pt.y - 15);
ctx.restore();
}
});
});
}
};
chartInstances[canvasId] = new Chart(G(canvasId).getContext('2d'), {
type: 'line',
data: { datasets: datasets, labels: xLabels },
options: {
responsive: true,
maintainAspectRatio: false,
plugins: {
annotation: { annotations: annotations },
tooltip: {
callbacks: {
title: function(ctx){ return 'Tanggal: ' + fmtDate(ctx[0].raw.dateLabel); },
label: function(ctx){
var p = ctx.raw;
return ctx.dataset.label.split(' ')[0] + ' = ' + p.rawValue + ' (Z=' + p._originalZ + ')';
}
}
},
legend: { position: 'top' }
},
scales: {
x: {
type: 'linear',
title: { display: true, text: 'Tanggal' },
min: 0,
max: Math.max(0, xDates.length - 1),
ticks: {
stepSize: 1,
callback: function(v){
var i = Math.round(v);
return (i >= 0 && i < xLabels.length) ? xLabels[i] : '';
},
maxRotation: 60,

```

```

minRotation: 45,
autoSkip: false,
font: { size: 9 }
},
grid: { color: 'rgba(226,232,240,0.5)', lineWidth: 1 }
},
y: {
title: { display: true, text: 'Z-Score (SD)' },
min: -4,
max: 4,
ticks: {
stepSize: 1,
callback: function(v){ return v + 'SD'; }
}
}
},
plugins: [ruleLabelPlugin]
});
}

```

```

//  PERBAIKAN v9.6: cekSigmaWestgard dengan filter tambahan (Permintaan #2,3,4,5,8)
function cekSigmaWestgard(){
  if(!lastGraphData){toast('Tampilkan grafik dulu','warning');return;}
  sl('Menghitung sigma...');
  var siklusPME=G('grafSiklusPME').value || null;
  var tahunSiklus=G('grafTahunSiklus').value || null;
  var periodeCSVal=G('grafPeriodeCS').value;
  var periodeCVOptVal=G('grafPeriodeCVOpt').value;
  var periodeCS=null;
  if(periodeCSVal){var
parts=periodeCSVal.split('|');if(parts.length===2)periodeCS={start:parts[0],end:parts[1]};}
  var periodeCVOpt=null;
  if(periodeCVOptVal){var
parts2=periodeCVOptVal.split('|');if(parts2.length===2)periodeCVOpt={start:parts2[0],end:parts2[1]}
;}
}

```

```

google.script.run.withSuccessHandler(function(r){
  hl();if(!r || !r.ok){toast(r?.msg:'Gagal','error');return;}
  lastSigmaBasedData=r;
  G('ljSigmaCard').style.display="";
  G('cardInterpSigmaBased').style.display="";
  // PERBAIKAN: Baris G('btnResetGrafik').style.display=""; DIHAPUS karena elemen tidak ada di
HTML

```

```

renderSigmaInfoBox(r);
drawLJUnifiedChart('ljChartSigma',r,true,false);

G('grafSigmaSourceInfo').innerHTML=<strong>Sumber Sigma:</strong> '+escH(r.sigmaSource)+'
— <strong>Sigma Terkecil:</strong> '+ (r.smallestSigma!=null?<span
class="" +sigmaClass(r.smallestSigma)+">'+r.smallestSigma+'</span>':'N/A')+' —
<strong>Mode:</strong> '+escH(r.activeRulesMode | '-' );
renderInterpSigmaBased(r);

setTimeout(function(){G('ljSigmaCard').scrollIntoView({behavior:'smooth',block:'start'});},200);
toast('Sigma-based aktif!', 'success',4000);
}).withFailureHandler(function(e){
hl();toast('Error: '+ (e.message | |e | |'Terjadi kesalahan'),'error');
}).getSigmaBasedGraphData({
paramID:G('grafParam').value,lotID:G('grafLot').value,
startDate:G('grafStart').value,endDate:G('grafEnd').value,
sumber:G('grafSumber').value | |'Manufaktur',
sigmaSource:G('grafSumberSigma').value,
siklusPME:siklusPME,tahunSiklus:tahunSiklus,
periodeCS:periodeCS,periodeCVOpt:periodeCVOpt
},getActiveUsername(),getActiveRole());
}
// FUNGSI BARU v9.5: Reset/Refresh halaman Grafik
function resetGrafikPage(){
console.log('Reset grafik page...');

G('grafikResult').style.display='none';
G('btnCekSigma').style.display='none';
G('ljSigmaCard').style.display='none';
G('cardInterpSigmaBased').style.display='none';

if(G('grafKopContainer')){
G('grafKopContainer').style.display='none';
G('grafKopContainer').innerHTML=";
}

lastGraphData=null;
lastSigmaBasedData=null;

// Destroy semua chart
['ljChartUnified','ljChartSigma','chartSigmaMetric'].forEach(function(id){
destroyChart(id);
});

// Clear konten

```



```

if(G('grafStatsBody'))G('grafStatsBody').innerHTML="";
if(G('grafQGIBody'))G('grafQGIBody').innerHTML="";
if(G('grafWGBody'))G('grafWGBody').innerHTML="";
if(G('sigmaInfoBox'))G('sigmaInfoBox').innerHTML="";
if(G('interpSigmaBasedBody'))G('interpSigmaBasedBody').innerHTML="";
if(G('grafSumberInfo'))G('grafSumberInfo').innerHTML="";
if(G('grafSigmaSourceInfo'))G('grafSigmaSourceInfo').innerHTML="";
if(G('grafCatatan'))G('grafCatatan').value="";

```

```
// Reset filter
```

```

G('grafStart').value="";
G('grafEnd').value="";
G('grafBidang').value="";
G('grafParam').innerHTML='<option value="">Pilih</option>';
G('grafLot').innerHTML='<option value="">Pilih Lot</option>';
G('grafSumber').value='Manufaktur';
G('grafSumberSigma').value='Sigma Terkecil Terhitung';
G('grafSiklusPME').value="";
G('grafTahunSiklus').value="";
G('grafPeriodeCS').value="";
G('grafPeriodeCVOpt').value="";

```

```
toast('Halaman grafik di-reset','info');
```

```
}
```

```

function renderSigmaInfoBox(r){var sigma=r.smallestSigma;var html='<h4><i class="fas fa-info-circle"></i> Sigma-Based QC Selection</h4><div style="margin-bottom:8px"><span class="sigma-mode">'+escH(r.activeRulesMode|'|N/A')+'</span>';if(sigma!==null)html+='<span><strong>Sigma Terkecil: </strong><span class="'+sigmaClass(sigma)+'" style="font-size:1.2rem;font-weight:800">'+sigma+'</span> ('+sigmaLabel(sigma)+'</span>';html+='</div><div class="rules-active"><strong>Aturan Aktif: </strong>';(r.activeRules|'|[]').forEach(function(rule){html+='<code>'+rule+'</code>';});html+='</div>';if(r.warning)html+='<div class="sigma-warning"><i class="fas fa-exclamation-triangle"></i> '+escH(r.warning)+'</div>';G('sigmaInfoBox').innerHTML=html;}

function renderInterpSigmaBased(r){var sigma=r.smallestSigma;var html='<table style="width:100%;font-size:.88rem"><thead style="background:var(--bg-soft)"><tr><th>Aspek</th><th>Detail</th></tr></thead><tbody>';html+='<tr><td><strong>Sumber</strong></td><td>'+escH(r.sigmaSource)+'</td></tr>';html+='<tr><td><strong>Sigma Terkecil</strong></td><td class="'+sigmaClass(sigma)+'" style="font-size:1.1rem;font-weight:800">'+(sigma!==null?sigma+' ('+sigmaLabel(sigma)+'):'N/A')+'</td></tr>';html+='<tr><td><strong>Mode</strong></td><td>'+escH(r.activeRulesMode|'|')+'</td></tr>';html+='<tr><td><strong>Aturan Aktif</strong></td><td>';(r.activeRules|'|[]').forEach(function(rule){html+='<code>'+rule+'</code>';});html+='</td></tr>';html+='</tbody></table>';G('interpSigmaBased').innerHTML=html;}

```

```

weight:600">'+rule+'</code>'}};html+='</td></tr></tbody></table>';G('interpSigmaBasedBody').innerHTML=html;}

function drawSigmaMetricChart(stats){destroyChart('sigmaMetric');var
labels=[],values=[],bgColors=[];['L1','L2','L3'].forEach(function(lv){if(!stats[lv])return;labels.push(lv);v
values.push(stats[lv].sigma);bgColors.push(sigmaColor(stats[lv].sigma));});if(!labels.length)return;char
tInstances.sigmaMetric=new
Chart(G('chartSigmaMetric').getContext('2d'),{type:'bar',data:{labels:labels,datasets:[{label:'Sigma',d
ata:values,backgroundColor:bgColors,borderRadius:6}]},options:{responsive:true,maintainAspectRat
io:false,plugins:{annotation:{annotations:{line6:{type:'line',yMin:6,yMax:6,borderColor:'#047857',bo
rderDash:[5,5],label:{display:true,content:'6σ',position:'end'}},line3:{type:'line',yMin:3,yMax:3,borde
rColor:'#f59e0b',borderDash:[5,5],label:{display:true,content:'3σ',position:'end'}}}}},scales:{y:{beginA
tZero:true,title:{display:true,text:'Sigma'}}}}});}

function renderGrafStats(stats){var html='<div class="table-
wrap"><table><thead><tr><th>Lvl</th><th>N</th><th>Mean</th><th>SD</th><th>CV%</th><th>B
ias%</th><th>TE%</th><th>Unc</th><th>σ</th><th>Kinerja</th></tr></thead><tbody>';['L1','L2','L
3'].forEach(function(lv){var
s=stats[lv];if(!s)return;html+='<tr><td>'+lv+'</td><td>'+s.n+'</td><td>'+round(s.mean,4)+'</td><td>
'+round(s.sd,4)+'</td><td>'+round(s.cv)+'</td><td>'+round(s.bias)+'</td><td>'+round(s.te)+'</td><t
d>'+round(s.unc,4)+'</td><td class="'+sigmaClass(s.sigma)+'">'+round(s.sigma)+'</td><td
class="'+sigmaClass(s.sigma)+'">'+sigmaLabel(s.sigma)+'</td></tr>'}};html+='</tbody></table></div
>';G('grafStatsBody').innerHTML=html;}

function renderQGIBoxes(qgiData,stats){var wrap=G('grafQGIbody');if(!wrap)return;var
html='',qgiVals=[],hasAny=false;
html+='<div class="qgi-mini-grid">';
['L1','L2','L3'].forEach(function(lv){var qd=qgiData[lv];if(!qd || qd.qgi===null)return;var
s=stats[lv];if(!s)return;if(s.sigma!==null&& s.sigma>=4)return;hasAny=true;var
qgi=qd.qgi;qgiVals.push(qgi);var barPct=Math.min(100,Math.abs(qgi)*50);var
barColor=qgi>1.2?'#2563eb':qgi<0.8?'#ef4444':'#f59e0b';html+='<div class="qgi-mini-box"><h5><i
class="fas fa-crosshairs"></i> QGI '+lv+'</h5><div class="qgi-mini-value"
style="color: '+barColor+'">'+round(qgi,3)+'</div><div class="qgi-mini-bar"><div class="qgi-mini-bar-
fill" style="width: '+barPct+'%;background: '+barColor+'"></div></div><div class="qgi-mini-
interp">'+escH(qd.interp)+'</div></div>'}};
html+='</div>';
if(!hasAny){html='<p style="color:var(--success);font-weight:600"><i class="fas fa-check-circle"></i>
Semua level σ≥4 — QGI tidak diperlukan.</p>'}}
else if(qgiVals.length){var avgQGI=qgiVals.reduce(function(a,b){return a+b;},0)/qgiVals.length;var
interp=avgQGI>1.2?'Imprecision dominant — perbaiki presisi':avgQGI<0.8?'Inaccuracy dominant —
perbaiki akurasi':Balanced — perbaiki keduanya';html+='<div class="qgi-conclusion"><strong><i
class="fas fa-chart-pie"></i> Kesimpulan QGI rata-rata:</strong>
'+round(avgQGI,3)+'<br>'+escH(interp)+'<br><small style="color:var(--text-secondary)">QGI>1.2:
Imprecision | QGI<0.8: Inaccuracy | 0.8-1.2: Balanced</small></div>'}}
wrap.innerHTML=html;
}

function renderGrafWG(data,useSigmaFilter){var html="";['L1','L2','L3'].forEach(function(lv){var
meta=data.levelMeta[lv];if(!meta)return;var

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ljd=data.ljDataUnified[lv];if(!ljd||!ljd.length){html+=''<p><strong>'+lv+':</strong> <em>Tidak ada
data</em></p>';return;}var sigma=meta.sigma;var ruleNote=' <small style="color:var(--text-
secondary)">(σ='+ (sigma!=null?sigma:'-')+')</small>';var
allViolations=[];Object.keys(meta.westgard).forEach(function(idx){var
viols=meta.westgard[idx]||[];viols.forEach(function(v){var ptIdx=parseInt(idx);var
pt=ljd[ptIdx];if(pt)allViolations.push({rule:v.rule,type:v.type,desc:v.desc,category:v.category||'Lainn
ya',categoryDesc:v.categoryDesc||'',ignored:v.ignored,pointDate:pt.date,pointValue:pt.value});});v
ar seenKeys={};var uniqueViolations=allViolations.filter(function(v){var
key=v.rule+'_'+v.pointDate+'_'+v.pointValue;if(seenKeys[key])return false;seenKeys[key]=true;return
true;});
if(uniqueViolations.length===0){html+=''<div style="padding:10px;background:var(--success-
light);border-left:4px solid var(--success);border-radius:6px;margin-
bottom:10px"><strong>'+lv+':</strong> <span style="color:#047857;font-weight:600"><i class="fas
fa-check-circle"></i> Tidak ada pelanggaran</span>'+ruleNote+'</div>';}
else{var byCat={};uniqueViolations.forEach(function(v){var
cat=useSigmaFilter&&v.ignored?'Diabaikan':v.category;if(!byCat[cat])byCat[cat]=[];byCat[cat].push(v
);});html+=''<div style="padding:12px;background:#fff;border-left:4px solid var(--danger);border-
radius:6px;margin-bottom:10px;border:1px solid var(--border-soft)"><strong>'+lv+':</strong>
'+uniqueViolations.length+' pelanggaran'+ruleNote;Object.keys(byCat).forEach(function(cat){var
catColor=cat==='Random Error'?'badge-random':cat==='Systematic Error'?'badge-
systematic':cat==='Warning'?'badge-warning':'badge-secondary';html+=''<div style="margin-
top:8px"><span class="badge '+catColor+'" style="font-size:.78rem">'+escH(cat)+'
'+byCat[cat].length+'</span><ul style="margin-left:20px;margin-top:4px;font-
size:.85rem">';byCat[cat].forEach(function(v){var cls=v.type==='warning'?'warning':'danger';var
ignoreMark=v.ignored&&useSigmaFilter?' <small style="text-decoration:line-
through">(diabaikan)</small>':'';html+=''<li><span class="wg-rule-badge
'+(cls==='warning'?'warning':'')+'" style="font-size:.7rem">'+v.rule+'</span> '+escH(v.desc)+'
<small>—
'+fmtDate(v.pointDate)+'</small>'+ignoreMark+'</li>');});html+=''</ul>';if(byCat[cat].length>0&&byCa
t[cat][0].categoryDesc)html+=''<div style="font-size:.78rem;color:var(--text-secondary);margin-
left:20px;font-style:italic"><i class="fas fa-info-circle"></i>
'+escH(byCat[cat][0].categoryDesc)+'</div>';html+=''</div>';});html+=''</div>';}
});if(data.wgAcross&&data.wgAcross.length){html+=''<hr style="margin:14px 0"><h5>Across
Levels</h5>';data.wgAcross.forEach(function(w){var
cls=w.ignored&&useSigmaFilter?'secondary':'danger';html+=''<p><span class="badge badge-
'+cls+'">'+w.rule+'</span> '+escH(w.desc)+'</p>');}
html+=''<hr style="margin:14px 0"><div style="font-size:.78rem;display:flex;gap:14px;flex-
wrap:wrap"><span><span class="badge badge-random">RE</span> Random</span><span><span class="badge badge-systematic">SE</span> Systematic</span><span><span class="badge badge-warning">W</span> Warning</span></div>';
G('grafWGBody').innerHTML=html||'<p style="color:var(--text-secondary)">Tidak ada data</p>';}
function
renderGrafTTD(){G('grafTTDBody').innerHTML=buildTTDHTML(CD.kop,'grafQR');setTimeout(function
){var kop=CD.kop||{};renderQR('grafQR1','didiQC|'+(kop.namaRS||'Lab')+'|

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```

'+(kop.kepalaRuangan | | ''));renderQR('grafQR2','didiQC | '+ (kop.namaRS | | 'Lab')+' |
'+(kop.dokterPJ | | ''));},200);}
function saveGrafCatatan(){var
paramID=G('grafParam').value,lotID=G('grafLot').value;if(!paramID | | !lotID){toast('Pilih parameter
dan
lot','warning');return;}sl('Menyimpan...');google.script.run.withSuccessHandler(function(r){hl();if(r.ok
)toast('Catatan disimpan','success');else
toast(r.msg,'error');}).withFailureHandler(function(e){hl();toast('Error: '+ (e.message | | e | | 'Terjadi
kesalahan'),'error');}).saveCatatanLaporan({filterKey:paramID+'_'+lotID,paramID:paramID,lotID:lotID,
namaAlat:'',catatan:G('grafCatatan').value},getActiveUsername(),getLogUser());}
var GRAFIK_OPTIONS=[{id:'all',label:'a. Semua',desc:'Seluruh komponen grafik dan
analisis'},{id:'b',label:'b. Grafik LJ + Statistik + Catatan + TTD',desc:'Grafik LJ, statistik, catatan dokter
dan tandatangan'},{id:'c',label:'c. Grafik LJ + LJ Sigma + Statistik + Catatan + TTD',desc:'Grafik LJ,
Grafik LJ Sigma-Based, statistik, catatan dokter dan tandatangan'},{id:'d',label:'d. Grafik LJ + LJ Sigma
+ Statistik + Interp Sigma + Westgard + Catatan + TTD',desc:'Grafik LJ, LJ Sigma-Based, statistik,
interpretasi sigma, analisa westgard, catatan dan tandatangan'},{id:'e',label:'e. Grafik LJ + Statistik +
Westgard + Catatan + TTD',desc:'Grafik LJ, statistik, analisa westgard, catatan dan
tandatangan'},{id:'f',label:'f. Grafik LJ (tanpa marker/label) + Catatan + TTD',desc:'Grafik LJ bersih
tanpa marker dan label, catatan dokter dan tandatangan'}];
function buildGrafikMenuHTML(action){return GRAFIK_OPTIONS.map(function(opt){return '<button
onclick="execGrafikExport(\''+action+\'\',\''+opt.id+\'\'")"><div class="option-
title">'+opt.label+</div><div class="option-desc">'+opt.desc+</div></button>'}}.join("");}
function toggleGrafikExportMenu(action){['print','pdf','ss'].forEach(function(a){var
menuId='grafik'+(a==='ss'? 'SS':a.charAt(0).toUpperCase()+a.slice(1))+ 'Menu';var
menu=G(menuId);if(menu)menu.classList.remove('show');});var
menuId='grafik'+(action==='ss'? 'SS':action.charAt(0).toUpperCase()+action.slice(1))+ 'Menu';var
menu=G(menuId);if(menu){menu.innerHTML=buildGrafikMenuHTML(action);menu.classList.add('sh
ow');}}
function execGrafikExport(action,optId){['print','pdf','ss'].forEach(function(a){var
menuId='grafik'+(a==='ss'? 'SS':a.charAt(0).toUpperCase()+a.slice(1))+ 'Menu';var
menu=G(menuId);if(menu)menu.classList.remove('show');});applyGrafikVisibility(optId);G('grafKopC
ontainer').style.display='';setTimeout(function(){if(action==='print')doGrafikPrint();else
if(action==='pdf')doGrafikPdf();else doGrafikScreenshot();},300);}
function
applyGrafikVisibility(optId){['grafLJCard','ljSigmaCard','grafStatsCard','grafSigmaMetricCard','grafQGIC
ard','cardInterpSigmaBased','grafWGCard','grafCatatanCard','grafTTDCard'].forEach(function(id){va
r
el=G(id);if(el)el.style.display='';});G('grafKopContainer').style.display='';if(optId==='b'){G('ljSigmaCard'
).style.display='none';G('cardInterpSigmaBased').style.display='none';G('grafWGCard').style.display='
none';G('grafSigmaMetricCard').style.display='none';G('grafQGICard').style.display='none';}else
if(optId==='c'){G('cardInterpSigmaBased').style.display='none';G('grafWGCard').style.display='none';
G('grafSigmaMetricCard').style.display='none';G('grafQGICard').style.display='none';if(!lastSigmaBase
dData)G('ljSigmaCard').style.display='none';}else
if(optId==='d'){G('grafSigmaMetricCard').style.display='none';G('grafQGICard').style.display='none';if(
!lastSigmaBasedData){G('ljSigmaCard').style.display='none';G('cardInterpSigmaBased').style.display='

```

```

none'}}else
if(optId==='e'){G('ljSigmaCard').style.display='none';G('cardInterpSigmaBased').style.display='none';G('grafSigmaMetricCard').style.display='none';G('grafQGICard').style.display='none';}else
if(optId==='f'){G('ljSigmaCard').style.display='none';G('cardInterpSigmaBased').style.display='none';G('grafWGCard').style.display='none';G('grafStatsCard').style.display='none';G('grafSigmaMetricCard').style.display='none';G('grafQGICard').style.display='none';if(lastGraphData)drawLJUnifiedChart('ljChartUnified',lastGraphData,false,true);}else{G('grafQGICard').style.display='none';}}
function
restoreGrafikVisibility(){[ 'grafLJCard','grafStatsCard','grafSigmaMetricCard','grafQGICard','grafWGCard','grafCatatanCard','grafTTDCard'].forEach(function(id){var
el=G(id);if(el)el.style.display='';});G('ljSigmaCard').style.display=lastSigmaBasedData?':none';G('cardInterpSigmaBased').style.display=lastSigmaBasedData?':none';G('grafKopContainer').style.display='none';if(lastGraphData)drawLJUnifiedChart('ljChartUnified',lastGraphData,false,false);}
function getCardHTMLWithChartImages(cardId){var el=G(cardId);if(!el)return '';var
clone=el.cloneNode(true);var origCanvases=el.querySelectorAll('canvas');var
clonedCanvases=clone.querySelectorAll('canvas');for(var i=0;i<clonedCanvases.length;i++){try{var
origCanvas=origCanvases[i];if(origCanvas){var dataUrl=origCanvas.toDataURL('image/png');var
img=document.createElement('img');img.src=dataUrl;img.style.maxWidth='100%';img.style.height='
auto';img.style.display='block';clonedCanvases[i].parentNode.replaceChild(img,clonedCanvases[i]);}}
catch(e){}}var origTextareas=el.querySelectorAll('textarea');var
clonedTextareas=clone.querySelectorAll('textarea');for(var
j=0;j<origTextareas.length;j++){if(clonedTextareas[j]&&origTextareas[j]){clonedTextareas[j].innerHTML=esch(origTextareas[j].value);clonedTextareas[j].setAttribute('value',origTextareas[j].value);var
div=document.createElement('div');div.style.cssText='white-space:pre-wrap;word-break:break-
word;padding:8px 10px;background:#fef9c3;border:1px solid #fde68a;border-radius:6px;font-
size:.88rem;min-height:20px;font-family:Inter,sans-
serif';div.textContent=origTextareas[j].value;clonedTextareas[j].parentNode.replaceChild(div,cloned
Textareas[j]);}}return clone.innerHTML;}
function doGrafikPrint(){var cards=G('grafikResult');if(!cards)return;var
w=window.open('','_blank');var kopHTML=G('grafKopContainer').innerHTML;var
contentHTML='';[ 'grafLJCard','ljSigmaCard','grafStatsCard','grafSigmaMetricCard','grafQGICard','cardInterpSigmaBased','grafWGCard','grafCatatanCard','grafTTDCard'].forEach(function(id){var
el=G(id);if(el&&el.style.display!=='none'){contentHTML+=''<div style="margin-bottom:2px;page-
break-inside:avoid">'+getCardHTMLWithChartImages(id)+'</div>'}});w.document.write('<!DOCTYPE
html><html><head><title>Grafik QC</title><style>{*{box-sizing:border-
box;margin:0;padding:0}body{font-family:Inter,sans-serif;padding:1mm 2mm;font-
size:9pt;color:#1a1a1a;max-width:210mm;margin:0 auto}table{width:100%;border-
collapse:collapse}th,td{padding:2px 4px;border:1px solid #ddd;text-align:left;font-
size:8pt}canvas,img{max-width:100%!important;height:auto!important}.card{margin-
bottom:2px!important;border:none!important;box-shadow:none!important}.card-
header{background:#f0f0f0;padding:3px 6px;font-weight:700;font-size:9pt;border-bottom:2px solid
#2563eb;margin-bottom:1px}.card-body{padding:1px 2px}.sigma-info-
box{background:#fff7ed;padding:4px;border-radius:4px;border:1px solid #fed7aa;margin-
bottom:4px}p{margin:0 0 1px 0!important;font-size:8pt}@page{margin:2mm 3mm;size:A4}.ttd-
area{display:flex;justify-content:space-between;margin-top:10px}.img-result-

```

```

footer{display:flex;justify-content:space-between;align-items:flex-start;margin-
top:10px!important;gap:10px}.ttd-col{text-align:center;width:45%}.ttd-col img,.ttd-col
canvas{width:60px!important;height:60px!important}.ttd-name{margin-top:16px!important;border-
top:1px solid #333;padding-top:3px}.img-result-header{margin-bottom:4px!important;padding-
bottom:4px!important}.img-electronic{margin-top:4px!important}h3{font-size:9.5pt;margin:1px
0!important}h4{font-size:9pt;margin:2px 0!important}h5{font-size:8.5pt;margin:1px
0!important}</style></head><body>'+kopHTML+contentHTML+'</body></html>');w.document.clos
e();setTimeout(function(){w.print();},600);}
function doGrafikPdf(){var cards=G('grafikResult');if(!cards)return;sl('Membuat PDF...');var
kopHTML=G('grafKopContainer').innerHTML;var
container=document.createElement('div');container.style.position='fixed';container.style.left='-
9999px';container.style.top='0';container.style.width='794px';container.style.background='#fff';cont
ainer.style.padding='2px 3px';container.style.fontSize='9pt';var
contentHTML="";['grafLJCard','ljSigmaCard','grafStatsCard','grafSigmaMetricCard','grafQGICard','cardI
nterpSigmaBased','grafWGCard','grafCatatanCard','grafTTDCard'].forEach(function(id){var
el=G(id);if(el&&el.style.display!=='none'){contentHTML+=''<div style="margin-
bottom:2px">'+getCardHTMLWithChartImages(id)+'</div>'}});container.innerHTML='<div
style="font-family:Inter,sans-
serif">'+kopHTML+contentHTML+'</div>';document.body.appendChild(container);setTimeout(funci
on(){html2canvas(container.firstElementChild,{scale:2,useCORS:true,allowTaint:true,backgroundCol
or:'#ffffff'}).then(function(canvas){document.body.removeChild(container);hl();var pdf=new
jspdf.jsPDF('p','mm','a4');var pW=pdf.internal.pageSize.getWidth()-
6,pH=pdf.internal.pageSize.getHeight()-6;var
cH=canvas.height*pW/canvas.width;if(cH<=pH){pdf.addImage(canvas.toDataURL('image/png'),'PNG'
,3,3,pW,cH);}else{var rem=cH,sY=0;while(rem>0){var sH=Math.min(pH,rem);var
sc=document.createElement('canvas');sc.width=canvas.width;sc.height=sH*canvas.width/pW;sc.get
Context('2d').drawImage(canvas,0,sY,canvas.width,sc.height,0,0,sc.width,sc.height);if(sY>0)pdf.addP
age();pdf.addImage(sc.toDataURL('image/png'),'PNG',3,3,pW,sH);sY+=sc.height;rem-
=sH;}}pdf.save('grafik_qc_'+new
Date().getTime()+'.pdf');restoreGrafikVisibility();}).catch(function(e){document.body.removeChild(co
ntainer);hl();restoreGrafikVisibility();toast('Error: '+e.message || e || 'Terjadi
kesalahan'),'error');});},500);}
function doGrafikScreenshot(){var cards=G('grafikResult');if(!cards)return;sl('Screenshot...');var
kopHTML=G('grafKopContainer').innerHTML;var
container=document.createElement('div');container.style.position='fixed';container.style.left='-
9999px';container.style.top='0';container.style.width='794px';container.style.background='#fff';cont
ainer.style.padding='2px 3px';var
contentHTML="";['grafLJCard','ljSigmaCard','grafStatsCard','grafSigmaMetricCard','grafQGICard','cardI
nterpSigmaBased','grafWGCard','grafCatatanCard','grafTTDCard'].forEach(function(id){var
el=G(id);if(el&&el.style.display!=='none'){contentHTML+=''<div style="margin-
bottom:2px">'+getCardHTMLWithChartImages(id)+'</div>'}});container.innerHTML='<div
style="font-family:Inter,sans-
serif">'+kopHTML+contentHTML+'</div>';document.body.appendChild(container);setTimeout(funci
on(){html2canvas(container.firstElementChild,{scale:2,useCORS:true,allowTaint:true,backgroundCol
or:'#ffffff'}).then(function(canvas){document.body.removeChild(container);hl();var

```

```

link=document.createElement('a');link.download='grafik_qc_'+new
Date().getTime()+'.png';link.href=canvas.toDataURL();link.click();toast('Screenshot
tersimpan','success');restoreGrafikVisibility();}).catch(function(e){document.body.removeChild(conta
iner);hl();restoreGrafikVisibility();toast('Error: '+e.message+' | e | 'Terjadi
kesalahan'),'error');});},500);}

```

```

// ===== EXPORT PANEL (Form Pilihan Cetak) v10.0 =====
var _exportPanelState={menu:'',action:''};
function openExportPanel(menu,action){
  _exportPanelState={menu:menu,action:action};
  var overlay=G('exportPanelOverlay');
  var titleEl=G('exportPanelTitle');
  var bodyEl=G('exportPanelBody');
  var iconMap={print:<i class="fas fa-print"></i>,pdf:<i class="fas fa-file-pdf"></i>,ss:<i class="fas
fa-camera"></i>};
  var titleMap={grafik:'Grafik & Analisis',laporan:'Laporan QC'};
  var actionLabel=action==='ss'? 'Screenshot':action==='pdf'? 'PDF': 'Print';
  titleEl.innerHTML=(iconMap[action] || '')+' '+actionLabel+' — '+ (titleMap[menu] || menu);
  var options=[];
  if(menu==='grafik')options=GRAFIK_OPTIONS;
  else if(menu==='laporan')options=LAPORAN_OPTIONS;
  var html="";
  options.forEach(function(opt,i){
    html+<div class="export-option-card" + (i===0? ' selected':'') + ""
    onclick="selectExportOption(this,\'"+opt.id+"\')">';
    html+<div class="export-option-radio"></div>';
    html+<div><div class="export-option-label">+escH(opt.label)+</div>';
    if(opt.desc)html+<div class="export-option-desc">+escH(opt.desc)+</div>';
    html+</div></div>';
  });
  html+<div style="margin-top:16px;text-align:center"><button class="btn btn-primary"
  onclick="executeExportPanel()" style="min-width:200px;padding:10px 24px;font-size:1rem;font-
weight:700"><i class="fas fa-check"></i> '+actionLabel+' Sekarang</button></div>';
  bodyEl.innerHTML=html;
  overlay.classList.add('show');
}
function selectExportOption(cardEl,optId){
  QA('.export-option-card').forEach(function(c){c.classList.remove('selected');});
  cardEl.classList.add('selected');
  _exportPanelState.optId=optId;
}
function closeExportPanel(){
  G('exportPanelOverlay').classList.remove('show');
}
function executeExportPanel(){

```

```

var
menu=_exportPanelState.menu,action=_exportPanelState.action,optId=_exportPanelState.optId;
if(!optId){var first=QA('.export-option-
card')[0];if(first){first.click();optId=_exportPanelState.optId;}else{closeExportPanel();return;}}
closeExportPanel();
if(menu==='grafik'){
applyGrafikVisibility(optId);G('grafKopContainer').style.display="";
setTimeout(function(){if(action==='print')doGrafikPrint();else if(action==='pdf')doGrafikPdf();else
doGrafikScreenshot();},300);
}else if(menu==='laporan'){
applyLaporanVisibility(optId);
setTimeout(function(){if(action==='print')doLaporanPrint();else if(action==='pdf')doLaporanPdf();else
doLaporanScreenshot();},300);
}
}

```

//  PERBAIKAN v9.6: renderLaporan dengan Data QC di atas Statistik + Catatan Dokter (Permintaan #9, #10)

```

function renderLaporan(r){
var kop=r.kop || {};
var html='<div class="card lap-kop-block"><div class="card-body">'+buildKopHTML(kop,true)+'<h3
style="text-align:center;margin:14px 0;font-weight:800">LAPORAN QUALITY CONTROL</h3>';
var filterStr="";
if(r.filter.startDate || r.filter.endDate)filterStr+='Periode: '+fmtDateIndo(r.filter.startDate || '-')+' s/d
'+fmtDateIndo(r.filter.endDate || '-')+' | ';
if(r.filter.bidang)filterStr+='Bidang: '+r.filter.bidang;
html+='<p style="text-align:center;font-size:.85rem;color:var(--text-
secondary)">'+escH(filterStr)+'</p></div></div>';
if(!r.data || !r.data.length){html+='<div class="card"><div class="card-body"><p style="text-
align:center;color:var(--text-secondary)">Tidak ada data</p></div></div>';}
else{r.data.forEach(function(item,idx){html+='<div class="card lap-section '+ (idx===0?'lap-first-
section':'')+' "><div class="card-body"><h4 style="background:linear-gradient(135deg,var(--
primary),var(--primary-dark));color:#fff;padding:10px 14px;border-radius:8px;margin-
bottom:4px">'+(idx+1)+' . Grafik Levey-Jennings (3 Level) — '+escH(item.parameter)+'</h4><p
style="font-size:.84rem;color:var(--text-secondary);margin-bottom:10px">Periode QC:
'+escH(r.filter.startDate || '-')+' s/d '+escH(r.filter.endDate || '-')+' | Parameter:
'+escH(item.parameter)+' | Nama Alat: '+escH(item.namaAlat)+' | No. Lot:
'+escH(item.noLot)+(item.expiredDate?' | ED: '+escH(item.expiredDate):')+'</p>';

```

//  DATA QC DI ATAS (Perbaikan #10)

```

html+='<div class="lap-qcdata-section"><h5 style="margin-top:14px;color:var(--primary)"><i
class="fas fa-table"></i> Data QC</h5><div class="table-wrap"><table style="font-
size:.82rem"><thead style="background:var(--bg-
soft)"><tr><th>No</th><th>Tgl</th><th>L1</th><th>L2</th><th>L3</th><th>Status</th><th>Valid
ator</th><th>Catatan</th></tr></thead><tbody>';

```



```

(item.qcData || []).forEach(function(q,qi){var
flagsHtml='';if(item.wgFlags&&item.wgFlags[q.qcID]){var
flagsObj=item.wgFlags[q.qcID];['L1','L2','L3'].forEach(function(lv){if(flagsObj[lv])flagsObj[lv].forEach(f
unction(f){var cn=f.category==='Random Error'?'random':'f.category==='Systematic
Error'?'systematic':'warning-flag';flagsHtml+=''<span class="wg-cell-flag
'+cn+'>'+lv+' '+f.rule+'</span> '});});var sb=q.validated?'<span class="badge badge-
success">Valid</span>':'<span class="badge badge-
warning">Pending</span>';html+=''<tr><td>'+(qi+1)+'</td><td>'+fmtDate(q.tanggal)+(flagsHtml?'<br
>'+flagsHtml:'')+'</td><td>'+(q.level1 || '-')+'</td><td>'+(q.level2 || '-')+'</td><td>'+(q.level3 || '-
')+'</td><td>'+sb+'</td><td><small>'+escH(q.validatedBy || '-
')+'</small></td><td><small>'+escH(q.catatanValidasi || '-')+'</small></td></tr>');};
html+=''</tbody></table></div></div>;
//  STATISTIK DI BAWAH (Perbaikan #10)
html+=''<div class="lap-stats-section"><h5 style="margin-top:14px;color:var(--primary)"><i class="fas
fa-chart-bar"></i> Statistik</h5><table style="width:100%;margin-top:6px;font-size:.84rem"><thead
style="background:var(--bg-
soft)"><tr><th>Lvl</th><th>N</th><th>Mean</th><th>SD</th><th>CV%</th><th>Bias%</th><th>T
E%</th><th> $\sigma$ </th><th>Kinerja</th></tr></thead><tbody>;
['L1','L2','L3'].forEach(function(lv){var
s=item.stats[lv];if(!s)return;html+=''<tr><td>'+lv+'</td><td>'+s.n+'</td><td>'+round(s.mean,4)+'</td
><td>'+round(s.sd,4)+'</td><td>'+round(s.cv)+'</td><td>'+round(s.bias)+'</td><td>'+round(s.te)+'<
/td><td class="'+sigmaClass(s.sigma)+'">'+round(s.sigma)+'</td><td
class="'+sigmaClass(s.sigma)+'">'+sigmaLabel(s.sigma)+'</td></tr>');};
html+=''</tbody></table>;
['L1','L2','L3'].forEach(function(lv){var ip=item.interpretasi?item.interpretasi[lv]:null;if(!ip)return;var
cls=ip.statusColor;html+=''<div class="laporan-interp '+cls+'><h5>'+lv+' —
'+ip.status.toUpperCase()+'</h5><ul>';ip.notes.forEach(function(n){html+=''<li>'+escH(n)+'</li>');};ht
ml+=''</ul></div>');};
html+=''</div></div></div>');};}
html+=''<div class="card lap-ttd-section"><div class="card-
body">'+buildTTDHTML(kop,'lapQR')+'</div></div>;
G('laporanContent').innerHTML=html;
//  Tampilkan card catatan dokter (Perbaikan #9)
if(r.data&&r.data.length>0){
G('laporanCatatanDokterCard').style.display="";
var firstItem=r.data[0];
loadLaporanCatatanDokter(firstItem.paramID,firstItem.lotID,firstItem.bidang);
}else{G('laporanCatatanDokterCard').style.display='none';}
setTimeout(function(){renderQR('lapQR1',didiQC | '+(kop.namaRS || 'Lab')+' |
'+(kop.kepalaRuangan || '');renderQR('lapQR2',didiQC | '+(kop.namaRS || 'Lab')+' |
'+(kop.dokterPJ || '');});,300);
}

//  FUNGSI BARU v9.6: Save & Load Catatan Dokter (Permintaan #9)
function saveLaporanCatatanDokter(){

```

```

var paramID=G('lapParam').value,lotID=G('lapLot').value,bidang=G('lapBidang').value;
var catatan=G('laporanCatatanDokter').value;
if(!paramID || !lotID){toast('Pilih parameter dan lot dulu','warning');return;}
var filterKey=bidang+'_'+paramID+'_'+lotID;
sl('Menyimpan catatan dokter...');
google.script.run.withSuccessHandler(function(r){hl();if(r.ok)toast('Catatan dokter
disimpan','success');else toast(r.msg,'error');});
.withFailureHandler(function(e){hl();toast('Error: '+e.message | e | 'Terjadi kesalahan'),'error');});
.saveCatatanDokter({filterKey:filterKey,bidang:bidang,paramID:paramID,lotID:lotID,catatan:catatan},
getActiveUsername());
}
function loadLaporanCatatanDokter(paramID,lotID,bidang){
if(!paramID || !lotID)return;
var filterKey=bidang+'_'+paramID+'_'+lotID;
google.script.run.withSuccessHandler(function(data){
var found=(data | []).find(function(c){return c.filterKey===filterKey;});
G('laporanCatatanDokter').value=found?(found.catatan | "):";
}).withFailureHandler(function(e){hl();toast('Error: '+e.message | e | 'Terjadi
kesalahan'),'error');}).getCatatanDokter(getActiveUsername(),filterKey);
}

var LAPORAN_OPTIONS=[{id:'all',label:'A. Semua'},{id:'b',label:'B.
Kop+DataQC+Stats+Catatan+TTD'},{id:'c',label:'C. Kop+DataQC+Catatan+TTD'},{id:'d',label:'D.
Kop+DataQC(bersih)+Catatan+TTD'},{id:'e',label:'E. Kop+DataQC+Stats(bersih)+Catatan+TTD'}];
function buildLaporanMenuHTML(action){return LAPORAN_OPTIONS.map(function(opt){return
'<button onclick="execLaporanExport(\'"+action+"\',\'"+opt.id+"\')"><div class="option-
title">'+opt.label+'</div></button>';}).join("");}
function toggleLaporanExportMenu(action){['print','pdf','ss'].forEach(function(a){var
menuId='laporan'+(a==='ss'?SS:a.charAt(0).toUpperCase()+a.slice(1))+ 'Menu';var
menu=G(menuId);if(menu)menu.classList.remove('show');});var
menuId='laporan'+(action==='ss'?SS:action.charAt(0).toUpperCase()+action.slice(1))+ 'Menu';var
menu=G(menuId);if(menu){menu.innerHTML=buildLaporanMenuHTML(action);menu.classList.add('s
how');}}
function execLaporanExport(action,optId){['print','pdf','ss'].forEach(function(a){var
menuId='laporan'+(a==='ss'?SS:a.charAt(0).toUpperCase()+a.slice(1))+ 'Menu';var
menu=G(menuId);if(menu)menu.classList.remove('show');});applyLaporanVisibility(optId);setTimeou
t(function(){if(action==='print')doLaporanPrint();else if(action==='pdf')doLaporanPdf();else
doLaporanScreenshot();},300);}
function applyLaporanVisibility(optId){QA('.lap-stats-section,.lap-qcdata-section,.wg-cell-
flag').forEach(function(el){el.style.display="";});if(optId==='c')QA('.lap-stats-
section').forEach(function(el){el.style.display='none';});else if(optId==='d'){QA('.lap-stats-
section').forEach(function(el){el.style.display='none';});QA('.wg-cell-
flag').forEach(function(el){el.style.display='none';});};else if(optId==='e')QA('.wg-cell-
flag').forEach(function(el){el.style.display='none';});}

```

```

function restoreLaporanVisibility(){QA('.lap-stats-section,.lap-qcdata-section,.wg-cell-flag').forEach(function(el){el.style.display="";});}
function doLaporanPrint(){QA('.page').forEach(function(p){p.classList.remove('print-active');});G('pageLaporan').classList.add('print-active');setTimeout(function(){window.print();setTimeout(restoreLaporanVisibility,600);},300);}
function
doLaporanPdf(){sl('PDF...');html2canvas(G('laporanContent'),{scale:2,useCORS:true}).then(function(c
anvas){hl();var pdf=new jsPDF('p','mm','a4');var w=pdf.internal.pageSize.getWidth()-20;var
h=canvas.height*w/canvas.width;var pH=pdf.internal.pageSize.getHeight()-
20;if(h<=pH)pdf.drawImage(canvas.toDataURL('image/png'),'PNG',10,10,w,h);else{var
rem=h,sY=0;while(rem>0){var sH=Math.min(pH,rem);var
sc=document.createElement('canvas');sc.width=canvas.width;sc.height=sH*canvas.width/w;sc.getCo
ntext('2d').drawImage(canvas,0,sY,canvas.width,sc.height,0,0,sc.width,sc.height);if(sY>0)pdf.addPag
e();pdf.drawImage(sc.toDataURL('image/png'),'PNG',10,10,w,sH);sY+=sc.height;rem-
=sH;}}pdf.save('laporan_'+new
Date().getTime()+'.pdf');restoreLaporanVisibility();}).catch(function(e){hl();restoreLaporanVisibility();
toast('Error: '+e.message | |e| |'Terjadi kesalahan'),'error');});}
function
doLaporanScreenshot(){sl('SS...');html2canvas(G('laporanContent'),{scale:2,useCORS:true}).then(func
tion(canvas){hl();var l=document.createElement('a');l.download='laporan_'+new
Date().getTime()+'.png';l.href=canvas.toDataURL();l.click();toast('Tersimpan','success');restoreLapora
nVisibility();}).catch(function(e){hl();restoreLaporanVisibility();toast('Error: '+e.message | |e| |'Terjadi
kesalahan'),'error');});}
function loadLaporan(){var
payload={startDate:G('lapStart').value || null,endDate:G('lapEnd').value || null,bidang:G('lapBidang').v
alue || null,paramID:G('lapParam').value || null,lotID:G('lapLot').value || null,namaAlat:G('lapAlat').valu
e.trim() || null};sl('Memuat
laporan...');google.script.run.withSuccessHandler(function(r){hl();if(!r || !r.ok){toast(r?r.msg:'Gagal','e
rror');return;}lastLaporanData=r;renderLaporan(r);G('laporanResult').style.display="";showExportBtns
('laporan');}).withFailureHandler(function(e){hl();toast('Error: '+e.message | |e| |'Terjadi
kesalahan'),'error');}).getLaporanData(payload,getActiveUsername(),getActiveRole());}
function resetTrendFilter() {
var now = new Date().getFullYear();
G('trendTahun').value = now;
G('trendBulanAwal').value = '1';
G('trendBulanAkhir').value = '12';
G('trendBidang').value = "";
G('trendParam').value = "";
G('trendLot').innerHTML = '<option value="">Semua</option>';
G('trendSiklusPME').value = "";
G('trendTahunSiklus').value = "";
G('trendResult').style.display = 'none';
G('trendContent').innerHTML = "";
}
function loadTrendAnalysis() {

```

```

sl('Memuat trend...');
google.script.run.withSuccessHandler(function(r) {
hl();
if (!r || !r.ok) { toast(r ? r.msg : 'Gagal', 'error'); return; }
G('trendResult').style.display = "";
showExportBtns('trend');
renderTrendAnalysis(r.data);
if(r.data && r.data.instrumentCompare){
var ic=r.data.instrumentCompare;
var html='<div class="card"><div class="card-header"><h3><i class="fas fa-balance-scale"></i>
Perbandingan Alat</h3></div><div class="card-body"><div class="table-
wrap"><table><thead><tr><th>Alat</th><th>Avg Sigma</th><th>Avg CV%</th><th>Avg
Bias%</th><th>TEa%</th><th>N Data</th><th>Interpretasi Sigma</th><th>Interpretasi
CV</th><th>Interpretasi Bias</th></tr></thead><tbody>';
ic.forEach(function(row){
html+='<tr><td>'+escH(row.namaAlat)+'</td>';
html+='<td class="'+sigmaClass(row.avgSigma)+'">'+round(row.avgSigma,2)+'</td>';
html+='<td>'+round(row.avgCV,2)+'%</td>';
html+='<td>'+round(row.avgBias,2)+'%</td>';
html+='<td>'+(row.tea || '-')+'</td>';
html+='<td>'+(row.nData || '-')+'</td>';
html+='<td>'+escH(row.interpSigma || '-')+'</td>';
html+='<td>'+escH(row.interpCV || '-')+'</td>';
html+='<td>'+escH(row.interpBias || '-')+'</td></tr>';
});
html+='</tbody></table></div></div>';
html+='<div class="card"><div class="card-header"><h3><i class="fas fa-lightbulb"></i> Interpretasi
Detail</h3></div><div class="card-body">'+(r.data.interpretasiDetail || '<p>Belum ada
interpretasi</p>')+'</div></div>';
html+='<div class="card"><div class="card-header"><h3><i class="fas fa-clipboard-check"></i>
Rekomendasi Alat</h3></div><div class="card-body">'+(r.data.rekomendasiAlat || '<p>Belum ada
rekomendasi</p>')+'</div></div>';
var tc=G('trendContent');if(tc)tc.innerHTML+=html;
}
}).withFailureHandler(function(e) { hl(); toast('Error', 'error'); }).getTrendAnalysisData({
tahun: G('trendTahun').value || null,
bulanAwal: G('trendBulanAwal').value || '1',
bulanAkhir: G('trendBulanAkhir').value || '12',
bidang: G('trendBidang').value || null,
paramID: G('trendParam').value || null,
lotID: G('trendLot').value || null,
siklusPME: G('trendSiklusPME').value || null,
tahunSiklus: G('trendTahunSiklus').value || null
}, getActiveUsername(), getActiveRole());
}

```

```

function renderTrendAnalysis(data){var html="";var months=data.months||[];var
labels=months.map(function(m){return
m.label;});if(!labels.length){G('trendContent').innerHTML=<p style="color:var(--text-
secondary)">Tidak ada data</p>;return;}
html+=<p style="font-size:.88rem;color:var(--text-secondary);margin-
bottom:16px"><strong>Parameter:</strong> '+escH(data.paramInfo||'')+</p>;
var sigmaSources=[{key:'terhitung',title:'Sigma Terhitung'},{key:'pme',title:'Sigma
PME'},{key:'pmecv',title:'Sigma PME CV'},{key:'cvopt',title:'Sigma CV Optional'}];
html+=<div class="chart-grid-4">;sigmaSources.forEach(function(src,idx){html+=<div class="chart-
box"><h5 style="font-size:.88rem;margin-bottom:8px">'+src.title+</h5><canvas
id="trendSigma_'+idx+" style="max-height:240px"></canvas></div>;});html+=</div>;
html+=<div class="chart-box"><h5 style="margin-bottom:8px">Trend TE vs TEa</h5><canvas
id="trendTE" style="max-height:280px"></canvas></div>;
html+=<h4 style="margin-top:16px"><i class="fas fa-lightbulb"></i> Interpretasi</h4>;
sigmaSources.forEach(function(src){var
interp=data.interpretasi[src.key];if(!interp)return;html+=<div class="trend-interp-
box"><h5>'+src.title+</h5><ul>;interp.notes.forEach(function(n){html+=<li>'+escH(n)+</li>;});htm
l+=</ul>;if(interp.estimasiError!==null)html+=<div class="error-estimate"><i class="fas fa-
exclamation-triangle"></i> Error per 100 pasien: ~'+interp.estimasiError+</div>;html+=</div>;});
var teInterp=data.interpretasi.te;if(teInterp){html+=<div class="trend-interp-box"><h5>Trend TE vs
TEa</h5><ul>;teInterp.notes.forEach(function(n){html+=<li>'+escH(n)+</li>;});html+=</ul></div>
;}}
G('trendContent').innerHTML=html;
setTimeout(function(){sigmaSources.forEach(function(src,idx){var
sigData=data.sigmaTrend[src.key];if(!sigData)return;drawTrendLine('trendSigma_'+idx,labels,[{label:'
σ L1',data:sigData.L1,color:'#2563eb'},{label:'σ L2',data:sigData.L2,color:'#f59e0b'},{label:'σ
L3',data:sigData.L3,color:'#ef4444'}],[{y:6,label:'6σ',color:'#047857'},{y:3,label:'3σ',color:'#f59e0b'}]);
});var teData=data.teTrend;if(teData){var teaArr=teData.tea.filter(function(v){return v!==null;});var
teaMean=teaArr.length?teaArr.reduce(function(a,b){return a+b;},0)/teaArr.length:null;var
refs=teaMean!==null?[y:teaMean,label:'TEa'+round(teaMean),color:'#ef4444']:[];drawTrendLine('
trendTE',labels,[{label:'TE L1',data:teData.L1,color:'#2563eb'},{label:'TE
L2',data:teData.L2,color:'#f59e0b'},{label:'TE L3',data:teData.L3,color:'#ef4444'}],refs);},300);
}
function drawTrendLine(cld,labels,datasets,refLines){var el=G(cld);if(!el)return;destroyChart(cld);var
ds=datasets.map(function(d){return{label:d.label,data:d.data,borderColor:d.color,backgroundColor:
d.color+'22',tension:0.3,fill:false,pointRadius:4,borderWidth:2;}});var
ann={};(refLines||[]).forEach(function(r,i){ann['ref'+i]={type:'line',yMin:r.y,yMax:r.y,borderColor:r.col
or,borderDash:[5,5],label:{display:true,content:r.label,position:'end',font:{size:10}};});chartInstance
s[cld]=new
Chart(el.getContext('2d'),{type:'line',data:{labels:labels,datasets:ds},options:{responsive:true,maintai
nAspectRatio:false,plugins:{annotation:{annotations:ann}},scales:{y:{beginAtZero:true}}});}
var TREND_PRINT_OPTIONS=[{id:'grafik',label:'Print Grafik Saja'},{id:'all',label:'Print Semua'}];
var TREND_PDF_OPTIONS=[{id:'grafik',label:'PDF Grafik Saja'},{id:'all',label:'PDF Semua'}];
var TREND_SS_OPTIONS=[{id:'grafik',label:'SS Grafik Saja'},{id:'all',label:'SS Semua'}];

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function toggleTrendExportMenu(action){['print','pdf','ss'].forEach(function(a){var
menuId='trend'+(a==='print'?Print:a==='pdf'?Pdf:'SS')+'Menu';var
menu=G(menuId);if(menu)menu.classList.remove('show');});var
menuId='trend'+(action==='print'?Print:a==='pdf'?Pdf:'SS')+'Menu';var menu=G(menuId);var
opts=action==='print'?TREND_PRINT_OPTIONS:action==='pdf'?TREND_PDF_OPTIONS:TREND_SS_OP
TIONS;if(menu){menu.innerHTML=opts.map(function(opt){return '<button
onclick="execTrendExport(\'"+action+\'\',\'"+opt.id+\'"\')"><div class="option-
title">'+opt.label+</div></button>'}).join("");menu.classList.add('show');}}
function execTrendExport(action,optId){['print','pdf','ss'].forEach(function(a){var
menuId='trend'+(a==='print'?Print:a==='pdf'?Pdf:'SS')+'Menu';var
menu=G(menuId);if(menu)menu.classList.remove('show');});if(action==='print')execTrendPrint(optId
);else if(action==='pdf')pdfTrend();else ssTrend();}
function
execTrendPrint(optId){G('trendPrintMenu').classList.remove('show');if(optId==='grafik')QA('.trend-
interp-
box').forEach(function(el){el.style.display='none';});QA('.page').forEach(function(p){p.classList.remov
e('print-active');});G('pageTrendanalysis').classList.add('print-
active');setTimeout(function(){window.print();setTimeout(function(){QA('.trend-interp-
box').forEach(function(el){el.style.display='';});,600);},300);}
function
pdfTrend(){sl('PDF...');html2canvas(G('trendContent'),{scale:2,useCORS:true}).then(function(canvas){
hl();var pdf=new jsPDF.jsPDF('p','mm','a4');var w=pdf.internal.pageSize.getWidth()-20;var
h=canvas.height*w/canvas.width;var pH=pdf.internal.pageSize.getHeight()-
20;if(h<=pH)pdf.addImage(canvas.toDataURL('image/png'),'PNG',10,10,w,h);else{var
rem=h,sY=0;while(rem>0){var sH=Math.min(pH,rem);var
sc=document.createElement('canvas');sc.width=canvas.width;sc.height=sH*canvas.width/w;sc.getCo
ntext('2d').drawImage(canvas,0,sY,canvas.width,sc.height,0,0,sc.width,sc.height);if(sY>0)pdf.addPag
e();pdf.addImage(sc.toDataURL('image/png'),'PNG',10,10,w,sH);sY+=sc.height;rem-
=sH;}}pdf.save('trend_'+new Date().getTime()+'pdf');}).catch(function(e){hl();toast('Error:
'+(e.message|e|'Terjadi kesalahan'),'error');});}
function
ssTrend(){sl('SS...');html2canvas(G('trendContent'),{scale:2,useCORS:true}).then(function(canvas){hl()
;var l=document.createElement('a');l.download='trend_'+new
Date().getTime()+'png';l.href=canvas.toDataURL();l.click();toast('Tersimpan','success');}).catch(funci
on(e){hl();toast('Error: '+e.message|e|'Terjadi kesalahan'),'error');});}
function loadInstrCompare(){if(!CU||!CU.username){toast('Sesi belum
aktif','warning');return;}sl('Membandingkan...');google.script.run.withSuccessHandler(function(r){hl()
;if(!r||!r.ok){toast(r?r.msg:'Gagal','error');return;}renderInstrCompare(r.data);}).withFailureHandler(
function(e){hl();toast('Error: '+e.message|e|'Terjadi
kesalahan'),'error');}).getInstrumentCompare(getActiveUsername(),getActiveRole(),{paramID:G('icPa
ramFilter').value||null});}
function renderInstrCompare(data){var
body=G('instrCompareBody');if(!data||!data.length){body.innerHTML='<p style="color:var(--text-
secondary)">Tidak ada data</p>';return;}var
maxS=Math.max.apply(null,data.map(function(d){return d.avgSigma||0;}));if(maxS<1)maxS=6;var

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medals=['\U0001f947','\U0001f948','\U0001f949'];var html='';html+='\<h4 style="margin-
bottom:12px"><i class="fas fa-trophy"></i> Peringkat
Instrumen</h4>';data.forEach(function(alat,i){var medal=i<3?medals[i]:(i+1)+'.';var
barPct=Math.min(100,(alat.avgSigma/maxS)*100);html+='\<div class="rank-card"><div
class="rank-medal">'+medal+'</div><div class="rank-bar-wrap"><div style="display:flex;justify-
content:space-between;margin-bottom:4px"><strong>'+escH(alat.namaAlat)+'</strong><span
class="'+sigmaClass(alat.avgSigma)+'">'+sigmaLabel(alat.avgSigma)+'</span></div><div class="rank-
bar"><div class="rank-bar-fill"
style="width:'+barPct+'%;background:'+sigmaColor(alat.avgSigma)+'"></div></div><small>'+alat.n+'
parameter</small></div><div class="rank-sigma">'+round(alat.avgSigma)+'\u03c3</div></div>');});
// Detail Table
html+='\<div style="margin-top:24px"><h4 style="margin-bottom:12px"><i class="fas fa-table"></i>
Detail Perbandingan Alat</h4><div class="table-wrap" style="max-height:400px;overflow-
y:auto"><table><thead><tr><th>No</th><th>Nama Alat</th><th>Rata-rata
\u03c3</th><th>Interpretasi \u03c3</th><th>Rata-rata CV%</th><th>Interpretasi
CV</th><th>Rata-rata Bias%</th><th>Interpretasi Bias</th><th>QGI</th><th>Jml
Param</th></tr></thead><tbody>';
data.forEach(function(alat,i){
  var cvs=[],biases=[],tes=[];
  (alat.params||[]).forEach(function(p){
    if(p.stats){
      [1,2,3].forEach(function(lv){var
s=p.stats['L'+lv];if(s){if(s.cv)cvs.push(s.cv);if(s.bias!==undefined&&s.bias!==null)biases.push(s.bias);}}
);
    }
  });
  var avgCV=cvs.length?parseFloat((cvs.reduce(function(a,b){return
a+b;},0)/cvs.length).toFixed(2)):null;
  var avgBias=biases.length?parseFloat((biases.reduce(function(a,b){return
a+b;},0)/biases.length).toFixed(2)):null;
  function is(s){if(s===null||s===undefined)return'N/A';if(s>=6)return'Sangat
Baik';if(s>=5)return'Baik';if(s>=4)return'Cukup';if(s>=3)return'Marginal';return'Buruk';}
  function ic(c){if(c===null||c===undefined)return'N/A';if(c<=2)return'Sangat
Baik';if(c<=5)return'Baik';if(c<=10)return'Cukup';return'Buruk';}
  function ib(b){if(b===null||b===undefined)return'N/A';if(b<=1)return'Sangat
Baik';if(b<=2)return'Baik';if(b<=5)return'Cukup';return'Buruk';}
  var qgiVal=null,qgiInterp=";

if(avgCV&&avgBias&&avgCV>0){qgiVal=parseFloat((avgBias/(1.5*avgCV)).toFixed(3));qgiInterp=qgiV
al>1.2?'Imprecision':qgiVal<0.8?'Inaccuracy':'Balanced';}
  html+='\<tr><td>'+(i+1)+'</td><td><strong>'+escH(alat.namaAlat)+'</strong></td><td
class="'+sigmaClass(alat.avgSigma)+'">'+(alat.avgSigma!==null?alat.avgSigma:'N/A')+'</td><td>'+is(a
lat.avgSigma)+'</td><td>'+(avgCV!==null?avgCV+'%':'N/A')+'</td><td>'+ic(avgCV)+'</td><td>'+(avgB
ias!==null?avgBias+'%':'N/A')+'</td><td>'+ib(avgBias)+'</td><td>'+(qgiVal!==null?qgiVal+'
('+qgiInterp+')':'N/A')+'</td><td>'+alat.n+'</td></tr>';

```

```

});
html+='/tbody></table></div></div>';
// Per-Parameter Detail
html+='/div style="margin-top:24px"><h4 style="margin-bottom:12px"><i class="fas fa-microscope"></i> Detail Perbandingan per Parameter</h4><div class="table-wrap" style="max-height:400px;overflow-y:auto"><table><thead><tr><th>Parameter</th><th>Alat</th><th>No. Lot</th><th>\u03c3 (Avg)</th><th>CV%</th><th>Bias%</th><th>Interpretasi</th></tr></thead><tbody>;
data.forEach(function(alat){(alat.params || []).forEach(function(p){var
sigStr=p.avgSigma!==null?p.avgSigma:'N/A';var cvStr="";var biasStr="";var interpStr="";if(p.stats){var
cvArr=[],biasArr=[];[1,2,3].forEach(function(lv){var
s=p.stats['L'+lv];if(s){if(s.cv)cvArr.push(s.cv);if(s.bias!==undefined&&s.bias!==null)biasArr.push(s.bias
)});cvStr=cvArr.length?parseFloat((cvArr.reduce(function(a,b){return
a+b;},0)/cvArr.length).toFixed(2))+'%':'-';
biasStr=biasArr.length?parseFloat((biasArr.reduce(function(a,b){return
a+b;},0)/biasArr.length).toFixed(2))+'%':'-';
interpStr=p.avgSigma>=5?'Excellent':p.avgSigma>=4?'Good':p.avgSigma>=3?'Marginal':'Poor';html
+='/tr><td>'+escH(p.parameter)+'</td><td>'+escH(p.namaAlat || alat.namaAlat)+'</td><td>'+escH(p.
noLot)+'</td><td>'+sigmaClass(p.avgSigma)+'</td><td>'+sigStr+'</td><td>'+cvStr+'</td><td>'+biasStr+'</td><td>'+inte
rpStr+'</td></tr>'}});
html+='/tbody></table></div></div>';
// Interpretasi Detail
html+='/div style="margin-top:24px"><h4 style="margin-bottom:12px"><i class="fas fa-clipboard-check"></i> Interpretasi Detail</h4><div style="background:var(--bg-soft);padding:16px;border-radius:var(--radius-sm);white-space:pre-wrap;font-size:.88rem;line-height:1.6">;
if(data.length<2){html+='/div><div style="background:var(--bg-soft);padding:16px;border-radius:var(--radius-sm);white-space:pre-wrap;font-size:.88rem;line-height:1.6">Hanya tersedia data untuk '+data.length+' instrumen. Perbandingan minimal memerlukan 2 instrumen.\n';if(data.length===1){var d0=data[0];html+='/div><div style="background:var(--bg-soft);padding:16px;border-radius:var(--radius-sm);white-space:pre-wrap;font-size:.88rem;line-height:1.6">Kinerja '+d0.namaAlat+' : \u03c3=' +round(d0.avgSigma)+' ['+(d0.avgSigma>=5?'Excellent':d0.avgSigma>=4?'Good':d0.avgSigma>=3?'Marginal':'Poor')+'], '+d0.n+' parameter dievaluasi.';}}
else{var best=data[0],worst=data[data.length-1];html+='/div><div style="background:var(--bg-soft);padding:16px;border-radius:var(--radius-sm);white-space:pre-wrap;font-size:.88rem;line-height:1.6">\U0001f4ca RINGKASAN PERBANDINGAN INSTRUMEN\n\n';html+='/div><div style="background:var(--bg-soft);padding:16px;border-radius:var(--radius-sm);white-space:pre-wrap;font-size:.88rem;line-height:1.6">Instrumen Terbaik: '+best.namaAlat+' (\u03c3=' +round(best.avgSigma)+' , '+best.n+' parameter)\n';html+='/div><div style="background:var(--bg-soft);padding:16px;border-radius:var(--radius-sm);white-space:pre-wrap;font-size:.88rem;line-height:1.6">Instrumen Terendah: '+worst.namaAlat+' (\u03c3=' +round(worst.avgSigma)+' , '+worst.n+' parameter)\n';var
sigDiff=best.avgSigma&&worst.avgSigma?parseFloat((best.avgSigma-
worst.avgSigma).toFixed(2)):0;html+='/div><div style="background:var(--bg-soft);padding:16px;border-radius:var(--radius-sm);white-space:pre-wrap;font-size:.88rem;line-height:1.6">Selisih Sigma: '+sigDiff+' point\n\n';html+='/div><div style="background:var(--bg-soft);padding:16px;border-radius:var(--radius-sm);white-space:pre-wrap;font-size:.88rem;line-height:1.6">\U0001f4cb DETAIL PER INSTRUMEN:\n';data.forEach(function(alat,i){html+='/div><div style="background:var(--bg-soft);padding:16px;border-radius:var(--radius-sm);white-space:pre-wrap;font-size:.88rem;line-height:1.6">\n'+(i+1)+' . '+alat.namaAlat+' : \n';html+='/div><div style="background:var(--bg-soft);padding:16px;border-radius:var(--radius-sm);white-space:pre-wrap;font-size:.88rem;line-height:1.6">- Rata-rata Sigma: '+round(alat.avgSigma)+' \u2192 '+alat.avgSigma>=6?'World Class':alat.avgSigma>=5?'Excellent':alat.avgSigma>=4?'Good':alat.avgSigma>=3?'Marginal':'Poor')+'\n';var
cvs2=[],biases2=[];(alat.params || []).forEach(function(p){if(p.stats){[1,2,3].forEach(function(lv){var
s=p.stats['L'+lv];if(s){if(s.cv)cvs2.push(s.cv);if(s.bias!==undefined&&s.bias!==null)biases2.push(s.bias
)});});});var ac2=cvs2.length?parseFloat((cvs2.reduce(function(a,b){return
a+b;},0)/cvs2.length).toFixed(2)):null;var

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ab2=biases2.length?parseFloat((biases2.reduce(function(a,b){return
a+b;},0)/biases2.length).toFixed(2)):null;html+=' - Rata-rata CV:
'+(ac2!==null?ac2+'%':'N/A')+'\n';html+=' - Rata-rata Bias: '+ (ab2!==null?ab2+'%':'N/A')+'\n';html+='
- Jumlah Parameter: '+alat.n+'\n';});html+='\n\U0001f4a1
REKOMENDASI:\n';if(best.avgSigma>=5)html+='1. '+best.namaAlat+' sudah Excellent/World Class.
Pertahankan kondisi alat.\n';else html+='1. Evaluasi '+best.namaAlat+' untuk peningkatan
kinerja.\n';if(worst.avgSigma<3)html+='2. PERHATIAN: '+worst.namaAlat+' memiliki sigma <3 (Poor).
Evaluasi metode, kalibrasi ulang, atau pertimbangkan penggantian.\n';else
if(worst.avgSigma<4)html+='2. '+worst.namaAlat+' Marginal. Perlu perbaikan presisi dan/atau
akurasi.\n';if(sigDiff>2)html+='3. Terdapat perbedaan signifikan (>2\U003c3) antar alat. Investigasi
penyebab (metode, reagen, kalibrasi).\n';else if(sigDiff>1)html+='3. Perbedaan antar alat cukup
terlihat. Monitoring berkala diperlukan.\n';else html+='3. Kinerja antar alat cukup merata.
Pertahankan konsistensi.\n';
html+='\n</div></div>';
body.innerHTML=html;
function initTabChecklist(){filterTabParams();var sel=G('tabTahun'),nowY=new
Date().getFullYear();sel.innerHTML="";for(var y=nowY;y>=nowY-5;y--){sel.innerHTML+='\n<option
value="'+y+'"'>'+y+'</option>';G('tabBulanAwal').value='01';G('tabBulanAkhir').value=String(new
Date().getMonth()+1).padStart(2,'0');}
function filterTabParams(){var bidang=G('tabBidangFilter').value;var
html="";(CD.params || []).forEach(function(p){if(bidang&&(p.bidang | 'Lainnya')!=bidang)return;html
+='\n<label class="checkbox-item"><input type="checkbox" class="tab-param-chk"
value="'+p.paramID+'" checked>
'+escH(p.parameter)+'</label>');});G('tabParamChecklist').innerHTML=html;
function tabCheckAll(c){QA('tab-param-chk').forEach(function(cb){cb.checked=c;});}
function loadTabulasi(){if(!CU || !CU.username){toast('Sesi belum aktif','warning');return;}var
th=G('tabTahun').value,bA=G('tabBulanAwal').value,bK=G('tabBulanAkhir').value;var
pids=[];QA('tab-param-
chk:checkbox').forEach(function(cb){pids.push(cb.value);});if(!pids.length){toast('Pilih min 1
parameter','warning');return;}var sd=th+'-'+bA+'-01';var lastDay=new
Date(parseInt(th),parseInt(bK),0).getDate();var ed=th+'-'+bK+'-
'+String(lastDay).padStart(2,'0');sl('Memuat...');google.script.run.withSuccessHandler(function(r){hl()
;if(!r || !r.ok){toast(r?r.msg:'Gagal','error');return;}G('tabulasiResult').style.display="";showExportBtns(
'tabulasi');G('tabulasiKopContainer').innerHTML=buildKopHTML(r.kop,true)+'<h3 style="text-
align:center;margin:14px 0;font-weight:800">TABULASI REKAP QC</h3><p style="text-
align:center;font-size:.85rem;color:var(--text-secondary)">Periode: '+fmtDateIndo(sd)+' s/d
'+fmtDateIndo(ed)+'</p>';renderTabulasiTable(r.data);renderTabulasiPME(r.pmeRekapDetail || []);re
nderTabulasiInterpretasi(r.data);
if(r.catatan)G('tabCatatan').value=r.catatan.catatan || '';G('tabCatatan').dataset.periodKey=r.periodK
ey || '';}).withFailureHandler(function(e){hl();toast('Error: '+e.message || e || 'Terjadi
kesalahan','error');}).getTabulasiData(getActiveUsername(),getActiveRole(),{paramIDs:pids,startDat
e:sd,endDate:ed,bidang:G('tabBidangFilter').value || null});}
function renderTabulasiTable(data){var
html='\n<table><thead><tr><th>Parameter</th><th>Bidang</th><th>Alat</th><th>N</th><th>Lvl</t
h><th>Mean</th><th>SD</th><th>CV%</th><th>Bias%</th><th>TE%</th><th>σ</th><th>TEa</th>

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<th>Kinerja</th></tr></thead><tbody>;(data || []).forEach(function(d){var
first=true;['L1','L2','L3'].forEach(function(lv){var
s=d.stats?d.stats[lv]:null;if(!s)return;html+=<tr>;if(first){var rs=['L1','L2','L3'].filter(function(l){return
d.stats&& d.stats[l]}).length;html+=<td rowspan=""+rs+"">'+escH(d.parameter)+'</td><td
rowspan=""+rs+""><span class="badge badge-info">'+escH(d.bidang)+'</span></td><td
rowspan=""+rs+"">'+escH(d.namaAlat)+'</td><td
rowspan=""+rs+"">d.nQC+'</td>;first=false;}}html+=<td>'+lv+'</td><td>'+round(s.mean,4)+'</td><
td>'+round(s.sd,4)+'</td><td>'+round(s.cv)+'</td><td>'+round(s.bias)+'</td><td>'+round(s.te)+'</td>
<td class="'+sigmaClass(s.sigma)+'">'+round(s.sigma)+'</td><td>'+(s.tea || d.tea || '-')+'</td><td
class="'+sigmaClass(s.sigma)+'">'+sigmaLabel(s.sigma)+'</td></tr>});if(!data || !data.length)html+
=<tr><td colspan="13" style="text-align:center">Tidak ada
data</td></tr>;html+=</tbody></table>;G('tabRekapTable').innerHTML=html;}
function
renderTabulasiPME(pmeData){if(!pmeData || !pmeData.length){G('tabPMECard').style.display='none'
;return;}G('tabPMECard').style.display="";var html=<div style="overflow-
x:auto"><table><thead><tr><th>Parameter</th><th>TEa%</th><th>Siklus</th><th>Tahun
Siklus</th><th>Alat</th><th>Hasil L1</th><th>Hasil L2</th><th>Hasil L3</th><th>Hasil P
L1</th><th>Hasil P L2</th><th>Hasil P L3</th><th>Bias L1%</th><th>Bias L2%</th><th>Bias
L3%</th><th>Sigma L1</th><th>Sigma L2</th><th>Sigma
L3</th></tr></thead><tbody>;pmeData.forEach(function(d){html+=<tr>;html+=<td>'+escH(d.Para
meter || '-')+'</td>;html+=<td>'+(d.TEa || '-')+'</td>;html+=<td>'+escH(d.Siklus || '-'
)+'</td>;html+=<td>'+escH(d.TahunSiklus || '-')+'</td>;html+=<td>'+escH(d>NamaAlat || '-'
)+'</td>;html+=<td>'+(d.HasilL1!==null?d.HasilL1:'-
')+'</td>;html+=<td>'+(d.HasilL2!==null?d.HasilL2:'-
')+'</td>;html+=<td>'+(d.HasilL3!==null?d.HasilL3:'-
')+'</td>;html+=<td>'+(d.HasilPL1!==null?d.HasilPL1:'-
')+'</td>;html+=<td>'+(d.HasilPL2!==null?d.HasilPL2:'-
')+'</td>;html+=<td>'+(d.HasilPL3!==null?d.HasilPL3:'-
')+'</td>;html+=<td>'+(d.BiasL1!==null?d.BiasL1+'%':'-
')+'</td>;html+=<td>'+(d.BiasL2!==null?d.BiasL2+'%':'-
')+'</td>;html+=<td>'+(d.BiasL3!==null?d.BiasL3+'%':'-')+'</td>;html+=<td
class="'+sigmaClass(d.SigmaL1)+'">'+(d.SigmaL1!==null?d.SigmaL1:'-')+'</td>;html+=<td
class="'+sigmaClass(d.SigmaL2)+'">'+(d.SigmaL2!==null?d.SigmaL2:'-')+'</td>;html+=<td
class="'+sigmaClass(d.SigmaL3)+'">'+(d.SigmaL3!==null?d.SigmaL3:'-
')+'</td>;html+=</tr>});if(!pmeData.length)html+=<tr><td colspan="18" style="text-
align:center">Tidak ada data
PME</td></tr>;html+=</tbody></table></div>;G('tabPMETable').innerHTML=html;}
function renderTabulasiInterpretasi(data){var
html=",good=0,marginal=0,poor=0;(data || []).forEach(function(d){['L1','L2','L3'].forEach(function(lv){
var s=d.stats?d.stats[lv]:null;if(!s || s.sigma===null)return;if(s.sigma>=4)good++;else
if(s.sigma>=3)marginal++;else poor++;});var total=good+marginal+poor;html+=<div
style="display:grid;grid-template-columns:repeat(auto-fill,minmax(150px,1fr));gap:12px;margin-
bottom:16px"><div style="text-align:center;padding:14px;background:var(--success-light);border-
radius:10px"><div style="font-size:1.6rem;font-
weight:800;color:#047857">'+good+'</div><small>Good</small></div><div style="text-

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align:center;padding:14px;background:var(--warning-light);border-radius:10px"><div style="font-
size:1.6rem;font-weight:800;color:#b45309">'+marginal+'</div><small>Marginal</small></div><div
style="text-align:center;padding:14px;background:var(--danger-light);border-radius:10px"><div
style="font-size:1.6rem;font-
weight:800;color:#b91c1c">'+poor+'</div><small>Poor</small></div></div>';if(total>0)html+=<p>D
ari '+total+' pengukuran, <strong>'+((good/total*100).toFixed(0))+'%</strong> memenuhi
 $\sigma \geq 4$ .</p>';G('tabInterpretasi').innerHTML=html;
function saveTabCatatan(){var
pk=G('tabCatatan').dataset.periodKey,c=G('tabCatatan').value;if(!pk){toast('Jalankan rekap
dulu','warning');return;}sl('Menyimpan...');google.script.run.withSuccessHandler(function(r){hl();if(r.
ok)toast('Disimpan','success');else
toast(r.msg,'error')}).withFailureHandler(function(e){hl();toast('Error: '+e.message | |e| |'Terjadi
kesalahan'),'error')}).saveCatatanTabulasi(pk,c,CU.username,getLogUser());
function printTabulasi(){QA('page').forEach(function(p){p.classList.remove('print-
active')});G('pageTabulasi').classList.add('print-active');window.print();}
function
pdfTabulasi(){sl('PDF...');html2canvas(G('tabulasiResult'),{scale:2,useCORS:true}).then(function(c){hl(
);var pdf=new jsPDF('l','mm','a4');var w=pdf.internal.pageSize.getWidth()-
20,h=c.height*w/c.width,pH=pdf.internal.pageSize.getHeight()-
20;if(h<=pH)pdf.drawImage(c.toDataURL('image/png'),'PNG',10,10,w,h);else{var
rem=h,sY=0;while(rem>0){var sH=Math.min(pH,rem);var
sc=document.createElement('canvas');sc.width=c.width;sc.height=sH*c.width/w;sc.getContext('2d').
drawImage(c,0,sY,c.width,sc.height,0,0,sc.width,sc.height);if(sY>0)pdf.addPage();pdf.drawImage(sc.to
DataURL('image/png'),'PNG',10,10,w,sH);sY+=sc.height;rem-=sH;}}pdf.save('tabulasi_'+new
Date().getTime()+'.pdf');}).catch(function(e){hl();toast('Error: '+e.message | |e| |'Terjadi
kesalahan'),'error')});}
function
screenshotTabulasi(){sl('SS...');html2canvas(G('tabulasiResult'),{scale:2,useCORS:true}).then(function
(c){hl();var l=document.createElement('a');l.download='tabulasi_'+new
Date().getTime()+'.png';l.href=c.toDataURL();l.click();toast('Tersimpan','success')}).catch(function(e){
hl();toast('Error: '+e.message | |e| |'Terjadi kesalahan'),'error')});}
function loadOPSspecs(){if(!CU || !CU.username){toast('Sesi belum
aktif','warning');return;}sl('Analisis...');google.script.run.withSuccessHandler(function(r){hl();if(!r || !r.
ok){toast(r?.msg:'Gagal','error');return;}G('opsSpecsResult').style.display="";G('opsSpecsKopContainer')
.innerHTML=buildKopHTML(r.kop,true)+'<h3 style="text-align:center;margin:14px 0;font-
weight:800">ANALISIS
OPSPECS</h3>';renderOPSspecsChart(r.data);renderCritErrorChart(r.data);renderOPSspecsTable(r.dat
a);renderOPSspecsKesimpulan(r.data);showExportBtns('opsSpecs');if(r.opsAnalisisDetail){var
ad=G('opsAnalisisDetail');if(ad)ad.innerHTML=r.opsAnalisisDetail.join('<br>');if(r.opsCriticalErrorDat
a){renderCriticalErrorAnalysis(r.opsCriticalErrorData);}if(r.opsKesimpulan){var
kc=G('opsKesimpulan');if(kc)kc.innerHTML='<pre style="white-space:pre-wrap;font-
family:Inter,sans-serif;font-size:.85rem;line-
height:1.7;margin:0">'+escH(r.opsKesimpulan)+'</pre>'}}.withFailureHandler(function(e){hl();toast(
'Error: '+e.message | |e| |'Terjadi
kesalahan'),'error')}).getOPSspecsData(getActiveUsername(),getActiveRole(),{bidang:G('opsBidangFil

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ter').value || null,paramID:G('opsParamFilter').value || null,startDate:G('opsStartDate').value || null,endDate:G('opsEndDate').value || null));}
function renderOPSspecsChart(data){destroyChart('opspecs');if(!data || !data.length)return;var
pts=data.map(function(d){return{x:d.tea?d.cv/d.tea:0,y:d.tea?d.bias/d.tea:0,label:d.parameter,sigma
a:d.sigma}});var l13=[],l12=[],lmr=[];for(var cx=0;cx<=1;cx+=0.02){l13.push({x:cx,y:Math.max(0,1-
3*cx)});l12.push({x:cx,y:Math.max(0,1-2*cx)});lmr.push({x:cx,y:Math.max(0,1-
1.65*cx)});}chartInstances.opspecs=new
Chart(G('chartOPSspecs').getContext('2d'),{type:'scatter',data:{datasets:[{label:'Parameter',data:pts,b
ackgroundColor:pts.map(function(p){return sigmaColor(p.sigma)}),pointRadius:8},{label:'1-
3s',data:l13,type:'line',borderColor:'#ef4444',borderWidth:2,pointRadius:0,fill:false,borderDash:[8,4]
},{label:'1-
2s',data:l12,type:'line',borderColor:'#f59e0b',borderWidth:2,pointRadius:0,fill:false,borderDash:[5,5]
},{label:'Multi-
rule',data:lmr,type:'line',borderColor:'#047857',borderWidth:2,pointRadius:0,fill:false}}],options:{res
ponsive:true,maintainAspectRatio:false,plugins:{tooltip:{callbacks:{label:function(c){var
p=c.raw;return p.label?(p.label+' (σ='+p.sigma+'|-|)'):''}}}},scales:{x:{title:{display:true,text:'CV/TEa'},min:0,max:1},y:{title:{display:true,text:'Bias/TEa'},min:0
,max:1}}}});}
function renderCritErrorChart(data){destroyChart('criterror');if(!data || !data.length)return;var
pts=data.map(function(d){return{x:d.sec | 0,y:d.rec | 0,label:d.parameter,sigma:d.sigma}});chartInst
ances.criterror=new
Chart(G('chartCritError').getContext('2d'),{type:'scatter',data:{datasets:[{label:'Parameter',data:pts,b
ackgroundColor:pts.map(function(p){return
sigmaColor(p.sigma)}),pointRadius:8}}],options:{responsive:true,maintainAspectRatio:false,plugins:{
annotation:{annotations:{zA:{type:'box',xMin:3,xMax:10,yMin:0,yMax:3,backgroundColor:'rgba(16,1
85,129,.1)'},zK:{type:'box',xMin:1,xMax:3,yMin:0,yMax:3,backgroundColor:'rgba(245,158,11,.1)'},zB:{
type:'box',xMin:0,xMax:1,yMin:0,yMax:10,backgroundColor:'rgba(239,68,68,.1)'},IA:{type:'label',xVal
ue:6,yValue:1,content:'AMAN',font:{size:14,weight:'bold'},color:'#047857'},IK:{type:'label',xValue:2,y
Value:1.5,content:'KRITIS',font:{size:14,weight:'bold'},color:'#b45309'},IB:{type:'label',xValue:0.5,yVa
lue:5,content:'BAHAYA',font:{size:14,weight:'bold'},color:'#b91c1c'}}}},scales:{x:{title:{display:true,te
xt:'ΔSec'},min:0},y:{title:{display:true,text:'ΔREc'},min:0}}}});}
function renderOPSspecsTable(data){var
html=<table><thead><tr><th>Parameter</th><th>Alat</th><th>CV%</th><th>Bias%</th><th>TEa<
/th><th>σ</th><th>Ped</th><th>Pfr</th><th>Zona</th></tr></thead><tbody>';(data || []).forEach(f
unction(d){var z=d.sigma>=4?'<span class="badge badge-
success">AMAN</span>':d.sigma>=2?'<span class="badge badge-warning">KRITIS</span>':<span
class="badge badge-
danger">BAHAYA</span>;html+=<tr><td>'+escH(d.parameter)+'</td><td>'+escH(d.namaAlat)+'</t
d><td>'+round(d.cv)+'</td><td>'+round(d.bias)+'</td><td>'+(d.tea || '-|')+'</td><td>
class="'+sigmaClass(d.sigma)+'">'+round(d.sigma)+'</td><td>'+(d.ped!==null?round(d.ped)+'%':-
|)+'</td><td>'+(d.pfr!==null?round(d.pfr)+'%':-
|)+'</td><td>'+z+'</td></tr>');html+=</tbody></table>;G('opsTableWrap').innerHTML=html;}
function
renderCriticalErrorAnalysis(criticalErrorData){if(!criticalErrorData || !criticalErrorData.length){return;}
var opsAnalysisDetailDiv=document.getElementById('opsAnalysisDetail');if(opsAnalysisDetailDiv){var

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prev=opsAnalysisDetailDiv.nextElementSibling;while(prev&&prev.getAttribute&&prev.getAttribute('
data-cep')!=null){var rm=prev;prev=prev.nextElementSibling;rm.remove();}var html='<div
class="card" style="margin-top:16px" data-cep="1"><div class="card-header"><h3><i class="fas fa-
exclamation-triangle"></i> Analisis Detail Critical-Error ( $\Delta$ SEc &  $\Delta$ REc)</h3></div><div class="card-
body"><div class="table-wrap" style="max-height:400px;overflow-
y:auto"><table><thead><tr><th>Parameter</th><th>Alat</th><th> $\Delta$ SEc (SD)</th><th>Interpretasi
 $\Delta$ SEc</th><th> $\Delta$ REc (CV)</th><th>Interpretasi
 $\Delta$ REc</th><th>Ped</th><th> $\sigma$ </th></tr></thead><tbody>;criticalErrorData.forEach(function(c){var
secCls=c.deltaSEc>=3?'badge-success':c.deltaSEc>=2?'badge-warning':'badge-danger';var
recCls=c.deltaREc>=2?'badge-success':c.deltaREc>=1?'badge-warning':'badge-danger';var
secInterp=c.deltaSEc>=3?'Deteksi SE Sangat Baik':c.deltaSEc>=2?'Deteksi SE Cukup':'Deteksi SE
Kurang';var recInterp=c.deltaREc>=2?'Deteksi RE Sangat Baik':c.deltaREc>=1?'Deteksi RE
Cukup':'Deteksi RE
Kurang';html+=<tr><td>'+escH(c.parameter)+'</td><td>'+escH(c.namaAlat)+'</td><td>
class="'+secCls+'" style="font-
weight:700">'+(c.deltaSEc!=null?c.deltaSEc:'N/A')+'</td><td><small>'+escH(secInterp)+'</small></
td><td class="'+recCls+'" style="font-
weight:700">'+(c.deltaREc!=null?c.deltaREc:'N/A')+'</td><td><small>'+escH(recInterp)+'</small></
td><td>'+(c.ped!=null?c.ped+'%':'-')+'</td><td>
class="'+sigmaClass(c.sigma)+'">'+(c.sigma!=null?c.sigma:'N/A')+'</td></tr>');html+=</tbody></t
able></div><div style="margin-top:14px;padding:12px;background:#f0f9ff;border-
radius:8px;border:1px solid #bae6fd"><strong><i class="fas fa-info-circle"></i> Panduan
Interpretasi:</strong><ul style="margin-top:6px;margin-left:20px;font-
size:.85rem"><li><strong> $\Delta$ SEc (Critical Systematic Error)</strong> = [(TEa - |Bias|) / CV] - 1.65.
Menunjukkan berapa SD pergeseran sistematik yang dapat dideteksi sebelum melampaui TEa. Makin
besar makin baik (target  $\geq 3$  SD).</li><li><strong> $\Delta$ REc (Critical Random Error)</strong> = [(TEa -
|Bias|) / (1.65  $\times$  CV)] - 1. Menunjukkan berapa kali peningkatan SD yang dapat dideteksi sebelum
melampaui TEa. Makin besar makin baik (target  $\geq 1$ ).</li><li> $\Delta$ SEc besar +  $\Delta$ REc kecil = Sistem QC
sangat sensitif terhadap error sistematik maupun acak.</li><li> $\Delta$ SEc kecil +  $\Delta$ REc besar = Sistem QC
kurang sensitif. Perlu perbaikan presisi dan/atau
akurasi.</li></ul></div></div></div>if(opsAnalysisDetailDiv){opsAnalysisDetailDiv.insertAdjacentHT
ML('afterend',html);}
function
renderOPSpecsKesimpulan(data){if(!data || !data.length){G('opsKesimpulan').innerHTML='<p>Tidak
ada data</p>';return;}var a=data.filter(function(d){return d.sigma>=4;}).length;var
k=data.filter(function(d){return d.sigma>=2&& d.sigma<4;}).length;var
b=data.filter(function(d){return d.sigma<2;}).length;G('opsKesimpulan').innerHTML='<p>Total:
'+data.length+' — <strong>AMAN: '+a+'</strong>, KRITIS: '+k+' , BAHAYA: '+b+'</p>';}
function printOPSpecs(){QA('page').forEach(function(p){p.classList.remove('print-
active');});G('pageOpspecs').classList.add('print-active');window.print();}
function
pdfOPSpecs(){sl('PDF...');html2canvas(G('opspecsResult'),{scale:2,useCORS:true}).then(function(c){hl(
);var pdf=new jsPDF('p','mm','a4');var w=pdf.internal.pageSize.getWidth()-
20,h=c.height*w/c.width,pH=pdf.internal.pageSize.getHeight()-
20;if(h<=pH)pdf.addImage(c.toDataURL('image/png'),'PNG',10,10,w,h);else{var

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rem=h,sY=0;while(rem>0){var sH=Math.min(pH,rem);var
sc=document.createElement('canvas');sc.width=c.width;sc.height=sH*c.width/w;sc.getContext('2d').
drawImage(c,0,sY,c.width,sc.height,0,0,sc.width,sc.height);if(sY>0)pdf.addPage();pdf.drawImage(sc.to
DataURL('image/png'),'PNG',10,10,w,sH);sY+=sc.height;rem-=sH;}}pdf.save('opspecs_'+new
Date().getTime()+'.pdf');}).catch(function(e){hl();toast('Error: '+e.message || e || 'Terjadi
kesalahan'),'error');});}
function
screenshotOPSpecs(){sl('SS...');html2canvas(G('opspecsResult'),{scale:2,useCORS:true}).then(function
(c){hl();var l=document.createElement('a');l.download='opspecs_'+new
Date().getTime()+'.png';l.href=c.toDataURL();l.click();toast('Tersimpan','success');}).catch(function(e){
hl();toast('Error: '+e.message || e || 'Terjadi kesalahan'),'error');});}
function loadUsers(){if(!CU || !CU.username){toast('Sesi belum
aktif','warning');return;}if(CU.role!=='superadmin')return;sl('Memuat...');google.script.run.withSucce
ssHandler(function(r){hl();if(!r || !r.ok)return;var users=r.data || [];var
pending=users.filter(function(u){return
u.status==='pending';});if(pending.length){G('pendingUsersWrap').style.display='';G('badgePendingU
sers').textContent=pending.length;G('badgePendingUsers').style.display='';var
ph='';pending.forEach(function(u){ph+=''<div style="display:flex;align-
items:center;gap:10px;padding:12px;border:1px solid var(--border);border-radius:10px;margin-
bottom:8px"><div><strong>'+escH(u.fullName)+'</strong> ('+escH(u.username)+'</div><div
style="margin-left:auto;display:flex;gap:6px"><button class="btn btn-success btn-xs"
onclick="approveUserAction(\''+u.username+\'','\'approve\')"><i class="fas fa-
check"></i></button><button class="btn btn-danger btn-xs"
onclick="approveUserAction(\''+u.username+\'','\'reject\')"><i class="fas fa-
times"></i></button></div></div>});G('pendingUsersList').innerHTML=ph;}}else{G('pendingUsersWr
ap').style.display='none';G('badgePendingUsers').style.display='none';}var
html='';users.forEach(function(u){var sb=u.status==='active'?<span class="badge badge-
success">Active</span>:u.status==='pending'?<span class="badge badge-
warning">Pending</span>:<span class="badge badge-
danger">'+escH(u.status)+'</span>;html+=<tr><td>'+escH(u.username)+'</td><td>'+escH(u.fullNa
me)+'</td><td>'+escH(u.email)+'</td><td><span class="badge badge-
info">'+escH(u.role)+'</span></td><td>'+sb+'</td><td>'+fmtDate(u.expiryDate)+'</td><td>'+(u.img
AnalAccess?'<i class="fas fa-check" style="color:var(--success)"></i>:'-')+</td><td class="table-
actions"><button class="btn btn-info btn-xs"
onclick="openUserPwdModal(\''+u.username+\'','\''+escH(u.fullName).replace(/'/g,'\\')+\'')"
title="Kelola Password"><i class="fas fa-key"></i></button></td><td class="table-actions"><button
class="btn btn-warning btn-xs" onclick="editUser(\''+u.username+\'')"><i class="fas fa-
edit"></i></button> <button class="btn btn-danger btn-xs" onclick="delUser(\''+u.username+\'')"><i
class="fas fa-
trash"></i></button></td></tr>});G('usersTableBody').innerHTML=html;}).withFailureHandler(func
tion(e){hl();toast('Error: '+e.message || e || 'Terjadi
kesalahan'),'error');}).getUsers(CU.username,CU.role);}
function approveUserAction(u,a){var exp='';if(a==='approve'){var d=new
Date();d.setFullYear(d.getFullYear()+1);exp=d.getFullYear()+'-
'+String(d.getMonth()+1).padStart(2,'0')+'-

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'+String(d.getDate()).padStart(2,'0');});sl('Memproses...');google.script.run.withSuccessHandler(function(r){hl();if(r.ok){toast('User '+a,'success');loadUsers();}else toast(r.msg,'error');}).withFailureHandler(function(e){hl();toast('Error: '+e.message+' | e | 'Terjadi kesalahan'),'error');}).approveUser(u,a,CU.username,exp);}

function
openUserModal(d){G('mUserIsNew').value=d?'0':'1';G('mUserUsername').value=d?d.username:'';G('mUserUsername').disabled=!d;G('mUserFullName').value=d?d.fullName:'';G('mUserEmail').value=d?d.email:'';G('mUserPassword').value='';G('mUserRole').value=d?d.role:'user';G('mUserStatus').value=d?d.status:'active';G('mUserExpiry').value=d?d.expiryDate:'';G('mUserImgAccess').checked=d?d.imgAnalAccess:false;G('modalUserTitle').textContent=d?'Edit User':'Tambah User';opM('modalUser');}

function
editUser(u){sl('Memuat...');google.script.run.withSuccessHandler(function(r){hl();if(!r || !r.ok)return;var u2=(r.data || []).find(function(x){return x.username===u;});if(u2)openUserModal(u2);}).withFailureHandler(function(e){hl();toast('Error: '+e.message+' | e | 'Terjadi kesalahan'),'error');}).getUsers(CU.username,CU.role);}

function saveUserForm(){var
p={isNew:G('mUserIsNew').value==='1',username:G('mUserUsername').value.trim(),fullName:G('mUserFullName').value.trim(),email:G('mUserEmail').value.trim(),password:G('mUserPassword').value || null,role:G('mUserRole').value,status:G('mUserStatus').value,expiryDate:G('mUserExpiry').value,imgAnalAccess:G('mUserImgAccess').checked};if(!p.username || !p.fullName){toast('Username dan nama wajib','warning');return;}sl('Menyimpan...');google.script.run.withSuccessHandler(function(r){hl();if(r.ok){toast('Disimpan','success');clM('modalUser');loadUsers();}else toast(r.msg,'error');}).withFailureHandler(function(e){hl();toast('Error: '+e.message+' | e | 'Terjadi kesalahan'),'error');}).saveUser(p,CU.role);}

function delUser(u){if(u===CU.username){toast('Tidak bisa hapus sendiri','warning');return;}cfm('Hapus User','Lanjutkan?','🗑️','Hapus',function(y){if(!y)return;sl('Menghapus...');google.script.run.withSuccessHandler(function(r){hl();if(r.ok){toast('Dihapus','success');loadUsers();}).withFailureHandler(function(e){hl();toast('Error: '+e.message+' | e | 'Terjadi kesalahan'),'error');}).deleteUser(u,CU.role);});}

// — RESET & TAMPILKAN INPUT QC TABS —————
function resetInputQCTabs(){
  G('tabBulk').style.display='none';
  G('tabSmart').style.display='none';
  G('btnSubmitBulk').style.display='none';
  G('btnSubmitSmart').style.display='none';
  G('btnClearSmart').style.display='none';
  G('bulkTableWrap').innerHTML='';
  G('smartTableWrap').innerHTML='<p style="padding:24px;text-align:center;color:var(--text-secondary)">Pilih dulu lalu klik Tampilkan</p>';
  ['bulkBidang','bulkParam','bulkLot','bulkStart','bulkEnd','bulkPwd',
   'smartBidang','smartParam','smartLot','smartStart','smartEnd','smartPwd'].forEach(function(id){var el=G(id);if(el)el.value='';});
  toast('Bulk & Smart input direset','info');
}

function tampilkanInputQCTabs(){

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G('tabBulk').style.display="";
G('tabSmart').style.display="";
toast('Bulk & Smart input ditampilkan','success');
}
// ——— MULTI-PASSWORD MANAGEMENT ———
function openUserPwdModal(username,fullName){
  G('mPwdTargetUser').value=username;
  G('mPwdTargetLabel').textContent=fullName+' ('+username+')';
  G('mPwdNew').value="";G('mPwdNama').value="";G('mPwdUsername').value="";
  G('mPwdNote').value="";G('mPwdEditOld').value="";

  G('mPwdAddBtn').style.display="";G('mPwdUpdateBtn').style.display='none';G('mPwdCancelEditBtn')
  .style.display='none';
  opM('modalUserPwd');
  loadUserPasswords(username);
}
function loadUserPasswords(username){
  var tb=G('userPwdTableBody');tb.innerHTML='<tr><td colspan="9" style="text-align:center">Memuat...</td></tr>';
  google.script.run.withSuccessHandler(function(r){
    if(!r || !r.ok){tb.innerHTML='<tr><td colspan="9" style="text-align:center;color:var(--danger)">'+escH(r.msg || 'Gagal')+'</td></tr>';return;}
    var data=r.data || [];
    if(!data.length){tb.innerHTML='<tr><td colspan="9" style="text-align:center;color:var(--text-secondary)">Belum ada password tambahan</td></tr>';return;}
    var html="";
    data.forEach(function(p,i){
      var stBadge=p.status==='active'?<span class="badge badge-success">Active</span>:'<span class="badge badge-secondary">Inactive</span>';
      var stToggle=p.status==='active'?'inactive':'active';
      var stBtnCls=p.status==='active'?'btn-warning':'btn-success';
      var stIcon=p.status==='active'?'fa-ban':'fa-check';
      html+=<tr><td>'+(i+1)+'</td><td><code>'+escH(p.loginUsername || '-')+'</code></td><td><code>
style="background:var(--bg-soft);padding:2px 8px;border-radius:4px;font-size:.82rem">'+escH(p.password)+'</code></td><td>'+escH(p.nama || '-')+'</td><td>'+escH(p.note || '-')+'</td><td>'+stBadge+'</td><td>'+escH(p.createdBy)+'</td><td>'+escH(p.createdDate)+'</td><td
class="table-actions"><button class="btn btn-info btn-xs"
onclick="editUserPwd(\''+escH(p.username)+'\',\''+escH(p.password).replace(/'/g,'\\\'')+\'\',\''+escH(
p.nama).replace(/'/g,'\\\'')+\'\',\''+escH(p.note).replace(/'/g,'\\\'')+\'\',\''+escH(p.loginUsername || '\')
.replace(/'/g,'\\\'')+\'\'') title="Edit"><i class="fas fa-edit"></i></button> <button class="btn
'+stBtnCls+' btn-xs"
onclick="togglePwdStatus(\''+escH(p.username)+'\',\''+escH(p.password).replace(/'/g,'\\\'')+\'\',\''+st
Toggle+\'\'') title="Ubah Status"><i class="fas '+stIcon+'"></i></button> <button class="btn btn-
danger btn-xs"

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onclick="delUserPwd(\"+escH(p.username)+'\\'+escH(p.password).replace(/'/g,\"\\'")+'\")"
title="Hapus"><i class="fas fa-trash"></i></button></td></tr>';
});
tb.innerHTML=html;
}).withFailureHandler(function(e){tb.innerHTML='<tr><td colspan="9" style="text-align:center;color:var(--danger)">Error</td></tr>';}).getUserPasswords(username,CU.role);
}
function addUserPwdAction(){
var target=G('mPwdTargetUser').value;
var pwd=G('mPwdNew').value;
var nama=G('mPwdNama').value.trim();var note=G('mPwdNote').value.trim();
var loginUsername=G('mPwdUsername').value.trim();
if(!nama){toast('Nama wajib diisi','warning');return;}
if(!loginUsername){toast('Username wajib diisi','warning');return;}
if(!pwd || pwd.length<6){toast('Password min 6 karakter','warning');return;}

sl('Menambahkan...');google.script.run.withSuccessHandler(function(r){hl();if(r.ok){toast(r.msg,'success');G('mPwdNew').value="";G('mPwdNama').value="";G('mPwdNote').value="";G('mPwdUsername').value="";G('mPwdEditOld').value="";loadUserPasswords(target);}else
toast(r.msg,'error');}).withFailureHandler(function(e){hl();toast('Error: '+e.message || e || 'Terjadi kesalahan'),'error');}).addUserPassword({targetUsername:target,newPassword:pwd,nama:nama,note:note,loginUsername:loginUsername},CU.role,CU.username);
}
function delUserPwd(username,password){
cfm('Hapus Password','Hapus password tambahan ini?','🗑️','Hapus',function(y){

if(!y)return;sl('Menghapus...');google.script.run.withSuccessHandler(function(r){hl();if(r.ok){toast(r.msg,'success');loadUserPasswords(username);}else
toast(r.msg,'error');}).withFailureHandler(function(e){hl();toast('Error: '+e.message || e || 'Terjadi kesalahan'),'error');}).deleteUserPassword(username,password,CU.role,CU.username);
});
}
function editUserPwd(username,password,nama,note,loginUsername){
G('mPwdEditOld').value=password;
G('mPwdNew').value=password;
G('mPwdNama').value=nama || "";
G('mPwdNote').value=note || "";
G('mPwdUsername').value=loginUsername || "";
G('mPwdAddBtn').style.display='none';
G('mPwdUpdateBtn').style.display="";
G('mPwdCancelEditBtn').style.display="";
toast('Mode Edit: Ubah data lalu klik Update','info');
}
function cancelEditPwd(){
G('mPwdEditOld').value="";

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G('mPwdNew').value="";
G('mPwdNama').value="";
G('mPwdNote').value="";
G('mPwdUsername').value="";
G('mPwdAddBtn').style.display="";
G('mPwdUpdateBtn').style.display='none';
G('mPwdCancelEditBtn').style.display='none';
}
function updateUserPwdAction(){
    var target=G('mPwdTargetUser').value;
    var oldPwd=G('mPwdEditOld').value;
    var pwd=G('mPwdNew').value;
    var nama=G('mPwdNama').value.trim();
    var note=G('mPwdNote').value.trim();
    var loginUsername=G('mPwdUsername').value.trim();
    if(!nama){toast('Nama wajib diisi','warning');return;}
    if(!loginUsername){toast('Username wajib diisi','warning');return;}
    if(!oldPwd){toast('Data edit tidak valid','warning');return;}

    sl('Memperbarui...');google.script.run.withSuccessHandler(function(r){hl();if(r.ok){toast(r.msg,'succes
s');cancelEditPwd();loadUserPasswords(target);}else
    toast(r.msg,'error');}).withFailureHandler(function(e){hl();toast('Error: '+e.message+' ||e|| 'Terjadi
kesalahan'),'error');}).editUserPassword({targetUsername:target,oldPassword:oldPwd,newPassword:
pwd,nama:nama,note:note,loginUsername:loginUsername},CU.role,CU.username);
}
function togglePwdStatus(username,password,newStatus){
    var label=newStatus==='active'?'mengaktifkan':'menonaktifkan';
    cfm('Ubah Status',label+' password
ini?', '🔑',label.charAt(0).toUpperCase()+label.slice(1),function(y){
        if(!y)return;sl('Mengubah
status...');google.script.run.withSuccessHandler(function(r){hl();if(r.ok){toast(r.msg,'success');loadUs
erPasswords(username);}else toast(r.msg,'error');}).withFailureHandler(function(e){hl();toast('Error:
'+e.message+' ||e|| 'Terjadi
kesalahan'),'error');}).toggleUserPasswordStatus(username,password,newStatus,CU.role,CU.userna
me);
    });
}
function loadKopSurat(){var
k=CD.kop||{};G('kopNamaRS').value=k.namaRS||"";G('kopAlamat').value=k.alamat||"";G('kopTelepon
').value=k.telepon||"";G('kopEmail').value=k.email||"";G('kopDokterPJ').value=k.dokterPJ||"";G('kopE
mailDokterPJ').value=k.emailDokterPJ||"";G('kopKepalaRuangan').value=k.kepalaRuangan||"";G('kop
EmailKepalaRuangan').value=k.emailKepalaRuangan||"";G('kopLogo').value=k.logo||"";G('kopLogo2').
value=k.logo2||"";G('kopPEMDA').value=k.pemda||"";if(k.logo)G('kopLogoPreview').innerHTML='<im
g src="'+k.logo+'" style="max-height:70px;border-radius:8px">';else
G('kopLogoPreview').innerHTML="";if(k.logo2)G('kopLogo2Preview').innerHTML='';else
G('kopLogo2Preview').innerHTML="";}
function handleLogoUpload(e){var f=e.target.files[0];if(!f)return;var r=new
FileReader();r.onload=function(ev){G('kopLogo').value=ev.target.result;G('kopLogoPreview').innerHT
ML='';r.readAsDataURL(f);}
function handleLogo2Upload(e){var f=e.target.files[0];if(!f)return;var r=new
FileReader();r.onload=function(ev){G('kopLogo2').value=ev.target.result;G('kopLogo2Preview').inner
HTML='';r.readAsDataURL(f);}
function saveKopSuratForm(){var
p={namaRS:G('kopNamaRS').value.trim(),alamat:G('kopAlamat').value.trim(),telepon:G('kopTelepon'
).value.trim(),email:G('kopEmail').value.trim(),dokterPJ:G('kopDokterPJ').value.trim(),emailDokterPJ:
G('kopEmailDokterPJ').value.trim(),kepalaRuangan:G('kopKepalaRuangan').value.trim(),emailKepalaR
uangan:G('kopEmailKepalaRuangan').value.trim(),logo:G('kopLogo').value,logo2:G('kopLogo2').value,
pemda:G('kopPEMDA').value.trim()};sl('Menyimpan...');google.script.run.withSuccessHandler(function
(r){hl();if(r.ok){toast('Kop disimpan', 'success');CD.kop=p;}else
toast(r.msg, 'error');}).withFailureHandler(function(e){hl();toast('Error: '+e.message || e || 'Terjadi
kesalahan', 'error');}).saveKopSurat(p, getActiveUsername(), getLogUser());}
function loadSettingsPage(){var
s=CD.settings || {};G('setThemeColor').value=s.theme_primary || '#2563eb';G('setSidebarColor').value
=s.sidebar_bg || '#0f172a';G('setSbFontColor').value=s.sb_font_color || '#f1f5f9';G('setSbFontSize').val
ue=s.sb_font_size || '14';G('setSbFontWeight').value=s.sb_font_weight || '500';G('setAutoLogout').val
ue=s.auto_logout || '180';G('setAutoBackup').checked=CD._backupTriggerActive;if(G('setAppTheme')
)G('setAppTheme').value=s.app_theme || 'light';G('setLoginBrightness')&&G('setLoginBrightness').re
move();G('setLoginBgColor').value=s.login_bg_color || '#1e293b';
if(G('setLoginFontSize'))G('setLoginFontSize').value=s.login_font_size || '16';
if(G('setLoginFontColor'))G('setLoginFontColor').value=s.login_font_color || '#ffffff';
if(G('setLoginFontWeight'))G('setLoginFontWeight').value=s.login_font_weight || '400';
var brightVal=s.login_brightness || '0.8';var activeBtn=document.querySelector('.brightness-btn[data-
val="'+brightVal+'"]');G('loginBrightVal').textContent=activeBtn?activeBtn.getAttribute('data-
label') || brightVal:brightVal;
document.querySelectorAll('.brightness-
btn').forEach(function(b){b.classList.toggle('active',b.getAttribute('data-
val')===brightVal)});G('themePresets').innerHTML=[{c:'#2563eb'},{c:'#10b981'},{c:'#8b5cf6'},{c:'#ef4
444'},{c:'#f59e0b'},{c:'#06b6d4'}].map(function(p){return '<div class="color-preset"
style="background:'+p.c+'"'
onclick="G('setThemeColor').value='"+p.c+"'></div>'}).join("");G('sidebarPresets').innerHTML=[{c
:'#0f172a'},{c:'#1e3a8a'},{c:'#064e3b'},{c:'#4c1d95'},{c:'#18181b'}].map(function(p){return '<div
class="color-preset" style="background:'+p.c+'"'
onclick="G('setSidebarColor').value='"+p.c+"'></div>'}).join("");G('loginBgPresets').innerHTML=[{
c:'#1e293b'},{c:'#0f172a'},{c:'#1e3a5f'},{c:'#2d1b69'},{c:'#3b1a1a'},{c:'#1a3b2d'}].map(function(p){ret
urn '<div class="color-preset" style="background:'+p.c+'"'
onclick="G('setLoginBgColor').value='"+p.c+"'></div>'}).join("");google.script.run.withSuccessHan
dler(function(r){G('emailQuotaDisplay').textContent=r.ok?r.remaining+'

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tersisa':'N/A';}).withFailureHandler(function(e){hl();toast('Error: '+e.message | |e | |'Terjadi
kesalahan'),'error');}).getEmailQuota();}
function saveAppSettings(){
var s={
theme_primary:G('setThemeColor').value,
sidebar_bg:G('setSidebarColor').value,
sb_font_color:G('setSbFontColor').value,
sb_font_size:G('setSbFontSize').value,
sb_font_weight:G('setSbFontWeight').value,
auto_logout:G('setAutoLogout').value,
backup_auto:G('setAutoBackup').checked?'true':'false',
app_theme:G('setAppTheme').value || 'light',
login_font_size:G('setLoginFontSize')?G('setLoginFontSize').value:'16',
login_font_color:G('setLoginFontColor')?G('setLoginFontColor').value:'#ffffff',
login_font_weight:G('setLoginFontWeight')?G('setLoginFontWeight').value:'400',
login_brightness:document.querySelector('.brightness-
btn.active')?document.querySelector('.brightness-btn.active').getAttribute('data-val'):'0.8',
login_bg_color:G('setLoginBgColor').value
};
sl('Menyimpan...');
google.script.run.withSuccessHandler(function(r){
hl();
if(r.ok){
toast('Tersimpan', 'success');
CD.settings=Object.assign(CD.settings | | {},s);
CD._backupTriggerActive=s.backup_auto==='true';
applyTheme();
applyUserTheme();
startAutoLogout();
// Auto setup/remove backup trigger berdasarkan setting
if(s.backup_auto==='true'){
google.script.run.withSuccessHandler(function(){toast('Auto Backup trigger
diaktifkan','info');}).withFailureHandler(function({}).setupAutoBackupTrigger());
}else{

google.script.run.withSuccessHandler(function({}).withFailureHandler(function({}).removeAutoBac
kupTrigger());
}
}
}).withFailureHandler(function(e){hl();toast('Error: '+e.message | |e | |'Terjadi
kesalahan'),'error');}).saveSettings(s);
}
function
initSheetsManual(){sl('Init...');google.script.run.withSuccessHandler(function(r){hl();toast(r.msg | |'OK

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    'success'));).withFailureHandler(function(e){hl();toast('Error: '+e.message+' | e | 'Terjadi
kesalahan'),'error');}).initializeSheets();}
function confirmResetDB(){cfm('RESET DATABASE','Semua data akan
dihapus!','🗑️','Reset!',function(y){if(!y)return;sl('Mereset...');google.script.run.withSuccessHandler(f
unction(r){hl();toast(r.msg,'success');loadInitData();}).withFailureHandler(function(e){hl();toast('Error
: '+e.message+' | e | 'Terjadi kesalahan'),'error');}).resetDatabase(CU.username,CU.role,'RESET
CONFIRM');});}
function
triggerAutoArchive(){sl('Arsip...');google.script.run.withSuccessHandler(function(){hl();toast('Selesai','
success');}).withFailureHandler(function(){hl();}).autoArchiveYearly();}
function
doBackupSheets(){cfm('Backup','Lanjutkan?','📁','Backup',function(y){if(!y)return;sl('Backup...');googl
e.script.run.withSuccessHandler(function(r){hl();toast(r.msg,'success',5000);}).withFailureHandler(fu
nction(){hl();}).backupAllSheets(CU.username);});}
function doBackupAppProject(){cfm('Backup
App','Lanjutkan?','📁','Backup',function(y){if(!y)return;sl('Backup...');google.script.run.withSuccessHand
ler(function(r){hl();toast(r.msg,'success',5000);}).withFailureHandler(function(){hl();}).backupAppProj
ect(CU.username);});}
function openRestoreModal(){opM('modalRestore');loadBackupList();}
function loadBackupList(){G('restoreList').innerHTML=<div class="spinner" style="margin:20px
auto;display:block"></div>;google.script.run.withSuccessHandler(function(r){if(!r || !r.ok){G('restore
List').innerHTML=<p>Gagal</p>;return;}var data=r.data || {};var
keys=Object.keys(data).sort();if(!keys.length){G('restoreList').innerHTML=<p>Tidak ada
backup</p>;return;}var html="";keys.forEach(function(folder){var
files=data[folder] || [];if(!files.length)return;var uid='rf_'+folder.replace(/^[^a-zA-Z0-
9]/g,'_');html+=<div class="restore-folder"><div class="restore-folder-header"
onclick="toggleRestoreFolder(\"'+uid+'\")><span><i class="fas fa-folder"></i> '+escH(folder)+'
('+files.length+')</span></div><div class="restore-folder-files"
id="'+uid+'">;files.forEach(function(f){var sizeKB=(f.size/1024).toFixed(1);var dt=new
Date(f.date);html+=<div class="restore-file-item"><div>'+escH(f.name)+'<br><span class="file-
meta" style="font-size:.72rem;color:var(--text-muted)">'+fmtDate(dt)+' •
'+sizeKB+'KB</span></div>;if(folder!='_AppProject'&&(f.name.indexOf('.json')>-
1 || f.name.indexOf('.csv')>-1))html+=<button class="btn btn-warning btn-xs"
onclick="confirmRestoreFile(\"'+f.id+'\",''+escH(folder)+'',''+escH(f.name)+'\")><i class="fas fa-
undo"></i></button>;html+=</div>;});html+=</div></div>;});G('restoreList').innerHTML=html;}).w
ithFailureHandler(function(){G('restoreList').innerHTML=<p>Error</p>;}).listBackups();}
function toggleRestoreFolder(uid){var el=G(uid);if(el)el.classList.toggle('show');}
function confirmRestoreFile(fileId,sheetName,filename){cfm('Restore','Sheet "' +sheetName+'"' akan
ditimpa?','📄','Restore',function(y){if(!y)return;sl('Restore...');google.script.run.withSuccessHandler(f
unction(r){hl();if(r.ok){toast(r.msg,'success');loadInitData();clM('modalRestore');}else
toast(r.msg,'error');}).withFailureHandler(function(){hl();}).restoreSheetFromBackup(fileId,sheetNam
e,CU.role);});}
function
setupBackupTrigger(){sl('Setup...');google.script.run.withSuccessHandler(function(){hl();toast('Trigger
aktif','success');}).withFailureHandler(function(){hl();}).setupAutoBackupTrigger();}

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function doExportSheet(format){var name=G('exportSheetName').value;if(!name){toast('Pilih sheet','warning');return;}sl('Export...');google.script.run.withSuccessHandler(function(r){hl();if(!r || !r.ok)return;var blob=new Blob([r.content],[{type:r.mime}]);var url=URL.createObjectURL(blob);var a=document.createElement('a');a.href=url;a.download=r.filename;document.body.appendChild(a);a.click();setTimeout(function(){URL.revokeObjectURL(url);document.body.removeChild(a);},100);toast(r.filename+' diunduh','success');}).withFailureHandler(function(){hl();}).exportSheet(name,format);}

function doTestEmail(){var to=G('testEmailTo').value;if(!to){toast('Email wajib','warning');return;}sl('Mengirim...');google.script.run.withSuccessHandler(function(r){hl();toast(r.msg || 'Ter kirim','success');}).withFailureHandler(function(){hl();}).testEmail(to,CU.username);}

function loadLogActivity(){if(!CU || !CU.username){toast('Sesi belum aktif','warning');return;}sl('Memuat log...');google.script.run.withSuccessHandler(function(data){hl();var html='';(data || []).forEach(function(l){html+<tr><td>'+escH(l.timestamp)+'</td><td>'+escH(l.userName)+'</td><td><span class="badge badge-info">'+escH(l.action)+'</span></td><td><small>'+escH(l.detail)+'</small></td></tr>');if(!data || !data.length)html='<tr><td colspan="4" style="text-align:center">Tidak ada log hari ini</td></tr>';G('logTableBody').innerHTML=html;}).withFailureHandler(function(){hl();}).getLogActivity(getActiveUsername(),getActiveRole());}

function clearLog(){cfm('Bersihkan','Lanjutkan?','🗑️','Bersihkan',function(y){if(!y)return;sl('Bersih...');google.script.run.withSuccessHandler(function(){hl();toast('Bersih','success');loadLogActivity();}).withFailureHandler(function(){hl();}).clearLogActivity(getActiveUsername(),getActiveRole());});}

function unlockHapusData(){if(G('hapusGatePwd').value!='didikqc'){toast('Password salah','error');return;}hapusUnlocked=true;G('hapusGate').style.display='none';G('hapusContent').style.display='';loadHapusData();}

function loadHapusData(){if(!CU || !CU.username){toast('Sesi belum aktif','warning');return;}if(!hapusUnlocked)return;switchHapusTab('inputqc');}

function switchHapusTab(tab){QA('#pageHapusdata .tab-btn').forEach(function(b,i){b.classList.toggle('active',(tab==='inputqc'&&i===0) || (tab==='lotqc'&&i===1) || (tab==='parameter'&&i===2));});G('hapusTabInputqc').classList.toggle('active',tab==='inputqc');G('hapusTabLotqc').classList.toggle('active',tab==='lotqc');G('hapusTabParameter').classList.toggle('active',tab==='parameter');if(tab==='inputqc')loadHapusInputQC();else if(tab==='lotqc')loadHapusLotQC();else loadHapusParam();}

function loadHapusInputQC(){var startDate=G('hapusQCStart').value || null;var endDate=G('hapusQCEnd').value || null;var paramIDs=[];QA('.hapus-param-chk:checked').forEach(function(cb){paramIDs.push(cb.value);});if(!paramIDs.length){G('hapusInputQCTable').innerHTML='<p style="text-align:center;color:var(--text-secondary);padding:20px">Pilih minimal 1 parameter</p>';return;}sl('Memuat...');google.script.run.withSuccessHandler(function(data){hl();var html='<table><thead><tr><th><input type="checkbox">onchange="toggleHapusAll(this,\'hqc\')"></th><th>Tgl</th><th>Parameter</th><th>Alat</th><th>L1</th><th>L2</th><th>L3</th></tr></thead><tbody>';(data || []).reverse().slice(0,200).forEach(function(q){html+<tr><td><input type="checkbox" class="hqc-chk" value="" + q.qcID + ""></td><td>'+fmtDate(q.tanggal)+'</td><td>'+escH(q.parameter)+'</td><td>'+escH(q.namaAlat)+'</td><td>'+(q.level1 || '-')+'</td><td>'+(q.level2 || '-')+'</td><td>'+(q.level3 || '-')

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')+</td></tr>'}};html+=</tbody></table>;G('hapusInputQCTable').innerHTML=html;toast((data || []).length+' data ditemukan','info'));}).withFailureHandler(function(){hl();toast('Error: '+e.message | e | 'Terjadi kesalahan'),'error');}).getInputQCForHapus(getActiveUsername(),{startDate:startDate,endDate:endDate,paramIDs:paramIDs});}
function loadHapusLotQC(){var html=<table><thead><tr><th><input type="checkbox" onchange="toggleHapusAll(this,\'hlot\')"></th><th>Parameter</th><th>No.Lot</th></tr></thead><tbody>;(CD.lots || []).forEach(function(l){html+=<tr><td><input type="checkbox" class="hlot-chk" value="'+l.lotID+'"></td><td>'+escH(getParamName(l.paramID))+'</td><td>'+escH(l.noLot)+'</td></tr>'}};html+=</tbody></table>;G('hapusLotQCTable').innerHTML=html;}
function loadHapusParam(){var html=<table><thead><tr><th><input type="checkbox" onchange="toggleHapusAll(this,\'hpar\')"></th><th>Parameter</th><th>Bidang</th></tr></thead><tbody>;(CD.params || []).forEach(function(p){html+=<tr><td><input type="checkbox" class="hpar-chk" value="'+p.paramID+'"></td><td>'+escH(p.parameter)+'</td><td>'+escH(p.bidang || '')+'</td></tr>'}};html+=</tbody></table>;G('hapusParamTable').innerHTML=html;}
function toggleHapusAll(m,p){QA('.'+p+'-chk').forEach(function(cb){cb.checked=m.checked;});}
function submitHapusData(){var alasan=G('hapusAlasan').value.trim();if(alasan.length<5){toast('Alasan min 5 karakter','warning');return;}var type="",ids=[];if(G('hapusTabInputqc').classList.contains('active')){type='inputqc';QA(''.hqcchk:checked').forEach(function(c){ids.push(c.value);});}else if(G('hapusTabLotqc').classList.contains('active')){type='lotqc';QA(''.hlotchk:checked').forEach(function(c){ids.push(c.value);});}else{type='parameter';QA(''.hparchk:checked').forEach(function(c){ids.push(c.value);});}if(!ids.length){toast('Pilih data','warning');return;}cfm('Hapus '+ids.length+' data','Permanen!','', 'Hapus!',function(y){if(!y)return;sl('Menghapus...');google.script.run.withSuccessHandler(function(r){hl();if(r.ok){toast(r.count+' dihapus','success');loadHapusData();refreshParams();refreshLots();}else toast(r.msg,'error');}).withFailureHandler(function(){hl();}).hapusDataPrivat(type,ids,getActiveUsername(),alasan,'didikqc',getLogUser());});}
function handleImportFile(e){var f=e.target.files[0];if(!f)return;var r=new FileReader();r.onload=function(ev){var lines=ev.target.result.trim().split(/\r?\n/);importData=[];var html=<div class="table-wrap"><table><thead><tr><th>#</th><th>Parameter</th><th>NoLot</th><th>Tgl</th><th>L1</th><th>L2</th><th>L3</th></tr></thead><tbody>;lines.forEach(function(line,i){if(i===0&&line.toLowerCase().indexOf('parameter')>-1)return;var cols=line.split(/[,;\t | ]/);if(cols.length<4)return;var row={parameter:cols[0]?cols[0].trim():'',noLot:cols[1]?cols[1].trim():'',tanggal:cols[2]?cols[2].trim():'',level1:cols[3]?cols[3].trim():'',level2:cols[4]?cols[4].trim():'',level3:cols[5]?cols[5].trim():''};importData.push(row);html+=<tr><td>'+(i+1)+'</td><td>'+escH(row.parameter)+'</td><td>'+escH(row.noLot)+'</td><td>'+escH(row.tanggal)+'</td><td>'+escH(row.level1)+'</td><td>'+escH(row.level2)+'</td><td>'+escH(row.level3)+'</td></tr>'}};html+=</tbody></table></div>;G('importPreview').innerHTML=html;G('importPreview').style.display="";G('btnImportSubmit').style.display=importData.length?'':'none';toast(importData.length+' baris','info');});r.readAsText(f);}

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function submitImport(){if(!importData.length){toast('Tidak ada
data','warning');return;}sl('Import...');google.script.run.withSuccessHandler(function(r){hl();if(r.ok){to
ast(r.count+'
imported','success');importData=[];G('importPreview').style.display='none';G('btnImportSubmit').styl
e.display='none';}else
toast(r.msg,'error');}).withFailureHandler(function(){hl();}).smartImportQC({rows:importData,smartP
assword:G('importPwd').value},getActiveUsername());}
function handleMultimgUpload(e,slotNum){var
f=e.target.files[0];if(!f)return;if(f.size>5*1024*1024){toast('Maks 5MB','warning');return;}var
reader=new FileReader();reader.onload=function(ev){var img=new
Image();img.onload=function(){var MAX_W=800,MAX_H=800,MAX_SIZE=200*1024;var
w=img.width,h=img.height;if(w>MAX_W||h>MAX_H){if(w>h){h=Math.round(h*MAX_W/w);w=MAX
_W;}else{w=Math.round(w*MAX_H/h);h=MAX_H;}}var
c=document.createElement('canvas');c.width=w;c.height=h;var
ctx=c.getContext('2d');ctx.drawImage(img,0,0,w,h);var q=0.85;var
base64=c.toDataURL('image/jpeg',q);while(base64.length>MAX_SIZE&&q>0.3){q-
=0.1;base64=c.toDataURL('image/jpeg',q);}G('mImgURL'+(slotNum===1?'':slotNum)).value=base64;v
ar slot=G('imgSlot'+slotNum);slot.innerHTML='<div class="slot-label">Gambar
'+slotNum+'</div><input type="file" id="imgFileInput'+slotNum+'"
accept="image/*" style="display:none"
onchange="handleMultimgUpload(event,'+slotNum+')"><button class="remove-img"
onclick="event.stopPropagation();removeImgSlot(''+slotNum+'")>&times;</button>';slot.classList.ad
d('has-image');img.src=ev.target.result;reader.readAsDataURL(f);}
function removeImgSlot(slotNum){G('mImgURL'+(slotNum===1?'':slotNum)).value='';var
slot=G('imgSlot'+slotNum);slot.innerHTML='<div class="slot-label">Gambar '+slotNum+'</div><i
class="fas fa-image"></i><p>Klik upload</p><input type="file" id="imgFileInput'+slotNum+'"
accept="image/*" style="display:none"
onchange="handleMultimgUpload(event,'+slotNum+')"><button class="remove-img"
onclick="event.stopPropagation();removeImgSlot(''+slotNum+'")>&times;</button>';slot.classList.re
move('has-image');}
function getIdImgData(type){var cap=type.charAt(0).toUpperCase()+type.slice(1);var
filter={startDate:G('img'+cap+'Start')?G('img'+cap+'Start').value:'',endDate:G('img'+cap+'End')?G('im
g'+cap+'End').value:'',noRM:G('img'+cap+'RM')?G('img'+cap+'RM').value:'',nama:G('img'+cap+'Nama'
)?G('img'+cap+'Nama').value:''};sl('Memuat...');google.script.run.withSuccessHandler(function(r){hl();
if(!r||!r.ok)return;imgDataCache[type]=r.data||[];renderImgTable(type,r.data);}).withFailureHandler
(function(){hl();}).getImgData(type,getActiveUsername(),filter);}
function resetImgFilter(type){var
cap=type.charAt(0).toUpperCase()+type.slice(1);['Start','End','RM','Nama'].forEach(function(f){if(G('i
mg'+cap+f))G('img'+cap+f).value=''});getIdImgData(type);}
function renderImgTable(type,data){var
body=G('img'+type.charAt(0).toUpperCase()+type.slice(1)+'Body');if(!body)return;var
sumCol=type==='hemato'?'Kesan':(type==='malaria'||type==='bta')?'HasilMikro':'Interpretasi';var
html='';(data||[]).forEach(function(d,i){var sv=d[sumCol]||'-
';html+='<tr><td>'+(i+1)+'</td><td>'+fmtDate(d.TglPeriksa)+'</td><td>'+escH(d.NoRM)+'</td><td>
'+escH(d>Nama)+'</td><td>'+escH(d.JK)+'</td><td>'+escH(d.Dokter)+'</td><td><small>'+escH(String

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(sv.substring(0,50))+</small></td><td class="table-actions"><button class="btn btn-info btn-xs"
onclick="viewAndPrintImg(\'"+type+\'\',\'"+d.ID+\' \')"><i class="fas fa-print"></i></button> <button
class="btn btn-warning btn-xs" onclick="editImgData(\'"+type+\'\',\'"+d.ID+\' \')"><i class="fas fa-
edit"></i></button> <button class="btn btn-danger btn-xs"
onclick="deleteImgDataConfirm(\'"+type+\'\',\'"+d.ID+\' \')"><i class="fas fa-
trash"></i></button></td></tr>');});if(!data || !data.length)html=<tr><td colspan="8" style="text-
align:center">Belum ada data</td></tr>';body.innerHTML=html;
function
openImgModal(type,data){G('mImgType').value=type;G('mImgID').value=data?data.ID:'';[ 'NoRM','Na
ma','TglLahir','Umur','TglPeriksa','TglHasil','Alamat','Dokter','Diagnosis','Asal'].forEach(function(f){var
el=G('mImg'+f);if(el)el.value='';});G('mImgJK').value='';G('mImgURL').value='';G('mImgURL2').value='';
G('mImgURL3').value='';[1,2,3].forEach(function(n){removeImgSlot(n);});[ 'Eritrosit','Leukosit','Tromb
osit','Kesan','Saran'].forEach(function(f){var
el=G('mImg'+f);if(el)el.value='';});if(G('mImgInterpretasiUrin'))G('mImgInterpretasiUrin').value='';if(G(
'mImgHasilMikroMalaria'))G('mImgHasilMikroMalaria').value='';if(G('mImgSpesies'))G('mImgSpesies'
).value='';if(G('mImgKepadatanMalaria'))G('mImgKepadatanMalaria').value='';if(G('mImgHasilMikroB
TA'))G('mImgHasilMikroBTA').value='';if(G('mImgKepadatanBTA'))G('mImgKepadatanBTA').value='';if
(G('mImgInterpretasiLain'))G('mImgInterpretasiLain').value='';G('imgFieldsHemato').style.display=typ
e==='hemato'?':none';G('imgFieldsUrin').style.display=type==='urin'?':none';G('imgFieldsMalaria').
style.display=type==='malaria'?':none';G('imgFieldsBTA').style.display=type==='bta'?':none';G('img
FieldsLain').style.display=type==='lain'?':none';
if(data){G('mImgNoRM').value=data.NoRM || '';G('mImgNama').value=data.Nama || '';G('mImgTglLahi
r').value=data.TglLahir || '';G('mImgUmur').value=data.Umur || '';G('mImgJK').value=data.JK || '';G('mI
mgTglPeriksa').value=data.TglPeriksa || '';G('mImgTglHasil').value=data.TglHasil || '';G('mImgAlamat').
value=data.Alatat || '';G('mImgDokter').value=data.Dokter || '';G('mImgDiagnosis').value=data.Diagn
osis || '';G('mImgAsal').value=data.Asal || '';
if(data.ImageURL){G('mImgURL').value=data.ImageURL;var s1=G('imgSlot1');s1.innerHTML=<div
class="slot-label">Gambar 1</div><img src="" +data.ImageURL+""><input type="file"
id="imgFileInput1" accept="image/*" style="display:none"
onchange="handleMultiImgUpload(event,1)"><button class="remove-img"
onclick="event.stopPropagation();removeImgSlot(1)">&times;</button>;s1.classList.add('has-
image');}
if(data.ImageURL2){G('mImgURL2').value=data.ImageURL2;var s2=G('imgSlot2');s2.innerHTML=<div
class="slot-label">Gambar 2</div><img src="" +data.ImageURL2+""><input type="file"
id="imgFileInput2" accept="image/*" style="display:none"
onchange="handleMultiImgUpload(event,2)"><button class="remove-img"
onclick="event.stopPropagation();removeImgSlot(2)">&times;</button>;s2.classList.add('has-
image');}
if(data.ImageURL3){G('mImgURL3').value=data.ImageURL3;var s3=G('imgSlot3');s3.innerHTML=<div
class="slot-label">Gambar 3</div><img src="" +data.ImageURL3+""><input type="file"
id="imgFileInput3" accept="image/*" style="display:none"
onchange="handleMultiImgUpload(event,3)"><button class="remove-img"
onclick="event.stopPropagation();removeImgSlot(3)">&times;</button>;s3.classList.add('has-
image');}

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if(type==='hemato'){G('mImgEritrosit').value=data.Eritrosit| |";G('mImgLeukosit').value=data.Leukosit| |";G('mImgTrombosit').value=data.Trombosit| |";G('mImgKesan').value=data.Kesan| |";G('mImgSaran').value=data.Saran| |";}
else if(type==='urin')G('mImgInterpretasiUrin').value=data.Interpretasi| |";
else
if(type==='malaria'){G('mImgHasilMikroMalaria').value=data.HasilMikro| |";G('mImgSpesies').value=data.Spesies| |";G('mImgKepadatanMalaria').value=data.Kepadatan| |";}
else
if(type==='bta'){G('mImgHasilMikroBTA').value=data.HasilMikro| |";G('mImgKepadatanBTA').value=data.Kepadatan| |";}
else if(type==='lain')G('mImgInterpretasiLain').value=data.Interpretasi| |";
G('modallmgTitle').textContent='Edit Data';
}else{G('mImgTglPeriksa').value=todayISO();G('mImgTglHasil').value=todayISO();G('modallmgTitle').textContent='Tambah Data';}
opM('modallmg');
}
function editlmgData(type,id){var d=(imgDataCache[type]| |[]).find(function(x){return x.ID===id;});if(d)openlmgModal(type,d);else toast('Tidak ditemukan','error');}
function calcImgUmur(){var t=G('mImgTglLahir').value;if(!t)return;var b=new Date(t),n=new Date();var a=n.getFullYear()-b.getFullYear();var m=n.getMonth()-b.getMonth();if(m<0| |(m===0&&n.getDate()<b.getDate()))a--;G('mImgUmur').value=a+' tahun';}
function saveImgForm(){var type=G('mImgType').value;var p={ID:G('mImgID').value| |null,NoRM:G('mImgNoRM').value.trim(),Nama:G('mImgNama').value.trim(),TglLahir:G('mImgTglLahir').value,Umur:G('mImgUmur').value.trim(),JK:G('mImgJK').value,Alamat:G('mImgAlamat').value.trim(),Dokter:G('mImgDokter').value.trim(),TglPeriksa:G('mImgTglPeriksa').value,TglHasil:G('mImgTglHasil').value,Diagnosis:G('mImgDiagnosis').value.trim(),Asal:G('mImgAsal').value.trim(),ImageUrl:G('mImgURL').value,ImageUrl2:G('mImgURL2').value,ImageUrl3:G('mImgURL3').value};if(!p.Nama){toast('Nama wajib','warning');return;}
if(type==='hemato'){p.Eritrosit=G('mImgEritrosit').value;p.Leukosit=G('mImgLeukosit').value;p.Trombosit=G('mImgTrombosit').value;p.Kesan=G('mImgKesan').value;p.Saran=G('mImgSaran').value;}
else if(type==='urin')p.Interpretasi=G('mImgInterpretasiUrin').value;
else
if(type==='malaria'){p.HasilMikro=G('mImgHasilMikroMalaria').value;p.Spesies=G('mImgSpesies').value;p.Kepadatan=G('mImgKepadatanMalaria').value;}
else
if(type==='bta'){p.HasilMikro=G('mImgHasilMikroBTA').value;p.Kepadatan=G('mImgKepadatanBTA').value;}
else if(type==='lain')p.Interpretasi=G('mImgInterpretasiLain').value;
sl('Menyimpan...');google.script.run.withSuccessHandler(function(r){hl();if(r.ok){toast('Disimpan','success');clM('modallmg');ldlmgData(type);}else toast(r.msg,'error');}).withFailureHandler(function(){hl();}).saveImgData(type,p,getActiveUsername(),getLogUser());
}
function
deletelmgDataConfirm(type,id){cfm('Hapus','Lanjutkan?','□','Hapus',function(y){if(!y)return;sl('Meng

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hapus...');google.script.run.withSuccessHandler(function(r){hl();if(r.ok){toast('Dihapus','success');Idl
mgData(type);}}).withFailureHandler(function(){hl();}).deleteImgData(type,id,getActiveUsername(),g
etLogUser());});
function viewAndPrintImg(type,id){var d=(imgDataCache[type]||[]).find(function(x){return
x.ID===id;});if(!d){toast('Tidak ditemukan','error');return;}var
kop=CD.kop||{};G('imgViewContent').innerHTML=generateImgReportHTML(type,d,kop,true);opM('
modallmgView');setTimeout(function(){renderQR('imgQR1','didiQC|'+(kop.namaRS||'Lab')+'|
'+(kop.kepalaRuangan||''));renderQR('imgQR2','didiQC|'+(kop.namaRS||'Lab')+'|
'+(kop.dokterPJ||''));},300);}
function getQRDataURL(text){var
tmp=document.createElement('div');tmp.style.position='fixed';tmp.style.left='-
9999px';document.body.appendChild(tmp);try{new
QRCode(tmp,{text:text||'didiQC',width:80,height:80,colorDark:'#0f172a',colorLight:'#fff',correctLeve
l:QRCode.CorrectLevel.L});var cv=tmp.querySelector('canvas');if(cv){var
url=cv.toDataURL('image/png');document.body.removeChild(tmp);return url;}var
img=tmp.querySelector('img');document.body.removeChild(tmp);return
img?img.src:'';catch(e){document.body.removeChild(tmp);return ''}}
function generateImgReportHTML(type,d,kop,includeImages){var tl={hemato:'HASIL HEMATOLOGI
SEL',urin:'HASIL URIN SEDIMEN',malaria:'HASIL MALARIA',bta:'HASIL BTA',lain:'HASIL PEMERIKSAAN
LAB'};var html='<div id="imgPrintArea">'+buildKopHTML(kop,true)+'<h3 style="text-
align:center;margin-bottom:16px;font-size:1rem">'+(tl[type]||'')+'</h3><div class="img-result-
body"><table style="width:100%;margin-bottom:16px"><tr><td>No. RM</td><td>
'+escH(d.NoRM||'-')+'</td><td>Tgl Periksa</td><td>
'+fmtDateIndo(d.TglPeriksa)+'</td></tr><tr><td>Nama</td><td>'+escH(d>Nama||'-
')+'</td><td>Tgl Hasil</td><td>'+fmtDateIndo(d.TglHasil)+'</td></tr><tr><td>Tgl Lahir</td><td>
'+fmtDateIndo(d.TglLahir)+'</td><td>Umur</td><td>'+escH(d.Umur||'-
')+'</td></tr><tr><td>JK</td><td>'+(d.JK==='L'? 'Laki-laki':d.JK==='P'? 'Perempuan':-
')+'</td><td>Dokter</td><td>'+escH(d.Dokter||'-')+'</td></tr><tr><td>Alamat</td><td>
'+escH(d.Alatat||'-')+'</td><td>Asal</td><td>'+escH(d.Asal||'-
')+'</td></tr><tr><td>Diagnosis</td><td colspan="3">'+escH(d.Diagnosis||'-
')+'</td></tr></table><h4 style="border-bottom:2px solid #333;padding-bottom:4px;margin-
bottom:12px">HASIL PEMERIKSAAN</h4>;
if(type==='hemato')html+='<table
style="width:100%"><tr><td>Eritrosit</td><td>'+escH(d.Eritrosit||'-
').replace(/\\n/g,'<br>')+'</td></tr><tr><td>Leukosit</td><td>'+escH(d.Leukosit||'-
').replace(/\\n/g,'<br>')+'</td></tr><tr><td>Trombosit</td><td>'+escH(d.Trombosit||'-
').replace(/\\n/g,'<br>')+'</td></tr><tr><td>Kesan</td><td><strong>'+escH(d.Kesan||'-
').replace(/\\n/g,'<br>')+'</strong></td></tr><tr><td>Saran</td><td>'+escH(d.Saran||'-
').replace(/\\n/g,'<br>')+'</td></tr></table>;
else if(type==='urin')html+='<div style="padding:10px;border:1px solid #ddd;border-
radius:8px"><strong>Interpretasi:</strong><br>'+escH(d.Interpretasi||'-
').replace(/\\n/g,'<br>')+'</div>;
else if(type==='malaria')html+='<table><tr><td>Hasil Mikro</td><td>'+escH(d.HasilMikro||'-
')+'</td></tr><tr><td>Spesies</td><td>'+escH(d.Spesies||'-
')+'</td></tr><tr><td>Kepadatan</td><td>'+escH(d.Kepadatan||'-')+'</td></tr></table>;

```

```

else if(type==='bta')html+<table><tr><td>Hasil Mikro</td><td>'+escH(d.HasilMikro|'|'-
')+</td></tr><tr><td>Kepadatan</td><td>'+escH(d.Kepadatan|'|'-')+</td></tr></table>;
else if(type==='lain')html+<div style="padding:10px;border:1px solid #ddd;border-
radius:8px"><strong>Interpretasi:</strong><br>'+escH(d.Interpretasi|'|'-
').replace(/\\n/g,'<br>')+</div>;
if(includeImages!<=false){var imgs = [d.ImageURL, d.ImageURL2,
d.ImageURL3].filter(function(u){return u;});if(imgs.length){html+<div style="margin-top:16px;text-
align:center"><h5>Gambar Mikroskopis</h5><div style="display:flex;gap:8px;flex-wrap:wrap;justify-
content:center">;imgs.forEach(function(u){html+;});html+</div></div>;}
var qrImg1=getQRDataURL('didiQC | '+<div><div class="img-result-footer" style="display:flex;justify-content:space-between;align-items:flex-
start;margin-top:30px;gap:20px"><div class="ttd-col" style="text-align:center;width:45%;flex-
shrink:0"><p style="font-weight:600;margin-bottom:4px">Kepala Ruangan Lab</p>'+(qrImg1?'':<div
style="width:80px;height:80px;margin:8px auto"></div>')+<p class="ttd-name" style="margin-
top:30px;border-top:1px solid #333;padding-top:4px;font-
weight:700">'+escH(kop.kepalaRuangan|'|'____')+</p></div><div class="ttd-col" style="text-
align:center;width:45%;flex-shrink:0"><p style="font-weight:600;margin-bottom:4px">Dokter
PJ</p>'+(qrImg2?'':<div style="width:80px;height:80px;margin:8px auto"></div>')+<p class="ttd-
name" style="margin-top:30px;border-top:1px solid #333;padding-top:4px;font-
weight:700">'+escH(kop.dokterPJ|'|'____')+</p></div></div><p class="img-electronic" style="text-
align:center;font-size:.75rem;color:#666;margin-top:10px">Dokumen ini dihasilkan secara elektronik
oleh didiQCsyst</p></div>;return html;}
```

```

function toggleImgPrintMenu(btn){var
menu=btn.nextElementSibling;document.querySelectorAll('.print-options-
menu').forEach(function(m){if(m!<=menu)m.classList.remove('show');});menu.classList.toggle('show'
);}
```

```

function printImgFromList(type,mode){var
data=imgDataCache[type];if(!data||!data.length){toast('Tidak ada data','warning');return;}var
kop=CD.kop|'|{};var
allHtml="";data.forEach(function(d,i){allHtml+=generateImgReportHTML(type,d,kop,mode==='withIm
g');allHtml+<div style="page-break-after:always"></div>;});var
w=window.open('','_blank');w.document.write('<!DOCTYPE
html><html><head><title>Hasil</title><style>body{font-family:Inter,sans-serif;padding:10px
12px;font-size:11pt}.img-result-header{margin-bottom:12px!important;padding-
bottom:10px!important}.img-result-header img{max-height:80px}table{width:100%;border-
collapse:collapse}td{padding:4px 8px;vertical-align:top}.img-result-footer{display:flex;justify-
content:space-between;align-items:flex-start;margin-top:24px;gap:20px}.ttd-col{text-
align:center;width:45%;flex-shrink:0}.img-mikroskopis{max-width:280px;border:1px solid
#ddd;border-radius:4px;margin:4px}.ttd-name{margin-top:24px;border-top:1px solid #333;padding-
```

```
top:4px;font-weight:700}@page{margin:8mm
10mm;size:A4}</style></head><body>'+allHtml+'</body></html>');w.document.close();setTimeout(
function(){w.print();},500);}
```

```
function pdfImgFromList(type,mode){var
data=imgDataCache[type];if(!data || !data.length){toast('Tidak ada data','warning');return;}var
kop=CD.kop | {};var
container=document.createElement('div');container.style.position='fixed';container.style.left='-
9999px';container.style.top='0';container.style.width='800px';container.style.background='#fff';var
allHtml='<div
id="imgPrintArea">';data.forEach(function(d,i){allHtml+=generateImgReportHTML(type,d,kop,mode
==='withImg');allHtml+=<div style="page-break-
after:always"></div>;});allHtml+=</div>';container.innerHTML=allHtml;document.body.appendChil
d(container);sl('Membuat
PDF...');setTimeout(function(){html2canvas(container.firstElementChild,{scale:2,useCORS:true,allow
Taint:true}).then(function(canvas){document.body.removeChild(container);hl();var pdf=new
jspdf.jsPDF('p','mm','a4');var pW=pdf.internal.pageSize.getWidth()-
16,pH=pdf.internal.pageSize.getHeight()-16;var cH=canvas.height*pW/canvas.width;var
totalPages=Math.ceil(cH/pH);for(var pg=0;pg<totalPages;pg++){if(pg>0)pdf.addPage();var
sc=document.createElement('canvas');sc.width=canvas.width;sc.height=Math.min(pH,canvas.height-
pg*pH*canvas.width/pW);sc.getContext('2d').drawImage(canvas,0,pg*pH*canvas.width/pW,canvas.
width,sc.height,0,0,sc.width,sc.height);var
sH=Math.min(pH,canvas.height*pW/canvas.width);pdf.addImage(sc.toDataURL('image/png'),'PNG',
10,10,pW,sH);pdf.save(type+'_hasil_'+new
Date().getTime()+'.pdf');}).catch(function(e){document.body.removeChild(container);hl();toast('Error
: '+e.message || e || 'Terjadi kesalahan'),'error');});},500);}
```

```
function printImgForm(mode){var type=G('mImgType').value;var
payload=buildImgFormPayload();if(!payload>Nama){toast('Simpan data dulu atau isi
Nama','warning');return;}var kop=CD.kop | {};var
html=generateImgReportHTML(type,payload,kop,mode==='withImg');var
w=window.open('', '_blank');w.document.write('<!DOCTYPE
html><html><head><title>Hasil</title><style>body{font-family:Inter,sans-serif;padding:10px
12px;font-size:11pt}.img-result-header{margin-bottom:12px!important;padding-
bottom:10px!important}.img-result-header img{max-height:80px}table{width:100%;border-
collapse:collapse}td{padding:4px 8px;vertical-align:top}.img-result-footer{display:flex;justify-
content:space-between;align-items:flex-start;margin-top:24px;gap:20px}.ttd-col{text-
align:center;width:45%;flex-shrink:0}.img-mikroskopis{max-width:280px;border:1px solid
#ddd;border-radius:4px;margin:4px}.ttd-name{margin-top:24px;border-top:1px solid #333;padding-
top:4px;font-weight:700}@page{margin:8mm
10mm;size:A4}</style></head><body>'+html+'</body></html>');w.document.close();setTimeout(fu
nction(){w.print();},500);}
```

```
function pdfImgForm(mode){var type=G('mImgType').value;var
payload=buildImgFormPayload();if(!payload>Nama){toast('Simpan data dulu atau isi
```

```

Nama,'warning');return;}var kop=CD.kop| |{};var
html=generateImgReportHTML(type,payload,kop,mode==='withImg');var
container=document.createElement('div');container.style.position='fixed';container.style.left='-
9999px';container.style.top='0';container.style.width='800px';container.style.background='#fff';cont
ainer.innerHTML=html;document.body.appendChild(container);sl('Membuat
PDF...');setTimeout(function(){html2canvas(container,{scale:2,useCORS:true,allowTaint:true}).then(f
unction(canvas){document.body.removeChild(container);hl();var pdf=new
jspdf.jsPDF('p','mm','a4');var pW=pdf.internal.pageSize.getWidth()-
16,pH=pdf.internal.pageSize.getHeight()-16;var
cH=canvas.height*pW/canvas.width;if(cH<=pH)pdf.drawImage(canvas.toDataURL('image/png'),'PNG',
10,10,pW,cH);else{var rem=cH,sY=0;while(rem>0){var sH=Math.min(pH,rem);var
sc=document.createElement('canvas');sc.width=canvas.width;sc.height=sH*canvas.width/pW;sc.get
Context('2d').drawImage(canvas,0,sY,canvas.width,sc.height,0,0,sc.width,sc.height);if(sY>0)pdf.addP
age();pdf.drawImage(sc.toDataURL('image/png'),'PNG',10,10,pW,sH);sY+=sc.height;rem-
=sH;}}pdf.save(type+'_hasil_'+new
Date().getTime()+'.pdf');}).catch(function(e){document.body.removeChild(container);hl();toast('Error
: '+e.message| |e| |'Terjadi kesalahan'),'error');});,500);}

```

```

function buildImgFormPayload(){var
p={NoRM:G('mImgNoRM').value.trim(),Nama:G('mImgNama').value.trim(),TglLahir:G('mImgTglLahir')
.value,Umur:G('mImgUmur').value.trim(),JK:G('mImgJK').value,Alamat:G('mImgAlamat').value.trim(),
Dokter:G('mImgDokter').value.trim(),TglPeriksa:G('mImgTglPeriksa').value,TglHasil:G('mImgTglHasil')
.value,Diagnosis:G('mImgDiagnosis').value.trim(),Asal:G('mImgAsal').value.trim(),ImageUrl:G('mImg
URL').value,ImageUrl2:G('mImgURL2').value,ImageUrl3:G('mImgURL3').value;if(G('mImgEritrosit'))
p.Eritrosit=G('mImgEritrosit').value;if(G('mImgLeukosit'))p.Leukosit=G('mImgLeukosit').value;if(G('mI
mgTrombosit'))p.Trombosit=G('mImgTrombosit').value;if(G('mImgKesan'))p.Kesan=G('mImgKesan').v
alue;if(G('mImgSaran'))p.Saran=G('mImgSaran').value;if(G('mImgInterpretasiUrin'))p.Interpretasi=G('
mImgInterpretasiUrin').value;if(G('mImgHasilMikroMalaria'))p.HasilMikro=G('mImgHasilMikroMalari
a').value;if(G('mImgSpesies'))p.Spesies=G('mImgSpesies').value;if(G('mImgKepadatanMalaria'))p.Kep
adatan=G('mImgKepadatanMalaria').value;if(G('mImgHasilMikroBTA'))p.HasilMikro=G('mImgHasilMi
kroBTA').value;if(G('mImgKepadatanBTA'))p.Kepadatan=G('mImgKepadatanBTA').value;if(G('mImgInt
erpretasiLain'))p.Interpretasi=G('mImgInterpretasiLain').value;return p;}

```

```

function printPatologiForm(mode){var
payload=buildPatologiFormPayload();if(!payload>NamaPasien){toast('Isi Nama Pasien
dulu','warning');return;}var kop=CD.kop| |{};var
html=generatePatologiReportHTML(payload,kop,mode==='withImg');var
w=window.open('','_blank');w.document.write('<!DOCTYPE html><html><head><title>Hasil
Patologi</title><style>body{font-family:Inter,sans-serif;padding:10px 12px;font-
size:11pt}table{width:100%;border-collapse:collapse}td{padding:4px 8px;vertical-align:top}img{max-
width:100%;max-height:400px}.ttd-name{margin-top:24px;border-top:1px solid #333;padding-
top:4px;font-size:.9rem}.img-result-header{margin-bottom:12px!important;padding-
bottom:10px!important}@page{margin:8mm
10mm;size:A4}</style></head><body>'+html+'</body></html>');w.document.close();setTimeout(fu
nction(){w.print();},500);}

```

```

function pdfPatologiForm(mode){var
payload=buildPatologiFormPayload();if(!payload>NamaPasien){toast('Isi Nama Pasien
dulu','warning');return;}var kop=CD.kop| |{};var
html=generatePatologiReportHTML(payload,kop,mode=== 'withImg');var
container=document.createElement('div');container.style.position='fixed';container.style.left='-
9999px';container.style.top='0';container.style.width='800px';container.style.background='#fff';cont
ainer.innerHTML=html;document.body.appendChild(container);sl('Membuat
PDF...');setTimeout(function(){html2canvas(container,{scale:2,useCORS:true,allowTaint:true}).then(f
unction(canvas){document.body.removeChild(container);hl();var pdf=new
jspdf.jsPDF('p','mm','a4');var pW=pdf.internal.pageSize.getWidth()-
16,pH=pdf.internal.pageSize.getHeight()-16;var
cH=canvas.height*pW/canvas.width;if(cH<=pH)pdf.drawImage(canvas.toDataURL('image/png'),'PNG',
10,10,pW,cH);else{var rem=cH,sY=0;while(rem>0){var sH=Math.min(pH,rem);var
sc=document.createElement('canvas');sc.width=canvas.width;sc.height=sH*canvas.width/pW;sc.get
Context('2d').drawImage(canvas,0,sY,canvas.width,sc.height,0,0,sc.width,sc.height);if(sY>0)pdf.addP
age();pdf.drawImage(sc.toDataURL('image/png'),'PNG',10,10,pW,sH);sY+=sc.height;rem-
=sH;}}pdf.save('patologi_'+new
Date().getTime()+'.pdf');}).catch(function(e){document.body.removeChild(container);hl();toast('Error
: '+e.message| |e| |'Terjadi kesalahan'),'error');});,500);}

```

```

function
buildPatologiFormPayload(){return{NoRM:G('mPatologiNoRM').value,NoPA:G('mPatologiNoPA').valu
e>NamaPasien:G('mPatologiNama').value,TglLahir:G('mPatologiTglLahir').value,Umur:G('mPatologiU
mur').value,JK:G('mPatologiJK').value,Alamat:G('mPatologiAlamat').value,TglTerima:G('mPatologiTgl
Terima').value,TglJawab:G('mPatologiTglJawab').value,AsalRuangan:G('mPatologiAsalRuangan').valu
e,AsalRujukan:G('mPatologiAsalRujukan').value>StatusBiaya:G('mPatologiStatusBiaya').value,DokterP
engirim:G('mPatologiDokterPengirim').value,Diagnosis:G('mPatologiDiagnosis').value,JenisPemeriksa
an:G('mPatologiJenisPemeriksaan').value,Makroskopis:G('mPatologiMakroskopis').value,Mikroskopis
:G('mPatologiMikroskopis').value,Kesan:G('mPatologiKesan').value,Saran:G('mPatologiSaran').value,
Catatan:G('mPatologiCatatan').value,Topografi:G('mPatologiTopografi').value,Morfologi:G('mPatolog
iMorfologi').value,ImageURL:G('mPatologiURL1').value,ImageURL2:G('mPatologiURL2').value,ImageU
RL3:G('mPatologiURL3').value,ImageURL4:G('mPatologiURL4').value,ImageURL5:G('mPatologiURL5').
value}};

```

```

function generatePatologiReportHTML(d,kop,includeImages){var html='<div
id="imgPrintArea">'+buildKopHTML(kop,true)+'<h3 style="text-align:center;margin-
bottom:16px;font-size:1rem">HASIL PEMERIKSAAN PATOLOGI ANATOMI</h3><table
style="width:100%;margin-bottom:16px"><tr><td>No. RM</td><td>'+escH(d.NoRM| |'-
')</td><td>Tgl Terima</td><td>'+fmtDateIndo(d.TglTerima)</td></tr><tr><td>No.
PA</td><td>'+escH(d.NoPA| |'-')</td><td>Tgl Jawab</td><td>
'+fmtDateIndo(d.TglJawab)</td></tr><tr><td>Nama</td><td>'+escH(d>NamaPasien| |'-
')</td><td>Umur</td><td>'+escH(d.Umur| |'-')</td></tr><tr><td>Tgl Lahir</td><td>
'+fmtDateIndo(d.TglLahir)</td><td>JK</td><td>'+(d.JK==='L'? 'Laki-laki':d.JK==='P'? 'Perempuan':'-
')</td></tr><tr><td>Alamat</td><td>'+escH(d.Alatam| |'-')</td><td>Dokter Pengirim</td><td>

```

```

'+escH(d.DokterPengirim || '-')+'</td></tr><tr><td>Asal Ruangan</td><td>:
'+escH(d.AsalRuangan || '-')+'</td><td>Status Biaya</td><td>: '+escH(d.StatusBiaya || '-'
')+'</td></tr><tr><td>Diagnosis</td><td colspan="3"> '+escH(d.Diagnosis || '-'
')+'</td></tr><tr><td>Jenis Pemeriksaan</td><td colspan="3"> '+escH(d.JenisPemeriksaan || '-'
')+'</td></tr></table>;var
fields=[['Makroskopis',d.Makroskopis],['Mikroskopis',d.Mikroskopis],['Kesan',d.Kesan],['Saran',d.Sara
n],['Catatan',d.Catatan],['Topografi',d.Topografi],['Morfologi',d.Morfologi]];fields.forEach(function(f){
if(f[1]){html+=''<h4 style="border-bottom:1px solid #333;padding-bottom:4px;margin:12px 0
8px">'+f[0]+'</h4><div style="white-space:pre-
wrap">'+escH(f[1]).replace(/\\n/g,'<br>')+'</div>'}});if(includeImages){var
imgs=[d.ImageURL,d.ImageURL2,d.ImageURL3,d.ImageURL4,d.ImageURL5].filter(function(u){return
u});if(imgs.length){html+=''<div style="margin-top:16px;text-align:center"><h5>Gambar</h5><div
style="display:flex;gap:8px;flex-wrap:wrap;justify-
content:center">imgs.forEach(function(u){html+='''}});html+=''</div></div>'}}var
pqrlmg1=getQRDataURL('didiQC | '+ (kop.namaRS || 'Lab')+' | '+ (kop.kepalaRuangan || ''));var
pqrlmg2=getQRDataURL('didiQC | '+ (kop.namaRS || 'Lab')+' | '+ (kop.dokterPJ || ''));html+=''<div
class="img-result-footer" style="display:flex;justify-content:space-between;align-items:flex-
start;margin-top:30px;gap:20px"><div class="ttd-col" style="text-align:center;width:45%;flex-
shrink:0"><p style="font-weight:600;margin-bottom:4px">Kepala Ruangan
Lab</p>'+(pqrlmg1?'':<div style="width:80px;height:80px;margin:8px auto"></div>')+<p class="ttd-
name" style="margin-top:30px;border-top:1px solid #333;padding-top:4px;font-
weight:700">'+escH(kop.kepalaRuangan || '___')+'</p></div><div class="ttd-col" style="text-
align:center;width:45%;flex-shrink:0"><p style="font-weight:600;margin-bottom:4px">Dokter
PJ</p>'+(pqrlmg2?'':<div style="width:80px;height:80px;margin:8px auto"></div>')+<p class="ttd-
name" style="margin-top:30px;border-top:1px solid #333;padding-top:4px;font-
weight:700">'+escH(kop.dokterPJ || '___')+'</p></div></div><p style="text-align:center;font-
size:.75rem;color:#666;margin-top:10px">Dokumen ini dihasilkan secara elektronik oleh
didiQCsys</p></div>;return html;}

```

```

function printImgResult(){var c=G('imgPrintArea');if(!c)return;var
w=window.open('', '_blank');w.document.write('<!DOCTYPE
html><html><head><title>Hasil</title><style>body{font-family:Inter,sans-serif;padding:10px
12px;font-size:11pt}.img-result-header{margin-bottom:12px!important;padding-
bottom:10px!important}.img-result-header img{max-height:80px}table{width:100%;border-
collapse:collapse}td{padding:4px 8px;vertical-align:top}.img-result-footer{display:flex;justify-
content:space-between;align-items:flex-start;margin-top:24px;gap:20px}.ttd-col{text-
align:center;width:45%;flex-shrink:0}.ttd-name{margin-top:24px;border-top:1px solid #333;padding-
top:4px;font-weight:700}.img-mikroskopis{max-width:280px;border:1px solid #ddd;border-
radius:4px;margin:4px}@page{margin:8mm
10mm;size:A4}</style></head><body>');w.document.write(c.innerHTML);w.document.write('</bod
y></html>');w.document.close();setTimeout(function(){w.print();},500);}

```



```
function pdfImgResult(){var
c=G('imgPrintArea');if(!c)return;sl('PDF...');setTimeout(function(){html2canvas(c,{scale:2,useCORS:true,allowTaint:true}).then(function(canvas){hl();var pdf=new jsPDF('p','mm','a4');var
pW=pdf.internal.pageSize.getWidth()-16,pH=pdf.internal.pageSize.getHeight()-16;var
cH=canvas.height*pW/canvas.width;if(cH<=pH)pdf.addImage(canvas.toDataURL('image/png'),'PNG',
10,10,pW,cH);else{var rem=cH,sY=0;while(rem>0){var sH=Math.min(pH,rem);var
sc=document.createElement('canvas');sc.width=canvas.width;sc.height=sH*canvas.width/pW;sc.get
Context('2d').drawImage(canvas,0,sY,canvas.width,sc.height,0,0,sc.width,sc.height);if(sY>0)pdf.addP
age();pdf.addImage(sc.toDataURL('image/png'),'PNG',10,10,pW,sH);sY+=sc.height;rem-
=sH;}}pdf.save('hasil_'+new Date().getTime()+'.pdf');}).catch(function(err){hl();toast('Error:
'+(err.message||err||'Terjadi kesalahan'),'error');});,500);}
```

```
function printImgResultWithImg(){printImgResultOption(true);}
function printImgResultNoImg(){printImgResultOption(false);}
function printImgResultOption(includeImg){var
c=G('imgPrintArea');if(!c){c=G('patPrintArea');}if(!c)return;var
w=window.open('', '_blank');w.document.write('<!DOCTYPE
html><html><head><title>Hasil</title><style>body{font-family:Inter,sans-serif;padding:10px
12px;font-size:11pt}.img-result-header{margin-bottom:12px!important;padding-
bottom:10px!important}.img-result-header img{max-height:80px}table{width:100%;border-
collapse:collapse}td{padding:4px 8px;vertical-align:top}.img-result-footer{display:flex;justify-
content:space-between;align-items:flex-start;margin-top:24px;gap:20px}.ttd-col{text-
align:center;width:45%;flex-shrink:0}.img-mikroskopis{max-width:280px;border:1px solid
#ddd;border-radius:4px;margin:4px}.ttd-name{margin-top:24px;border-top:1px solid #333;padding-
top:4px;font-weight:700}@page{margin:8mm
10mm;size:A4}</style></head><body>');if(!includeImg){var
temp=c.cloneNode(true);temp.querySelectorAll('img').forEach(function(img){if(img.src&&img.src.ind
exOf('qr')<0)img.remove();});w.document.write(temp.innerHTML);}else{w.document.write(c.innerHT
ML);w.document.write('</body></html>');w.document.close();setTimeout(function(){w.print();},50
0);}
function pdfImgResultWithImg(){pdfImgResultOption(true);}
function pdfImgResultNoImg(){pdfImgResultOption(false);}
function pdfImgResultOption(includeImg){var
c=G('imgPrintArea');if(!c){c=G('patPrintArea');}if(!c)return;if(!includeImg){var
temp=c.cloneNode(true);temp.querySelectorAll('img').forEach(function(img){if(img.src&&img.src.ind
exOf('qr')<0)img.remove();});c=temp;}sl('Membuat
PDF...');setTimeout(function(){html2canvas(c,{scale:2,useCORS:true,allowTaint:true}).then(function(
canvas){hl();var pdf=new jsPDF('p','mm','a4');var pW=pdf.internal.pageSize.getWidth()-
16,pH=pdf.internal.pageSize.getHeight()-16;var
cH=canvas.height*pW/canvas.width;if(cH<=pH)pdf.addImage(canvas.toDataURL('image/png'),'PNG',
10,10,pW,cH);else{var rem=cH,sY=0;while(rem>0){var sH=Math.min(pH,rem);var
sc=document.createElement('canvas');sc.width=canvas.width;sc.height=sH*canvas.width/pW;sc.get
Context('2d').drawImage(canvas,0,sY,canvas.width,sc.height,0,0,sc.width,sc.height);if(sY>0)pdf.addP
age();pdf.addImage(sc.toDataURL('image/png'),'PNG',10,10,pW,sH);sY+=sc.height;rem-
```

```

=sh;}}pdf.save('hasil_'+new Date().getTime()+'.pdf');}).catch(function(e){hl();toast('Error:
'+(e.message|e|'Terjadi kesalahan'),'error');});,500));
document.addEventListener('click',function(e){if(!e.target.closest('.print-options-
wrap')){document.querySelectorAll('.print-options-
menu.show').forEach(function(m){m.classList.remove('show');});QA('.print-options-
menu').forEach(function(m){m.classList.remove('show');});});
document.addEventListener('DOMContentLoaded',function(){
google.script.run.withSuccessHandler(function({}).withFailureHandler(function(e){hl();toast('Error:
'+(e.message|e|'Terjadi kesalahan'),'error');}).initializeSheets();
G('inputTanggal').value=todayISO();
});
var tablnited=false;
var origGoPage=goPage;
goPage=function(p){origGoPage(p);if(p==='tabulasi'&&!tablnited&&CD.params&&CD.params.length)
{initTabChecklist();tablnited=true;}};
/* === PATOLOGI ANATOMI v9.7 === */
function IdImgPatologiData(){var
filter={startDate:G('imgPatologiStart').value|'|',endDate:G('imgPatologiEnd').value|'|',noRM:G('img
PatologiRM').value|'|',nama:G('imgPatologiNama').value|'|',noPA:G('imgPatologiPA').value|'|'};sl('
Memuat...');google.script.run.withSuccessHandler(function(r){hl();if(!r||!r.ok){if(r&&r.msg)toast(r.m
sg,'error');else if(!r)toast('Gagal memuat
data','error');return;}}imgDataCache['patologi']=r.data|'|[];renderImgPatologiTable(r.data);}).withFail
ureHandler(function(e){hl();toast('Error: '+e.message|e|'Terjadi
kesalahan'),'error');}).getImgPatologi(getActiveUsername(),filter);}

function
renderImgPatologiTable(data){if(!data||!data.length){G('imgPatologiBody').innerHTML='<tr><td
colspan="9" style="text-align:center;color:var(--text-secondary)">Belum ada
data</td></tr>';return;}var
html='';data.forEach(function(d,i){html+='<tr><td>'+(i+1)+'</td><td>'+fmtDate(d.TglTerima)+'</td><
td>'+escH(d.NoRM)+'</td><td>'+escH(d.NoPA)+'</td><td>'+escH(d>NamaPasien)+'</td><td>'+escH(
d.JK)+'</td><td>'+escH(d.JenisPemeriksaan)+'</td><td>'+escH(d.Diagnosis)+'</td><td><div
class="btn-group"><button class="btn btn-xs btn-outline" onclick="viewImgPatologiDetail(''+i+'')"><i
class="fas fa-eye"></i></button><button class="btn btn-xs btn-outline"
onclick="openImgPatologiModal(imgDataCache['\x27patologi\x27'][''+i+''])"><i class="fas fa-
edit"></i></button><button class="btn btn-xs btn-danger" onclick="dellmgPatologi(''+i+'')"><i
class="fas fa-trash"></i></button></div></td></tr>');G('imgPatologiBody').innerHTML=html;}

function
resetImgPatologiFilter(){G('imgPatologiStart').value='';G('imgPatologiEnd').value='';G('imgPatologiR
M').value='';G('imgPatologiNama').value='';G('imgPatologiPA').value='';IdImgPatologiData();}

function openImgPatologiModal(data){var m=G('modalImgPatologi');if(!m){toast('Modal tidak
ditemukan','error');return;}}['NoRM','NoPA','Nama','TglLahir','Umur','Alamat','TglTerima','TglJawab','
AsalRuangan','AsalRujukan','StatusBiaya','DokterPengirim','Diagnosis'].forEach(function(f){var

```

```

el=G('mPatologi'+f);if(el)el.value="";G('mPatologiJK').value="";G('mPatologiJenisPemeriksaan').value=
="";['Makroskopis','Mikroskopis','Kesan','Saran','Catatan','Topografi','Morfologi'].forEach(function(f){
var
el=G('mPatologi'+f);if(el)el.value="";[1,2,3,4,5].forEach(function(n){G('mPatologiURL'+n).value="";va
r slot=G('patologiSlot'+n);if(slot){slot.innerHTML='<div class="slot-label">Gambar '+n+'</div><input
type="file" accept="image/*"
onchange="handlePatologiImgUpload(event,'+n+')">';slot.classList.remove('has-
image');}});if(data){G('mPatologiNoRM').value=data.NoRM||";G('mPatologiNoPA').value=data.NoPA
||";G('mPatologiNama').value=data>NamaPasien||";G('mPatologiTglLahir').value=data.TglLahir||";G
('mPatologiUmur').value=data.Umur||";G('mPatologiJK').value=data.JK||";G('mPatologiAlamat').val
ue=data.Alatat||";G('mPatologiTglTerima').value=data.TglTerima||";G('mPatologiTglJawab').value
=data.TglJawab||";G('mPatologiAsalRuangan').value=data.AsalRuangan||";G('mPatologiAsalRujukan
').value=data.AsalRujukan||";G('mPatologiStatusBiaya').value=data.StatusBiaya||";G('mPatologiDok
terPengirim').value=data.DokterPengirim||";G('mPatologiDiagnosis').value=data.Diagnosis||";G('mP
atologiJenisPemeriksaan').value=data.JenisPemeriksaan||";['Makroskopis','Mikroskopis','Kesan','Sar
an','Catatan','Topografi','Morfologi'].forEach(function(f){var
el=G('mPatologi'+f);if(el&&data[f])el.value=data[f];});if(data.ImageURL){G('mPatologiURL1').value=d
ata.ImageURL;setPatologiImgSlot(1,data.ImageURL);if(data.ImageURL2){G('mPatologiURL2').value=
data.ImageURL2;setPatologiImgSlot(2,data.ImageURL2);if(data.ImageURL3){G('mPatologiURL3').val
ue=data.ImageURL3;setPatologiImgSlot(3,data.ImageURL3);if(data.ImageURL4){G('mPatologiURL4').
value=data.ImageURL4;setPatologiImgSlot(4,data.ImageURL4);if(data.ImageURL5){G('mPatologiURL
5').value=data.ImageURL5;setPatologiImgSlot(5,data.ImageURL5);G('mPatologiID').value=data.ID||
';G('modalPatologiTitle').textContent='Edit
Data';}else{G('mPatologiTglTerima').value=todayISO();G('mPatologiTglJawab').value=todayISO();G('m
PatologiID').value="";G('modalPatologiTitle').textContent='Tambah Data';}opM('modallmgPatologi');}

function setPatologiImgSlot(n,url){G('mPatologiURL'+n).value=url;var
slot=G('patologiSlot'+n);if(slot){slot.innerHTML='<div class="slot-label">Gambar '+n+'</div><input type="file" accept="image/*"
onchange="handlePatologiImgUpload(event,'+n+')" style="display:none"><button class="remove-
img" onclick="removePatologiImgSlot('+n+')">&times;</button>';slot.classList.add('has-image');}}

function removePatologiImgSlot(n){G('mPatologiURL'+n).value="";var
slot=G('patologiSlot'+n);if(slot){slot.innerHTML='<div class="slot-label">Gambar '+n+'</div><i
class="fas fa-image"></i><p>Klik upload</p><input type="file" id="patologiFileInput'+n+'"'
accept="image/*" style="display:none" onchange="handlePatologiImgUpload(event,'+n+')"><button
class="remove-img"
onclick="event.stopPropagation();removePatologiImgSlot('+n+')">&times;</button>';slot.classList.re
move('has-image');}}

function handlePatologiImgUpload(e,n){var
f=e.target.files[0];if(!f)return;if(f.size>5*1024*1024){toast('Maks 5MB','warning');return;}var
reader=new FileReader();reader.onload=function(ev){var img=new
Image();img.onload=function(){var MAX_W=800,MAX_H=800,MAX_SIZE=200*1024;var
w=img.width,h=img.height;if(w>MAX_W||h>MAX_H){if(w>h){h=Math.round(h*MAX_W/w);w=MAX

```

```

_W;}}else{w=Math.round(w*MAX_H/h);h=MAX_H;}}var
c=document.createElement('canvas');c.width=w;c.height=h;var
ctx=c.getContext('2d');ctx.drawImage(img,0,0,w,h);var q=0.85;var
base64=c.toDataURL('image/jpeg',q);while(base64.length>MAX_SIZE&&q>0.3){q-
=0.1;base64=c.toDataURL('image/jpeg',q);}G('mPatologiURL'+n).value=base64;var
slot=G('patologiSlot'+n);slot.innerHTML='<div class="slot-label">Gambar '+n+'</div><input type="file" id="patologiFileInput'+n+'" accept="image/*"
style="display:none" onchange="handlePatologiImgUpload(event,'+n+')"><button class="remove-
img"
onclick="event.stopPropagation();removePatologiImgSlot('+n+')">&times;</button>';slot.classList.ad
d('has-image');img.src=ev.target.result;reader.readAsDataURL(f);}

```

```

function saveImgPatologiForm(){var
payload={ID:G('mPatologiID').value||null,NoRM:G('mPatologiNoRM').value,NoPA:G('mPatologiNoPA
').value>NamaPasien:G('mPatologiNama').value,TglLahir:G('mPatologiTglLahir').value,Umur:G('mPato
logiUmur').value,JK:G('mPatologiJK').value,Alamat:G('mPatologiAlamat').value,TglTerima:G('mPatolo
giTglTerima').value,TglJawab:G('mPatologiTglJawab').value,AsalRuangan:G('mPatologiAsalRuangan').
value,AsalRujukan:G('mPatologiAsalRujukan').value,StatusBiaya:G('mPatologiStatusBiaya').value,Dok
terPengirim:G('mPatologiDokterPengirim').value,Diagnosis:G('mPatologiDiagnosis').value,JenisPemer
iksaan:G('mPatologiJenisPemeriksaan').value,Makroskopis:G('mPatologiMakroskopis').value,Mikrosk
opis:G('mPatologiMikroskopis').value,Kesan:G('mPatologiKesan').value,Saran:G('mPatologiSaran').val
ue,Catatan:G('mPatologiCatatan').value,Topografi:G('mPatologiTopografi').value,Morfologi:G('mPato
logiMorfologi').value,ImageURL:G('mPatologiURL1').value,ImageURL2:G('mPatologiURL2').value,Ima
geURL3:G('mPatologiURL3').value,ImageURL4:G('mPatologiURL4').value,ImageURL5:G('mPatologiUR
L5').value};if(!payload.NoRM||!payload>NamaPasien){toast('No RM dan Nama wajib
diisi','warning');return;}sl('Menyimpan...');google.script.run.withSuccessHandler(function(r){hl();if(r&
&r.ok){toast('Tersimpan','success');clM('modallmgPatologi');ldImgPatologiData();}else
toast('Gagal','error');}).withFailureHandler(function(e){hl();toast('Error: '+e.message|e|'Terjadi
kesalahan'),'error');}).saveImgPatologi(payload,getActiveUsername(),getLogUser());}

```

```

function delImgPatologi(idx){var data=imgDataCache['patologi'];if(!data||!data[idx])return;var
d=data[idx];cfm('Hapus data Patologi?','Data '+esch(d>NamaPasien)+' akan
dihapus.','Hapus','Batal',function(){sl('Menghapus...');google.script.run.withSuccessHandler(function(
r){hl();if(r&&r.ok){toast('Dihapus','success');ldImgPatologiData();}else
toast('Gagal','error');}).withFailureHandler(function(e){hl();toast('Error: '+e.message|e|'Terjadi
kesalahan'),'error');}).deleteImgPatologi(d.ID,getActiveUsername(),getLogUser());});}

```

```

function viewImgPatologiDetail(idx){var
data=imgDataCache['patologi'];if(!data||!data[idx])return;var d=data[idx];var html='<div
id="imgPrintArea" style="max-width:800px;margin:0 auto;padding:20px">';html+='<div style="text-
align:center;border-bottom:2px solid var(--primary);padding-bottom:12px;margin-
bottom:16px"><h2 style="margin:0">HASIL PEMERIKSAAN PATOLOGI
ANATOMI</h2></div>';html+='<table style="width:100%;border-collapse:collapse;font-
size:.9rem">';var rows=[['No. RM',d.NoRM],['No. PA',d.NoPA],['Nama',d>NamaPasien],['Tgl
Lahir',fmtDate(d.TglLahir)],['Umur',d.Umur],['JK',d.JK],['Alamat',d.Alatat],['Tgl

```

```

Terima',fmtDate(d.TglTerima)],['Tgl Jawab',fmtDate(d.TglJawab)],['Asal
Ruangan',d.AsalRuangan],['Asal Rujukan',d.AsalRujukan],['Status Biaya',d.StatusBiaya],['Dokter
Pengirim',d.DokterPengirim],['Diagnosis',d.Diagnosis],['Jenis
Pemeriksaan',d.JenisPemeriksaan]];rows.forEach(function(r){html+<tr style="border-bottom:1px
solid var(--border)"><td style="padding:6px 10px;font-
weight:600;width:180px">'+escH(r[0])+'</td><td style="padding:6px
10px">'+escH(r[1])+'</td></tr>'}};html+<table>';var
templates=[['Makroskopis',d.Makroskopis],['Mikroskopis',d.Mikroskopis],['Kesan',d.Kesan],['Saran',d.
Saran],['Catatan',d.Catatan],['Topografi',d.Topografi],['Morfologi',d.Morfologi]];templates.forEach(fu
nction(t){if(t[1]){html+<div style="margin-top:12px"><h4 style="color:var(--primary);margin-
bottom:4px">'+t[0]+'</h4><div style="background:var(--bg-soft);padding:10px;border-radius:var(--
radius-sm);white-space:pre-wrap">'+escH(t[1])+'</div></div>'}};[1,2,3,4,5].forEach(function(n){var
url=d['ImageURL'+(n>1?'n:')]';if(n===1)url=d.ImageURL;if(url){html+<div style="margin-
top:12px"></div>'}};html+<div style="margin-top:20px;text-
align:center;font-size:.8rem;color:var(--text-secondary)">Dicetak: '+new Date().toLocaleString('id-
ID')+'</div></div>';G('imgViewContent').innerHTML=html;opM('modalImgView');}

```

```

function printPatologiResult(){var c=G('patPrintArea');if(!c){toast('Tidak ada area
cetak','warning');return;}var w=window.open('','_blank');w.document.write('<!DOCTYPE
html><html><head><title>Hasil Patologi Anatomi</title><style>body{font-family:Inter,sans-
serif;padding:10px 12px;font-size:11pt}table{width:100%;border-collapse:collapse}td{padding:4px
8px;vertical-align:top}img{max-width:100%;max-height:400px}.img-result-header{margin-
bottom:12px!important;padding-bottom:10px!important}@page{margin:8mm
10mm;size:A4}</style></head><body>');w.document.write(c.innerHTML);w.document.write('</bod
y></html>');w.document.close();setTimeout(function(){w.print();},500);}

```

```

function pdfPatologiResult(){var c=G('patPrintArea');if(!c){toast('Tidak ada area
cetak','warning');return;}sl('Membuat
PDF...');setTimeout(function(){html2canvas(c,{scale:2,useCORS:true,allowTaint:true}).then(function(
canvas){hl();var pdf=new jsPDF('p','mm','a4');var pW=pdf.internal.pageSize.getWidth()-
16,pH=pdf.internal.pageSize.getHeight()-16;var
cH=canvas.height*pW/canvas.width;if(cH<=pH)pdf.drawImage(canvas.toDataURL('image/png'),'PNG',
10,10,pW,cH);else{var rem=cH,sY=0;while(rem>0){var sH=Math.min(pH,rem);var
sc=document.createElement('canvas');sc.width=canvas.width;sc.height=sH*canvas.width/pW;sc.get
Context('2d').drawImage(canvas,0,sY,canvas.width,sc.height,0,0,sc.width,sc.height);if(sY>0)pdf.addP
age();pdf.drawImage(sc.toDataURL('image/png'),'PNG',10,10,pW,sH);sY+=sc.height;rem-
=sH;}}pdf.save('patologi_anatomi_'+new Date().getTime()+'.pdf');}).catch(function(){hl();toast('Error
membuat PDF','error');});},500);}

```

```

function analyzePatologiAI(){var urls=[];[1,2,3,4,5].forEach(function(n){var
u=G('mPatologiURL'+n).value;if(u)urls.push(u);});if(!urls.length){toast('Upload minimal 1 gambar
dulu','warning');return;}var jp=G('mPatologiJenisPemeriksaan').value;if(!jp){toast('Pilih Jenis
Pemeriksaan','warning');return;}var
patientData={NamaPasien:G('mPatologiNama').value,JK:G('mPatologiJK').value,Umur:G('mPatologiU

```

```

mur').value,Diagnosis:G('mPatologiDiagnosis').value);sl('Menganalisis
gambar...');google.script.run.withSuccessHandler(function(r){hl();if(r&&r.ok&&r.data){if(r.data.makr
oskopisDesk&&G('mPatologiMakroskopis'))G('mPatologiMakroskopis').value=r.data.makroskopisDes
k;if(r.data.mikroskopisDesk&&G('mPatologiMikroskopis'))G('mPatologiMikroskopis').value=r.data.mi
kroskopisDesk;if(r.data.kesanDesk&&G('mPatologiKesan'))G('mPatologiKesan').value=r.data.kesanD
esk;if(r.data.saranDesk&&G('mPatologiSaran'))G('mPatologiSaran').value=r.data.saranDesk;if(r.data.
topografiDesk&&G('mPatologiTopografi'))G('mPatologiTopografi').value=r.data.topografiDesk;if(r.da
ta.morfologiDesk&&G('mPatologiMorfologi'))G('mPatologiMorfologi').value=r.data.morfologiDesk;to
ast('Analisis selesai','success');}else toast('Gagal
menganalisis','error');}).withFailureHandler(function(e){hl();toast('Error: '+e.message+' | e | 'Terjadi
kesalahan'),'error');}).analyzePatologiImage(urls,jp,patientData);}

```

```

/* Auto-calculate umur from TglLahir */
function calcPatologiUmur(){var v=G('mPatologiTglLahir').value;if(v){var dob=new Date(v);var
today=new Date();var umur=Math.floor((today-
dob)/(365.25*24*60*60*1000));G('mPatologiUmur').value=umur>0?umur:'';}}

```

```

/* Toggle template field visibility - for checkbox template items */
function togglePatTemplate(cb,fieldId){var
field=G(fieldId);if(field)field.style.display=cb.checked?'':'none';}

```

```

// Global error handler for GAS framework errors
window.onerror = function(msg, url, line, col, error) {
  console.error('Global error:', msg, 'Line:', line, error);
  if (msg && msg.indexOf('script') > -1) {
    toast('Terjadi kesalahan koneksi. Halaman akan dimuat ulang.','error');
    setTimeout(function(){ location.reload(); }, 3000);
  }
  return false;
};

```

```

// Handle GAS iframe communication errors
window.addEventListener('message', function(event) {
  if (event.data && event.data.type === 'iframeError') {
    console.error('GAS iframe error:', event.data);
  }
});

```

```

</script>

```

```

<!-- EXPORT PANEL MODAL v10.0 -->

```

```

<div class="export-panel-overlay" id="exportPanelOverlay"
onclick="if(event.target===this)closeExportPanel()">
<div class="export-panel">
<div class="export-panel-header"><h3 id="exportPanelTitle"><i class="fas fa-print"></i> Pilih Format
Cetak</h3><button class="export-panel-close"
onclick="closeExportPanel()">&times;</button></div>

```

```
<div class="export-panel-body" id="exportPanelBody"></div>
</div>
</body>
</html>
```